PHYSICS

IIT-JEE Advanced Revision Package

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PHYSICS - PART - I

EXERCISE # 01 TOPIC: MECHANICS

SECTION : (A) - Single Correct Options

1. The potential energy function for a particle executing simple harmonic motion is given by U(x) = $4+\frac{x^2}{4}$ joule. If the total energy of the particle is 5 J, then it would turn back from

(A) x = $\pm$1m (B) x = $\pm$ 2 m (C) x = $\pm$ 3 m (D) x = $\pm$ 4 m

2. A solid cube is placed on a rough horizontal surface. What should be the maximum value of coefficient of friction between them such that when the cube is given a horizontal velocity, it does not topple on the surface?

(A) 0.5 (B) $\sqrt{0.5}$ (C) 0.25 (D) 0.33

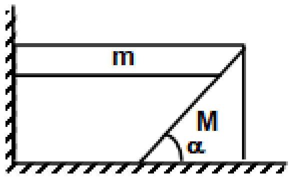
3. A particle of mass 3m is projected in a vertical x – y plane with some initial velocity at t = 0. At t = 2 sec it explodes into two particles having mass ratio 1 : 2. At t = 3 sec, the smaller mass is observed having a velocity $\left(4\vec{i}-16\vec{j}\right)$ m/sec and that of bigger mass is $\left(\vec{i}-\vec{j}\right)$ m/s. Then projection velocity of the original mass at t = 0 is (Take g = 10 m/s²)

(A) $2\vec{i}+16\vec{j}$ m/s (B) $2\vec{i}-6\vec{j}$ m/s (C) $2\vec{i}+24\vec{j}$ m/s (D) $2\vec{i}+14\vec{j}$ m/s

4. A dog of mass 15 kg runs with a speed of 10 m/sec at surface of a circular drum of radius 2 m. The drum can rotate about its own axis AB freely as shown in the figure. The motion of the dog causes the rotation of drum. The speed of rotation is such that the relative position of the dog is unaltered. The kinetic energy of rotation of dog is

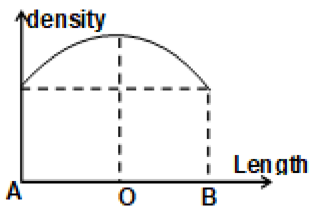
(A) 750 J (B) 37.5 J (C) 150 J (D) 60 J

5. A triangular wedge of mass M is placed on a smooth floor near a wall. A prism of mass m is kept as shown in figure. If the wall is also frictionless find the velocity of wedge as the prism falls through a height h.



(A) $\sqrt{\dfrac{2gh}{\tan^2\alpha+\dfrac{M}{m}}}$ (B) $\sqrt{\dfrac{2ghM}{m}}$ (C) $\sqrt{\dfrac{2ghM}{m\tan\alpha}}$ (D) $\sqrt{\dfrac{gh}{m}\tan\alpha}$

6. The density of a rod AB changes from A to B as shown in the figure. It's midpoint is O and its centre of mass is at C. Four axes pass through A, B, O and C, all perpendicular to the length of the rod. The moment of inertia of the rod about these axes are I_A, I_B, I_O and I_C respectively, then



(A) $I_A = I_B = I_O = I_C$ (B) $I_A = I_B > I_O = I_C$ (C) $I_A = I_B < I_O = I_C$ (D) $I_A < I_B < I_O = I_C$

7. A constant external force F is applied for a very small time interval Δt along a tangential direction of a small disc of m and radius r placed on a smooth horizontal surface, then

(A) the disc has pure translational motion

(B) the disc has pure rotational motion

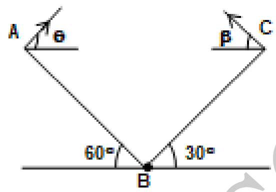
(C) the angular momentum of the disc will change by $Fr\,\Delta t$

(D) the kinetic energy of the disc will change by $\dfrac{2(Fr\Delta t)^2}{mr^2}$

8. A moving mass of 8 kg collides elastically with a stationary mass of 2 kg. If E be the initial kinetic energy of the moving mass, then the kinetic energy left with it after the collision will be

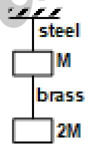
(A) 0.80E (B) 0.64E (C) 0.36E (D) 0.08E

9. Two inclined plane AB and BC are at inclination of 60° and 30° as shown in the figure. The two projectiles of same mass are thrown from A and C with speed 2 m/s & v_0 m/s respectively, such that each strikes at B with same speed. If length of AB is $\dfrac{1}{\sqrt{3}}$ m and BC is 1 m. Find the value of v_0



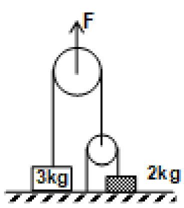
(A) 1/2 m/s (B) 1 m/s (C) 2 m/s (D) none of the above

10. If the ratio of lengths, radii and Young's Moduli of steel and brass wire in the figure are a, b and c respectively then the corresponding ratio of increase in their lengths is



(A) $2a^2c/b$ (B) $3a/2b^2c$ (C) $2ac/b^2$ (D) $3c/2ab^2$

11. What is the minimum value of applied force F so that both the block lifts up. Consider string to ideal & pulley to be massless & frictionless.



(A) 4g (B) 8g (C) 5g (D) 7g

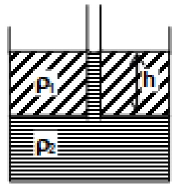
12. A block of mass m is resting on the rough horizontal ground. Minimum force required to move it is $\dfrac{4mg}{5}$. Then coefficient of friction is

(A) $\dfrac{1}{5}$ (B) $\dfrac{4}{5}$ (C) $\dfrac{1}{4}$ (D) $\dfrac{4}{3}$

13. A massless string of length ℓ, cross section area A & Young's modulus Y is tied up with a bob of mass m the bob is moving with constant angular speed ω in circular path of radius r as shown. Find elongation $\Delta\ell$ in the string assuming $\Delta\ell \ll \ell$.

(A) $\frac{mg\sqrt{\ell^2 - r^2}}{AY}$ (B) $\frac{mg}{AY}\frac{\ell^2}{\sqrt{\ell^2 - r^2}}$ (C) $\frac{mgr}{AY}$ (D) $\frac{mg}{AY}\frac{\ell^2}{r}$

14. A container contains two immiscible liquids of density ρ_1 & ρ_2 ($\rho_2 > \rho_1$). A capillary of radius r is inserted in the liquid so that its bottom reaches upto denser liquid. Denser liquid rises in capillary & attain height equal to h which is also equal to column length of lighter liquid. Assuming zero contact angle find surface tension of heaver liquid



(A) $\frac{r\rho_2 gh}{2}$ (B) $2\pi r\rho_2 gh$ (C) $\frac{r}{2}(\rho_2 - \rho_1)gh$ (D) $2\pi r(\rho_{2\cdot 1})gh$

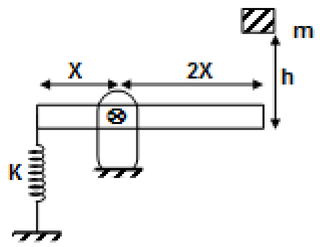
15. A ball is projected with a velocity 4 m/sec on the ground surface making an angle 30° with vertical. After collision with the ground, the velocity of ball becomes 2 m/sec and its direction is 60 with vertical, then the coefficient of friction between the ball and ground surface

(A) $\frac{(5\sqrt{3}-8)}{11}$ (B) $\frac{(5\sqrt{3}+8)}{11}$ (C) $\frac{(5\sqrt{3}+8\sqrt{2})}{11}$ (D) none of these

16. A car of mass 6 kg accelerates on a level road under the action of driving force 10N from a speed 4 m/sec to a higher speed V m/sec in a distance 8 m. If the engine develops a constant power output 38 J/sec, then the magnitude of V is

(A) 6 m/sec (B) 4 m/sec (C) 3 m/sec (D) 12 m/sec

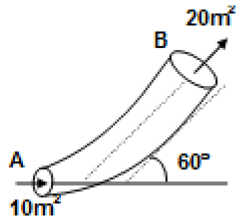
17. A crate of mass m falls from a height h onto the end of a platform, as shown in figure. The spring is initially unstretched and the mass of the plateform can be neglected. Assuming that there is no loss of energy, then the maximum elongation of spring is



(A) $\frac{2mg - \sqrt{4m^2g^2 + 2mghk}}{k}$ (B) $\frac{mg - \sqrt{m^2g^2 + 2mghk}}{k}$

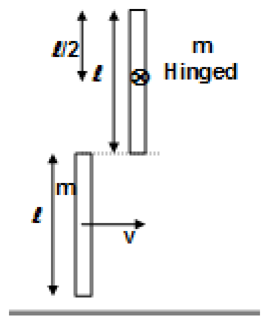
(C) $\frac{mg - \sqrt{m^2g^2 + 2mghk}}{k}$ (D) $\frac{2mg + \sqrt{4m^2g^2 + 2mghk}}{k}$

18. Liquid is flowing from A to B in the pipe and velocity of liquid at A is 40 m/sec and the pipe lie on a horizontal surface. If the specific gravity of the liquid is 4, then the force exerted by the pipe in Newton is



(A) $(\sqrt{3072})\times 10^{2}$ (B) $(\sqrt{3072})\times 10^{3}$ (C) $(\sqrt{3072})\times 10$ (D) $(\sqrt{3072})\times 10^{-2}$

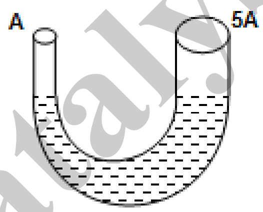
19. A bar of mass m and length ℓ is in pure translatory motion with its centre of mass moves with velocity v. It collides with a second identical bar which is initially at rest, and sticks to it the angular velocity of composite bar about hinged point will be



(A) $\frac{4}{3}\frac{v}{\ell}$ (B) $\frac{3}{4}\frac{v}{\ell}$

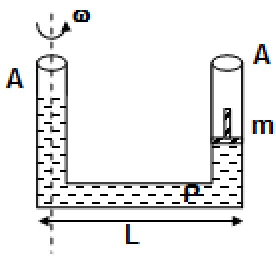
(C) $\frac{6}{7}\frac{v}{\ell}$ (D) $\frac{7}{6}\frac{v}{\ell}$

20. A U-tube contains m kg of a liquid of density ρ as shown in figure and is disturbed so that it oscillates back and forth from arm to arm. If we neglect friction, then the period of oscillation is



(A) $2\pi\sqrt{\frac{5m}{6\rho gA}}$ (B) $2\pi\sqrt{\frac{m}{6\rho gA}}$ (C) $2\pi\sqrt{\frac{m}{\rho gA}}$ (D) $2\pi\sqrt{\frac{6m}{5\rho gA}}$

21. Mass of the piston is m and its thickness is negligible and surface is frictionless. Then, find the angular velocity ω such that the level of liquid in both limb becomes equal. ($L \gg d$) d is the diameter of the limb



(A) $\left(\frac{4mg}{A\rho L^{2}}\right)^{1/2}$ (B) $\left(\frac{2mg}{A\rho L^{2}}\right)^{1/2}$ (C) $\left(\frac{mg}{A\rho L^{2}}\right)^{1/2}$ (D) $\left(\frac{4mg}{2A\rho L^{2}}\right)^{1/2}$

22. The breaking stress of aluminium 7.5×10^7 N/m^2. If the density of aluminium is 2.7 g/cc. Find the maximum length in km of the aluminium wire that could hang vertically without breaking (upto one decimal)
(A) 2.7 (B) 3.7 (C) 1.7 (D) 2.2

23. A car goes on a horizontal circular road of radius R, the speed increasing at a constant rate $\frac{dv}{dt} = a$. The friction coefficient between the road and tyre is μ. Then the speed at which the car will skid.
(A) $[(\mu^2g^2 + a^2)R^2]^{1/2}$ (B) $[(3\mu^2g^2 - a^2)R^2]^{1/4}$ (C) $[(\mu^2g^2 + a^2)R^2]^{1/4}$ (D) $[(\mu^2g^2 - a^2)R^2]^{1/4}$

24. A particle moves in the x-y plane with constant acceleration a directed along the negative y-axis. The equation of path of the particle has the form $y = bx - cx^2$, where b and c are positive constant. Then, the velocity of the particle at the origin is

(A) $\sqrt{\frac{a}{2c}(1-b^2)}$ (B) $\sqrt{\frac{a}{2c}(1+b^2)}$ (C) $\sqrt{\frac{a}{c}(1+b^2)}$ (D) $\sqrt{\frac{2a}{c}(1-b^2)}$

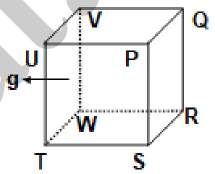
25. The relation between length and radius of a cylinder of given mass and density so that its moment of inertia about an axis through its centre of mass and perpendicular to its length may be minimum.

(A) $\sqrt{\frac{2}{3}}$ (B) 1 (C) $\frac{4}{\sqrt{3}}$ (D) $\sqrt{\frac{3}{2}}$

SECTION : (B) - Multiple Correct Options

26. A satellite revolves round the earth with orbital velocity v_0. Then
(A) The escape velocity at the planet is $v_0\sqrt{2}$.
(B) The escape velocity at the planet is $2v_0$
(C) Particle is projected radially outward from the earth.
(D) Particle is projected in any direction

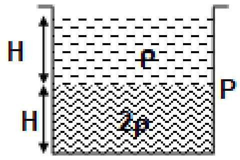
27. A closed hollow cube of side a is filled with a liquid of density ρ and it is accelerated with an acceleration g in the horizontal direction, then



(A) Force exerted by surface PQRS is $\frac{a^3}{2}\rho g$. (B) Force exerted by surface PQRS is $\frac{3}{2}a^3\rho g$

(C) Force exerted by surface UTWV is $\frac{a^3}{2}\rho g$ (D) Torque about T is $\frac{\rho g a^4}{6}$

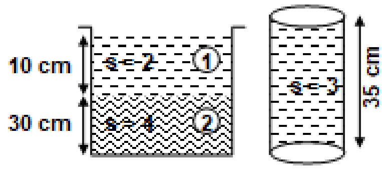
28. A tank is filled with two liquid of density 2ρ and ρ as shown in figure. A hole is punched in one of the walls, such that range becomes maximum, then



(A) The position of hole from P in 2ρ liquid is $\frac{H}{4}$ (B) The position of hole from P in 2ρ is $\frac{3H}{4}$

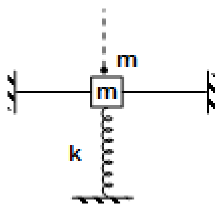
(C) The position of hole from P in ρ liquid is $\frac{H}{2}$ (D) The position of hole in ρ Liquid coincides with P

29. A tank is filled with two liquid of specific gravity s = 2 and s = 4 respectively as shown in figure. Then a solid cylinder of length 35 cm having specific gravity s = 3, dropped vertically in the liquid, then



(A) The length of cylinder immersed in liquid (2) is 21.5 cm.
(B) Then length of cylinder immersed in liquid (2) is 17.5 cm.
(C) The length of cylinder immersed in liquid (1) is 10 cm.
(D) The length of cylinder immersed in liquid (1) is 7.5 cm.

30. A particle of mass m is just placed on a block of mass m as shown in the figure. The block and particle stick firmly. Assume the oscillation are very small. So, that the strings of mass per unit length μ remains almost horizontal.



(A) The amplitude of small oscillation is mg/k

(B) Angular frequency of SHM is $\sqrt{\frac{k}{2m}}$

(C) Wavelength of the propagating wave in the string is $\frac{1}{2\pi}\sqrt{\frac{kT}{2\mu m}}$

(D) The oscillation of the string particles is forced oscillation

31. The schematic representations of the variation in acceleration vector for two particles P and Q during their motion from A to B are given as in figure I and figure II respectively
Choose the correct statement (s) [nature of variation of speed of A and B remains same throught the motion]

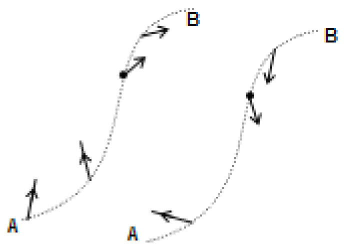


Fig-I Fig.-II

(A) speed of P increases with time
(B) speed of Q increases with time
(C) speed of P decreases with time
(D) speed of Q decreases with time

32. A satellite is revolving around the earth in a circular orbit. The universal gravitational constant suddenly becomes zero at time t = 0. Then
(A) The kinetic energy for t < 0 and t > 0 is same
(B) The angular momentum for t < 0 and t > 0 is same
(C) The angular momentum keeps changing for t < 0
(D) The angular momentum keeps changing for t > 0

33. A body is performing simple harmonic motion. Its:

(A) Average kinetic energy per cycle is equal to half of its maximum kinetic energy

(B) Root mean square velocity is $\frac{1}{\sqrt{2}}$ times of its maximum velocity.

(C) Mean velocity over a complete cycle is equal to $(2/\pi)$ times of its maximum velocity.

(D) Average acceleration for complete cycle is zero

34. Choose the correct statements (s)

(A) Viscous force is a non conservative force

(B) A denser fluid is always more viscous than a less dense fluid

(C) Any object can not have a velocity more than its terminal velocity in a given fluid

(D) If a body is moving in a fluid then the viscous force on it is zero

35. A tank full of water has a small hole at its bottom. Let t_1 be the time taken to empty first half of the tank and t_2 be the time needed to empty the rest half of the tank then

(A) $t_2 > t_1$ (B) $t_1 = t_2$ (C) $t_1 = \frac{t_2}{\sqrt{2}}$ (D) $t_1 = 0.414\ t_2$

36. A hollow sphere and a solid sphere of same radius and same material fall through a liquid from same height (neglect viscous force). Then, the correct option(s) from the following is/are:

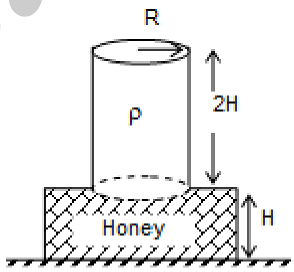
(A) The solid sphere reaches the ground earlier than the hollow sphere.

(B) The hollow sphere reaches the ground earlier than the solid sphere.

(C) Buoyant forces on both solid as well as hollow sphere are same.

(D) Buoyant forces on the solid and hollow spheres are different

37. A bottle is kept on the ground as shown in the figure. The bottle can be modelled as having two cylindrical zones. The lower zone of the bottle has a cross-sectional radius of R$\sqrt{2}$ and is filled with honey of density 2ρ. The upper zone of the bottle is filled with the water of density ρ and has a cross-sectional radius R. The height of the lower zone is H while that of the upper zone is 2H. If now the honey and the water parts are mixed together to form a homogeneous solution (Assume that total volume does not change)



(A) The pressure inside the bottle at the base will remain unaltered.

(B) The normal reaction on the bottle from the ground will remain unaltered.

(C) The pressure inside the bottle at the base will increase by an amount $(1/2)\rho gH$

(D) The pressure inside the bottle at the base will decrease by an amount $(1/4)\ \rho gH$

38. A ball is dropped in a tunnel across the earth's diameter from a height h above the surface. (Assume h is small)

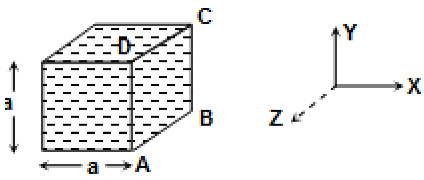
(A) Motion of the ball is simple harmonic.

(B) Motion is not simple harmonic.

(C) The period of motion is $4\sqrt{\frac{R}{g}}\sin^{-1}\left(\frac{R}{R+2h}\right)+4\sqrt{\frac{2h}{g}}$, (where R is the radius of earth)

(D) None of the above

39. Liquid is stored in a cubical box of side a. Now top face of the box is removed. The liquid has density ρ? (see figure)



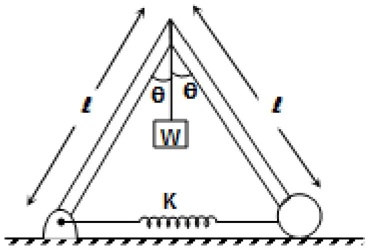
(A) The force exerted by a liquid on face ABCD is $\left(P_0 + \frac{\rho g h}{2}\right)a^2$

(B) The force exerted by a liquid on face ABCD is

(C) The force exerted by liquid on face ABCD will be along positive x-axis.

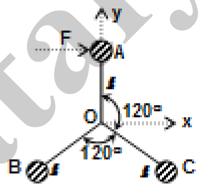
(D) The force exerted by liquid on face ABCD will be along negative x-axis.

40. Determine the values of θ for which the system will remain in equilibrium. Assume the rods to be massless and surface to be frictionless. Assume $W < 4k\ell$



(A) 0 (B) $\cos^{-1}\frac{W}{4k\ell}$ (C) $\sin^{-1}\frac{W}{2k\ell}$ (D) $\tan^{-1}\frac{W}{4k\ell}$

41. Each of the three balls of mass m and negligible radius is welded to a rigid frame of negligible mass and each rod having length ℓ. A force F is applied to one of the ball as shown in the figure. Then just at the instant of applying the force



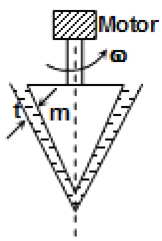
(A) net acceleration of point O is $\frac{F}{3m}\hat{i}$

(B) net acceleration of point A is $\frac{2F}{3m}\hat{i}$

(C) net acceleration of point B is $\frac{F}{6m}\hat{i} + \frac{F}{2\sqrt{3}m}\hat{j}$

(D) net acceleration of point C is $\frac{F}{6m}\hat{i} - \frac{F}{2\sqrt{3}m}\hat{j}$

42. The shown figure, there is a conical shaft rotating on a bearing of very small clearance t. The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity η. The shaft is having radius R and height h. If the external torque applied by the motor is τ and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω. Then



(A) $P = \frac{\pi\omega^2\eta R^3\sqrt{R^2+h^2}}{2t}$

(B) $\tau = \frac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{2t}$

(C) $P = \frac{\pi\omega^2\eta R^3 h}{2t}$

(D) $\tau = \frac{\pi\omega\eta R^3 h^2}{2t\sqrt{R^2+h^2}}$

43. A metal wire of length L, area of cross section A and Young modulus Y is stretched by a variable force F such that F is always slightly greater than the elastic forces of resistance in the wire. When the longation of the wire is ℓ, then choose the correct option(s).

(A) the work done by F is $\frac{1}{2}\frac{YA}{L}\ell^2$
(B) the elastic potential energy stored in the wire is $\frac{1}{2}\frac{YA}{L}\ell^2$
(C) no heat is produced during the elongation
(D) the work done by F is $\frac{YA}{L}\ell^2$

44. A particle is moving on a path whose trajectory is given by

$$x - a = A\sin(pt)$$
$$y - b = A\cos(pt)$$

where x and y represents the co-ordinates of the particle at any instant t and a, b and A are positive constants, then
(A) Speed of the particle is constant and is equal to pA
(B) The particle must be moving in clockwise sense.
(C) At $t = \frac{3\pi}{2p}$, net acceleration of the particle is p^2A towards positive x-axis.
(D) At $t = \frac{\pi}{p}$, velocity of the particle is pA towards positive x-axis

45. The potential energy of a particle of mass 2 kg, moving along the x-axis is given by $U = 16(x^2 - 2x)$J, where x is in metres. Its speed at x = 1m is 2 m/s
(A) The motion of the particle is uniformly acceleration.
(B) The motion of the particle is oscillatory force x = 0.5 m to x = 1.5 m
(C) The motion of the particle is simple harmonic.
(D) The period of oscillation of the particle is $\pi/2$ s.

46. Let v, T, L K and r denote the speed period, angular momentum, kinetic energy and radius of satellite in a circular orbit, then
(A) $v \propto r^{-1}$
(B) $L \propto r^{1/2}$
(C) $T \propto r^{3/2}$
(D) $K \propto r^{-4}$

47. A linear harmonic oscillator of force constant 2×10^6 N/m and amplitude 0.01 m has a total mechanical energy of 160 J. Its maximum
(A) P.E. is 100 J
(B) PE is 160 J
(C) KE is 100 J
(D) PE is OJ

48. When a bullet is fired horizontally into a stationary block, kept on a rough horizontal surface, then,
(A) linear momentum of the system may remain approximately conserved
(B) angular momentum of the system remains conserved about any point on the base of the block
(C) Energy of the system will not be conserved even if we neglect losses.
(D) all of the above

49. A body moves on a horizontal circular road of radius r with a tangential acceleration a_r. The coefficient of friction between the body and the road surface is μ. It begins to slip when its speed is v. then
(A) $v^2 = \mu gr$
(B) $\mu g = \frac{v^2}{r} + a_T$
(C) $\mu^2g^2 = \frac{v^4}{r^2} + a_T^2$
(D) The force of friction makes an angle $\tan^{-1}\left(\frac{V^2}{a_T r}\right)$ with the direction of motion at the point of slipping.

50. When a capillary is tube dipped in a liquid the liquid, rises to height h in the tube. The free liquid surface in side the tube is hemispherical in shape. The tube is now pushed down so that the height of the tube outside the liquid is less than h
(A) The liquid will come out of the tube like a small fountain
(B) The liquid will come out of the tube slowly.
(C) The liquid will fill the tube but not come out of its upper end
(D) The free liquid surface inside the tube will not be hemispherical.

51. A solid sphere of mass m is given initial linear speed v and angular speed ω as shown and kept on a rough incline. As soon as the sphere is kept on the incline, then the frictional force

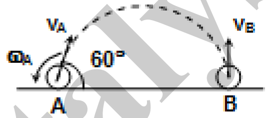
(A) acts up the incline
(B) acts down the incline
(C) is equal to $\mu mg \cos\theta$
(D) is equal to $mg \sin\theta$

52. There is a solid sphere of radius R & mass M the gravitational field g & potential V due to the sphere at a distance 'r' from its centre
(A) g & V both increases for r < R
(B) g increases & V decreases for r < R
(C) g & V both decreases, for $R < r < \infty$
(D) g decreases & V increases for $R < r < \infty$

53. A solid sphere of mass m and radius R is rolling on a rough surface. The work done by friction
(A) can never be positive
(B) may be positive
(C) may be negative
(D) may be zero

54. The function $x = A \sin \omega t + B$ represents an SHM for
(A) $B = 0, A \neq 0$
(B) any real values of A and B
(C) $A > 0, B \neq 0$
(D) $A > 0, B = 0$

55. A small rubber ball of radius R is thrown against a rough floor with a velocity $\vec{v}_A$ of magnitude v_0 and angular velocity $\vec{\omega}_A$ of magnitude ω_0. It is observed that ball bounce at B in vertically upward direction and zero angular velocity. Take collision is elastic then

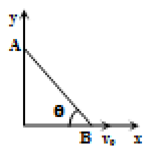


(A) $\omega_0 = \frac{5}{4}\frac{v_0}{R}$
(B) $\omega_0 = \frac{4}{5}\frac{v_0}{R}$
(C) $H_{max} = \frac{3}{8}\frac{v_0^2}{g}$
(D) $H_{max} = \frac{3}{2}\frac{v_0^2}{g}$

where H_{max} is maximum height attained by ball after first collision.

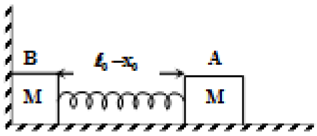
56. In a one dimensional collision between two identical particles B is stationary and A has momentum P before impact. During impact, A gives impulse J to B. Then
(A) The total momentum of A plus B system is P before and after the impact and (P-J) during the impact.
(B) during the impact B gives impulse J to A.
(C) the co-efficient of restitution is [(2J/P) – 1]
(D) the co-efficient of restitution is [(2J/P) + 1]

57. The end B of the rod AB which makes angle θ with the floor is being pulled with a constant velocity v_0 as shown. The length of the rod is ℓ at the instant when $\theta = 37°$



(A) velocity of end A is $\frac{4}{3}v_0$ downward
(B) Angular velocity rod is $\frac{5}{3}\frac{v_0}{\ell}$
(C) Angular velocity of rod is constant.
(D) velocity of end A is constant

58. If two block spring system is compressed towards rigid wall by x_0 as shown, over smooth surface and released. If spring constant is K. Then choose correct alternatives.

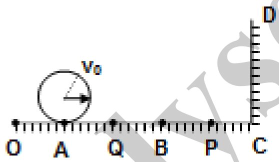


(A) Net force on the system till spring comes to natural length is kx_0
(B) Net force on the system if B leaves contact with wall is zero.
(C) Time when B leaves contact with wall is $\frac{\pi}{4}\sqrt{\frac{m}{K}}$.
(D) Time period of spring block system after B leaves contact is $2\pi\sqrt{\frac{M}{2k}}$

SECTION : (C) -Passage Type Questions

PASSAGE 01:

There is a frictionless wall CD on a horizontal surface OC. The surface OB has a friction coefficient 0.1 and the surface BC is frictionless. A sphere of uniform volume density of mass 0.14 kg and radius 4 m is kept at point A as shown in the figure. A sharp impulse is given to the sphere which provides a linear velocity v_0 = 14 m/s to the centre of mass of the sphere. If AQ = BQ = BP = PC = 192 m and collision with sphere and wall is perfectly elastic. Read the above passage carefully and answer the following questions. [Take g = 10 m/s^2]



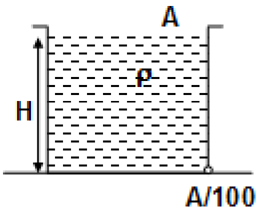
59. The sphere starts rolling without slipping at
(A) part Q (B) part B (C) part P (D) Any other point

60. The number of revolutions made by sphere during the times interval during its motion with rolling without slipping till just before collision is (approximately)
(A) 28.64 (B) 20.68 (C) 24.80 (D) none of these

61. The work done by the friction during the whole motion of sphere is
(A) 13.72 Joule (B) 1.8 Joule (C) 9.8 Joule (D) 11.92 Joule

PASSAGE 02:

A tank is placed on a frictionless surface having cross sectional area A and height H and the density of liquid is ρ. A hole is punched in one of the walls as shown in figure having area A/100, then



62. The time when height of liquid column becomes H/2.
(A) $100(\sqrt{2}-1)\sqrt{\frac{H}{g}}$ (B) $100(\sqrt{2}+1)\sqrt{\frac{H}{g}}$ (C) $200(\sqrt{2}-1)\sqrt{\frac{H}{g}}$ (D) $50(\sqrt{2}-1)\sqrt{\frac{H}{g}}$

63. The distance travelled by tank when height of the liquid column becomes zero.
(A) 200 H (B) 100 H (C) 400 H (D) 50 H

64. The acceleration of tank is
(A) g/5 (B) g/100 (C) g/50 (D) none of these

PASSAGE 3:

A vertical spring carries a 5 kg body and is hanging in equilibrium. An additional force is applied so that the spring is further stretched. When released from this position it performs 50 complete oscillation in 25 seconds, with an amplitude of 5 cm, then

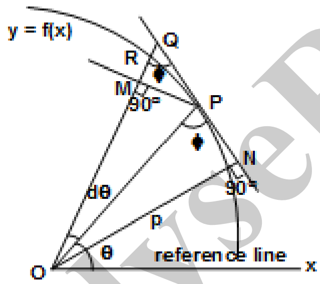
65. The value of the additional force applied (upto one decimal) in Newton is
(A) 39.5 (B) 38.5 (C) 31. 4 (D) 32.5

66. Force exerted by the spring on the body when it is at the lowest point in Newton (upto one decimal)
(A) 86.9 (B) 80.9 (C) 8 (D) 47.4

67. Force exerted by the spring on the body when it is at the middle point in Newton (upto one decimal)
(A) 86.9 (B) 80.9 (C) 8 (D) 47.4

PASSAGE 4:

If a point P moves in plane along a given curve y = f(x), the angular velocity of point P about a fixed point O in the plane is the rate of change of the angle that OP line makes with a fixed direction $[OX - \text{line}]$ in the plane



Let OP = r at t = t sec

PM = r$d\theta$ = PQ sinϕ, But if $d\theta$ is very small then. $PQ \cong \widehat{PR} = ds$ (arc length)

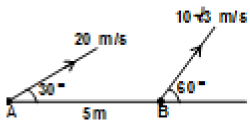
$\Rightarrow$ rd = ds sin ϕ

$$\Rightarrow \frac{d\theta}{dt} = \frac{1}{r}\frac{ds}{dt}\sin\phi = \frac{v\sin\phi}{r} = \frac{\text{Magnitude of component of velocity of point perpendicular to radius vector}}{\text{Magnitude of radius vector}}$$

$$\Rightarrow \frac{d\theta}{dt} = \frac{vp}{r^2}$$

68. $r^2\frac{d\theta}{dt}$ represents
(A) rate at which radius vector sweeps out area
(B) angular momentum
(C) moment of velocity about origin
(D) rate of increase of sectional area as P moves along curve

69. If two particles A and B are having speed 10√3 m/s and 20 m/s at a particular instant as shown in the figure, then the angular velocity of A with respect to B at the same instant is



(A) 1 rad/s clockwise
(B) 1 rad/s anticlockwise
(C) 2 rad/s clockwise
(D) 2 rad/s anticlockwise

70. If point P moves on parabolic path $y^2 = 4(x + 1)$, where x and y are in meter with constant speed 2 m/s. Its angular velocity about focus at an instant when it makes angle 60° at focus with x-axis is [all angles are measured in anticlockwise direction with positive x-axis]

(A) 0.25 rad/s (B) 0.50 rad/s (C) 0.12 rad/s (D) none of the above

PASSAGE 05

If the container filled with liquid gets accelerated horizontally or vertically, pressure in liquid gets changed. In case of horizontally accelerated liquid (a_x), the free surface has the slope $\frac{a_x}{g}$. In case of vertically accelerated liquid (a_y) for calculation of pressure, effective g is used. A closed box with horizontal base 6m by 6m and a height 2m is half filled with liquid. It is given a constant horizontal acceleration g/2 and vertical downward acceleration g/2.

71. The angle of the free surface with the horizontal is equal to -

(A) 30° (B) $\tan^{-1}\frac{2}{3}$ (C) $\tan^{-1}\frac{1}{3}$ (D) 45°

72. Length of exposed portion of top of box is equal to-

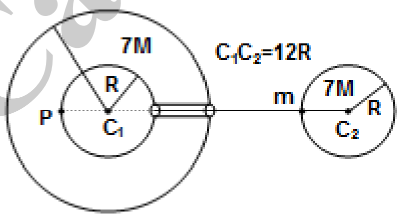
(A) 2m (B) 3m (C) 4m (D) 2.5 m

73. What is the value of vertical acceleration of box for given horizontal acceleration (g/2), so that no part of bottom of box is exposed :

(A) g/2 upward (B) g/4 downward (C) g/4 upward (D) not possible

PASSAGE 06:

Two planets having radii R and 2R with equal mass 7M respectively. The bigger planet has concentric cavity of radius R. A narrow hole is drilled through the bigger planet as shown in the figure [Assume that the process of drilling does not change the mass of bigger planet] A point mass of mass m is given a velocity v_0 from point Q at smaller planet so that it will reach to point P of bigger planet, assume there is no other force in the space except their mutual attraction of gravity



74. The distance of point of journey of particle of mass m from c_1 where it will experiences no force is

(A) 6 R (B) 8 R (C) 4 R (D) none of the above

75. The minimum value of v_0 so that it will reach to point P

(A) $\sqrt{\frac{35}{33}\left(\frac{GM}{R}\right)}$ (B) $\sqrt{\frac{211}{33}\left(\frac{GM}{R}\right)}$ (C) $\sqrt{\frac{350}{33}\left(\frac{GM}{R}\right)}$ (D) $\sqrt{\frac{21}{33}\left(\frac{GM}{R}\right)}$

76. Potential at P due to hollow sphere only

(A) $\frac{-GM}{2R}$ (B) $\frac{-7GM}{R}$ (C) $\frac{-9GM}{2R}$ (D) none of the above

PASSAGE 07:

One of the ways to approach gravitational behaviour of light follows from the observation that, although a photon has no rest mass, it nevertheless interacts with electrons as through it has the inertial mass given by

$$m = p/v \qquad \text{. . . (i)}$$

where p is the momentum of the photon and v, its velocity. According to the principle of equivalence, gravitational mass is always equal to inertial mass, so a photon of frequency ν ought to act gravitationally like a particle of mass given by equation (i).

Gravitational behaviour of light can be demonstrated in the laboratory. When we drop a stone of mass m from a height H near the earth's surface, it gains an energy mgH on the way to ground, and its final speed becomes $\sqrt{2gH}$.

All photons travel with the speed of light so can not go any faster. However a photon that falls through a height H can manifest the increase of mgH in its energy by an increase in its frequency from ν to ν'. Because the frequency change is extremely small on a laboratory scale experiment, we can neglect the corresponding change in photon's 'mass'. An interesting astronomical effect is suggested by the gravitational behaviour of light. If the frequency associated with a photon moving towards the earth increases, then the frequency of a photon moving away from it should decrease. Suppose a star of mass M and radius R emits a photon of frequency ν, then we can express the total energy of the photon as the summation of its quantum energy ($h\nu$) and its gravitational potential energy ($-\frac{GMm}{R}$) as $h\nu - \frac{GMm}{R}$

where m can be written as $m = \frac{h\nu}{c^2}$.

Using the energy conservation, we can find its frequency (ν') as the photon escapes to infinity from the surface of the star. It must be noted that a photon will be able to escape to infinity only if its total energy is positive. If the photon is not able to escape to infinity it means its total energy is negative, we have what are known as 'black hole stars'. The phenomenon of reduction of frequencies of photons emitted from the star is quantified by a term called gravitational red shift.(GRS). We define GRS = $\Delta\nu/\nu = (\nu - \nu')/\nu$ where ν' is the frequency of the escaped photon and ν is the frequency of emitted photon.

Now answer the following questions:

77. If ν' is the frequency of a photon which escapes to ∞ from a star and ν is its frequency when it was emitted from the surface of star of radius R then

(A) $\frac{\nu}{\nu'} = 1 + \frac{GM}{c^2R}$ (B) $\frac{\nu'}{\nu} = 1 - \frac{GM}{c^2R}$ (C) $\frac{\nu'}{\nu} = 1 - \sqrt{\frac{GM}{c^2R}}$ (D) $\frac{\nu'}{\nu} = 1 - \left(\frac{GM}{c^2R}\right)^2$

78. According to the passage the Gravitational red shift of a photon escaped to ∞ from a star's surface is

(A) $\sqrt{\frac{GM}{c^2R}}$ (B) $\frac{G^2M^2}{c^4R^2}$ (C) $\frac{GM}{c^2R}$ (D) $\frac{c^2R}{GM}$

79. As suggested by the passage, for a black hole:

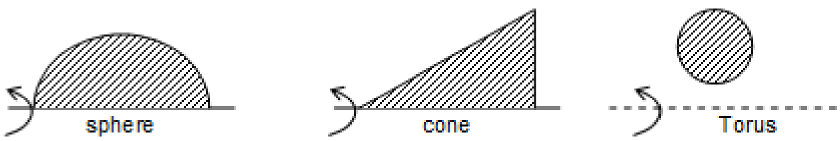
(A) $\frac{GM}{c^2R} > \sqrt{2}$ (B) $\frac{GM}{c^2R} > 1$ (C) $\frac{c^2R}{GM} < \frac{1}{2}$ (D) $\frac{GM}{c^2R} < 1$

PASSAGE 8

Theorems of Pappus-Guldinus

These theorems, which were first formulated by the Greek Geometer Pappus during the third century A.D. and later related by the Swiss mathematician Guldinus, or Guldin, (1577-1643) deal with the surfaces and bodies of revolutions. A surface of revolution is a surface which can be generated by rotating a plane curve about a fixed axis. For example the surface of a sphere can be generated by rotating a semicircular arc about its diameter. Similarly, a cone can be generated by rotating a line segment about an axis passing through one of its end points. The surface of a torus of a ring can be generated by rotating the circumference of a circle about a non - intersecting axis.

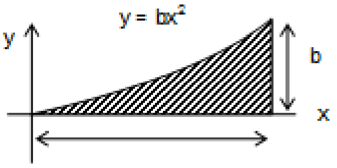
A body of revolution is a body which can be generated by rotating a plane area about a fixed axis. As shown in the figure, a sphere, a cone and a torus can each be generated by rotating the appropriate shape about the indicated axis



Now we shall state the two theorems of Pappus Guldinus.

Theorem – 1
The area of a surface of revolution is equal to the length of the generating curve times the distance travelled by the centroid centre of mass of the curve if the curve is made of a material of uniform linear density) of the curve while the surface is being generated.
Theorem – 2
The volume of a body of revolution is equal to the generating area times the distance travelled by the centroid of the area while the body is being generated. Now answer the following questions.

80. The below figure shows an area (known as parabolic spandrel). The location of centre of mass of the spandrel is

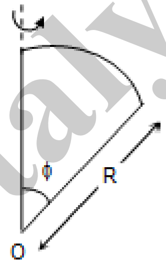


(A) $\left(\frac{3}{5}, \frac{3}{10}b\right)$ (B) $\left(\frac{3}{4}, \frac{1}{3}b\right)$ (C) $\left(\frac{3}{4}, \frac{3}{10}b\right)$ (D) $\left(\frac{4}{5}, \frac{1}{3}b\right)$

81. If the spandrel in the previous question was given a turn of 360° about y axis, the volume of the body thus generated will be

(A) $\frac{\pi b}{3}$ (B) $\frac{\pi b}{2}$ (C) $\frac{\pi b^2}{2}$ (D) $\frac{\pi b}{\sqrt{2}}$

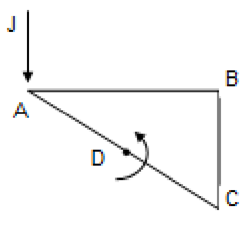
82. It is known that volume of revolution of a sector of a circle about the shown axis is $(4/3)\pi R^3 \sin^2(\phi/2)$. The distance of the centre of mass of the sector from point O must be



(A) $\frac{4R}{3\phi}\sin^2\phi$ (B) $\frac{4R}{3\phi}\sin^2(\phi/2)$ (C) $\frac{R}{3\phi}\sin^2(\phi/2)$ (D) $\frac{2R}{3\phi}\sin^2(\phi/2)$

PASSAGE 09

The figure shows the top view of a uniform solid prism. The sides of the prism are AB = 4 cm, BC = 3 cm and AC = 5 cm. the thickness of the prism (perpendicular to the plane of the paper) is t = 1 cm. The prism is mounted on a frictionless axis passing through D (which is the mid point of AC) and perpendicular to the plane of the paper. An impulse J = 1 Ns is imparted at point A of the prism, perpendicular to the edge AB of the prism. (The impulse vector lies in the plane of the paper). It was found that after the impulse was imparted, the prism took 1 second to undergo one complete rotation about the axis.



83. The moment of Inertia of the prism about the given axis is:

(A) $\frac{10^{-2}}{\pi}kgm^2$ (B) $\frac{10^{-2}}{2\pi}kgm^2$ (C) $\frac{2\times10^{-2}}{\pi}kgm^2$ (D) $\frac{\sqrt{2}\times10^{-2}}{\pi}kgm^2$

84. The mass of the prism is

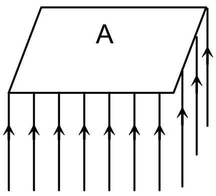
(A) $\frac{96}{\pi}$kg (B) $\frac{48}{\pi}$kg (C) $\frac{24}{\pi}$kg (D) $\frac{12}{\pi}$kg

85. If all the dimension of the prism were doubled while maintaining the same material ($\therefore$ AB = 8 cm, BC = 6 cm and AC = 10 cm, t = 2 cm) and an impulse of 1 Ns is applied at the point A once again. The time taken by the prism to complete one full rotation will be:

(A) 8 s (B) 32 s (C) 16 s (D) 64 s

PASSAGE 10:

A small metallic plate of area A and mass m is held in mid-air by striking a water jet from beneath. ρ is the density of water and v is the speed with which the water jet strikes the plate. Assume that the water jet strikes the plate normally.



86. If the water particles come to rest after colliding with the plate then the values of v for which the plate remains is equilibrium is

(A) $\sqrt{\frac{mg}{2\rho A}}$ (B) $\sqrt{\frac{mg}{\rho A}}$ (C) $\frac{1}{2}\sqrt{\frac{mg}{\rho A}}$ (D) $\sqrt{\frac{2mg}{\rho A}}$

87. What would be the value of v for the plate to be a equilibrium if after striking the plate water particles rebound with the same speed?

(A) $\sqrt{\frac{mg}{2\rho A}}$ (B) $\sqrt{\frac{mg}{\rho A}}$ (C) $\frac{1}{2}\sqrt{\frac{mg}{\rho A}}$ (D) $\sqrt{\frac{2mg}{\rho A}}$

88. The plate is held in equilibrium by striking the water jet. The plate is now given a sharp impulse at the centre of the plate in the downward direction

(A) The plate will perform SHM.

(B) The plate will perform oscillatory motion.

(C) The plate will keep moving in the downward direction.

(D) The plate will start rotating.

PASSAGE 11:

In this passage a brief idea is given of the motion of the rolling bodies on an inclined plane.

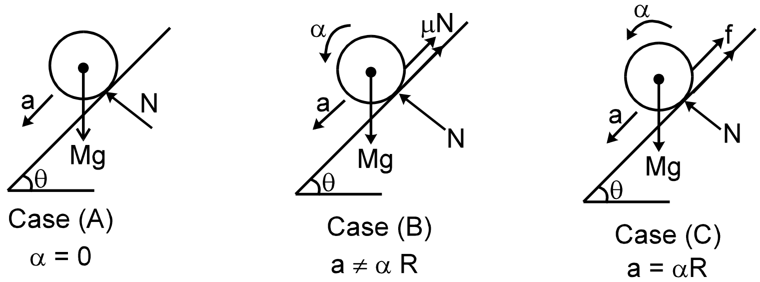
We will consider three cases : Objects are released on an incline plane

Case A : which is smooth. ;

Case B : where friction is insufficient to provide pure rolling.

Case C : where friction is sufficient to provide pure rolling.

Force diagram for three cases are as follows : (where symbols have their usual meanings)



Equations for case (C) :

$Mg\sin\theta - f = Ma$
$fR = (Mk^2)\alpha$; where k = radius of gyration and f is force of friction.
$a = \alpha R$
on solving the above equations we will get

$$a = \frac{g\sin\theta}{\left(1+\frac{k^2}{R^2}\right)}$$

Object	Ring	Disc	Hollow sphere	Solid sphere
k	R	$\frac{R}{\sqrt{2}}$	$\sqrt{\frac{2}{3}}R$	$\sqrt{\frac{2}{5}}R$

To decide the minimum friction coefficient to provide pure rolling put
$f = \mu Mg\cos\theta$

And we will get $\mu_{min} = \dfrac{\tan\theta\left(\frac{k^2}{R^2}\right)}{\left(1+\frac{k^2}{R^2}\right)}$

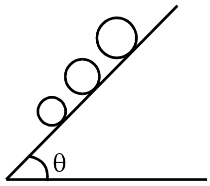
Equations for case (B) :

$Mg\sin\theta - \mu N = Ma$
$\mu NR = Mk^2\alpha$
$N = Mg\cos\theta$

The K.E. of a rolling body can be expressed as :

$$K.E. = \frac{1}{2}MV_{CM}^2 + \frac{1}{2}I_{CM}\omega^2$$

89. Three solid uniform spheres are released on an inclined plane as shown. The distance between the spheres remains constant during motion in :



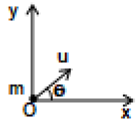
(A) all three cases
(B) case 'A' & 'B'
(C) case 'C'
(D) depends on the mass of the spheres.

90. We have four objects : a solid sphere, a hollow sphere, a ring & a disc, all of same radius. When these are released on an inclined plane, it may happen that all of them do not perform pure rolling. But from the information of pure rolling, if one object can be confirmed to be purely rolling then it can be said that rest all will perform pure rolling. This object whose pure rolling confirms pure rolling of all other objects is :
(A) Hollow sphere (B) solid sphere (C) ring (D) disc

91. If the four objects given in the above question are of same mass, same radius having the same friction coefficient & are released from the same height, then at the bottom the object which will have least K.E. for case 'B' will be the :
(A) Hollow sphere (B) solid sphere (C) ring (D) disc

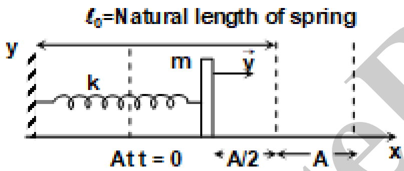
SECTION : (D) - Matrix Match

92. A particle of mass m is projected from origin at angle θ to x-axis with initial speed u as shown in the figure.



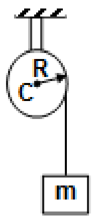
Column A	Column B
(A) Distance travel by particle in the time interval t_1 to t_2	(i) positive
(B) Instantaneous power at time t	(ii) negative
(C) Average power developed by gravity in the time interval t_1 to t_2	(iii) Zero
(D) $\dfrac{\text{Work done by gravity in time interval } t_1 \text{ to } t_2}{t_2 - t_1}$	(iv) $mg\left[\frac{1}{2}g(t_2+t_1) - u\sin\theta\right]$

93. A spring mass-system perform simple harmonic motion as shown in the figure, where $\vec{v}$, ℓ_0 and A are the velocity of particle natural length of spring and amplitude of motion respectively. If $\vec{x}$ and $\vec{a}$ represents displacement of particle from mean position and acceleration of particle respectively. [Assume x = A sin ($\omega t + \phi$)]



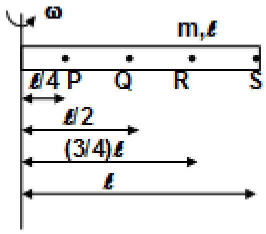
(A) Initial phase of motion	(i) Negative
(B) Phase of particle of motion when particle is at mean position first time.	(ii) $\dfrac{11\pi}{6}$
(C) $\vec{x} \times \vec{a}$ for the motion at time t = t sec is	(iii) $\vec{a} \times \vec{v}$
(D) $\vec{a} \cdot \vec{x}$ for the motion at time t = t sec is	(iv) zero

94. The string is wrapped on the given cylinder and the cylinder can rotate about the horizontal axis. Then, match the acceleration of the block in list II with list I



List I	List II
(A) Cylinder is solid of mass m and there is sufficient friction between rope and cylinder.	(i) g
(B) Cylinder is hollow of mass m and there is no friction between the rope and cylinder	(ii) (2/3)g
(C) Cylinder is massless	(iii) g/3
(D) Cylinder is hollow and mass of the cylinder is m and there is sufficient friction between the block and the cylinder	(iv) g/2

95. A rod of mass m and length ℓ is rotating about a vertical axis with a constant angular velocity ω. Then, match the following:



List I	List II
(A) Tension at P	(i) $\frac{3m\omega^2\ell}{8}$
(B) Tension at Q	(ii) $\frac{7m\omega^2\ell}{32}$
(C) Tension at R	(iii) zero
(D) Tension at S	(iv) $\frac{15m\omega^2\ell}{32}$

96. A particle is moving according to the displacement time relation $x = 3t^2 - \frac{t^3}{2}$ (where x is in meters and t is in seconds). Match the condition of column I with time interval and instant of column II

Column A	Column B
(A) Velocity and acceleration will be in same direction	(i) At t = 0 and t = 6 sec
(B) particle will be at origin	(ii) 0 < t < 2 sec and t > 4 sec
(C) particle will retard	(iii) at t = 0 and t = 4 sec
(D) velocity is zero	(iv) 2 < t < 4

97. A bob is attached to a string of length ℓ whose other end is fixed and is given horizontal velocity at the lowest point of the circle so that the bob moves in a vertical plane. Match the velocity given at the lowest point of circle in column I with tension and velocity at the highest point of the circle corresponding to velocity of column I of column II

Column A	Column B
(A) $\sqrt{2g\ell}$	(i) V = 0 and $T \neq 0$
(B) $\sqrt{g\ell}$	(ii) T = 0 and $v \neq 0$
(C) $\sqrt{3g\ell}$	(iii) T = 0 and $v = \sqrt{g\ell}$
(D) $\sqrt{5g\ell}$	(iv) T = 0 and v = 0

98. If the position vector of a particle at point P moving in space and distance traveled from a fixed point on the path are given by $\vec{r}$ and s respectively. Three vectors are defined as follows

$\vec{N} = \frac{d\vec{r}}{ds}$, $\vec{T} = R\frac{d\vec{N}}{ds}$ and $\vec{B} = \vec{N} \times \vec{T}$ where R is the radius of curvature at point P and $\vec{r}$ is a non-zero vector. Now match the following on the basis of above concept

Column A	Column B
(A) $\vec{T}$	(i) Unit vector
(B) $\vec{N}$	(ii) Zero
(C) $\vec{N} \cdot \vec{T}$	(iii) $\hat{v}$ (unit vector along the direction of velocity)
(D) $\vec{B}$	(iv) $\frac{v\vec{a} - \vec{v}\, a_T}{\lvert\vec{v} \times \vec{a}\rvert}$ where $\vec{a}$ and a_T are acceleration and magnitude of tangential acceleration of particle P

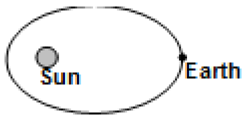
99. In this question $\vec{r}$ represents the position vector of a particle and and a are some constants. Now match the appropriate options: (constant magnitude includes a zero magnitude)

Column I	**Column II**
(A) $\vec{r} = \cos\omega t\hat{i} + \sin\omega t\hat{j} + at\hat{k}$	(1) Constant speed
(B) $\vec{r} = 5\cos\omega t\hat{i} + 5(\sin\omega t + t)\hat{j}$	(2) Helical motion
(C) $\vec{r} = 5at^2\hat{i} + 2at\hat{j}$	(3) Constant magnitude of tangential acceleration.
(D) $\vec{r} = 5[(\cos\omega t^2)\hat{i} + (\sin\omega t^2)\hat{j}]$	(4) Constant magnitude of total acceleration

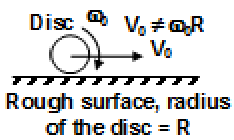
100. Choose the right choice.

System

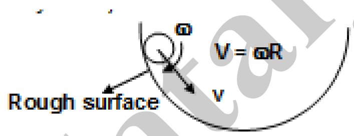
(A) Earth moving in an elliptical orbit (Only earth in system) direction



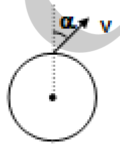
(B) A disc having translation and rotation motion both (with slipping) on a rough surface. (only disc in system along a specific direction



(C) A sphere rolling without slipping on a curved surface (Only the sphere in system) in the space



(D) Projection of a particle from the surface of earth. (Only particle in system)



Conservation principle

(i) Conservation of linear momentum along any

(ii) Conservation of linear momentum

(iii) conservation of angular momentum about any point

(iv) Conservation of angular momentum about a specific point in the space

(v) Conservation of mechanical energy.

101. In list I the axes of rotation of rotation of circular disc of mass m and radius R are stated. Match these with the corresponding expressions of moment of inertia given in

(A) Through its centre and normal to its plane	(i) $\frac{5}{4}MR^2$
(B) Through any diameter.	(ii) $\frac{1}{4}MR^2$
(C) Through tangent in the plane of the disc	(iii) $\frac{3}{2}MR^2$
(D) Through tangent perpendicular to the plane of the disc.	(iv) $\frac{1}{2}MR^2$

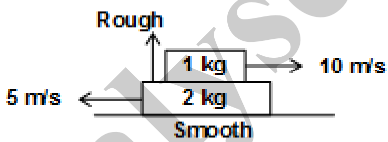
102. Match the following, considering R to be radius of earth, M mass of earth and G as universal gravitational constant

List – I	List – II
(A) Orbital velocity at height R from earth's surface	(i) $\sqrt{\frac{2GM}{R}}$
(B) Orbital velocity at near earth's surface	(ii) $\sqrt{\frac{GM}{R}}$
(C) Escape velocity from earth's surface	(iii) $\sqrt{\frac{2GM}{3R}}$
(D) Orbital velocity in a particular orbit	(iv) $\sqrt{\frac{GM}{2R}}$

103. For a particle of mass m moving in straight line, match the following

List – I	List - II
(A) Acceleration and velocity	(i) May be equal at all time
(B)Instantaneous speed and magnitude of instantaneous velocity	(ii) Must be equal at all time
(C) Magnitude of displacement and distant	(iii) Must be along the same line at all time
(D) Displacement and velocity	(iv) May be along the same line at all time

104. In a two block system shown in the figure. Match the following. (Assume 2 kg block is long enough)



List – I	List – II
A.Velocity of centre of mass	(i) Keep on changing all the time
B.Momentum of centre of mass	(ii) First decrease then become constant
C.Momentum of 1 kg block	(iii)Zero
D.Kinetic energy of 2 kg block	(iv)Constant

105. Density of a planet is two times the density of earth. Radius of this planet is half. Match the following (As compared to earth)

List – I	List – II
(A) Acceleration due to gravity on this planets surface	(i) Half
(B) Gravitational potential on the surface	(ii) Same
(C) Gravitational potential at centre	(iii) Two times
(D) Gravitational field strength at centre	(iv) Four times

106. There is hemisphere of radius 'R' & Mass M with base in x–z plane then match the following

List – I	List – II
(A) Moment of inertia about XX' of solid hemisphere.	(i) $\frac{2}{5}MR^2$
(B) Moment of inertia about YY' solid hemisphere	(ii) $\frac{1}{5}MR^2$
(C) Moment of inertia about CM parallel to XX' of hollow hemisphere	(iii) $\frac{5}{12}MR^2$
(D) Moment of inertia about ax is passing through centre of mass of solid hemisphere parallel to XX'.	(iv) $\left(\frac{83}{320}\right)MR^2$

107. List – I / List – II

List – I	List – II
(A) Area under the curve a-x	(i) Instaneous velocity
(B) Total work done per unit mass	(ii) Change in kinetic energy per mass
(C) Area under the net force verses time (R)	(iii) Average acceleration.
(D) Slope of line joining two points on velocity verses time graph	(iv) Change in momentum.

108. List – I / List – II

List – I	List – II
(A) Equation of continuity (ρAv = constant)	(i) non-viscous flow
(B) Bernoulli's equation	(ii) Steady flow
(C) Velocity of a given point is constant with time	(iii) incompressible flow
(D) Ideal flow	(iv) irrotational flow

109. Let there be two bodies with masses m_1 and m_2 moving with velocities u_1 and u_2 in same direction. They collide at an instant and acquire velocities v_1 and v_2. The coefficient of restitution of collision is e ($0 \le e \le 1$). Match the following

List – I	List – II
(A) $m_1 = m_2$, $u_1 > u_2$, e = 1	(i) $v_1 > v_2$
(B) $m_1 \ll m_2$, $u_1 < u_2$, e = 1	(ii) $v_1 < v_2$
(C) $m_1 \gg m_2$, $u_1 > u_2$, e = 1	(iii) $v_1 = v_2$
(D) If collision is perfectly inelastic	(iv) Total momentum of the system is conserved.

110. A particle of mass 1 kg has velocity $\vec{v}_1 = 2t\hat{i}$ and another particle of mass 2 kg has velocity $\vec{v}_2 = t^2\hat{j}$. Match the following

List – I	List – II
(A) Net force on centre of mass at t = 2 sec.	(i) 20/9 unit
(B) Velocity of centre of mass at t = 2 sec.	(ii) $\sqrt{68}$ unit
(C) Displacement of centre of mass at t = 2 sec.	(iii) $\sqrt{80}/3$ unit
(D) Magnitude of linear momentum of centre of mass at t = 2 sec.	(iv) $\sqrt{80}$ unit

111. A boy is experimenting in lab with simple pendulum. He denotes the four – physical quantities with $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$. He knows that either of them denotes force, displacement, velocity and acceleration. He obtained few results from the above data: as

(1) $\vec{C}\cdot\vec{B}$ and $\vec{A}\cdot\vec{B}$ are always negative in an SHM.

(2) $\vec{A}\cdot\vec{C}$ is always positive in an SHM.

(3) $\vec{A}\times\vec{C}$, $\vec{D}\times\vec{B}$, $\vec{C}\times\vec{B}$ and $\vec{A}\times\vec{B}$ are always zero in an SHM.

(4) Average of all the quantities (individually) for one time-period in an SHM is zero.

Match the following

List – I	List – II
(A) $\vec{A}$	(i) displacement vector
(B) $\vec{B}$	(ii) velocity vector
(C) $\vec{C}$	(iii) acceleration vector
(D) $\vec{D}$	(iv) force vector

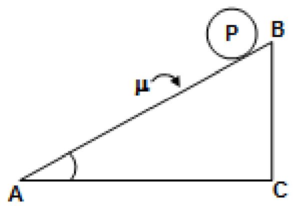
112. List – I / List – II

List – I	List – II
(A) The third significant digit in 14.745 after rounding it off to three significant digits is	(i) 2
(B) The third significant digit in 14.650 after rounding it off to there significant digits is	(ii) 5
(C) The number of significant digits in 0.00023 is	(iii) 6
(D) The number of significant digits in 1.0023 is	(iv) 7

113.

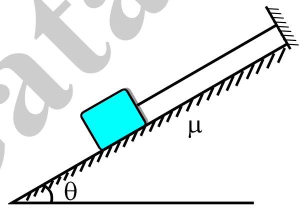
List – I	**List – II**
(A) Gravitational potential at the centre of a uniform solid sphere of radius R.	(i) Directly proportional to R
(B) Gravitational field at a distance R from the centre of a uniform solid sphere of radius ‘a’ (R < a)	(ii) Depends on ‘a’
(C) Gravitational potential at a distance ‘a’ from the centre of a thin spherical shell of radius R(a<R)	(iii) Directly proportional to R^{-1}
(D) Gravitational potential at a distance R from the centre of a uniform solid sphere of radius a (R > a)	(iv) Increases as R increases

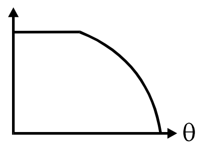
114. A body P is rolling without slipping on the rough inclined surface as shown in the figure. The frictional force acting on the body is listed in list II. Match the following lists

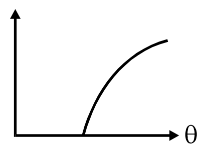


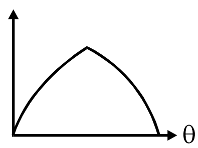
List – I	**List – II**
(A) For ring	(i) mg sin θ (2/5)
(B) For solid sphere	(ii) $\dfrac{mg \sin \theta}{3}$
(C) For solid cylinder	(iii) (2/7) mg sin θ
(D) For hollow sphere	(iv) (mg sin θ)/2

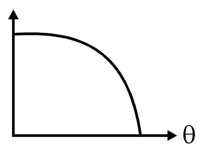
115. A block of mass m is put on a rough inclined plane of inclination θ, and is tied with a light thread shown. Inclination θ is increased gradually from $\theta = 0°$ to $\theta = 90°$. Match the column according to corresponding curve.



(i) Tension in the thread versus θ (P) 

(ii) Normal reaction between the block and the incline versus θ (Q) 

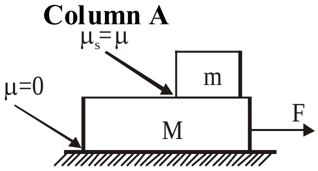
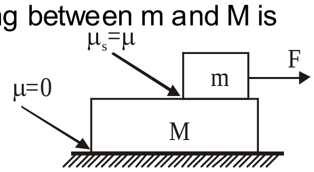
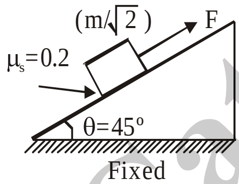
(iii) friction force between the block and the incline versus θ (R) 

(iv) Net interaction force between the block and the incline versus θ (S) 

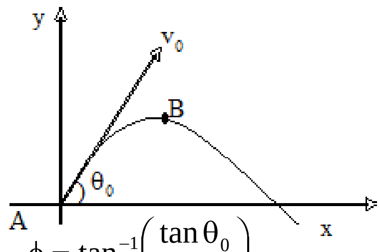
116. The velocity of an aircraft as seen by the driver of a car is 5 m/s upwards. A passenger in a train simultaneously sees the car to move southwards with 5 m/s. The conductor of a bus feels that the train is moving north with a velocity of 10 m/s. A dacoit running towards the bus feels it moving 6 m/s eastwards. A police jeep chasing the dacoit feels him to be moving westwards with 3 m/s. A person standing on the ground sees the police jeep moving north-west with $15\sqrt{2}$ m/s.

(a) Velocity of the aircraft as seen by the conductor	(P) $-3\hat{i}-5\hat{j}-5\hat{k}$
(b) Velocity of the conductor as seen by the police	(Q) $5\hat{j}+5\hat{k}$
(c) Velocity of the aircraft	(R) $3\hat{i}$
(d) Velocity of police as seen by the pilot of aircraft	(S) $-12\hat{i}+20\hat{j}+5\hat{k}$

117.

	Column A		Column B
(A)	 In the above case the condition of no slipping between m and M is	(P)	$F \le \mu mg\left(1+\dfrac{m}{M}\right)$
(B)	 In the above case the condition of no slipping between m & M is	(Q)	$F \le \mu\,(m + M)\,g$
(C)	 In the above case the condition no slipping is	(R)	$F > 0.6\,mg$
(D)	 In the above case the value of F so that the small block can move up the inclined plane is	(S)	$F \le \mu\,mg$

118. In oblique projection of a particle:



(a) Radius of curvature at A (p) $\dfrac{2g\cos\theta_0 \sin 2\theta}{v_0 \sin^2\theta_0}$

(b) Radius of curvature at B (q) $\dfrac{g^2 \sin 2\phi}{v_0^2 \sin 2\theta_0}$, where $\phi = \tan^{-1}\left(\dfrac{\tan\theta_0}{2}\right)$

(c) ω_{BA} (r) $\dfrac{v_0^2}{g\cos\theta_0}$

(d) α_{BA} (s) $\dfrac{v_0^2\cos^2\theta_0}{g}$

where ω_{BA} and α_{BA} are angular velocity and angular acceleration of the particle at the highest point B relative to the point of projection A.

119. (a) A man of mass m is seen holding a rope hanging from roof. The tension in the string can be(man is also in air) — (p) mg

(b) A projectile experiences an average force during its motion in air (neglecting air resistance) — (q) > mg

(c) A pendulum bob of mass m swings by an inextensible string when hanging from a fixed support.Tension in the string at the extreme position — (r) < mg

(d) When you jump from a building holding a bag of mass m, the reaction force of the bag on you is is — (s) zero

120. Match the enteries of Column I with the entries of Column II.

Column I	**Column II**
(a) Young's modulus	(p) Shear strain
(b)Modulus rigidity	(q) Normal strain
(c) Bulk modulus	(r) Transverse strain
(d) Poisson ratio	(s) Volumetric strain

121. The velocity of an aircraft as seen by the driver of a car is 5 m/s upwards. A passenger in a train simultaneously sees the car to move southwards with 5 m/s. The conductor of a bus feels that the train is moving north with a velocity of 10 m/s. A dacoit running towards the bus feels it moving 6 m/s eastwards. A police jeep chasing the dacoit feels him to be moving westwards with 3 m/s. A person standing on the ground sees the police jeep moving north-west with $15\sqrt{2}$ m/s.

(a) Velocity of the aircraft as seen by the conductor — (p) $-3\hat{i}-5\hat{j}-5\hat{k}$

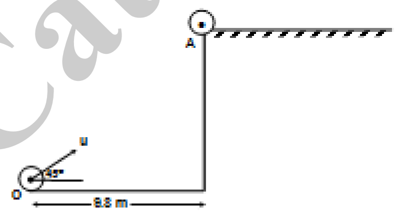
(b) Velocity of the conductor as seen by the police — (q) $5\hat{j}+5\hat{k}$

(c) Velocity of the aircraft — (r) $3\hat{i}$

(d) Velocity of police as seen by the pilot of aircraft — (s) $-12\hat{i}+20\hat{j}+5\hat{k}$

SECTION : (E) - Integer Type

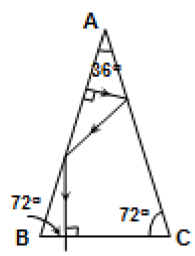
122. A uniform sphere is projected at an angle of 45° above horizontal from a point O behind 9.8 m from a rough horizontal elevated surface as shown in the figure. It is observed that the lowermost point of the sphere lands on the surface at A horizontally and the surface can sustain rolling without sliding. The ratio of the speed with which the sphere lands at A to the speed of sphere after pure rolling starts is X/Y. Find the value of |X-Y| (take g = 10 m/s²)



123. A uniform spring of mass m = 0.0100 kg length L = 1.0000 m and spring constant K = 4.00 N/m is kept in a gravity free space. What will be the speed of longitudinal waves on the spring in m/s?

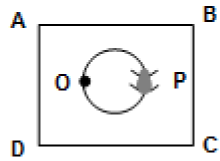
124. A thin ring has mass M = 4 kg and radius R = 1 m. Find its moment of inertia about an axis passing through its centre which makes an angle of exactly $\pi/4$ radians with the normal of the plane containing the ring.

125. A room is in the shape of a prism as shown in the figure. The front view of the room is as shown in the figure. A ball is projected normally from the wall ABFE and it strikes the floor normally after 2 reflection as shown in the figure. Gravitational force and friction are assumed to be absent. The coefficient of restitution between the ball and any wall is e. Find the value of e for which the ball will strike the floor normally

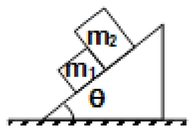


126. A uniform rod AB of mass 3m and length 2a has a small smooth ring of mass m attached at A. The ring is threaded on a horizontal fixed wire. Initially the rod is held horizontally alongside the wire and is released from rest from this position. Find the angular velocity (in rad/sec) of the rod when AB become vertical. (Assume the rod can rotate freely about A and $g = 10 m/s^2$).

127. A uniform square laminar plate ABCD having moment of inertia $250\,\pi$ kgm^2 is placed on a horizontal smooth surface. The plate is having a circular groove of radius 2.5 m, whose centre is concentric to the centre of the plate. The plate is free to rotate about a point O on the groove fixed to the horizontal surface, as shown in the figure. A tortoise of mass 8 kg starts moving along the groove from a point P on the groove, which is diametrically opposite to O. The angle (in radian) upto one decimal place rotated by the plate ABCD is X/10, when the tortoise comes back to point P after a complete rotation on the groove. Find X (Assume that plate is much heavier than the tortoise. Such that the velocity of the plate is much smaller than the velocity of the tortoise)



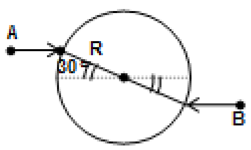
128. Two blocks of masses m_1 and m_2 are kept touching each other on an incline plane of inclination θ as shown in figure. If the coefficient of friction between the block m_1 and the plane is μ_1 and that between block m_2 and the plane is $\mu_2(<\mu_1)$.



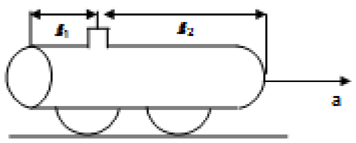
If minimum value of the inclination is θ at which the blocks just start sliding down the plane, then $\tan\theta$ is $\dfrac{X}{Y}$.

Then |X-Y| is:
[given $\mu_1 = 0.3$, $\mu = 0.2$, $m_1 = 30$ kg, $m_2 = 20$ kg]

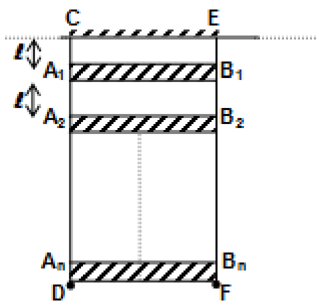
129. A uniform disc of mass m and radius 0.5 m is kept on a horizontal frictionless surface. Two point masses each of equal mass m and moving with speed of 30 m/s hit the disc as shown, and stick to it. The angular velocity (in rad/s) of the disc after collision is:



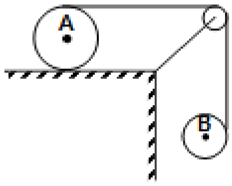
130. A closed cylindrical oil tanker filled with oil of density $\rho = 800\ kgm^{-3}$ has a stopper at the top. The acceleration a in horizontal direction at which the stopper will come out if it can support a pressure p = 0.05 atm is X/4 and the distance of the stopper is $\ell_1 = 0.1$ m from one end $\ell_2 = 1$m from the other end as shown in the figure. Find the value of X.



131. Two massless rods CD and EF are hinged at C and E respectively as shown. n rods, each of mass m and length are fixed to CD and EF at points A_1B_1 A_2B_2, …………., A_nB_n such that the separation between each rod is ℓ. The first rod A_1B_1 is at a distance ℓ from CE. The whole arrangement is at rest in the vertical equilibrium, Find the value of impulse (in Ns) to be given to the n^{th} rod so that the complete system just reaches the horizontal level. (Impulse given is perpendicular to the rod and is horizontal).
Take: n = 7; $g = 10\ m/s^2$; $\ell = 25$ cm; m = 1kg. The system is free to rotate about an axis passing through CE

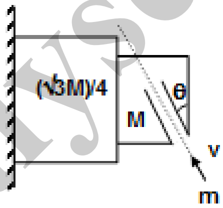


132. Two cylinder A&B of equal mass and radius, have an inextensible thread wound around them. The thread unwinds itself without slipping on the cylinders. Cylinder A is kept on a smooth horizontal surface. Cylinder B is hanging in the vertical plane. When the system is released, find the magnitude of acceleration (in m/s^2) of a point on the rim of the cylinder A when it is at the topmost point. Assume, frictionless pulley and sufficient length of string. (Take $g = 10\ m/s^2$)

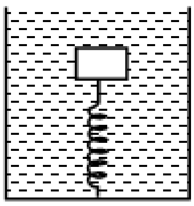


133. A particle moving with S.H.M on a straight line passes through two points A & B with same velocity. Time taken by particle to move from B to A is 1 sec & another 5 sec to return B from A. What is time period of S.H.M (in sec)

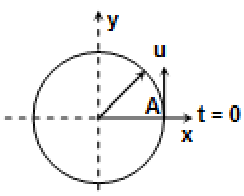
134. A block of mass M has a groove (see the figure) which gets 'n' particles per second each of mass 'm' moving with velocity v. The coefficient of friction between M and $\frac{\sqrt{3}}{4}M$ is zero. Assume $M \gg m$ and each particle gets stuck after hitting. M is found to be at rest. The value of coefficient of friction between the wall and $\frac{\sqrt{3}}{4}M$ so that the latter remains at rest is 1/X. The value of X is: [given $\theta = 60°$]



135. A block of density ρ_b and volume v is attached to a spring of spring constant k inside a liquid of density ρ_ℓ kept in a beaker is then accelerated by an acceleration a in vertical direction. The amplitude of oscillation ($\rho_b > \rho_\ell$) [Given $\rho_b = 2\rho_\ell$-, va = $\frac{k}{500\rho_\ell}$] is (in mm)

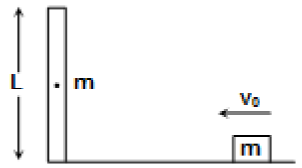


136. A particle is moving along a circular track of radius $R_O = 1$ m, the centre of which located at the origin of coordinate i.e. (0, 0). At time t = 0, the initial position of the particle is (1, 0). The speed of the particle at this instant is 4π m/s and the magnitude of tangential acceleration is $2\pi\ m/s^2$. The speed of the particle at 2.5 sec is $n\pi$ m/sec, then the value of n is



137. A satellite is revolving around a planet of mass M in an elliptic orbit such that the planet is at the focus and the maximum and minimum distance of the satellite from the planet are 450 km and 150 km. If the orbital speed v of the satellite when it is at a distance of 400 km from the planet is given by $v^2 = \frac{GM}{R}$ where R is constant and has dimensions of distance, then R = _______ km.

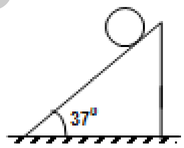
138. In the shown figure, a particle of mass m is moving with speed v_0 on a frictionless surface and collides with the uniform horizontal rod of length L and mass m. The collision is perfectly elastic. The rod rotates about its centre of mass. The rod deflects by angle $\theta(= /2)$ from initial position in time t. if t is $\frac{K\pi L}{24v_0}$, then the value of K is………………….



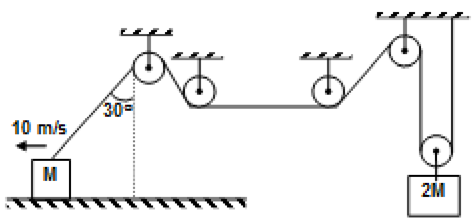
139. A solid sphere of mass 'm' and radius r is attached through a rod of mass 'm' and length ℓ and hinged at point 'P' vertical plane and a spring is attached as shown in the figure. The time period of oscillation (in second) is

given $\ell = \left[\frac{\sqrt{g}}{\pi\left(\sqrt{\frac{142}{75}} + \sqrt{\frac{71}{165}}\right)}\right]^2$

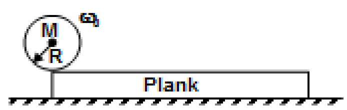
140. A solid metallic sphere of mass m and radius R is free to roll (without sliding) over inclined surface of wooden wedge of mass m. Wedge lies on a smooth horizontal floor. If the system is released from rest. The frictional force between sphere and wedge is $\frac{nmg}{9}$, then the value of n is …………



141. If at an instant, shown block B is moving towards left with speed of 10 m/s. The block of mass 2M is moving with speed V/2 m/s at that instant, then value of V is



142. A uniform solid spherical ball of mass m and radius R is given an angular velocity ω_0 in clockwise direction and placed gently on a plank of same mass M. The friction coefficient between plank and surface is zero and that of plank and ball is μ. The kinetic energy of the ball in Joule after infinite time is…………
[Assume plank is sufficiently long and M = 4.05 kg, R = 1m, ω_0 = 10 rad/s]

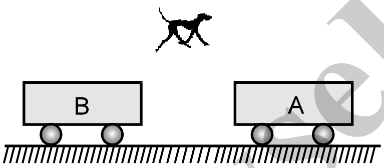


143. Two simple harmonic motions of the form $y = -\sqrt{669}\cos\left(\omega t - \frac{\pi}{6}\right)$m and $y = 2\sqrt{669}\sin\left(\omega t + \frac{2\pi}{3}\right)$m superimpose. The square of the resultant amplitude is ______m^2.

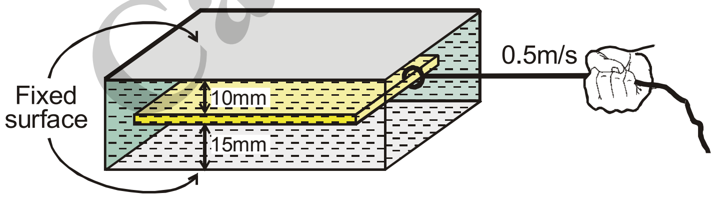
144. A liquid is filled in a large cylindrical vessel up to height 40 m and there are two identical holes, one at a depth 20 m from the free surface and other at the bottom of the vessel. The holes are initially plugged. At time t = 0 the first hole is unplugged and the liquid starts coming out. The second hole is unplugged when liquid reaches the height of the first hole. Area of the each hole is 2×10^{-3} m^2. Base area of the vessel is 6 m^2. The time taken to empty the tank is _________ minutes. [take g = 10 m/s^2]

145. An engineer works at a plant out of town. A car is sent for him from the plant every day that arrives at the railway station at the same time as the train he takes. One day the engineer arrived at the station half an hour before his usual time and without waiting for the car, started walking towards factory. On his way he met the car and reached his plant 20 minutes before the usual time. For how much time (in minute) did the engineer walk before he met the car ? The car moves with the same speed everyday.

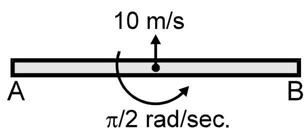
146. In a circus act, a 4 kg dog is trained to jump from B cart to A cart and then immediately back to the B cart. The carts each have a mass of 20 kg and they are initially at rest. In both cases the dog jumps at 6 m/s relative to the cart. If the cart moves along the same line with negligible friction, If the final magnitude of velocity of cart B with respect to the floor is X/36. Value of X is:



147. Glycerine is filled in 25 mm wide space between two large plane horizontal surfaces. Calculate the force required to drag a very thin plate 0.75 m^2 in area between the surfaces at a constant speed of 0.5m/s if it is at a distance of 10 mm from one of the surfaces in horizontal position ? Take coefficient of viscosity η = 0.5 Ns/m^2. Fill the value of X where X = 100 × force required to drag (in newton).

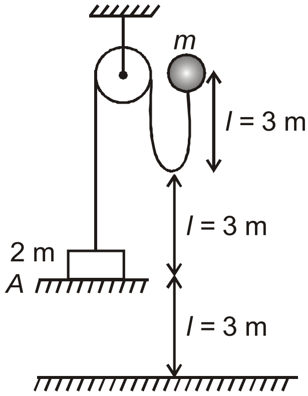


148. A uniform rod AB of length 4m and mass 12 kg is thrown such that just after the projection the centre of mass of the rod moves vertically upwards with a velocity 10 m/s and at the same time it is rotating with an angular velocity $\frac{\pi}{2}$ rad/sec about a horizontal axis passing through its mid point. Just after the rod is thrown it is horizontal and is as shown in the figure. Find the acceleration (in m/sec^2) of the point A in m/s^2 when the centre of mass is at the highest point. (Take g = 10m/s^2 and π^2 = 10)

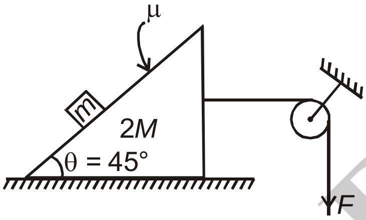


149. Two particles P and Q describe simple harmonic motions of same period, same amplitude along the same line about the same equilibrium position O. When P and Q are on opposite sides of O at the same distance from O they have the same speed of 1.2 m/s in the same direction, when their displacements are same they have the same speed of 1.6 m/s in opposite directions. Find the maximum velocity of P (in m/s).

150. A small ball having mass m is released as shown in figure. Find the maximum height attained by mass $2m$ (in metre) from the position A shown in figure.



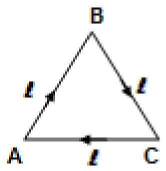
151. In the figure shown, the contact surface of prism and ground is smooth. If the value of F is $10x$ N so that there will be no contact force between m and inclined plane, then find the value of x. Take m = 1 kg, M = 2 kg, μ = 0.5 and g = 10 m/s^2.



EXERCISE # 02

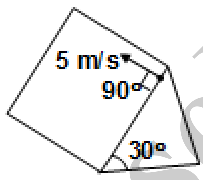
SECTION : (A) - Single Correct Options

152. A particle moves over the sides of an equilateral triangle of side ℓ with constant speed v as shown in figure. The magnitude of average acceleration as it moves from A to C is



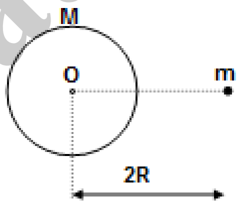
(A) $\frac{v^2}{\ell}$ (B) $\frac{\sqrt{3}}{2}\frac{v^2}{\ell}$ (C) $\frac{\sqrt{3}v^2}{\ell}$ (D) $\frac{v^2}{2\ell}$

153. A particle is given velocity 5 m/s on a fixed large inclined surface as shown in the figure. The radius of curvature of the path of the particle after 1 sec from the start is about(g = 10 m/s^2)



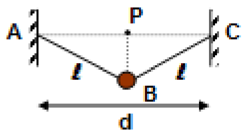
(A) 7 m (B) 14 m (C) 21 m (D) 28 m

154. A particle of mass m is at a distance 2R from the centre of a thin shell of mass M and having radius R as shown in figure. The gravitational field at the centre of shell is



(A) zero (B) $\frac{GM}{R^2}$ (C) $\frac{G(M+m)}{4R^2}$ (D) $\frac{Gm}{4R^2}$

155. A bob of mass m is in equilibrium with the help of two inextensible string connected to fixed support. The bob is slightly displaced perpendicular to the plane of figure and released. The time period of oscillation of bob is



(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{d}{g}}$ (C) $2\pi\sqrt{\frac{\ell+d}{g}}$ (D) $2\pi\sqrt{\frac{(\sqrt{4\ell^2-d^2})}{2g}}$

156. Ratio of lateral contraction strain to longitudinal elongation strain for a stretched copper wire, (assuming volume remains constant)

(A) 2 (B) 1/2 (C) 1 (D) 1/4

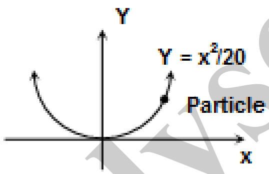
157. A small meteorite of mass m travelling toward centre of earth strikes earth at equator. Earth is uniform sphere of mass M radius R. Length of day is T seconds before meteorite strikes when meteorite strikes length of day would be increased by

(A) $\frac{m}{M}T$ sec (B) $\frac{5m}{2M}T$ sec (C) $\frac{2mT}{5M}$ sec (D) $\frac{M}{5mT}$ sec

158. A cube is floating in water having edge a with $\frac{a}{3}$ length dipped inside. A liquid of relative density ρ if filled till the block is completely submerged. The force of buoyancy acting on cube in second case in comparison to first case will

(A) remain same (B) increase (C) decrease (D) can't tell

159. The coefficient of static friction between the parabola surface and the particle is 0.8. If the equation of parabolic surface is $y = \frac{x^2}{20}$, then find the maximum x-coordinate the particle can have so that the particle does not slide down.



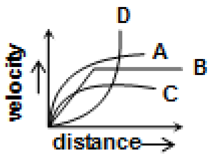
(A) 8 m (B) $4\sqrt{10}$ m (C) 12 m (D) 160 m

160. A body is dropped from a height h. The magnitude of angular momentum of the body, about a fixed point in space, during the fall of the body may

(A) Increase continuously (B) remains constant

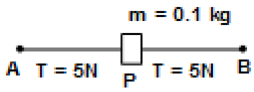
(C) Decrease continuously (D) Any of (A) or (B) may be true.

161. A small spherical ball is dropped in a viscous liquid. The graph of velocity verses distance is given by



(A) curve A (B) curve B (C) curve C (D) curve D

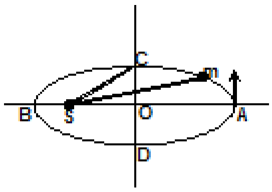
162. A particle of mass 0.1 kg is attached to a light wire which is stretched tightly between two fixed points A and B with tension 5N in each.



The whole system is kept on a horizontal frictionless surface. If AP = BP = $(1/\pi^2)$m and a very small transverse displacement is given to m so that it executes SHM. The frequency of oscillation is

(A) 5 Hz (B) 10 Hz (C) 20 Hz (D) None

163. A hypothetical planet having mass m is moving on an elliptical path around sun [mass of sun (M) $\gg$ m], which is situated at one of the focus as shown in the figure. The speed of planet, when it is at C, is [SC = r and AB = 2r]

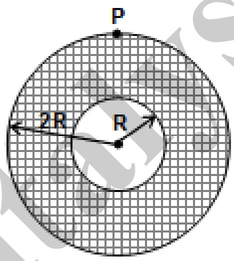


(A) $\sqrt{\frac{GM}{r}}$ (B) $\sqrt{\frac{3GM}{r}}$ (C) $2\sqrt{\frac{GM}{r}}$ (D) none of the above

164. Two satellites S_1 and S_2 revolve around a planet in coplanar circular orbits in the same sense. Their orbital radii are 10^4 km and 4×10^4 km respectively. The time period for S_1 is 1 hr. The absolute angular speed of S_2 as observed by an astronaut in S_1, when S_2 is closest to S_1, is

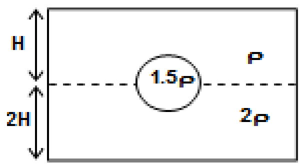
(A) $\frac{\pi}{3}$ rad/hr (B) $\frac{7\pi}{4}$ rad/hr (C) $\frac{9\pi}{4}$ rad/hr (D) π rad/hr

165. From a uniform circular disc of radius 2R a concentric disc of radius R is removed. The mass of the remaining portion is M. The disc is suspended through a small pin hole at point P as shown in the figure. Its time period of small oscillation will be



(A) $\pi\sqrt{\frac{13R}{g}}$ (B) $\pi\sqrt{\frac{5R}{g}}$ (C) $2\pi\sqrt{\frac{R}{g}}$ (D) $\pi\sqrt{\frac{7R}{g}}$

166. A cylindrical vessel with base area 'A' has two liquids with densities ρ and 2ρ. A solid sphere of density 1.5ρ is in equilibrium at the interface of liquid with half its volume immersed in one liquid as shown in the figure. If the sphere has volume V and heights of liquids are H and 2H, the force exerted on the bottom of the vessel is



(A) $\frac{(3V+10HA)\rho g}{2}$ (B) $5\rho gHA$ (C) $\frac{3}{2}V\rho g$ (D) data insufficient

167. A body weighed with a spring balance in a train at rest shows a weight W_0. When the train begins to move with a velocity v around the equator from west to east and if the angular velocity of earth is ω, then the weight recorded in the spring balance is approximately

(A) $W_0\left(1+\frac{2v\omega}{g}\right)$ (B) $W_0\left(1-\frac{2v\omega}{g}\right)$ (C) $W_0\left(1-\frac{v\omega}{g}\right)$ (D) $W_0\left(1+\frac{v\omega}{g}\right)$

168. A vessel of uniform cross-sectional area A is filled with liquid. At a depth h below the free surface of the liquid is a hole of cross sectional area a. The velocity with which the liquid comes out of the hole is

(A) $\sqrt{\frac{2ghA}{A-a}}$ (B) $\sqrt{2gh}$ (C) $\sqrt{\frac{2gh}{A^2+a^2}}$ (D) $A\sqrt{\frac{2gh}{A^2-a^2}}$

169. Drops of liquid of density σ are floating half immersed in an immissible liquid of density ρ. The surface tension of liquid is T then the radius of the drops are

(A) $\sqrt{\frac{3T}{g(2\sigma+\rho)}}$ (B) $\sqrt{\frac{3T}{g(2\sigma-\rho)}}$ (C) $\sqrt{\frac{3T}{g(\sigma-2\rho)}}$ (D) $\sqrt{\frac{3T}{2g(\sigma-\rho)}}$

170. A solid sphere, made from a material of specific gravity 27, has a concentric spherical cavity and just sinks in water. Then, the ratio of radius of the cavity to that of outer radius of the sphere must be

(A) $\frac{(13)^{1/3}}{3}$ (B) $\frac{(26)^{1/3}}{3}$ (C) $\frac{(28)^{1/3}}{3}$ (D) $\frac{(9)^{1/3}}{3}$

171. Two immiscible liquids of densities ρ_1 and ρ_2 ($>\rho_1$) are mixed in equal quantities and filled into a tank upto height h. The tank has a small hole drilled at the bottom of the right hand wall. The velocity of efflux is

(A) $\sqrt{\left(1+\frac{\rho_2}{\rho_1}\right)gh}$ (B) $\sqrt{\left(1+\frac{\rho_1}{\rho_2}\right)gh}$ (C) $\sqrt{\rho_1 gh}$ (D) $\sqrt{\rho_2 gh}$

172. A scalene triangular lamina of uniform mass density and negligible thickness has one of its vertices at the origin. The position vectors of its other two vertices are $\vec{a}$ and $\vec{b}$. The location of its centre of mass will be

(A) $\frac{\vec{a}+\vec{b}}{2}$ (B) $\frac{\vec{a}+\vec{b}}{3}$ (C) $\frac{2(\vec{a}+\vec{b})}{3}$ (D) $\frac{3(\vec{a}+\vec{b})}{4}$

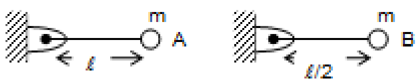
173. A particle moves with a constant speed u along the curve y = sin x. The magnitude of its acceleration at the point corresponding to x = $\pi/2$ is

(A) $\frac{u^2}{2}$ (B) $\frac{u^2}{\sqrt{2}}$ (C) u^2 (D) $\sqrt{2}\,u^2$

174. A circular disc of mass m and radius R is rotating on a rough surface having a coefficient of friction with an initial angular velocity ω. Assuming a uniform normal reaction on the entire contact surface, the time after which the disc comes to rest is

(A) $\frac{\omega R}{\mu g}$ (B) $\frac{3\omega R}{4\mu g}$ (C) $\frac{1}{2}\frac{\omega R}{\mu g}$ (D) $\frac{\sqrt{3}}{2}\frac{\omega R}{\mu g}$

175. Consider the two bobs as shown in the figure. The bobs are pivoted to the hinges through massless rods. If t_A be the time taken by the bob A to reach the lowest position and t_B be the time taken by the bob B to reach the lowest position. (Both bobs are released from rest from a horizontal position) then the ratio t_A/t_B is



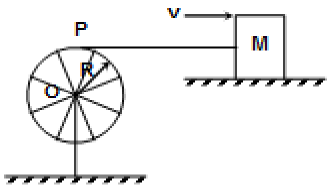
(A) $\sqrt{3}$ (B) $\sqrt{5}$ (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

176. A uniform solid cube of mass M has edge length a. The moment of inertia of the cube about its face diagonal will be

(A) $\frac{\sqrt{3}}{2}Ma^2$ (B) $\frac{1}{2}Ma^2$ (C) $\frac{5}{12}Ma^2$ (D) $\frac{7}{12}Ma^2$

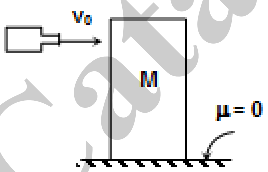
SECTION : (B) - More Than One Correct Options

177. A circular ring of mass M is hinged (by spokes of negligible mass and radius R) at the centre O on a firm vertical pillar and it can rotate freely about O. An inextensible, massless string has one end connected to the circumference of the ring at point P. Other end of the string is connected to a block of mass M at its centre of mass as shown. Initially the string is slack. The block is given sharp impulse such that it moves with speed v on a frictionless surface away from the ring. The string gets an impulse J.



(A) The impulse received by the string is Mv/2.
(B) Angular speed of rotation of ring is v/2R
(C) Loss of energy in the process is $\frac{1}{4}Mv^2$
(D) Work done by the impulse on the block is $-\frac{3}{8}Mv^2$

178. A block of mass M is kept stationary on a frictionless floor. A jet of water starts colliding horizontally at time t = 0 on the block with speed v_0. The area of C.S. of jet is a and the density of liquid is ρ. After colliding with the block, water always falls on the ground parallel to the vertical surface of the block. Assume the speed of the block is v at any time t. The acceleration of the block at that instant is



(A) $\frac{\rho}{M}(v_0 - v)av_0$ (B) $\frac{\rho a v_0}{M}(v_0 + v)$ (C) $\frac{\rho a v_0^2}{M}e^{-\frac{\rho a v_0 t}{M}}$ (D) $\frac{\rho a v_0^2}{M}t$

179. A brass ring of density ρ_r and mass m_r is tied with a piece of cork of mass m_C and density ρ_C. This arrangement floats completely immersed in an liquid of density ρ_ℓ.

(A) mass of liquid displaced is $\frac{1}{2}\left(\frac{m_r}{\rho_r} + \frac{m_C}{\rho_C}\right)\rho_C$

(B) Ratio of mass of cork and that of ring is $\left(\frac{1 + \rho_\ell/\rho_C}{1 + \rho_\ell/\rho_r}\right)$

(C) mass of liquid displaced is given by $\left(\frac{m_r}{\rho_r} + \frac{m_C}{\rho_C}\right)\rho_1$

(D) radio of mass of cork and that of ring is $\frac{(1 - \rho_\ell/\rho_r)}{(\rho_\ell/\rho_C - 1)}$

180. Surface tension of water in a pool is s = 7.2×10^{-2} Nm^{-1}. An insect or mass m = 10^{-6} kg rests on the free surface of water. Spherical base of its each of 6 legs has radius r = 1×10^{-5} m. Each leg shares equal weight of the insect.

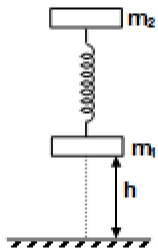
(A) Vertical component of force acting on each leg when q be the angle of contact is $2\pi rs \cos\theta$.

(B) Angle of contact, $\cos\theta = \dfrac{mg}{12\pi rs}$

(C) Q = 57°

(D) Q = 0.54°

181. The shown system is relaxed when at a height h form the ground. Coefficient of restitution between m_1 and ground is nil.



(A) When released the change in length of spring is given by $(m_1 + m_2)g/k$

(B) When released the change in length of spring is given by m_1g/k

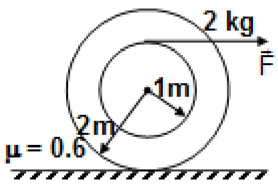
(C) maximum value of h so that system has a tendency to set rebound when released is $\dfrac{m_2g}{k}\left(\dfrac{m_1+m_2}{m_2}\right)$

(D) minimum value of h so that system has a tendency to get rebound when released is given by $\dfrac{m_1g}{k}\left(\dfrac{2m_2+m_1}{2m_2}\right)$

182. A mass m kg is subjected to a constant force F kgf which cause it move in t sec to a distance x m. The velocity acquired is v m/sec. then the distance covered x is given by

(A) $\dfrac{v^2m}{2Fg}$ (B) $\dfrac{2}{3}\dfrac{v^2m}{Fg}$ (C) $\dfrac{2}{3}\dfrac{Fgt^2}{m}$ (D) $\dfrac{1}{2}\dfrac{Fgt^2}{m}$

183. An annular disc of radii 2 m and 1 m, having mass 2 kg is kept on rough surface as shown in the figure and pulled by horizontal force FN. Choose the correct statement(s)



(A) Friction force acting on the body is forward direction.

(B) Friction force acting on the body is backward direction.

(C) If F = 150 N body will roll without slipping

(D) If F = 150 N body will roll with slipping

184. Suppose that the position vector function for a particle is given as

$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}$, where x(t) = t + 1 and y(t) = $\frac{t^2}{8} + 1$

choose correct statement(s)

(A) Average speed of the particle during the time interval t = 2.0 sec to 4.0 sec is 1.25 m/s

(B) Average speed of the particle during the time interval t = 2.0 sec to 4.0 sec is 2.50 m/s

(C) Speed of the particle at t = 2.0 sec is $\frac{\sqrt{5}}{2}$ m/s

(D) Speed of the particle at t = 2.0 sec is $\sqrt{5}$ m/s

185. A heavy particle is projected from a point at the foot of a fixed plane, inclined at an angle 45° to the horizontal, in the vertical plane containing the line of greatest slope through the point.

If ϕ (> 45°) is the inclination to the horizontal of the initial direction of projection then

(A) Particle will strike the plane horizontally if tan ϕ = 2

(B) Particle will strike the plane at right angle if tan ϕ = 3.

(C) Particle will strike the plane at right angle if tan ϕ = 2

(D) Particle will strike the plane horizontally if tan ϕ = 4

186. In a one dimensional collision between two particles, their relative velocity is $\vec{v}_1$, before the collision and $\vec{v}_2$ after the collision.

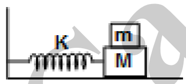
(A) $\vec{v}_1 = \vec{v}_2$, if the collision is elastic

(B) $\vec{v}_1 = -\vec{v}_2$, if the collision is elastic

(C) $\vec{v}_1 = -k\vec{v}_2$ in all cases, where k ≥ 1

(D) $|\vec{v}_2| = |\vec{v}_1|$ in all cases

187. The friction coefficient between the two blocks shown in the figure is μ and the horizontal plane is smooth. The system is slightly displaced horizontally and released. Which of the following statements is/are true?



(A) The time period of small oscillation is $2\pi\sqrt{\frac{M+m}{K}}$

(B) The magnitude of the frictional force between the blocks, when the displacement from the mean position is x, is $\frac{mk|x|}{M+m}$

(C) If the upper block does not slip relative to the lower block then the value of maximum amplitude is $\frac{\mu(M+m)g}{K}$

(D) If the upper block does not slip relative to the lower block then the value of maximum amplitude is $\frac{\mu m(M+m)g}{MK}$

188. A body totally immersed in water by a height h. The density of the body is d and the density of water is d_0 while the volume of the body is V and d > d_0. Which of the following statements will be true?

(A) the net work done on raising the body is Vdgh.

(B) the increase in the potential energy of the body is vgh (d – d_0)

(C) the potential energy of water must increase by raising the body

(D) the work done on the body by gravitational force is equal and opposite to the work done by the hydrostatic forces

189. Which of the following statements are true?

(A) when two quantities are subtracted, the absolute error in the final result is the difference of the absolute errors in the individual quantities taken in same order.

(B) when two quantities are multiplied, the fractional error or the relative error in the result is the sum of the relative errors of the two quantities.

(C) when two quantities are divided, the fractional error or the relative error in the result is the difference of the relative errors of the two quantities.

(D) When two quantities are multiplied, the percentage uncertainty of the final result is equal to the square of the sum of the squares of the percentage uncertainties of the original number.

190. A satellite revolves around a planet in circular orbit of radius R (much larger than the radius of the planet) with a time period of revolution T. If the satellite is stopped and then released in its orbit (Assume that the satellite experiences gravitational force due to the planet only).

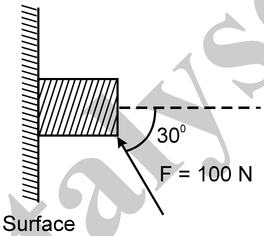
(A) It will fall into the planet

(B) The time of fall of the satellite is nearly $\frac{T}{\sqrt{8}}$

(C) The time of fall of the satellite into the planet is nearly $\frac{\sqrt{2}T}{8}$

(D) It cannot fall into the planet so time of fall of the satellite is meaningless

191. A force of 100 N is applied on a stationary block of mass 3kg as shown in figure. If the coefficient of friction between the surface and the block is 0.25 then :



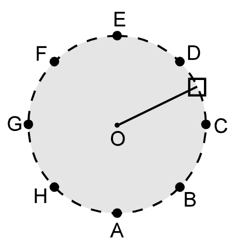
(A) The frictional force acting on the block is 20N downwards

(B) The normal reaction on the block is $50\sqrt{3}$.

(C) The friction force (kinetic) acting on the block is $\frac{25\sqrt{3}}{2}$N

(D) If coefficient of friction is changed to 0.35 then the friction force acting on the block is again 20 N downwards.

192. A machine in an amusement park, consist of a cage at the end of one arm hinged at O. The cage revolves along a vertical circle of radius r (ABCDEFGH) about its hinge O at constant linear speed $v = \sqrt{gr}$. The cage is so attached that the man of weight 'w' standing on a weighing machine inside the cage is always vertical then



(A) the reading of his weight on the machine is same at all positions

(B) the weight reading at A is greater than the weight reading at E by 2 w

(C) the weight reading at G = w

(D) the ratio of the weight reading at E to that at A = 0

(E) the ratio of the weight reading at A to that at C = 2.

193. In a tug of war, the team that exerts a larger tangential force on the ground wins. Consider the period in which a team is dragging the opposite team by applying a larger tangential force on the ground. Which of the following works are negative?

(A) work by the losing team on the winning team

(B) work by the ground on the winning team

(C) work by the ground on the losing team

(D) total external work on the two teams.

194. Which of the following is **incorrect** ?

(A) Total mechanical energy is always conserved.

(B) Work done by kinetic friction is always negative.

(C) Every conservative force is a constant force.

(D) Work done by all forces on a rigid body is the change in its kinetic energy.

195. A rigid body is in pure rotation.

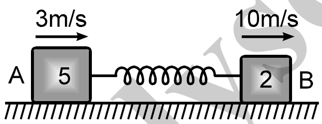
(A) You can find two points in the body in a plane perpendicular to the axis of rotation having same velocity.

(B) You can find two points in the body in a plane perpendicular to the axis of rotation having same acceleration.

(C) Speed of all the particles lying on the curved surface of a cylinder whose axis coincides with the axis of rotation is same.

(D) Angular speed of the body is same as seen from any point in the body.

196. Two blocks A (5kg) and B(2kg) attached to the ends of a spring constant 1120N/m are placed on a smooth horizontal plane with the spring undeformed. Simultaneously velocities of 3m/s and 10m/s along the line of the spring in the same direction are imparted to A and B then



(A) when the extension of the spring is maximum the velocities of A and B are zero.

(B) the maximum extension of the spring is 25cm.

(C) the first maximum compression occurs $3\pi/56$ seconds after start.

(D) maximum extension and maximum compression occur alternately.

197. A mass of 0.2kg is attached to the lower end of a massless spring of force-constant 200 N/m, the upper end of which is fixed to a rigid support. Which of the following statements is/are true?

(A) In equilibrium, the spring will be stretched by 1cm.

(B) If the mass is raised till the spring is unstretched state and then released, it will go down by 2cm before moving upwards.

(C) The frequency of oscillation will be nearly 5 Hz.

(D) If the system is taken to the moon, the frequency of oscillation will be the same as on the earth.

198. A particle is executing SHM between points $-X_m$ and X_m, as shown in figure-I. The velocity V(t) of the particle is partially graphed and shown in figure-II. Two points A and B corresponding to time t_1 and time t_2 respectively are marked on the V(t) curve.

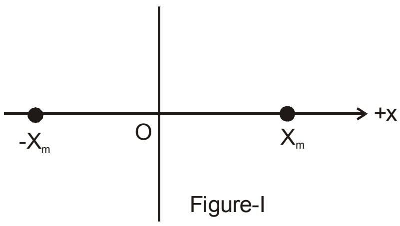


Figure-I

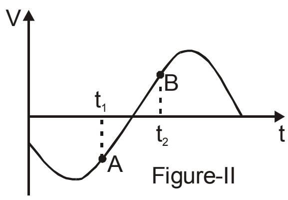


Figure-II

(A) At time t_1 , it is going towards X_m.

(B) At time t_1, its speed is decreasing.

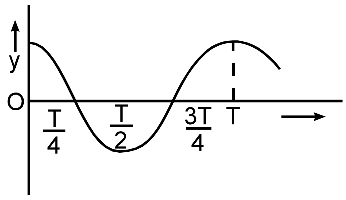
(C) At time t_2, its position lies in between $-X_m$ and O.

(D) The phase difference $\Delta\phi$ between points A and B must be expressed as $90° < \Delta\phi < 180°$.

199. A particle moves on the X-axis according to the equation $x = x_0 \sin^2 \omega t$. The motion is simple harmonic

(A) with amplitude x_0 (B) with amplitude $x_0/2$ (C) with time period $\frac{2\pi}{\omega}$ (D) with time period $\frac{\pi}{\omega}$

200. The displacement time graph of a particle executing S.H.M. is shown. Which of the following statements is/are true ?



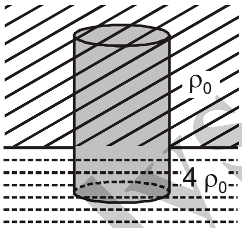
(A) the velocity is maximum at t = T/2

(B) the acceleration is maximum at t = T

(C) the force is zero at t = 3T/4

(D) the potential energy equals the total oscillation energy at t = T/2

201. A uniform solid cylinder of length l submerged partially in two immiscible liquids is in equilibrium as shown. If one-third length of the cylinder is submerged in liquid of density $4\rho_0$



(A) Density of the cylinder is $2\rho_0$

(B) Pressure difference between top and bottom of the cylinder is $2\rho_0 gl$

(C) For slight vertical displacement from the position shown, angular frequency of oscillation is $\sqrt{\frac{3g}{2l}}$

(D) For slight vertical displacement from the position shown, angular frequency of oscillation is $\sqrt{\frac{5g}{2l}}$

202. Consider a thin spherical shell of mass m and radius r. A particle of mass m is located at a distance $\frac{r}{2}$ from its centre. Choose the correct alternative(s).

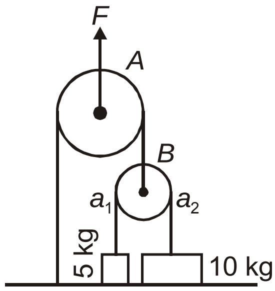
(A) Gravitational force by the particle on the shell is zero

(B) Gravitational interaction energy of the particle and shell is $\frac{-Gm^2}{r}$

(C) Gravitational potential energy of the system is $\frac{-Gm^2}{2r}$

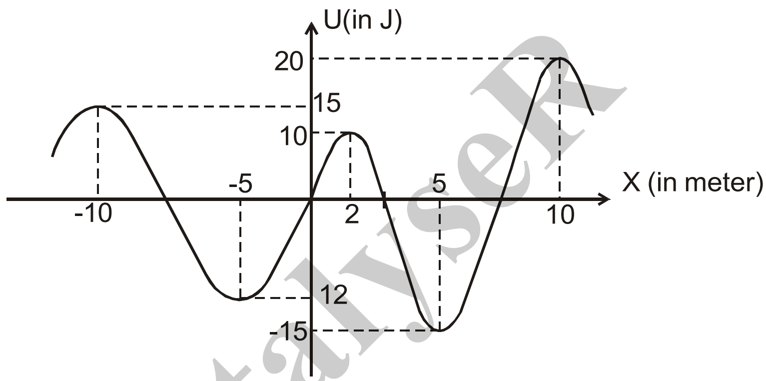
(D) Work done by external agent if the particle is moved to the shell centre may be zero or positive

203. The acceleration of block of masses 5 kg and 10 kg are a_1 and a_2 respectively, choose the correct alternative



(A) Both a_1 and a_2 are zero if F = 100 N
(B) a_1 = 5 m/s^2 and a_2 = 0 if F = 300 N
(C) a_1 = 15 m/s^2 and a_2 = 2.5 m/s^2 if F = 500 N
(D) Accelerations of blocks independent of F

204. In the figure the variation of potential energy of a particle of mass m = 2kg is represented w.r.t. its x-coordinate. The particle moves under the effect of this conservative force along the x-axis. Which of the following statements about the particle is true :



(A) If it is released at the origin it will move in negative x-axis.
(B) If it is released at x = 2 + Δ where $\Delta \to 0$ then its maximum speed will be 5 m/s and it will perform oscillatory motion
(C) If initially x = – 10 and $\vec{u} = \sqrt{6}\,\hat{i}$ then it will cross x = 10
(D) x = – 5 and x = + 5 are unstable equilibrium positions of the particle

205. A ball is dropped from a height h. At the same time another ball is thrown vertically up along the same line with a velocity u:

(A) they will strike in air if, u > $\sqrt{\dfrac{gh}{2}}$

(B) the maximum time elapse in between the collision will be $\sqrt{\dfrac{2h}{g}}$

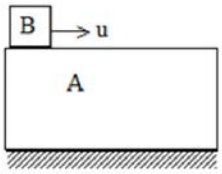
(C) they will strike in air if u < $\sqrt{\dfrac{gh}{2}}$

(D) if both the particles are travelling along parallel line they may meet twice

206. A tank is filled up to a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m, a small hole is punched at a distance y from the free surface of the liquid. Then

(A) $x_m = 2h$ (B) $x_m = 1.5h$ (C) $y = h$ (d) $y = 0.75$ h

207. A long block A is at rest on a smooth horizontal surface. A small block B, whose mass is half of mass of A, is placed on A at one end and projected along the surface of A with some velocity u. The coefficient of friction between the blocks is μ.



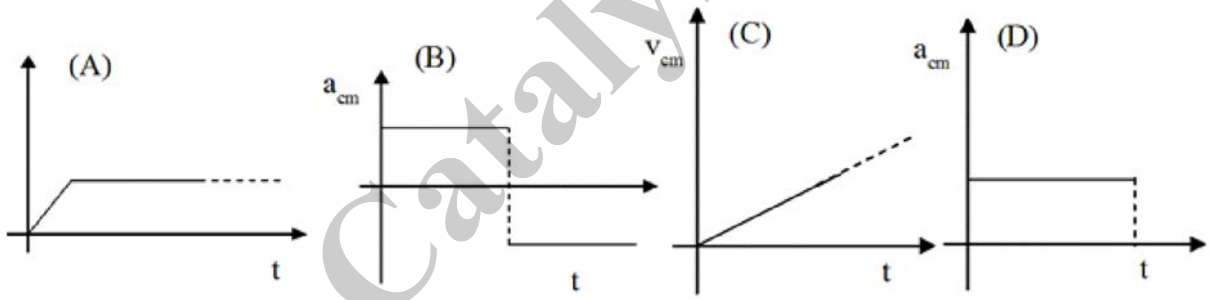
(A) The blocks will reach a final common velocity $\frac{u}{3}$.

(B) The work done against friction is two thirds of the intitial kinetic energy B.

(C) Before the blocks reach a common velocity the acceleration of A relative to B is $\frac{2}{3}\mu g$.

(D) Before the blocks reach a common velocity the acceleration of A relative to B is $\frac{3}{2}\mu g$.

208. A uniform disc of radius r is rotated clockwise with angular speed ω and kept vertically on rough surface. If v_{cm}, a_{cm} are velocity and acceleration of centre of mass, then which of the following graphs holds true?



SECTION : (C) -Passage Type Questions

PASSAGE 01

The concept of a black hole is one of the most interesting products of modern gravitational theory, yet the basic idea can be understood on the basis of Newtonian principles.

Using the theory of gravitation, we know that escape velocity from the surface of a star is given as

$v = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{8\pi G\rho R}{3}}$ (expression for escape velocity)

If we find the escape velocity from surface of the sun, it comes out to be about 6.18×10^5 m/s. This value is roughly $\frac{1}{500}$ times the velocity of light. Now consider various stars with same average density ρ as that of the sun and different radii R. The equation of escape velocity suggests that for a given value of density ρ, the escape speed v is directly proportional to R. In 1783, Rev. John Mitchell, an amateur astronomer, noted that if a body with same average density as sun had about 500 times the radius of the sun, the magnitude of its escape velocity would be greater than the speed of light c. With his statement " All light emitted from such a body would be made to return towards it", Mitchell became the first person to suggest the existence of what we now call a " black hole".

The expression for escape speed also suggests that a body of mass M will act as a black hole if its radius is less than or equal to a certain critical radius.

How can we determine this critical radius? You might think that you can find the answer by simply setting v = c in the expression for escape velocity. As a matter of fact, this does give the correct result but only because of two compensating errors. Kinetic energy of light is not ½ mc^2 and the gravitational potential energy near a black hole is not –GMm/r. An 1916, Karl Schwarzschild used Einstein's general theory of relativity (in part a generalization and extension of Newtonian gravitation theory) to derive an expression for the critical radius R_s, now called the Shwarzschild radius, which is given as $R_S = \frac{2GM}{c^2}$. The surface of the sphere with radius R_s surrounding a black hole is called the event horizon. Since light cannot escape from within that sphere, we can't see the events occurring inside.

If light cannot escape from a black hole, how can we know such things exist? The answer is that any gas or dust near the black hole tends to be pulled into an accretion disc that swirls around and in to the black hole, rather like a whirlpool. While there are some black holes with masses of the order of a few times the solar mass, there is also an evidence for super massive black holes.

One example is thought to lie at the centre of the milky was Galaxy, some 26000 light years away from earth in the direction of constellation Sagittarius. High resolution images of Galactic centre reveal stars moving at speeds greater than 1500 km/s about an unseen object that lies at the position of a source of radio waves called SgrA*. By analysing these motions astronomers can infer the period t and semi-major axis of each star's orbit. The mass m_x of the unseen object can now be calculated using Kepler's third law $T = \frac{2\pi a^{3/2}}{\sqrt{2Gm_x}}$. At points far away from a black hole, its gravitational effects are same as those of any normal body with same mass.

209. An object of mass m is located at a distance r form the centre of a black hole with Schwarzschild radius R_s. The attractive force exerted by the black hole on the body is

(A) $\frac{mc^2R_s}{r^2}$ (B) $\frac{2mc^2Rs}{r^2}$ (C) $\frac{\sqrt{2}mc^2R_s}{r^2}$ (D) $\frac{mc^2R_s}{2r^2}$

210. Astronomers have observed a small massive object at the centre of our Milky way galaxy. A ring of material orbits this massive object; the ring has a diameter of about 15 light years and an orbital speed of 200 km/s. What is the mass of the massive object at the centre of milky way? Given G = 6.67 × 10^{-11} Nm^2 kg^{-2} and 1 light year = 9.5 × 10^{15} m

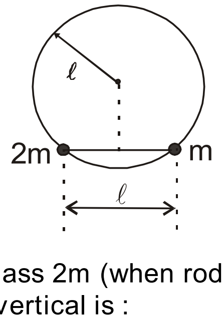
(A) Approximately 8.1 × 10^{37} kg (B) approximately 4.3 × 10^{37} kg
(C) Approximately 6 × 10^{37} kg (D) approximately 3 × 10^{37} kg

211. Many astronomers believe that the massive object at the centre of the Milky way galaxy (same as the one in previous question) is a black hole. If so, what must the Schwarzschild radius of this black hole be?

(A) 6.4 × 10^{10} m approx (B) 3.2 × 10^{10} m approx
(C) 9.6 × 10^{10} m approx (D) 3.2 × 10^{9} m approx

PASSAGE 02

Two beads of mass 2m and m, connected by a rod of length ℓ and of negligible mass are free to move in a smooth vertical circular wire frame of radius ℓ as shown. Initially the system is held in horizontal position (Refer figure)

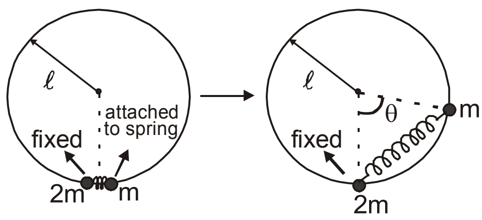


212. The velocity that should be given to mass 2m (when rod is in horizontal position) in counter-clockwise direction so that the rod just becomes vertical is :

(A) $\sqrt{\frac{5g\ell}{3}}$ (B) $\sqrt{\left(\frac{3\sqrt{3}-1}{3}\right)g\ell}$ (C) $\sqrt{\frac{3}{2}g\ell}$ (D) None of these

213. If the rod is replaced by a massless string of length ℓ and the system is released when the string is horizontal then :
(A) Mass 2m will arrive earlier at the bottom.
(B) Mass m will arrive earlier at the bottom.
(C) Both the masses will arrive together but with different speeds.
(D) Both the masses will arrive together with same speeds.

214. The string is now replaced by a spring of spring constant k and natural length ℓ. Mass 2m is fixed at the bottom of the frame. The mass m which has the other end ofthe spring attached to it is brought near the mass 2m and released as shown in figure. The maximum angle θ that the spring will substend at the centre will be : (Take k = 10 N/m, ℓ = 1 m, m = 1 kg and ℓ = r)



(A) 60° (B) 30° (C) 90° (D) None of these

PASSAGE 03

The velocity of a particle is varying with time according to the relation $v = v_0 \sin \omega t$ m/s
Where, v_0 ω are constants.

Considering the motion from time t = 0 till $t = \frac{\pi}{\omega}$, answer the following:

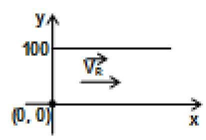
215. The acceleration of the particle follows a
(A) sinusoidal curve (B) straight line (C) semi-circular path (D) none of the above

216. The total distance travelled by the particle during this time is:
(A) $\frac{2v_0\pi}{\omega}$ (B) $\frac{2v_0}{\omega}$ (C) $\frac{v_0}{\omega}$ (D) $\frac{v_0}{2\omega}$

217. The acceleration of the particle at $t = \frac{\pi}{2\omega}$ is
(A) $v_0\omega$ (B) 0 (C) $2v_0\omega$ (D) $\pi v_0\omega$

PASSAGE 4

A river is flowing with a speed of 10 m/s along the x-axis as shown. The river is 100 m wide. A man in a boat starts from one bank of the river at the point (0, 0), (origin) and crosses the river with speed 20 m/s. One river bank is along the x-axis and the other is parallel to the x-axis along y = 100



218. For the boat to reach the point (0, 100) directly, the man should row the boat with a velocity:
(A) $10(\hat{i} + \sqrt{3}\hat{j})$ (B) $10(-\hat{i} + \sqrt{3}\hat{j})$ (C) $10(-\sqrt{3}\hat{i} + \hat{j})$ (D) $10(\sqrt{3}\hat{i} + \hat{j})$

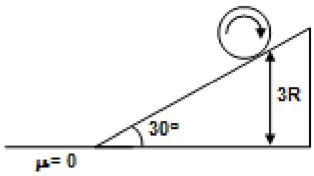
219. If the man rows the boat in such a way that the direction of velocity of boat is always perpendicular to the river velocity, then the man lands at the point:
(A) (50, 50) (B) (100, 100) (C) (50, 100) (D) (5, 100)

220. The minimum time required to cross the river is

(A) 5 sec (B) 10 sec (C) $5\sqrt{3}$ sec (D) none of these

PASSAGE 05

A cylinder of mass m and radius R is rotated about its axis with angular velocity ω_0 (as shown in the figure) and lowered on a rough inclined plane at an angle 30° with horizontal and $\mu = \frac{1}{\sqrt{3}}$. The point of initial contact of cylinder and incline is at a height of 3 R from horizontal



221. Find the time when cylinder comes at rest.

(A) $\frac{R\omega_0}{g}$ (B) $\frac{2R\omega_0}{g}$ (C) $\frac{R\omega_0}{2g}$ (D) none of these

222. Find the time t when cylinder reach the bottom of the incline.

(A) $\frac{2R\omega_0}{g}+6\sqrt{\frac{R}{g}}$ (B) $\frac{2R\omega_0}{g}+3\sqrt{\frac{R}{g}}$ (C) $\frac{R\omega_0}{g}+6\sqrt{\frac{R}{g}}$ (D) $\frac{R\omega_0}{2g}+6\sqrt{\frac{R}{g}}$

223. Find the work done by friction during t = 0 to t = t sec.

(A) $\frac{mR^2\omega^2}{2}$ (B) $\frac{mR^2\omega^2}{4}$ (C) $mR^2\omega^2$ (D) $\frac{mR^2\omega^2}{3}$

PASSAGE 06

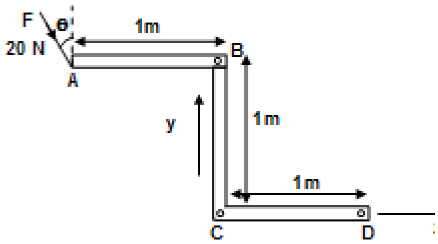
Torque of a force $\vec{F}$ about a point O is defined as $\vec{\tau} = \vec{r}_0 \times \vec{F}$, where $\vec{r}_0$ is the position vector of the point of application of force with respect to point O. The torque of force $\vec{F}$ about an axis having unit vector $\hat{p}$ passing through O will be the component of torque $\vec{\tau}_0$ along the axis.

$$\vec{\tau}_{Axis} = (\vec{\tau}\cdot\hat{p})\hat{p}$$

Hence $\vec{\tau}$ about an axis is zero if the axis is parallel to the force and torque about a point is zero if the line of action of force, when produced intersect the axis or passes through the point O. It can also be concluded that magnitude of torque about an axis is equal to the product of the magnitude of force and the length of the common perpendicular on the line of action of force as well as the axis

If A z-frame of rod of uniform mass density is kept in the x-y –plane as shown in the figure. Force F also lies in the x-y plane.

224. Choose the correct statement ($\vec{\tau}_B, \vec{\tau}_C, \vec{\tau}_D$ are torque about point B, C and D respectively)



(A) $\vec{\tau}_B = \vec{\tau}_C$ (B) $\vec{\tau}_B = \vec{\tau}_D$ (C) $\vec{\tau}_C = \vec{\tau}_D$ (D) none of these

225. If $\theta = 0$, the magnitude of torque of force about centre of mass of system is

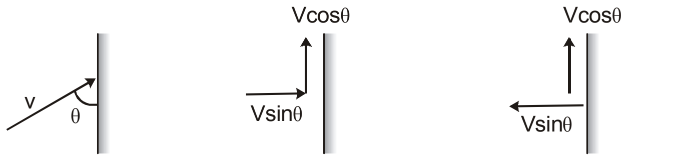
(A) 20 Nm (B) 40 Nm (C) 10 Nm (D) zero

226. If $\theta = 90°$, the magnitude of torque of force about an axis passing through centre of mass of system and parallel to the x-axis is

(A) 20 Nm (B) 40 Nm (C) 10 Nm (D) zero

PASSAGE 07

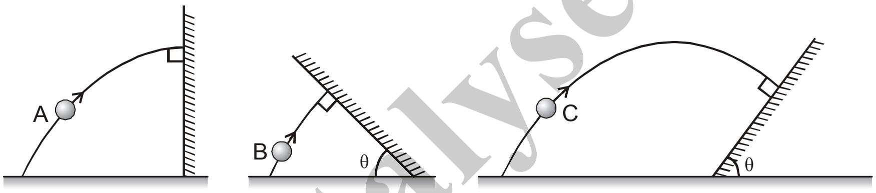
We know how by neglecting the air resistance, the problems of projectile motion can be easily solved and analysed. Now we consider the case of the collision of a ball with a wall. In this case the problem of collision can be simplified by considering the case of elastic collision only. When a ball collides with a wall we can divide its velocity into two components, one perpendicular to the wall and other parallel to the wall. If the collision is elastic then the perpendicular component of velocity of the ball gets reversed with the same magnitude.



Velocity just before collision | Components of velocity just before collision | Components of velocity just after collision

The other parallel component of velocity will remain constant if wall is given smooth.

Now let us take a problem. Three balls 'A' and 'B' & 'C' are projected from ground with same speed at same angle with the horizontal. The balls A,B and C collide with the wall during their flight in air and all three collide perpendicularly with the wall as shown in figure.



227. If the time taken by the ball A to fall back on ground is 4 seconds and that by ball B is 2 seconds. Then the time taken by the ball C to reach the inclined plane after projection will be :

(A) 6 sec. (B) 4 sec. (C) 3 sec. (D) 5 sec.

228. The maximum height attained by ball B from ground is :

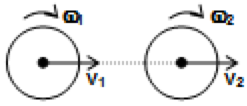
(A) 20 m (B) 5 m (C) 15 m (D) None of these

229. The vertical component of velocity of balls with which they are projected :

(A) 20 m/s (B) 10 m/s (C) $10\sqrt{3}$ m/s (D) Undeterminable

PASSAGE 08

Two disks of equal mass are kept in a horizontal plane on a smooth horizontal surface. The linear and angular speeds of disk 1 and disk 2 are v_1, w_1 and v_2, w_2 respectively as shown ($v_1 > v_2$). All surfaces are smooth and coefficient of restitution between disk 1 and 2 is 1



230. Let v_1' and v_2' be the final linear speeds after collision of the disks 1 and 2 respectively. Then,

(A) $v_1 = v_1'$ (B) $v_2 = v_2'$ (C) $v_1 v_1' = v_2 v_2'$ (D) $v_1 v_2' = v_2 v_1'$

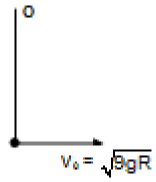
231. Let ω_1' and ω_2' be the final angular speeds after collision of the disks 1 & 2 respectively, then

(A) $\omega_1 = \omega_2'$ (B) $\omega_2 = \omega_1'$ (C) $\omega_1 \omega_1' = \omega_2 \omega_2'$ (D) $\omega_1 \omega_2' = \omega_2 \omega_1'$

232. The angular momentum of the second disk is conserved about
(A) only the center of the disk
(B) no point on the disk
(C) only the center and point of contact of disks during collision
(D) none of the above

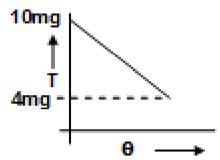
PASSAGE 09

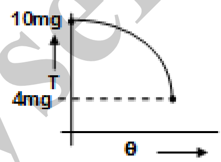
A ball is projected with horizontal velocity $\upsilon_0 = \sqrt{9gR}$ at the bottom most point attached with inextensible string of length R & fixed at O as shown. Give the answer of following questions

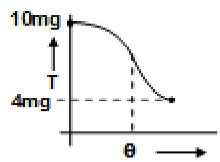


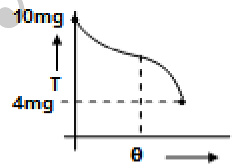
233. Tension in the string in horizontal position
(A) 10 mg (B) 7 mg (C) mg (D) 8 mg

234. Graph between tension vs angle θ rotated from shown position is best represented by
(for $\theta \in [0, \pi]$)

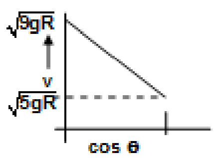
(A)


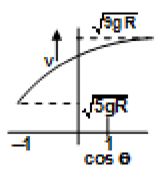
(B)


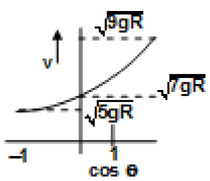
(C)


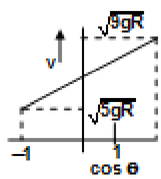
(D)


235. Graph between v vs cos θ is best represented by for θ ε [0, π]

(A)


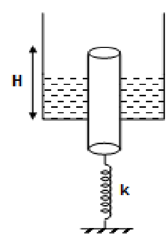
(B)


(C)


(D)


PASSAGE 10

A smooth vertical cylindrical rod just fits the hole at the bottom of the container as shown in the figure. The volume of the rod submerged is V. Mass of the rod is such that ρVg = mg ⇒ m = ρV, where ρ is the density of the liquid. Let K be the spring constant of the spring. Now answer the following questions:



236. The initial compression of the spring is

(A) $\frac{\rho Vg}{k}$ (B) 0 (C) $\frac{2\rho Vg}{k}$ (D) $\frac{\rho Vg}{2k}$

237. Let T be the time period of small oscillation of the rod, then

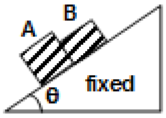
(A) $T > 2\pi\sqrt{\frac{m}{k}}$ (B) $T < 2\pi\sqrt{\frac{m}{k}}$ (C) $T = 2\pi\sqrt{\frac{m}{k}}$ (D) $T = \infty$

238. Net force applied by the liquid on the rod is

(A) mg (B) ρvg (C) 0 (D) $2\rho vg$

PASSAGE 11

In arrangement shown in figure mass of block A is m_1 and mass of block B is m_2 friction coefficient between m_1 & wedge is μ_1 and m_2 and wedge is μ_2. Answer the following questions.



239. If $m_1 = m_2 = m$ (say) and $\mu_1 > \mu_2 > \tan$ then select the correct statement

(A) contact force between the two blocks is $m\frac{(\mu_1 - \mu_2)}{2}g\tan\theta$

(B) contact force between the two blocks is $m\frac{(\mu_1 + \mu_2)}{2}g\tan\theta$

(C) contact force between the two blocks is $\left(\frac{\mu_1 - \mu_2}{2}\right)mg\sin\theta$

(D) contact force is zero

240. If $\mu_1 < \tan\theta$ and $\mu_2 > \tan\theta$ then select correct statement

(A) acceleration of both block is same and non zero

(B) acceleration of both block is zero

(C) acceleration of block A is $(g\sin\theta - \mu g\cos\theta$ and acceleration of block of B is $(g\sin - \mu_2 g\cos\theta)$.

(D) acceleration of block A is $(g\sin\theta - \mu_1 g\cos\theta)$ and acceleration of block B is zero.

SECTION : (D) - Matrix Match

241.

Column I		Column II
(a) $\int \tau dt$ about an axis	(P)	$\frac{L}{2m}$
(b) Aerial velocity	(Q)	Maximum with $\perp r$ force
(c) Torque	(R)	Pseudo force
(D) Centrifugal force	(S)	Change is angular momentum

242. In vertical circular motion, suppose v is the velocity of bob at bottommost point, then match the following:

Table-1		Table-2
(a) If $v = 2\sqrt{gR}$	(p)	Bob will complete the circle
(b) If $v = 3\sqrt{gR}$	(q)	Bob will oscillate
(C) If $v = 4\sqrt{gR}$	(r)	String will slack
(d) If $v = \sqrt{gR}$	(s)	String will break

243. In the following problem v_0 = orbital speed and v_{esc} = escape velocity

Match the following :

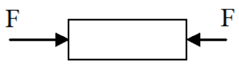
Table -1	**Table -2**	
(a) When $v < v_0$	(P)	The path of satellite is hyperbolic
(b) When $v = v_{esc} = \sqrt{2}(v_0)$	(Q)	The satellite may strike the earth
(c) When $v > v_{esc}$	(R)	The orbit of satellite is elliptical
(d) $v_{esc} > v > v_0$	(s)	The path of satellite is parabolic

244. **Match the following:**

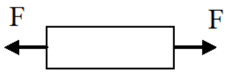
Table -1	**Table-2**	
(a) Surface tension	(P)	N/m^2
(b) coefficient of viscosity velocity	(Q)	J/m^3
(c) Modulus of elasticity	(R)	N/M
(d) Pressure	(S)	J/m^2

245. In column I, a uniform bar of uniform cross-section area under the application of forces is shown in the figure and in column II, some effects/phenomena are given. Match the entries of Column I with the entries of Column II.

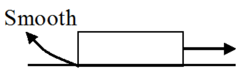
Column I **Column II**

(a) 

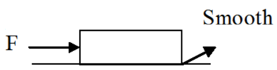
(p) Uniform stresses developed in the rod

(b) 

(q) Non-uniform stresses developed in the rod

(c) 

(r) Compressive stresses developed

(d) 

(s) Tensile stesses developed

246.

	Column I		*Column II*
A.	A body moving on a circular path	(P)	less than that at the point of projection
B.	In a projectile motion, radius of curvature at the point of projection	(Q)	friction lies between zero to limiting friction
C.	In a projectile motion, radius of curvature at the top of its motion	(R)	greater than that at the top of its motion
D.	A car moves on flat horizontal circular road with increasing speed	(S)	there must have radial acceleration

247. Four bodies each of mass m are moving on earth with equal speed u.

1^{st} one on equator along west to east, 2^{nd} one on equator along east to west,

3^{rd} one on north pole along 0° longitude, 4^{th} one on south pole along 180° longitude,

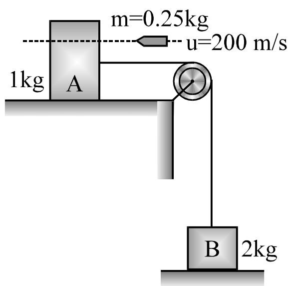
(Assume : Earth is sphere, ω = angular velocity of earth, R = radius of earth)

	Column I		*Column II*
(A)	Normal reaction on 1^{st} body due to earth	(P)	mg
(B)	Normal reaction on 2^{nd} body due to earth	(Q)	$mg + \frac{m(u+\omega R)^2}{R}$
(C)	Normal reaction on 3^{rd} body due to earth	(R)	$mg - \frac{m(\omega R - u)^2}{R}$
(D)	Normal reaction on 4^{th} body due to earth	(S)	$mg - \frac{m(\omega R + u)^2}{R}$

248. For the following statements, except gravity and contact force between the contact surfaces, no other force is acting on the body.

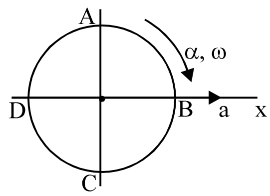
	Column A		Column B
(A)	When a sphere is in pure–rolling on a fixed horizontal surface.	(P)	Upward direction
(B)	When a cylinder is in pure rolling on a fixed inclined plane in upward direction then friction force acts in	(Q)	$v_{cm} > R\omega$
(C)	When a cylinder is in pure rolling down a fixed incline plane, friction force acts in	(R)	$v_{cm} < R\omega$
(D)	When a sphere of radius R is rolling with slipping on a fixed horizontal surface, the relation between v_{cm} and ω is	(S)	No frictional force acts.

249. A block A of mass M_A = 1 kg is kept on a smooth horizontal surface and attached by a light thread to another block B of mass M_B = 2 kg. Block B is resting on ground and thread and pulley are massless and frictionless. A bullet of mass m = 0.25 kg moving horizontally with velocity of u = 200 m/s penetrates through the block A and comes out with a velocity of 100 m/s.



	Column I		*Column II (Values are in their respective SI units)*
(a)	velocity of 2 kg block just after bullet comes out	(p)	50/3
(b)	max. disp. of 1 kg block in left direction	(q)	25
(c)	impulse by string on block B	(r)	25/3
(d)	Impulse by particle on block A	(s)	5.2

250. A disc of radius R is rolling with angular velocity ω, angular acceleration α and linear acceleration 'a', along x-direction. There are 4 points A, B, C and D on the disc as shown.



	Column I		*Column II*
(a)	Acceleration of point A	(p)	$\sqrt{(a+\omega^2 R)^2+(\alpha R)^2}$
(b)	Acceleration of point B	(q)	$\sqrt{(a+\alpha R)^2+(\omega^2 R)^2}$
(c)	Acceleration of point C	(r)	$\sqrt{(a-\alpha R)^2+(\omega^2 R)^2}$
(d)	Acceleration of point D	(s)	$\sqrt{(a-\omega^2 R)^2+(\alpha R)^2}$

251. A satellite is projected horizontally near the surface of a planet with a speed V. The value of acceleration of a freely falling body near this planet is found to be 4.9 m/s². Radius of the planet is 3200 km. For various values of V, the path of satellite can be predicted. Match the velocity of satellite with its respective path.

(Take $\sqrt{2} = 1.4$)

	Column I		***Column II***
(a)	V = 5 km/s	(P)	Hyperbola
(b)	V = 4 km/s	(Q)	Circular
(c)	V = 5.6 km/s	(R)	Ellipse
(d)	V = 6.6 km/s	(S)	Parabola

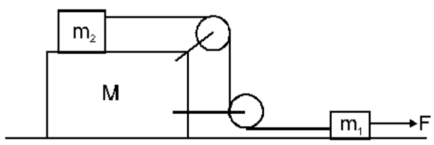
252. A particle is moving under the action of a variable force which is always pointing towards a fixed point.

	Column A		**Column B**
(a)	The particle may be in	(p)	Conserved
(b)	Total mechanical energy	(q)	May be conserved
(c)	Total angular momentum about the fixed point	(r)	Circular motion
(d)	Total kinetic energy	(s)	Rectilinear motion.

253.

	Column A		**Column B**
(a)	When two particles are executing SHM on parallel lines in such a way that each time particles crosses each other in opposite direction at a displacement A/2, where A is the amplitude of SHM, then the phase difference between the particles is	(p)	$\frac{\pi}{2}$
(b)	A particle starts SHM from the extreme position, the initial phase of the particle is	(q)	Transverse waves
(c)	The phenomenon of beats can take place for	(r)	$\frac{2\pi}{3}$
(d)	The waves created in metal can be	(s)	Longitudinal waves

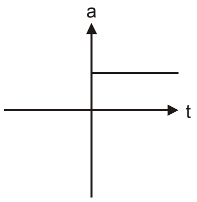
254. Three blocks of masses m_1, m_2 and M are arranged as shown in figure. All the surfaces are frictionless and string is inextensible. Pulleys are light. A constant force F is applied on block of mass m_1. All the pulley and string are light. Part of the string connecting both pulleys is vertical and part of the strings connecting pulleys with masses m_1 and m_2 are horizontal.

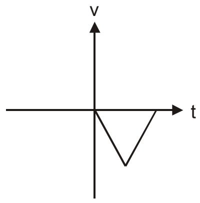


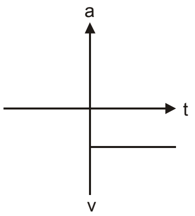
(A) Acceleration of mass m_1	(p) $\frac{F}{m_1}$
(B) Acceleration of mass m_2	(q) $\frac{F}{m_1 + m_2}$
(C) Acceleration of mass M	(r) zero
(D) Tension in the string	(s) $\frac{m_2 F}{m_1 + m_2}$

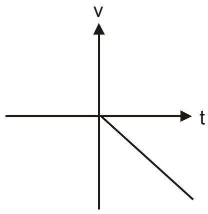
255. In each of the situations assume that particle was initially at rest at origin and there after it moved rectilinearly. Some of the graph in left column represent the same motion as represented by graphs in right column match these graphs.

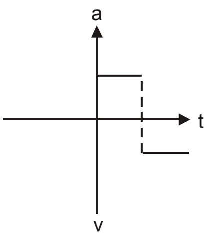
Column 1 | **Column 2**

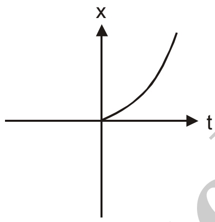
(A) 

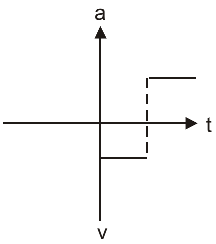
(p) 

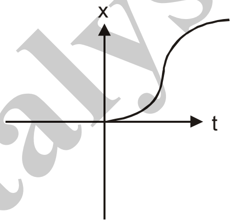
(B) 

(q) 

(C) 

(r) 

(D) 

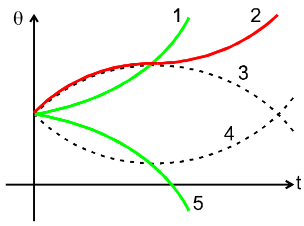
(s) 

256. For a particle moving in x-y plane initial velocity of particle is $\vec{u} = u_1\hat{i} + u_2\hat{j}$ and acceleration of particle is always $\vec{a} = a_1\hat{i} + a_2\hat{j}$ where u_1, u_2, a_1, a_2 are constants. Some parameters of motion is given in column-I, match the corresponding path given in column-II.

Column I	**Column II**
(A) If $u_1 \neq 0$, $u_2 = 0$, $a_1 \neq 0$, $a_2 \neq 0$	(p) path of particle must be parabolic
(B) If $u_1 = 0$, $u_2 \neq 0$, $a_1 \neq 0$, $a_2 \neq 0$	(q) path of particle must be straight line
(C) If $u_1 = 0$, $u_2 = 0$, $a_1 \neq 0$, $a_2 \neq 0$	(r) path of particle may be parabolic
(D) If $u_1 \neq 0$, $u_2 \neq 0$, $a_1 \neq 0$, $a_2 \neq 0$	(s) path of particle may be straight line

257. A particle is moving in circular motion around an axis. The motion of the particle in four different situations is described in the table. In the graph shown. Five curves are plotted and marked, and vertical axis gives angular position θ of the particle. Correctly match the curves with the situations to which they belong.

Situation	I	II	III	IV
Initial θ (rad)	+10	+10	+10	+10
Intial angular Velocity ω (rad/s)	+5	– 5	– 5	+ 5
Constant angular acceleration on α (rad/s^2)	+ 2	– 2	+ 2	– 2



Situation	**Curve**
(A) I	(P) 1
(B) II	(Q) 2
(C) III	(R) 3
(D) IV	(S) 4
	(T) 5

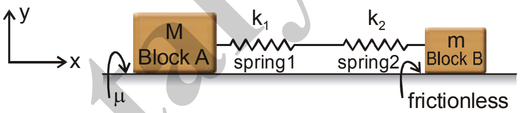
258. Motion of particle is described in column-I. In column-II, some statements about work done by forces on the particle from ground frame is given. Match the particle's motion given in column-I with corresponding possible work done on the particle in certain time interval given in column-II.

Column-I	**Column-II**
(A) A particle is moving in horizontal circle	(p) Total work done by all the forces may be positive
(B) A particle is moving in vertical circle with uniform speed	(q) Total work done by all the forces may be negative
(C) A particle is moving in air (projectile motion without any air resistance) under gravity	(r) Total work done by all the forces must be zero
(D) A particle is attached to roof of moving train on inclined surface.	(s) Total work done by gravity may be positive.

259. In the diagram shown in figure, all pulleys are smooth and massless and strings are light. Match the blocks in column-I with their motion in column-II.

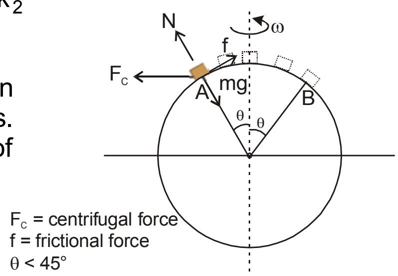
Column I	**Column II**
(A) 1 kg block	(p) will remain stationary
(B) 2 kg block	(q) will move down
(C) 3 kg block	(r) will move up
(D) 4 kg block	(s) has acceleration 5 m/s^2

260. Two blocks A and B of masses m and M are placed on a horizontal surface, both being interconnected with a horizontal series combination of two massless springs 1 and 2, of force constants k_1 and k_2 respectively as shown. Friction coefficient between block A and the surface is m and the springs are initially non-deformed. Now the block B is displaced slowly to the right by a distance x, and it is observed that block A does not slip on the surface. Block B is kept in equilibrium by applying an external force at that position. Match the required information in the left column with the options given in the right column.



Left column	**Right column**
(A) Friction force on block A by the surface	(p) $k_1 x (-\hat{i})$
(B) Force by spring 1 on block A	(q) $\mu Mg (-\hat{i})$
(C) Force exerted by spring 2 on spring 1.	(r) $\frac{k_1k_2x}{k_1+k_2}(\hat{i})$
(D) External force on block B.	(s) $\frac{k_1k_2x}{k_1+k_2}(-\hat{i})$

261. The block is placed at different position of earth from A to B as shown. Then the following parameters for different positions from A to B will vary as follows. Consider the effect of rotation of earth about its own axis. Neglect the effect of rotation of earth around the sun and assume earth as a perfect sphere.



Column I	**Column II**
(A) Gravity force	(p) first increases and then decreases
(B) Normal force	(q) first decreases and then increases
(C) Centrifugal force	(r) remains constant
(D) Frictional force	(s) will not act

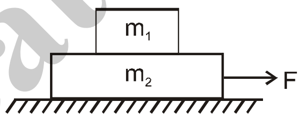
262. Two identical uniform solid spheres of mass m each approach each other with constant velocities such that net momentum of system of both spheres is zero. The speed of each sphere before collision is u. Both the spheres then collide. The condition of collision is given for each situation of column-I. In each situation of column-II information regarding speed of sphere(s) is given after the collision is over. Match the condition of collision in column-I with statements in column-II.

Column-I	Column-II
(A) Collision is perfectly elastic and head on	(p) speed of both spheres after collision is u
(B) Collision is perfectly elastic and oblique	(q) velocity of both spheres after collision is different
(C) Coefficient of restitution is e = $\frac{1}{2}$ and collision is head on	(r) speed of both spheres after collision is same but less than u.
(D) Coefficient of restitution is e = $\frac{1}{2}$ and collision is oblique	(s) speed of one sphere may be more than u.

263.

Column I	Colulmn II
(A) A constant force acting along the line of SHM affects	(p) the time period
(B) A constant torque acting along the arc of angular SHM affects	(q) the frequency
(C) A particle falling and sticking on the block (spring + mass system) executing SHM on a smooth horizontal plane when the later cross the mean position affects.	(r) the mean position
(D) A particle executing SHM on a smooth horizontal plane when kept on a uniformly accelerated car it changes	(s) the amplitude

264. A small block of mass m_1 lies over a long plank of mass m_2. The plank in turn lies over a smooth horizontal surface. The coefficient of friction between m_1 and m_2 is m. A horizontal force F is applied to the plank as shown in figure. Column-I gives four situation corresponding to the system given above. In each situation given in column-I, both bodies are initially at rest and subsequently the plank is pulled by the horizontal force F. Take length of plank to be large enough so that block does not fall from it. Match the statements in column-I with results in column-II.



Column-I	Column-II
(A) If there is no relative motion between the block and plank, the work done by force of friction acting on block in some time interval is	(p) positive
(B) If there is no relative motion between the block and plank, the work done by force of friction acting on plank is some time interval	(q) negative
(C) If there is relative motion between the block and plank, then work done by friction force acting on block plus work done by friction acting on plank is	(r) zero
(D) If there is no relative motion between the block and plank, then work done by friction force acting on block plus work done by friction acting on plank is	(s) is equal to negative of loss in mechanical energy of two block plus plank system.

265. **Match the following**

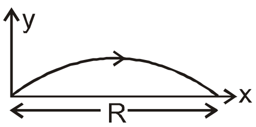
Column I	**Column II**
(a) Instantaneous speed	(P) is a vector quantity
(b) Instantaneous velocity	(Q) Its magnitude can decrease with time
(c) Average velocity	(R) Will remain constant for a particle moving uniformly in a circle
(d) Average speed	(S) Does not depend on the initial and final position only but depends on the motion in between

266. Consider motion of a particle in one dimension. Initially particle is at origin and has velocity towards positive x - direction. x, v, a and t denote displacement, velocity, acceleration and time respectively. Column II gives subsequent motion of the particle under the conditions in column I. Match the condition in Column I with the resultant motion in Column II

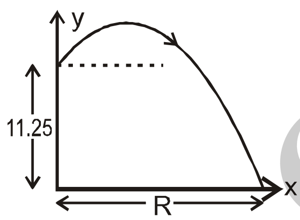
Column I		**Column II**
(A)	$a = -3\,v$	(p) Particle never stops
(B)	$v = 6 - 3t$	(q) Particle stops at least once
(C)	$x = 3 - 3\cos 2t$	(r) Particle travels finite distance before coming to rest first time.
(D)	$x = 3t + 6t^2$	(s) Particle comes back to origin at least once.

267. In the column-I, the path of a projectile (initial velocity 10 m/s and angle of projection with horizontal 60° in all cases) is shown in different cases. Rangle 'R' is to be matched in each case from column-II. Take $g = 10\ m/s^2$. Arrow on the trajectory indicates the direction of motion of projectile.

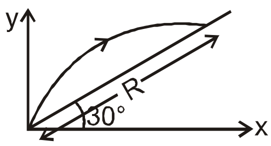
Column-I | **Column-II**

(A) 

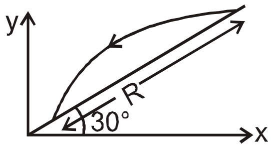
(p) $R = \frac{15\sqrt{3}}{2}$ m

(B) 

(q) $R = \frac{40}{3}$ m

(C) 

(r) $R = 5\sqrt{3}$ m

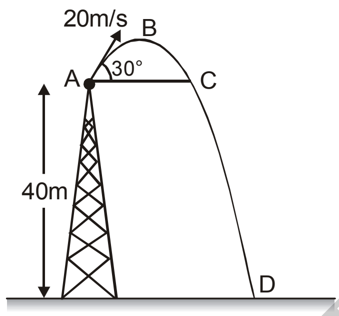
(D) 

(s) $R = \frac{20}{3}$ m

268. A particle is projected horizontally at time t = 0 from a given height above the ground level. Then match the physical quantities given in Column - I with the corresponding results given in Column - II. Consider all quantities in Column I from t = 0 and before the particle reaches the ground.

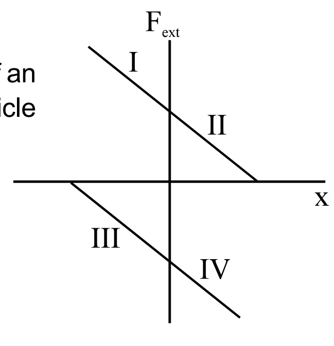
Column - I	Column - II
(A) Magnitude of acceleration	(p) remains constant
(B) Magnitude of average velocity from t = 0 to any time t	(q) decreases with time t
(C) Angle between acceleration and velocity vector	(r) increases with time t
(D) Distance of particle from its initial position.	(s) depends on initial velocity.

269. A projectile is fired from top of a 40 m high tower with velocity 20 m/s at an angle of 30° with the horizontal (see figure). g = 10 m/s^2.



(a) Ratio of time taken from A to D with time taken from A to C is equal to	(P) 1
(b) Ratio of vertical distance travelled from A to D with the maximum height from ground is less than.	(Q) 2
(c) Ratio of final speed at D with the initial speed at A is less than	(R) 3
(d) Ratio of horizontal displacement from A to D with height of tower is greater than	(S) 4

270. A block is executing SHM on a rough horizontal surface under the action of an external variable force. The force is plotted against the position x of the particle from the mean position.

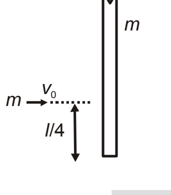


	Column I	Column II	
(A)	x positive , v positive	(P)	I
(B)	x positive , v negative	(Q)	II
(C)	x negative , v positive	(R)	III
(D)	x negative , v negative	(S)	IV

SECTION : (E) - Integer Type

271. The gravitational field in a region is given by $\vec{E} = (3\hat{i} + 2\hat{j})$ N/kg. Calculate the work done (in joule) by gravitational field when a particle (of mass 1 kg) is moved from $A\left(0, \frac{5}{2}\right)$ to $B(1, 1)$ along the line $2y + 3x = 5$

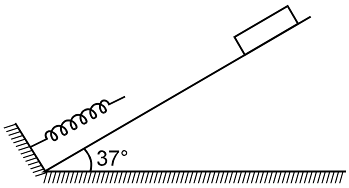
272. A rod of mass m = 1 kg and length l = 36 cm is hinged on a horizontal table as shown in figure. A ball of same mass, m = 1 kg is given a velocity v_0 = 4.3 m/s on the table to hit the rod at a distance $\frac{l}{4}$ from the lowest position of the rod. The ball after collision sticks to the rod. Angular speed, just after collision is $2n$ (rad/s). Find n.



273. A box weighing 2000 N is to be slowly slid through 20 m on a straight track having friction coefficient 0.2 with the box. A person is pulling the box with a chain at an angle θ with the horizontal. Find the work when the person has chosen a value of θ which ensures him the minimum magnitude of the force.

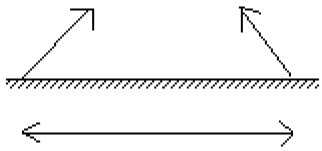
274. The US athlete Florence Griffith-Joyner won the 100 m sprint gold medal at Seol Olympic 1988 setting a new Olympic record of 10.54 s. Assume that she achieved her maximum speed in a very short time and then ran the race with that speed till she crossed the line. Take her mass to be 50 kg. Assume that the track, the wind etc. offered an average resistance of one tenth of her weight. What power Griffith-Joyner had to exert to maintain uniform speed ?

275. Figure shows a spring fixed at the bottom end of a rough incline of inclination 37°. A small block of mass 2 kg starts slipping down the incline from a point 4.8 m away from the spring. The block compresses the spring by 20 cm, stops momentarily and then rebounds through a distance of 1 m up the incline. Find the spring constant of the spring. Take g = 10 m/s^2.

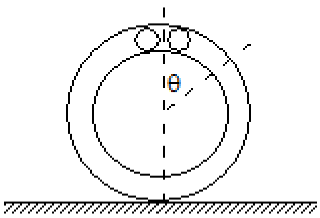


276. The limbs of a manometer consists of uniform capillary tubes of radii 1.44×10^{-3} m and 7.2×10^{-4} m. Find out the correct pressure difference (in N/m^{2}) if the level of the liquid (density 10^3 kg/m^3, surface tension 72×10^{-3} N/m) in the narrower tube stands 0.2 m above that in the broader tube

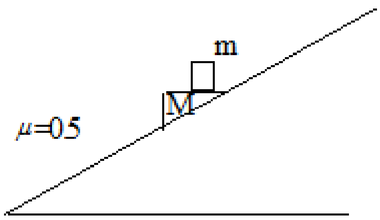
277. Two balls having masses 'M' each are launched at an angle of 45^0 above horizontal with a velocity of 10 m/s shown in the figure. For the elastic collision between the balls; find the value of X (initial separation between the balls). Take their velocity to be horizontal when they collide in air. Can the two balls collide for more than once ?



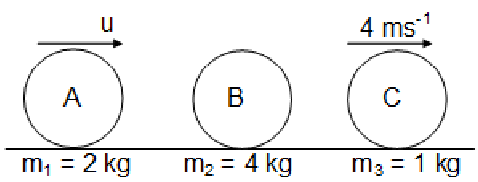
278. A circular tube of mass M is placed vertically on a horizntal surface as shown in the figure. Two small spheres, each of mass m and size such that they, just fit in the tube are released from the top, as shown n the figure. If θ gives the angle between the radius vector of either ball with the verticle, obtain the value of the ratio m/M for which the tube brakes its contact with ground when $\theta = 60^0$ (Ignore any friction)



279. A block of mass m is kept on the horizontal top surface of wedge of mass M which is kept on an incline plane of inclination θ ($\sin\theta = 3/5$) as shown in the figure. Coefficient of friction between the wedge and incline is 0.5. The minimum coefficient of friction between m and M so that m does not slip on M when the system is released from rest is X/Y. The value of Y-X is:

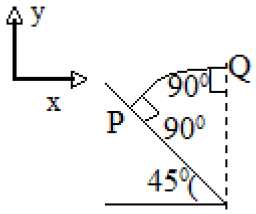


280. Three balls A, B, C are kept on a smooth surface. Balls A and C start moving with a constant velocity u & 4 ms^{-1} respectively as shown in the figure. If the collision between A and B is elastic then find out the value of u so that B can also collide with ball C after some time.



281. Two identical particles A and B of mass m are released from the positions shown in the figure. They collide elastically on horizontal portion MN. The ratio of the heights attained by the balls A and B after the collision is X/Y. Find the value of |X-Y|(Assume all the surfaces to be smooth)

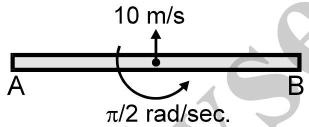
282. A ball is projected normally from point P (Which is on an inclined plane) with speed u= $10\sqrt{2}$ m/s. It strikes the vertical wall normally. If all the collisions are perfectly elastic, then find the time period (in seconds) of periodic motion.



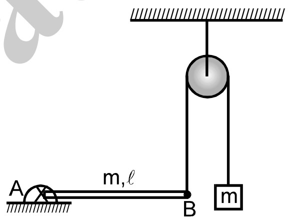
283. The potential energy of a particle is determined by the expression $U = \alpha(x^2 + y^2)$, where α is a positive constant. The particle begins to move from a point with the coordinates (2, 2) (m), only under the action of potential field force. Then its kinetic energy T at the instant when the particle is at a point with the coordinates (1, 1) (m) is n α .Find the value of n.

284. A uniform disc of radius R = 2 is given velocity 5m/s over a rough surface. After some time its kinetic energy becomes zero. Then find the initial angular velocity (in rad/sec).

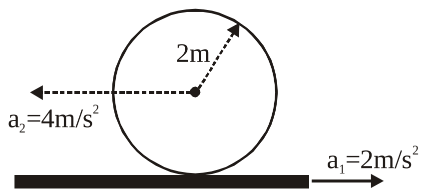
285. A uniform rod AB of length 4m and mass 12 kg is thrown such that just after the projection the centre of mass of the rod moves vertically upwards with a velocity 10 m/s and at the same time it is rotating with an angular velocity $\frac{\pi}{2}$ rad/sec about a horizontal axis passing through its mid point. Just after the rod is thrown it is horizontal and is as shown in the figure. Find the acceleration (in m/sec^2) of the point A in m/s^2 when the centre of mass is at the highest point. (Take g = 10m/s^2 and π^2 = 10)



286. Uniform rod AB is hinged at the end A in a horizontal position as shown in the figure. The other end is connected to a block through a massless string as shown. The pulley is smooth and massless. Masses of the block and the rod are same and are equal to ' m '. Then acceleration of the block just after release from this position is xg/8.Find the value of x.

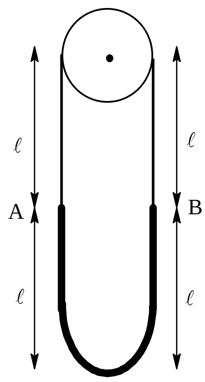


287. In the figure, a sphere of radius 2 metre rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is

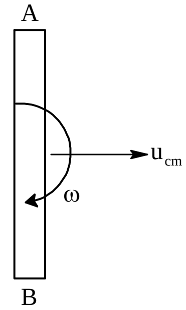


288. Two uniform ropes having linear mass densities m and 4m, length 2ℓ. Each are joined to form a closed loop. The loop is hanging over a fixed frictionless small pulley with the lighter rope above as shown in the figure (In the figure equilibhrium position is shown). Now if the point B (joint) is slightly displaced in downward direction and released. It is found that, the loop perform angular SHM with the period of the oscillation

$N \times 10$ sec. Find the value of N (take $\ell = \frac{150}{\pi^2}$ metre)



289. A uniform rod AB of length l travelling with linear velocity u_{cm} and rotating with angular velocity ω about its centre of mass such that $u_{cm} = \omega l/2$. The distance covered by the end B w.r.t ground when the rod completes one full rotation is xl. Find the value of x.

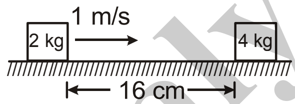


290. The position vector of a particle is given as $\vec{r} = (t^2 - 4t + 6)\hat{i} + (t^2)\hat{j}$. The time after which the velocity vector and acceleration vector becomes perpendicular to each other is equal to:

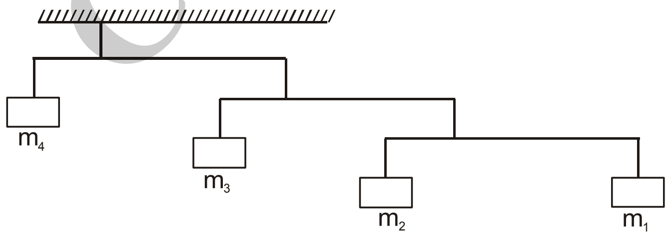
291. A juggler throws balls into air. He throws one whenever the previous one is at its highest point. He throws two balls per second. Find the height to which each ball goes (in centimeter). ($g = 10$ m/s^2)

292. A smooth right circular cone of semi vertical $\angle\alpha = \tan^{-1}(5/12)$ is at rest on a horizontal plane. A rubber ring of mass 2.5 kg which requires a force of 15 N for an extension of 10 cm is placed on the cone. Find the increase in the radius of the ring (in cm) when it is in equilibrium. ($\pi^2 = 10$)

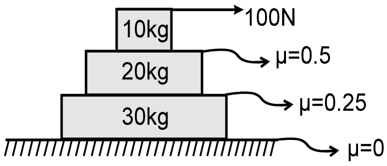
293. The friction coefficient between the horizontal surface and each of the blocks shown in figure is 0.20. The collision between the blocks is perfectly elastic. Find the separation between the two blocks (in cm) when they come to rest. Take $g = 10$ m/s^2.



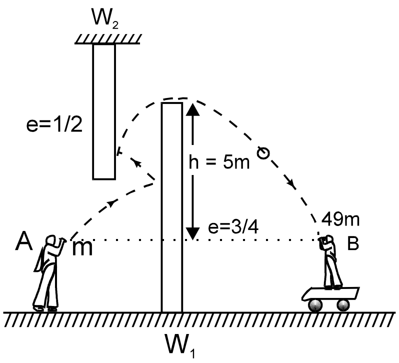
294. Figure shows an arrangement of masses hanging from a ceiling. In equilibrium, each rod is horizontal, has negligible mass and extends three times as far to the right of the wire supporting it as to the left. If mass m_4 is 48 kg then mass m_1 is equal to



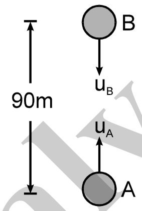
295. Three blocks are kept as shown in the figure. Acceleration (in SI units) of 20 kg block with respect to ground is:



296. Sachin Tendulkar (A) projects a ball of mass ' m ' towards Virendra Sehwag (B) of mass 48 m as shown in figure. The ball collides with the two vertical walls and when it just passes the wall W_1, its velocity is horizontal. Sehwag is standing on a cart of mass ' m ', catches the ball at the same level at which the ball is projected. After the catch, cart starts moving with a velocity 0.3 m/s horizontally towards right. Find X/Y if velocity of projection is $X\hat{i}+Y\hat{j}$. Walls are smooth and there is no friction between cart and ground. Sehwag remains fixed with respect to the cart. Coefficient of restitution of ball with the walls and the required height is shown in the figure.



297. Two particles A and B of masses 1 kg and 2 kg respectively are projected in the same vertical line as shown in figure with speeds u_A = 200 m/s and u_B = 85 m/s respectively. Initially they were 90 m apart. Find the maximum height (in metres) attained by the centre of mass of the system of particles A and B, from the initial position of centre of mass of the system. Assume that none of these particles collides with the ground in that duration Take g = 10 m/s^2.



298. An ideal spring of spring constant K= 4000 N/m and unstretched length ℓ_0 = 0.5 m is placed on a smooth table. One end of the spring is fixed at the centre of the table and other end is attached to a small block of mass m = 20 kg. The block is moving in a circle with constant speed 10 m/s. Find the tension (in kN) in the spring.

299. An insect starts from rest from point (3, 4) and moves with an acceleration $2\sqrt{2}$ m/s^2 in x-y plane along a line, equally inclined to both the axis. After 3 sec insect turns towards right in perpendicular direction without wasting any time and keeping speed same at the moment of turning. For the further motion acceleration is $2\sqrt{2}$ m/sec^2 in the direction of motion. The position of insect after 5 seconds from the starting is $\mathbf{X}\,\hat{i}\ \mathbf{-Y}\,\hat{j}$. The value of |X-Y| is:

300. A solid sphere moving with linear velocity 2 m/s and angular velocity 8 rad/s is rolling without slipping on a rough horizontal surface to collide eleastically with identical sphere at rest of mass 1 kg and radius R. There is no friction between them. Find the ratio of linear sphere of first sphere after it again starts rolling without slipping to the net angular impulse imparted to the second sphere by the external forces.

PHYSICS
PART - II
TOPIC: WAVES & THERMODYNAMICS
EXERCISE # 01

SECTION : (A) - Single Correct Options

301. Three one dimensional mechanical waves in a medium given as:

$y_1 = 3A \sin(\omega t - kx)$, $y_{-2} = A \sin(\omega t - kx + \pi)$ and $y_3 = 2A \sin(\omega t + kx)$ are superposed with each other. The maximum displacement amplitude of the medium particle would be

(A) 4 A (B) 3A (C) 2 A (D) A

302. A string of length 0.3 m and mass 10^{-2} kg is clamped at its ends. The tension in the string is 1.2 N. When a pulse travels along the string, the shape of the string is found to be the same at times t and $t + \Delta t$. The minimum value of Δt is

(A) 0.1 sec (B) 0.2 sec (C) 0.3 sec (D) 0.4 sec

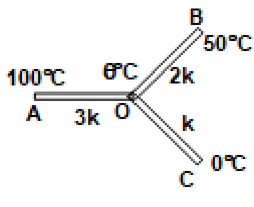
303. A sound wave is propagating unidirectionaly in a gaseous medium of bulk modulus 2×10^5 N/m^2 having displacement equation S (mm) = $(2 \times 10^{-5}\text{m}) \sin(800\pi t - 8\pi x)$. Find the density change amplitude of the medium.

(A) 2×10^{-5} m (B) $0.032\,\pi$ kg/m^3 (C) $0.016\,\pi$ kg/m^3 (D) can't be calculated.

304. One mole of an ideal diatomic gas is expanded. During expansion, volume and temperature of gas vary such that $\frac{V}{T^2} =$ constant. Find heat transfer during expansion if temperature of the gas increases by 50 K.

(A) 225 R (B) 125 R (C) 100 R (D) None of the above

305. Three rods of same dimensions having thermal conductivity as 8k, 2k and k are arranged as shown in the figure. The value of θ



(A) 60°C (B) (100/3)°C (C) (200/3)°C (D) None of these

306. A wave of frequency 1000 Hz has a wave velocity of 400 m/s. The distance between two consecutive points. Which are 60° out of the phase is

(A) $\frac{1}{15}$ m (B) $\frac{2}{15}$ m (C) $\frac{7}{3}$ m (D) $\frac{2}{3}$ m

307. The graphs of fall of temperature of a liquid (250 g) and water (200 g) (cooled under identical conditions having equal volumes) are shown in figure. Then the specific heat of the liquid (mass of the calorimeter is 100 gm and specific heat = 0.1)

(A) 0.24 (B) 2.24 (C) 1.24 (D) 0.14

308. A metal cylinder of inner radius R_1 and outer radius R_2 having length L carries a constant flow of hot water maintained at a temperature θ_1. The outer temperature of the surrounding is θ_2. If k is the thermal conductivity of the metal then the rate of heat loss through the walls of the cylinder in cal/s is

(A) $\dfrac{\pi kL(\theta_2-\theta_1)}{Ln(R_2/R_1)}$ (B) $\dfrac{2\pi kL(\theta_2-\theta_1)}{Ln(R_2/R_1)}$ (C) $\dfrac{\pi kL(\theta_2-\theta_1)}{2Ln(R_2/R_1)}$ (D) $\dfrac{4\pi kL(\theta_2-\theta_1)}{Ln(R_2/R_1)}$

309. A spherical black body with a radius of 10 cm radiates 450 watt power at 500 K. If the radius were halved and temperature doubled, the power radiated in watts would be

(A) 225 (B) 450 (C) 1200 (D) 1800

310. A tuning fork vibrates at 220 Hz. The length of the shortest closed organ pipe that will resonate with the tuning fork will be (Speed of sound in air = 340 m/s)

(A) 39 cm (B) 34 cm (C) 30 cm (D) 36 cm.

311. A balloon whose volume is 500 m^3 is to be filled with hydrogen at atmosphere pressure. If hydrogen is stored in cylinders of volume 0.05 m^3 at an absolute pressure of 15×10^5 Pa, then how many cylinders are required ($P_{atm} = 1.013 \times 10^5$ N/m^2, ρ_{air} = 1.3 kg/m^3 and $\rho_H = 9 \times 10^{-2}$ kg/m^3)

(A) 575 (B) 675 (C) 775 (D) 600

312. An organ pipe of cross sectional area 100 cm^2 resonate with a tuning fork of frequency 1000 Hz in fundamental tone. The minimum volume of water to be drain out so that the pipe again resonate with the same tuning fork is (take velocity of wave = 320 m/s)

(A) 800 cm^3 (B) 1200 cm^3 (C) 1600 cm^3 (D) 2000 cm^3

313. A wall of width ℓ and cross sectional area A is having variable coefficient of thermal conductivity given by $k = k_0 + \alpha x$ (where k_0 and α are positive constant) and x is measured from outer surface of wall. If the temperature of surroundings is T_0°C and the temperature of the room is maintained at T_r°C($T_0 > T_r$). The temperature of the wall at $x = \ell/2$ is (given $\propto\ell/K_0$ = 1 S.I. unit, T_0 = 45°C, and T_r = 20°C)

(A) 20°C (B) 25°C (C) 28°C (D) 30.37°C

314. A gaseous mixture has 0.5 gm of nitrogen and 1 gm of oxygen and is enclosed in a cylinder of 15 litre capacity at 27°C. The total pressure of mixture is

(A) 0.08054 atm (B) 0.05 atm (C) 0.03 atm (D) 0.02 atm

315. The volume of an air bubble is doubled as it rises from bottom to surface. The atmospheric pressure is H m of mercury and density of mercury is n times that of lake water the depth of lake is (assuming temperature of air to be constant)

(A) H/n (B) (nH)/2 (C) nH (D) 2nH

SECTION : (B) - Multiple Correct Options

316. The free ends of the rod of length (L = 0.5 m) form anti-nodes while rod is clamped at distance L/4 from one free end. The density of metal of rod and its Young's modulus are 8×10^3 kg/m^3 and 200×10^9 N/m^2 respectively. Choose the correct statements)

(A) There are three frequencies of longitudinal waves in rod in the range of 0 kHz to 50 kHz

(B) There are two frequencies of longitudinal waves in rod in the range of 0 kHz to 50 kHz

(C) Speed of longitudinal sound wave in rod is 4×10^3 m/s

(D) Speed of longitudinal sound wave in rod is 5×10^3 m/s

317. The solar constant for the earth is S. The surface temperature of sun is T K. The sun subtends an angle θ at the earth then

(A) $S \propto T^4$ (B) $S \propto T^2$ (C) $S \propto \theta^2$ (D) $S \propto \theta^4$

318. A plane progressive wave of frequency 25 Hz, amplitude 2.5×10^{-5} m and initial phase zero moves along the $-$x direction with velocity of 300 m/s. A and B are two points 6m apart on the line of propagation of the wave. At any instant the phase difference between A and B is ϕ. The magnitude maximum difference in the displacements at A and B is x.

(A) $\phi = \pi$ (B) $\phi = 0$ (C) x = 0 (D) x = 5×10^{-5} m

319. A one litre of perfect gas at a pressure of 72 cm of Hg, is compressed isothermally to a volume of 900 cc, then

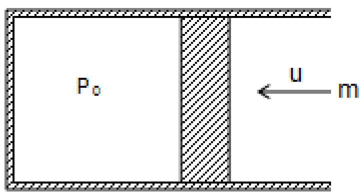
(A) Stress of the gas is 10.88×10^3 N/m^2

(B) Strain of the gas is 0.1

(C) Bulk modulus of gas at constant temperature is 10.88×10^3 N/m^2

(D) Bulk modulus of gas at constant temperature is 10.88×10^4 N/m^2

320. A horizontal piston cylinder arrangement encloses a gas at a pressure P_0. The atmospheric pressure is also P_0. Now a particle of mass m strikes the piston of identical mass m with a horizontal speed u and sticks to it. Subsequently the piston moves without friction and compresses the gas. The process of compression an be assumed to be adiabatic. In this process



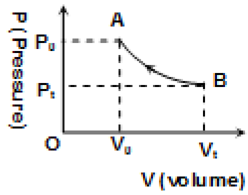
(A) when the piston comes to rest at maximum compression, the internal energy of gas would have increased by $\frac{1}{4}mu^2$

(B) During the time interval starting from the mass striking the piston to the maximum compression of the enclosed, gas the work done by the gas is $-\frac{1}{4}mu^2$.

(C) When the piston comes to rest after compressing the gas, the internal energy of the gas would have increased by the amount move than $\frac{1}{4}mu^2$.

(D) During the compression of the gas till the time when piston comes to rest, work done by the gas is less than $-\frac{1}{4}mu^2$.

321. The slope of the PV curve shown in the figure is P, where P is the pressure at the point on which slope is to be calculated. The value of slope of T-V curve between points A and B will be (Assume the gas undergoing compression to be ideal and n = number of moles of the gas)



(A) $\frac{P_0 e^{v_0 - v}}{nR}[1 - v]$ (B) $\frac{P_0 e^{v_0 - v}}{nR}\left[v^2 - 3v + 1\right]$ (C) $\frac{P_0 e^{v_0}}{nR}$ (D) None of these.

322. A furnace X at temperature θ_0 is connected to a body Y of heat capacity C, at an initial temperature of θ_1, with the help of a rod of uniform cross-section A, uniform thermal conductivity K and length L. In some time interval temperature of body Y increases to θ_2 from θ_1 such that $\theta_1 < \theta_2 < \theta_0$. Then choose the correct option(s).

(A) Rate of heat flow through the rod is not constant.

(B) Temperature of the body Y increases exponentially.

(C) Rate of change of temperature of body Y increases with time.

(D) Rate of change of temperature of body Y decreases with time.

323. The rates of fall of temperature of two identical solid spheres of different materials(behaves as black body) are equal at a certain temperature.

(A) Their specific heat capacities are equal.

(B) Their heat capacities are equal

(C) Their specific heat capacities are proportional to their densities.

(D) Their specific heat capacities are inversely proportional to their densities

324. A perfectly black body:

(A) absorbs radiations of all wavelengths and does not transmit any radiations.

(B) absorbs and transmit radiations of all wavelengths.

(C) when heated, emits radiations of more wavelengths than it absorbs.

(D) when heated still absorbs and emits radiations of all wavelengths.

325. Select the correct statements(s):

(A) In an adiabatic process, the heat gain is equal to heat loss and hence, there is no heat exchange

(B) An ideal black body absorbs and emits radiations of all wavelength.

(C) During melting of ice, the heat given to ice goes in increasing the internal energy

(D) All of the above

326. A black body kept in sunlight is maintained at constant temperature, then the block body

(A) absorbs all but emits radiations of a few wavelengths

(B) absorbs a few but emits radiations of all wavelengths.

(C) absorbs and emits radiations of all wavelengths.

(D) is in thermal equilibrium

327. The equation of a process of diatomic gas is $P^2 = \alpha^2 v$ where α is +ve constant. Then choose

(A) Work done by gas for a temperature change T is $\frac{2}{3}\alpha nRT$

(B) The change in internal energy is $\frac{5}{2}nRT$ for temperature change T

(C) The specific heat for the process is $\frac{19}{6}R$

(D) The change in internal energy for temperature change T is $\frac{5}{2}\alpha nRT$

328. Two spheres having temperature T_1 and T_2 touch each other and are kept in a room. Then heat transfer between the spheres takes place through

(A) conduction if $T_1 \neq T_2$

(B) convection

(C) radiation

(D) no heat transfer takes place if $T_1 = T_2$.

329. Plane harmonic waves of frequency 500 Hz are produced in air with displacement amplitude of 10μm. Given density of air is 1.29 kg/m^3 and speed of sound in air is 340 m/s

(A) Pressure amplitude is 13.8 N/m^2

(B) Energy density is 0.22×10^{-4} J/m^3

(C) The energy flux is 6.4×10^{-2} J/m^2-s

(D) None of these.

330. For a certain stretched string, three consecutive resonance frequencies are observed as 105, 175, 245 Hz respectively. The select the correct alternative.

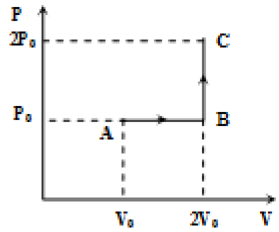
(A) The string is fixed at both ends.

(B) The string is fixed at one end

(C) The fundamental frequency is 35 Hz

(D) The fundamental frequency is 52.5 Hz

331. One mole of an ideal monoatomic gas is taken from A to C along the path ABC. The temperature of the gas at A is T_0. For the process ABC



(A) work done by the gas is RT_0

(B) change in internal energy of the gas is $\frac{11}{2}RT_0$

(C) heat absorbed by the gas is $\frac{11}{2}RT_0$

(D) heat absorbed by the gas is

(where R = universal gas constant)

332. A uniform rope of mass m and length ℓ hangs vertically from a rigid support. A transverse pulse of wavelength λ_0 is produced at the lower end. Here the speed of the pulse is v_0. At any time t, the pulse speed is v and wavelength of the pulse is λ. Then

(A) $\frac{v_0}{\lambda_0} = \frac{v}{\lambda}$

(B) acceleration of the wave pulse along the string is g/2.

(C) Acceleration of the wave pulse is independent of wavelength.

(D) None of the above.

333. It is usually more convenient to describe a sound wave in terms of pressure wave as compared to displacement wave because

(A) Two waves of same intensity but different frequencies have different displacement amplitude but same pressure amplitude.

(B) The human ear responds to the change in pressure and not to the displacement wave.

(C) The electronic detector does respond to the change in pressure but not to the displacement wave.

(D) None of the above

334. Which of the statements are correct about the nodes in a standing wave?

(A) The energy never propogates through nodes.

(B) At a time, when all particles pass through mean position, the energy can propagate in the form of kinetic energy.

(C) The nodes remain fixed because the two waves pass through simultaneously in opposite direction.

(D) The nodes remain fixed because the two waves pass through simultaneous in opposite phase.

335. The rate of heat energy emitted by a body at an instant depends upon

(A) area of the surface

(B) difference of temperature between the surface and its surroundings

(C) nature of the surface

(D) None of these

336. A partition divides a container having insulated walls into two compartments I and II. The same gas fills the two compartments whose initial parameters are given. The partition is a conducting wall which can move freely without friction. Which of the following statements is/are correct, with reference to the final equilibrium position?

P, V, T I	2P, 2V, T II

(A) The Pressure in the two compartments are equal.

(B) Volume of compartment I is $\frac{3V}{5}$

(C) Volume of compartment II is $\frac{12V}{5}$

(D) Final pressure in compartment I is $\frac{5P}{3}$

337. The length, tension, diameter and density of a wire B are double than the corresponding quantities for another stretched wire A. Then (both are fixed at the ends)

(A) Fundamental frequency of B is $\frac{1}{2\sqrt{2}}$ times that of A.

(B) The velocity of wave in B is $\frac{1}{\sqrt{2}}$ times that of velocity in A.

(C) The fundamental frequency of A is equal to the third overtone of B.

(D) The velocity of wave in B is half that of velocity in A.

338. In a stationary wave,

(A) all the particles of the medium vibrate in phase

(B) all the antinodes vibrate in phase

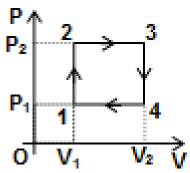
(C) the alternate antinodes vibrate in phase

(D) all the particles between consecutive nodes vibrate in phase

SECTION : (C) -Passage Type Questions

Passage 01:

An ideal monatomic gas [1 mole] is taken through a cyclic process, whose p-v graph is shown in the figure. If $P_2 = P_0$ and efficiency of cycle is 10%. If $v_2 = 2v_1$ and $v_1 = v_0$, now answer the following questions



339. The maximum temperature of the gas in the process is

(A) $\frac{P_0V_0}{R}$ (B) $\frac{P_0V_0}{2R}$ (C) $\frac{2P_0V_0}{R}$ (D) $\frac{4P_0V_0}{2R}$

340. The net heat supplied to gas in one cycle is

(A) $\frac{12P_0V_0}{17}$ (B) $\frac{5P_0V_0}{17}$ (C) $\frac{P_0V_0}{17}$ (D) $\frac{3P_0V_0}{17}$

341. The value of P_1 is

(A) $\frac{12P_0}{17}$ (B) $\frac{5P_0}{17}$ (C) $\frac{P_0}{17}$ (D) $\frac{3P_0}{17}$

PASSAGE 02:

The air column in a pipe closed at one end is made to vibrate in its third over tone by tuning fork of frequency 220 Hz. The speed of sound in air is 330 m/sec. End correction may be neglected. Let P_0 denote the mean pressure at any point in the pipe and ΔP_0 the maximum amplitude of pressure vibration.

342. Find the length of the air column

(A) 3.2 m (B) 2.625 m (C) 4.23 m (D) 1.16 m

343. What is the amplitude of pressure variation at the middle of the column

(A) $\frac{P_0}{\sqrt{2}}$ (B) $\frac{\sqrt{3}\Delta P_0}{2}$ (C) ΔP_0 (D) $\frac{\Delta P_0}{2}$

PASSAGE 03:

A cubical box of side 2 metre contains oxygen gas (atomic weight 32) at a pressure of 100 N/m^2. During an observation time of one second, an atom travelling with the root-mean square speed parallel to one of the edges of the cube, was found to make 500 hits with a particular wall, without any collision with the other atoms. ($R = \frac{25}{3}$ J/mol–K and $K = 1.38 \times 10^{-23}$ J/K)

344. What is the temperature of the gas

(A) 500 K (B) 4520 K (C) 5120 K (D) 3600 K

345. What is the average translational kinetic energy per atom

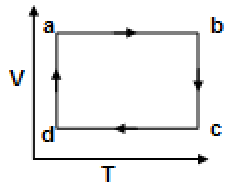
(A) 106×10^{-21} J (B) 96×10^{-21} J (C) 116×10^{-21} J (D) 112×10^{-21} J

346. Find the total mass of oxygen gas in the box.

(A) 30×10^{-4} kg (B) 2.4×10^{-4} kg (C) 6×10^{-4} kg (D) 15×10^{-4} kg

PASSAGE 04:

V–T graph of a monoatomic ideal gas is as shown in figure.



347. Sum of work done by the gas in process abcd is

(A) zero (B) Positive (C) negative (D) data is insufficient

348. Heat is supplied to the gas in process(s)

(A) da, ab and bc (B) da and ab only (C) da only (D) ab and bc only

349. Change in internal energy of the gas is zero in process

(A) da, ab and bc (B) da and bc only (C) da only (D) da and ab only

PASSAGE 05:

Internal energy: Every system (solid, liquid or gas) processes a certain amount of energy. This energy is called the internal energy and is usually denoted by the symbol U. The internal energy of a solid, liquid or gas consists of two parts (i) kinetic energy due to the motion (translational, rotational and vibrational) of the molecules, and (μ) potential energy due to the configuration of the molecules.

The internal energy of homogenous system depends on its thermodynamic state i.e., on its thermodynamic coordinates P, V and T. each definite state of the system possesses a define quantity of internal energy. A change in the internal energy can occur only if a transfer of energy between the system and surroundings is permitted. This can take place if (i) some work is performed on or by the system, and (ii) some heat is observed or given out by the system

350. In a given process on an ideal gas dW = 0 and dQ > 0. Then

(A) the temperature of the gas will decrease. (B) the temperature of the gas will remain constant.

(C) the internal energy of the gas will decrease. (D) the internal energy of the gas will increase.

351. The internal energy of an ideal gas consists of

(A) Kinetic energy of the particle of the gas.

(B) Potential energy of the particle of the gas.

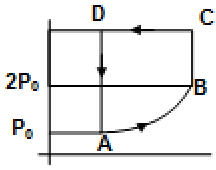
(C) Kinetic and potential energy of the particle of the gas.

(D) none of these.

PASSAGE 06:

A sample of ideal monatomic gas is expanded according to $P = \Delta v^2$ with initial volume $100m^3$ & temp 300k where ? = $3atm/m^6$ in AB. Then at constant volume (BC) till pressure becomes twice of pressure at B. CD is at constant pressure till initial volume and finally DA at constant volume as shown

352. What is molar heat capacity in process AB.



(A) $\frac{3R}{2}$ (B) $\frac{R}{2}$ (C) 13R (D) $\frac{13R}{6}$

353. Total work done in cyclic process ABCDA by Gas

(A) 65×10^6J (B) -65×10^6J (C) -72×10^6J (D) None

354. Heat absorbed by gas in DA process.

(A) -1.035×10^8J (B) Zero (C) -1.035×10^6J (D) None

PASSAGE 07:

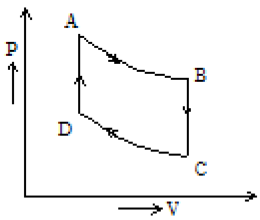
2 moles of an ideal monatomic gas undergoes in a cyclic process through the following changes

(i) Isothermal expansion from A to B from a volume $0.04\ m^3$ to $0.10\ m^3$ at 87^0 C

(ii) at constant volume from B to C cooling to 27^0 C.

(iii) Isothermal compression from C to D at 27^0C to $0.04\ m^3$

(iv) at constant volume from D to A heating to original pressure volume and temperature [In 2.5 = 0.92]



355. Heat supplied in the A B process is

(A) 5483.25 J (B) 1496.5 J (C) 6979.75 J (D) 913.85 J

356. Work done by the gas during the cyclic process is

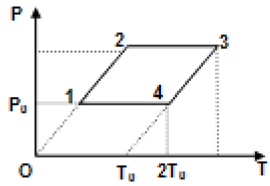
(A) 5483.25 J (B) 1496.5 J (C) 6979.75 J (D) 913.85 J

357. Heat absorbed by the gas during the cyclic process is

(A) 5483.25 J (B) 1496.5 J (C) 6979.75 J (D) 913.85 J

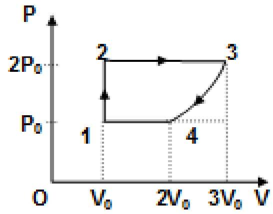
PASSAGE 08:

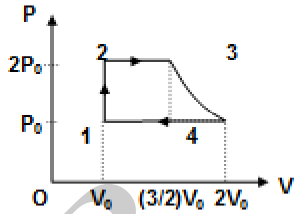
One mole of an ideal monoatomic gas is taken through the cycle process 1–2–3–4–1 as shown in the figure. The initial volume of gas is V_0.

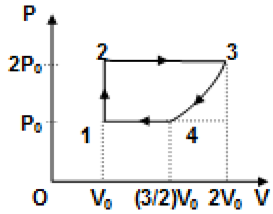


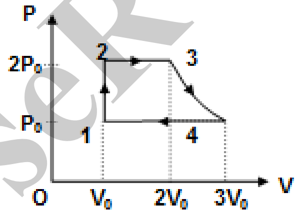
Now answer the following questions

358. The P–V diagram of the cyclic process will be

(A)


(B)


(C)


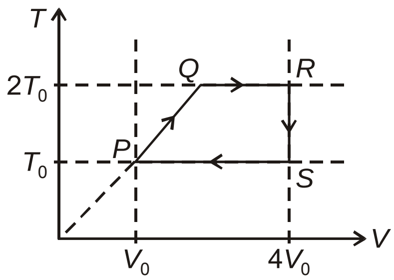
(D)


359. The work done by the gas in one cycle is

(A) $RT_0(3-\ln 2)$ (B) $RT_0\ln 2$ (C) $\frac{RT_0}{2}\ln 2$ (D) $\frac{RT_0}{2}(3-\ln)$

PASSAGE 09:

n moles of argon are undergoing a cyclic process *PQRSP* as shown by the *T-V* diagram here. *T* and *V* represent thermodynamic gas temperature and volume occupied by the gas respectively. Considering argon to be behaving ideally, answer the following questions.



360. Heat supplied to the gas during the cycle is

(A) nRT_0 (B) $1.5nRT_0$ (C) $2nRT_0$ (D) $2.5nRT_0$

361. Efficiency of the cycle is

(A) $\frac{1}{5+\ln 4}$ (B) $\frac{2}{5+\ln 16}$ (C) $\frac{4}{5+\ln 4}$ (D) $\frac{1}{5+\ln 2}$

SECTION : (D) - Matrix Match

362. Match the processes in column I with the possibilities in column II, where ΔQ and W are heat change and work done respectively

	Column A		**Column B**
(A)	Isobaric	(i)	$\Delta Q > O$
(B)	Isothermal	(ii)	$\Delta Q < O$
(C)	Isochoric	(iii)	$W > O$
(D)	Adiabatic	(iv)	$W < O$

363. For a organ pipe closed at one end. Match the following:

	Column A		**Column B**
(A)	Third overtone frequency is x times the fundamental frequencies. Here x is equal to	(i)	3
(B)	Number of nodes in second overtone	(ii)	4
(C)	Number of antinodes in second overtone	(iii)	5
(D)	Fifth harmonic frequency is x′ times the fundamental frequency. Here x′ is equal to	(iv)	none of these

364. For one mole of a monoatomic ideal gas match the following:

	Column A		**Column B**
(A)	Isothermal bulk modulus	(i)	$-\frac{RT}{V^2}$
(B)	Adiabatic bulk modulus	(ii)	$-\frac{5P}{3V}$
(C)	Slope of P–V graph in isothermal process	(iii)	$\frac{T}{V}$
(D)	Slope of PV graph in adiabatic process	(iv)	none of these

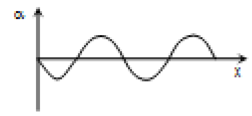
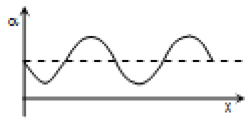
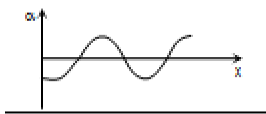
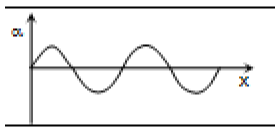
365. One mole of an ideal monoatomic gas undergoes a process $PV^{1/2} = C$, where C is a constant.

Match the following:

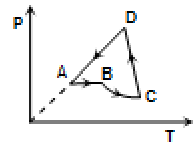
	Column – I		Column – II
(A)	Heat given to the gas to increase the temperature of the gas by 100°C.	(i)	150 R
(B)	Work done on the gas to change the volume from V_0 to $2V_0$.	(ii)	$\frac{7R}{2}$
(C)	Molar heat capacity of the gas	(iii)	350 R
(D)	Change in internal energy of gas when the temperature of the gas is increased by 100°C	(iv)	$2c\sqrt{v_0}\left[\sqrt{2}-1\right]$

366. The displacement equation of a standing wave in a homogeneous elastic medium is given by y = a cos (kx) cos (ωt)

Match the physical quantities α in the List (I) to the correct plot(s) in the List (II)

	List – I		List – II
(A)	Displacement y of the particles at t = (T/2)	(i)	
(B)	Velocity of the particles at t = (T/4)	(ii)	
(C)	Change in pressure of the medium at t = 0	(iii)	
(D)	Density of the medium at t = (T/2)	(iv)	

367. A hypothetical heat engine working with an ideal gas passes through the cycle A→B→C→D→A as shown in an P-T indicator diagram, where BC is an adiabatic process.



For the heat engine, match the physical quantities in List-II with the correct sign in List-I

	List – I		List – II
(A)	x > 0	(i)	W_{DA}
(B)	x = 0	(ii)	W_{CD}
(C)	x < 0	(iii)	W_{BC}
		(iv)	ΔQ_{BC}
		(v)	ΔQ_{CD}
		(vi)	ΔQ_{BC}

368. Source has frequency f. Source and observer both have same speed along same line. For the apparent frequency observed by observer

	List – I		List – II
A.	Observer is approaching the source but source is receding from the observer	(i)	More than f
B.	Observer and source both approaching towards each other	(ii)	Less than f
C.	Observer and source both receding from each other	(iii)	Equal to f
D.	Source is approaching but observer is receding		

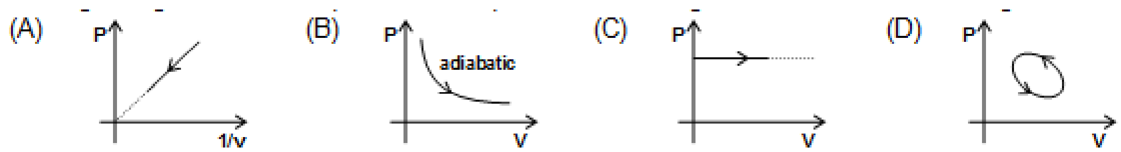
369. Match the processes in list I with the possibilities in list II, where, ΔQ and W are heat gained and work done by the system respectively

	List – I		List – II
(A)	Isobaric	(i)	ΔQ > O
(B)	Isothermal	(ii)	ΔQ < O
(C)	Isochoric	(iii)	W > O
(D)	Adiabatic	(iv)	W < O

370. **List – I** **List – II**

	List – I		List – II
(A)	Longitudinal waves	(i)	May be formed by two waves of same frequency
(B)	Transverse wave	(ii)	)May be formed by two waves of different frequency
(C)	Standing waves	(iii)	Cannot travel in free space
		(iv)	Can travel in free space

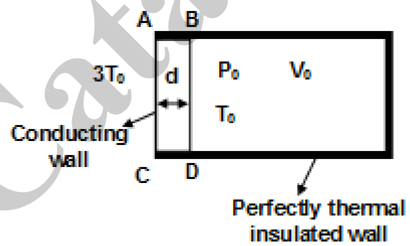
371. The figures given below depict different processes for a given amount for an ideal gas



	List – I		List – II
(A)	In Fig(i)	(i)	Heat is absorbed by the system
(B)	In Fig(ii)	(ii)	work is done on the system
(C)	In Fig(iii)	(iii)	Heat is rejected by the system
(D)	In Fig(iv)	(iv)	Work is done by the system

372. One mole of a monoatomic ideal gas is kept in a container of volume V_0. the conducting wall has surface area A, thickness surface area A, thickness d and thermal conductivity K. Assume that surrounding B at a constant temperature $3T_0$.

[given $\frac{2kA}{3Rd} = 1\ \text{sec}^{-1}$ and $T_0 = 100$ K]



	List I		List II
(A)	Rate of flow of heat in the gas	1.	$\frac{3KA}{d}(3T_0 - T)$
(B)	Rate of change of internal energy of gas	2.	$\frac{3}{2}R\frac{dT}{dt}$
(C)	Ratio of temperature of gas at t = ln2 sec to the temperature at t = 2 ln2 sec	3.	$\frac{4}{5}$
(D)	Ratio of change in internal energy of gas in the interval 0 to ln2 sec to change in the internal energy in the internal 0 to 2 ln2 sec.	4.	(2/3)

373. **List I** **List II**

	List I		List II
(A)	A mechanical wave propagates in a medium along the x-axis. The particles of the medium may move along	(i)	x-axis
(B)	A transverse wave travels along the z-axis. The particles of the medium may move along	(ii)	y-axis
(C)	A wave moving in a solid may be	(iii)	Transverse
(D)	A wave moving in a gas must be	(iv)	Longitudinal

374. For transverse wave on a string.

	Column I		Column II
(a)	If amplitude increases	(p)	maximum instantaneous power increases
(b)	If frequency increases	(q)	average power increases
(c)	If amplitude decreases,	(r)	maximum instantaneous power decreases
(d)	If frequency decreases,	(s)	average power decreases

375. Suppose a wave pulse has been created at free end of a taut string by moving the hand up and down once. The string is attached at its other end to distant wall.

	Column I		ColumnII
(a)	Moving hand more quickly but still up and down once by the same amount in different time,	(p)	the amplitude changes
(b)	Moving hand more quickly, but still up and down once changes by more amount in same time	(q)	time width of the pulse
(c)	Moving hand at same speed, but still up and down once by more amount	(r)	the wave speed changes
(d)	Moving hand more quickly, but still up and down once by less amount,	(s)	the particle speed changes

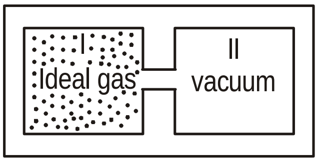
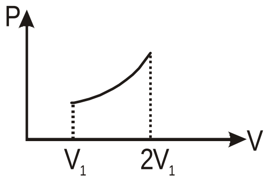
376.

	Column A		Column B
(A)	Temperature of a gas	(P)	Internal energy increases
(B)	Work done by the gas	(Q)	Intermolecular force decreases
(C)	Thermal expansion	(R)	Path function
(D)	Mechanical compression	(S)	State function

377. Two identical speakers emit sound waves of frequency 680 Hz uniformly in all directions with a total output of 1m W each. The speed of sound in air is 340 m/s. A point P is at a distance 2m from one speaker and 3m from the other. Match the following

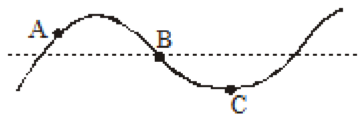
	Column-I		Column-II
(a)	The intensity due to a speaker at point P which is 2 m from point P	(p)	5.6×10^{-5} W/m^2
(b)	If the speakers are driven coherently and in phase, the intensity at point P will be due to both speakers	(q)	2.7×10^{-6} W/m^2
(c)	If they are driven coherently but out of phase by 180^{0}C, the intensity at point P will be due to both speakers	(r)	2.9×10^{-5} W/m^2
(d)	If the speakers are incoherent, the intensity(s) 2×10^{-5} w/m^2 at point P will be due to both speakers		

378. Column I contains a list of processes involving expansion of an ideal gas. Match this with Column II describing the thermodynamic change during this process. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

	Column-I		**Column-II**
(A)	An insulated container has two chambers separated by a valve Chamber I containsan ideal gas and the Chamber II has vacuum. The valve is opened. 	(P)	The temperature of the gas decreases
(B)	An ideal monoatomic gas expands to twice its original volume such that such that its pressure $p \propto (1/V^2)$, where V is the volume of the gas.	(Q)	The temperature of or remains constant
(C)	An ideal monoatomic gas expands to twice its original volume such that its pressure $p \propto (1/V^{4/3})$, where V is the volume.	(R)	The gas loses heat
(D)	An ideal monoatomic gas expands such that its pressure p and volume V follows the behaviour shown in the graph. 	(S)	The gas gains heat

379.

The figure shows a string at a certain moment as a transverse wave passes through it. Three particles A, B and C of the string are also shown. Match the physical quantities in the left column with the description in the column on the right.

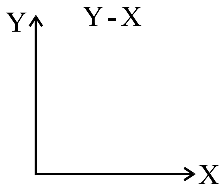


	Column A		**Column B**
(a)	Velocity of A	(p)	Downwards, if the wave is travelling towards right.
(b)	Acceleration of A	(q)	Downwards, if the wave is travelling towards left.
(c)	Velocity of B	(r)	Downwards, no matter which way the wave is travelling.
(d)	Velocity of C	(s)	Zero.

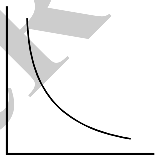
380.

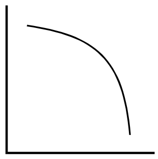
List – I		List – II	
(A)	Conduction	(i)	Through partial transfer of medium
(B)	Convection	(ii)	Without transfer of medium
(C)	Radiation	(iii)	Even in absence of medium.

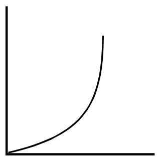
381. Respective graphs for adiabatic process taking first term on Y-axis

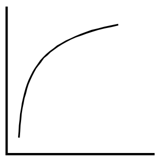


Column I | **Column II**

(A) Pressure - Temperature — (P) 

(B) Volume - Temperature — (Q) 

(C) Pressure - Volume — (R) 

(D) Pressure - Internal Energy — (S) 

SECTION : (E) - Integer Type

382. At very low temperature, the molar heat capacity of rock salt varies with temperature according to $C = K\frac{T^3}{Q^3}$ where K is 1940 J mol^{-1}K^{-1} and Q = 281 K. Then how much heat (in Joules) is required to raise the temperature of 2 moles of rock salt from 10 K to 50 K?

383. A sonometer wire fixed at one end has a solid mass M hanging from its other end to produce tension in it. It is found that 70 cm length of the wire produces a certain fundamental frequency when plucked. When the same mass is hanging in water, completely submerged in it, it is found that the length of the wire has to be changed by 5 cm if it is to produce the same fundamental frequency. The density (in in g/cc) of the material of the mass M hanging from the wire is $\frac{196}{X}$. The value of X is:

384. The equation of a transverse wave produced in a taut wire of length 64 cm and mass 50 gm is given as y = 2cm sin[200π (s^{-1})t–π/4 (cm^{-1})x]. What is the tension in the string?

385. A body cools down from 80°C to 70°C in 5 minute and from 70°C to 60°C in another 10 minutes. What is the temperature of the surrounding (in °C)?

386. Two wires are fixed on a sonometer. Their tensions are in the ratio 8 : 1, their lengths are in the ratio 36 : 35, the diameters are in the ratio 4 : 1 and densities are in the ratio 1 : 2. Find the frequencies of the beats produced (in per second) if the note of the higher pitch has a frequency of 360 per second.

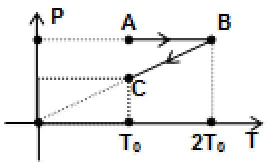
387. Air is filled in a bottle at atmospheric pressure and it is corked at 35°C. If the cork can come out at 4 atmospheric pressure. Then upto what temperature in °C should the bottle be heated in order to remove the cork (in °C)?

388. A steel wire of length 1.0 m and radius 1.0 mm is stretched without tension between two rigid supports at a temperature of 30°C. The tension in Newton in wire when the temperature falls to 20°C is X/2. Find the value of X. The coefficient of linear expansion and the Young's modules of steel are 1.1×10^{-5}/°C and 2.1×10^{11}N/m^2 respectively

389. A set of 65 tuning forks is so arranged that each gives 3 beats per second with the preceding one. If the last fork is the active of the first, find the frequency in Hz of the first tuning fork

390. A body of mass 10 kg is pulled on a rough horizontal surface with a constant speed of 1 m/s for 4 seconds. The rise in temperature of the body if 50% of the heat generated is absorbed by the body is: (Take, coefficient of friction = 0.5, specific heat of the body = 1J/kg/k and g = 10 m/s^2).

391. The P-T diagram for 1 mole of an ideal diatomic gas is as shown. Then the total work done (in KJ) during ABC is X/10. The value of X is: (Take, T_0 = 100 K, R = 8 J/mol/K)

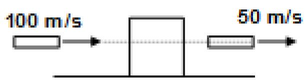


392. An ice cube of mass 0.1 kg at –10°C is kept in an isolated container. 90 gm of water at 10°C is poured into the container. Assume there is no heat gain or loss by the container. Find the equilibrium temperature of the mixture

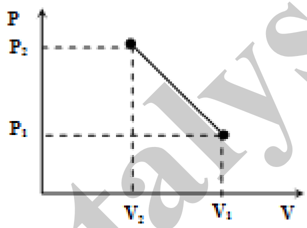
393. The heat added to a monoatomic ideal gas in a process is equal to the loss in its internal energy. When the temperature doubles then the volume becomes $\frac{n}{32}$ times its initial value. Find the value of n.

394. Two Strings S_1 & S_2 are made of silver & are kept under same tension. If S_1 has a radius twice that of S_2, what should be the length of S_2 in m. If first over tone of S_1 is equal fourth harmonic of S_2 & length of S_1 = 0.5 m.

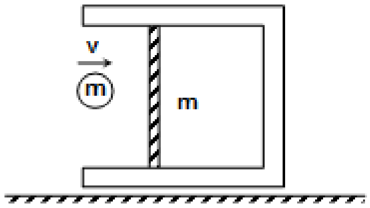
395. A bullet of mass 100 g moving with velocity 100 m/s hits an ice block of same mass at 0°C kept on a smooth horizontal surface. The bullet just comes out of the ice block and its velocity becomes 50 m/s. After some time 0.5g of the block melts. Assume all the energies are in the forms of kinetic and thermal energies. Given $L_{fusion} = 80$ cal/g, S_{water} = 1 cal/gK, mechanical equivalent of heat = 4.2 J/cal. Estimate the thermal energy in Joule going into the bullet



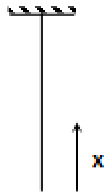
396. 20gm of helium in a cylinder under piston is transferred infinitely slowly from a state with a volume of v_1 = 30 litre and pressure P_1 = 5 atm to a state with volume v_2 = 10 litre and pressure P_2 = 15 atm. What maximum temperature in K (closest integer value) will the gas reach in this process if it is depicted on PV diagram as a straight line? [Given R = 0.082 line atm/mole-k



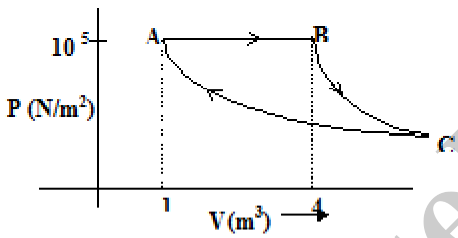
397. An adiabatic chamber has frictionless insulated piston of mass m. Mass of the remaining chamber is 4m. An ideal monoatomic gas is present inside the chamber in very small quantity i.e. the mass of gas is very small compared to the system (Take n mole) at atmospheric temperature and pressure. Piston is in relaxed state and confined to move horizontally along the length of chamber. A block of mass m moving horizontally with speed v, strikes elastically with the piston. The change in temperature of the gas is ΔT, when the compression of the piston is maximum. Then change in temperature of gas is $\frac{Kmv^2}{15nR}$, then the value of K is ………………(R is gas constant)



398. A rope of length L hangs vertically from a rigid support. The linear mass density varies as $\mu = \alpha x$ from the lower end, where x is the distance from lower end. A wave pulse is generated at the lower end. The time taken by the pulse to reach the upper end is $K\sqrt{\frac{2L}{g}}$, then the value of K is ....



399. A fixed mass of gas is taken through a process $A \rightarrow B \rightarrow C \rightarrow A$. Here $A \rightarrow B$ is isobaric. $B \rightarrow C$ is adiabatic and $C \rightarrow A$ is isothermal. The work done in the process is 4.9×10^n J, then the value of n is ....(take $\gamma = 1.5$)



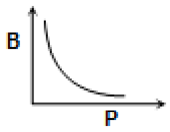
400. A uniform rope of length 12 m and mass 6 kg hangs vertically from a rigid support. A block of mass 2kg is attached to the free end of the rope. A transverse pulse of wavelength 0.06 m is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of rope is ____________ centimeter.

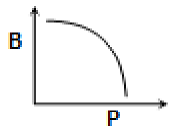
401. A body at temperature 40ºC is kept in a surrounding of constant temperature 20ºC. It is observed that its temperature falls to 35ºC in 10 minutes. Find how much time (in min) will it take for the body to attain a temperature of 30ºC by approximate method & report your answer in the nearest whole number.

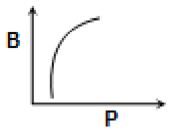
EXERCISE 02

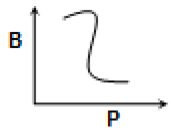
SECTION : (A) - Single Correct Options

402. Which graph correctly represent variation of $B = \frac{-\partial v / \partial p}{v}$ with p for an ideal gas at constant temperature

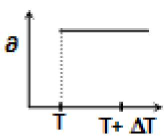
(A) 

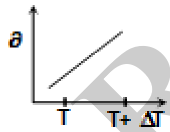
(B) 

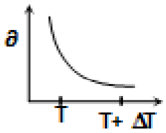
(C) 

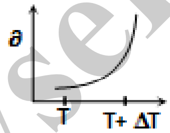
(D) 

403. An ideal gas is initially at temperature T, volume V, its volume is increased by Δv due to an increase in temperature ΔT, pressure remaining constant. The quantity $\delta = \frac{\Delta v}{v \Delta T}$ varies with temperature as

(A) 

(B) 

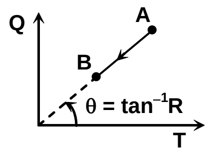
(C) 

(D) 

404. Gaseous hydrogen contained initially under standard condition in a sealed vessel of 5 litre was cooled by 55 K. The internal energy will change by

(A) 0.23 kJ (B) 0.5 kJ (C) 0.1 kJ (D) none of these

405. The heat input (Q) vs temperature (T) curve for a process for 1mole of monoatomic gas is



The corresponding PV equation for the process will be

(A) PV = constant (B) PV^3 = constant (C) $PV^{3/2}$ = constant (D) PV^5 = constant

406. The pressure and density of a diatomic gas is changed adiabatically from (p, d) to (p′, d′). If $\frac{d}{d'} = 32$. Find value of $\frac{P}{P'}$.

(A) $\frac{1}{128}$ (B) 32 (C) 128 (D) none of these

407. Two air column closed at one end with lengths 40 cm and 40.5 cm length are sounding their fundamental note at temperature of 36°C. They produce 14 beats in 5 seconds. The velocity of sound in air at 0°C is

(A) 340 m/s (B) 350 m/s (C) 360m/s (D) none

408. The explosion of a fire cracker in air at height of 40 m produces a 100 dB sound level at ground below. What is the instantaneous total radiated power assuming it radiates as point source.

(A) 151watt (B) 201 watt (C) 251 watt (D) 301 watt

409. A whistling train approaches a junction. An observer standing at junction observes frequency to be 2.2 kHz and 1.8 kHz of approaching and receding train. The speed of train is (speed of sound = 300 m/s)

(A) 25 m/s (B) 30 m/s (C) 35 m/s (D) 40 m/s

410. A spherical black body with radius 12 cm radiates 450 W power at 500 K. If radius were halved and temperature doubled power radiated in watt would be

(A) 225 (B) 450 (C) 900 (D) 1800

411. A closed cylinder contain acetylene at 27°C under pressure of 4.05 M pa. What will be the pressure in cylinder after half of mass of gas has been used up if temperature has there by fallen to 12°C.

(A) 1.94×10^6 Pa (B) 2.025×10^6 Pa (C) 1.025×10^6 Pa (D) 3.075×10^6 Pa

412. Apparent frequency is n_1, when a source approaches a stationary observer with speed u and is n_2 when observer approaches same stationary source with same speed. Then

(A) $n_2 > n_1$ (B) $n_1 > n_2$ (C) $n_1 = n_2$ (D) none of these

413. If R stands for the gas constant and C_p, C_v are specific heats per mole for a solid, then (for a solid) which of the following is correct?

(A) $C_p - C_v = R$ (B) $C_p - C_v < R$ (C) $C_p - C_v = 0$ (D) $C_p - C_v < 0$

414. If escape velocity from the earth is 11.1 km/s and the mass of one molecule of oxygen is 5.34×10^{-26} kg, the temperature at which the oxygen molecule will escape from earth, is

[Boltzmann constant k = 1.38×10^{-23} J/K]

(A) 1.6×10^5 K (B) 1.6×10^3 K (C) 1.6×10^2 K (D) none of these

415. Three resonant frequencies of a string are 75, 125 and 175 Hz. If the length of the string is 1 metre, the speed of transverse wave in the string is

(A) 50 m/s (B) 100 m/s (C) 150 m/s (D) none of these

416. An electric lamp draws a steady power P from a voltage source. If the emissivity of the filament is e and its surface area is A, Stefan's constant is σ, steady state temperature of the filament will be

(A) $\left(\frac{P}{Ae\sigma}\right)^{3/4}$ (B) $\left(\frac{P}{Ae\sigma}\right)^{5/4}$ (C) $\left(\frac{P}{Ae\sigma}\right)^{1/4}$ (D) $\left(\frac{P}{Ae\sigma}\right)^{1/2}$

SECTION : (B) - More Than One Correct Options

417. As a wave propagates :

(A) the wave intensity remains constant for a plane wave

(B) the wave intensity decreases as the inverse of the distance from the source for a spherical wave

(C) the wave intensity decreases as the inverse square of the distance from the source for a spherical wave.

(D) total power of the spherical wave over the spherical surface centered at the source remains constant at all times.

418. An ideal gas (γ = 1.5) undergoes a thermodynamic process in which temperature and density of the gas are related as $T\rho^2$ = constant. Choose the correct statement(s)

(A) Pressure is inversely proportional to volume during process

(B) Pressure is directly proportional to volume

(C) Molar specific heat capacity for the process is 2.5R

(D) Molar specific heat capacity for the process is 3R

419. The molar heat capacity for an ideal gas

(a) is zero for an adiabatic process

(b) is infinite for an isothermal process

(c) depends only on the nature of the gas for a process in which either volume or pressure is constant

(d) is equal to the product of the molecular weight and specific heat capacity for any process

420. For an ideal gas,

(a) the change in internal energy in a constant-pressure process from temperature T_1 to T_2 is equal to $nC_V(T_2-T_1)$. Where C_V is the molar heat capacity at constant volume and n is the number of moles of the gas

(b) the change in internal energy of the gas and the work done by the gas are equal in magnitude in an adiabatic process

(c) the internal energy does not change in an isothermal process

(d) no heat is added or removed in an adiabatic process

421. If the third harmonic of vibaration in an open air pipe equals the fifth harmonic of vibration in a closed pipe, then the possible values of the length of air colun in the closed pipe and open pipe, respectively, are:

(a) 100 cm, 120 cm (b) 60 cm, 72 cm (c) 120cm, 150 cm (d) 150 cm, 180 cm

422. The free ends of the rod of length (L= 0.5m) form anti-nodes while rod is clamped at distance L/4 from one free end. The density of metal of rod and its Young's modulus are 8×10^3 kg/m^3 and 200×10^9 N/m^2 respectively. Choose the correct statements

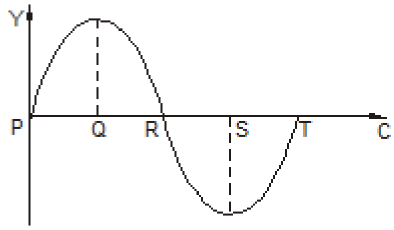
(a) There are three frequencies of longitudinal waves in rod in the range of 0 kHz to 50 kHz

(b) There are two frequencies of longitudinal waves in rod in the range of 0 kHz to 50 kHz

(c) Speed of longitudinal sound wave in rod is 4×10^3 m/s

(d) Speed of longitudinal sound wave in rod is 5×10^3 m/s

423. A sound wave is travelling along positive x-direction. Displacement (y) of particles from their mean positions at anytime t is shown in the figure.



(a) Particle located at S has zero velocity.
(b) Particle located at T has its velocity in the negative direction
(c) Change in pressure at S is zero
(d) Particles located near R under compression

424. During an experiment, an ideal gas is found to obey a condition $\frac{P^2}{\rho}$ = constant [ρ = density of the gas]. The gas is initially at temperature T, pressure P and density ρ. The gas expands such that density changes to $\rho/2$.

(a) The pressure of the gas changes to $\sqrt{2}P$.

(b) The temperature of the gas changes to $\sqrt{2}T$.

(c) The graph of the above process on the P-T diagram is parabola.

(d) The graph of the above process on the P-T diagram is hyperbola.

425. A wave is represented by the equation:

$$y = (lmm)\sin\left[\left(50s^{-1}\right)t + \left(2.0m^{-1}\right)x\right] + (1mm)\cos\left[\left(50s^{-1}\right)t - \left(2.0m^{-1}\right)x\right]$$

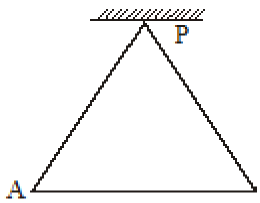
(a) The wave-velocity is zero, since it is a standing wave.

(b) A node is formed at $x = \frac{3\pi}{8}m$.

(c) The amplitude of the oscillation at the antinode is 2 mm.

(d) Energy transfer occurs along the positive x-axis.

426. A very light rod AB is initially hung from a point P by means of two identical copper wires of the same length as the rod as shown in the figure. Particles of masses 1 kg and 4 kg are then attached to the ends A and B of the rod. The ratio of the fundamental frequencies of vibration of the wires AB and BP, i.e., $\frac{f_A}{f_B} =$



(a) 4 (b) $\frac{1}{2}$ (c) 16 (d) 2

427. A metal cylinder of 5 kg is heated electrically by a 20 W heater in a room at 20ºC. The cylinder temperature rises uniformly to 30ºC in 5 minute and finally becomes constant at 45ºC. Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings :

(A) The rate of loss of heat of cylinder to surrounding at 25ºC is 4W

(B) The rate of loss of heat of cylinder to surrounding at 45ºC is 20W

(C) The rate of loss of heat of cylinder to surrounding at 25ºC is 8W

(D) The rate of loss of heat of cylinder to surrounding at 45ºC is zero

428. Monoatomic , diatomic and triatomic gases whose initial volume and pressure are same, each is compressed till their pressure becomes twice the initial pressure. Then :

(A) if the compression is isothermal , then their final volumes will be same

(B) if the compression is adiabatic , then their final volumes will be different

(C) if the compression is adiabatic , then triatomic gas will have maximum final volume

(D) if the compression is adiabatic , then monoatomic gas will have maximum final volume

429. Standing waves are produced on a stretched string of length L with fixed ends . When there is a node at a distance L/3 from one end, then :

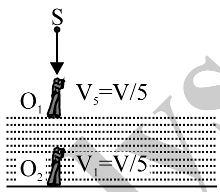
(A) minimum and next higher number of nodes excluding the ends are 2, 5 respectively

(B) minimum and next higher number of nodes excluding the ends are 2, 4 respectively

(C) frequency produced may be $\frac{V}{3L}$

(D) frequency produced may be $\frac{3V}{2L}$ [V = Velocity of waves in the string]

430. In the figure shown an observer O_1 floats (static) on water surface with ears in air while another observer O_2 is moving upwards with constant velocity V_1 = V/5 in water. The source moves down with constant velocity V_s = V/5 and emits sound of frequency 'f'. The velocity of sound in air is V and that in water is 4V. For the situation shown in figure.



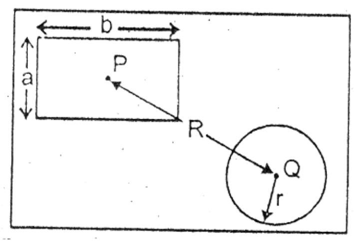
(A) The wavelength of the sound received by O_1 is $\frac{4V}{5f}$

(B) The wavelength of the sound received by O_1 is V/f

(C) The frequency of the sound received by O_2 is $\frac{21f}{16}$

(D) The wavelength of the sound received by O_2 is $\frac{16V}{5f}$

431. There is a rectangular metal plate in which two cavities in the shape of rectangle and circle are made, as shown with dimensions. P and Q are centres of these cavities. On heating the plate, which of the following quantities increase ?



(A) πr^2 (B) ab (C) R (D) b

432. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes

(1) isothermal expansion (2) adiabatic expansion

(3) free expansion (4) expansion at constant pressure

If E_1, E_2, E_3 and E_4 respectively are the changes in average kinetic energy of the molecules for the above four processes, then :

(A) $E_2 = E_3$ (B) $E_1 = E_3$ (C) $E_1 > E_4$ (D) $E_4 > E_3$

433. In Newton's law of cooling, $\frac{d\theta}{dt} = -k(\theta - \theta_0)$, the constant 'k' is proportional to :

(A) A, surface area of the body (B) S, specific heat of the body

(C) $\frac{1}{m}$, m being mass of the body (D) e, emmisivity of the body

434. A cube, a pyramid (with all four faces identical) and a sphere (all of them hollow) are made from the same material and have equal mass and bound equal volume. They are heated to the same temperature and then left to cool. After some time,

1. sphere will have the highest temperature 2. pryamid will have the highest temperature.

3. cube will have the lowest temperature. 4. sphere will have the lowest temperature.

5. pyramid will have the lowest temperature

Correct option will be :

(A) 2 and 3 (B) 3 and 1 (C) 1 and 5 (4) 2 and 4

435. Which of the following functions represent a stationary wave ? Here a, b and c are constants

(A) y = a cos (bx) sin (ct) (B) y = a sin (bx) cos (ct)

(C) y = a sin (bx + ct) (D) y = a sin (bx + ct) + a sin (bx − ct)

436. $Y(x, t) = \frac{0.8}{(4x+5t)^2+5}m$ represents a moving pulse, where x, y are in metre and t in second :

(a) It is a non-periodic travelling pulse (b) It is traveling along –ve x-axis.

(c) Maximum displacement is 0.16m (d) Velocity of motion of pulse is -1.25 ms^{-1}

437. A plane progressive wave of frequency 25 Hz, amplitude 2.5×10^{-5} m and initial phase zero moves along the negative x-direction with a velocity of 300 m/s. A and B are two points 6m apart on the line of propagation of the wave. At any instant the phase difference between A and B is ϕ. The maximum difference in the displacement of the particles at A and B is Δ, then

(a) $\phi = \pi$ (b) $\phi = 0$ (c) $\Delta = 0$ (d) $\Delta = 5\times10^{-5} m$

438. A heated body emits radiation which has maximum intensity at frequency υ_m. If the temperature of body is doubled:

(a) the maximum intensity radiation will be at frequency $2\upsilon_m$.

(b) the maximum intensity radiation will be at frequency $\upsilon_m / 2$.

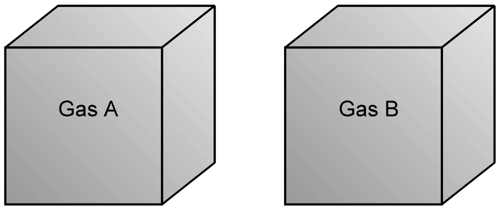
(c) the total emitted energy will increase by a factor 16.

(d) the total emitted energy will increase by a factor 2

SECTION : (C) -Passage Type Questions

PASSAGE 01:

Two closed identical conducting containers are found in the laboratory of an old scientist. For the verification of the gas some experiments are performed on the two boxes and the results are noted.



Experiment 1. When the two containers are weighed W_A = 225 g , W_B = 160 g and mass of evacuated container W_C = 100 g.

Experiment 2. When the two containers are given same amount of heat same temperature rise is recorded. The pressure change found are

$$\Delta P_A = 2.5 \text{ atm. } \Delta P_B = 1.5 \text{ atm.}$$

Required data for unknown gas :

Mono (molar mass)	He 4g	Ne 20g	Ar 40 g	Kr 84 g	Xe 131 g	Rd 222 g
Dia (molar mass)	H_2 2g	F_2 19 g	N_2 28g	O_2 32g	Cl_2 71 g	

439. Identify the gas filled in the container A and B.
(A) N_2, Ne (B) He, H_2 (C) O_2, Ar (D) Ar, O_2

440. Total number of molecules in 'A' (here N_A = avagadro number)
(A) $\frac{125}{64}N_A$ (B) 3.125 N_A (C) $\frac{125}{28}N_A$ (D) 31.25 N_A

441. The initial internal energy of the gas in container 'A', If the containers were at room temperature 300K initially
(A) 1406.25 cal (B) 1000 cal (C) 2812.5 cal (D) none of these

PASSAGE 02:

A pulse is started at a time t = 0 along the +x direction on a long, taut string. The shape of the pulse at t = 0 is given by function f(x) with

$$f(x) = \begin{cases} \frac{x}{4}+1 & \text{for} \quad -4 < x \le 0 \\ -x+1 & \text{for} \quad 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

here f and x are in centimeters. The linear mass density of the string is 50 g/m and it is under a tension of 5N,

442. The shape of the string is drawn at t = 0 and the area of the pulse enclosed by the string and the x-axis is measured. It will be equal to
(A) 2 cm^2 (B) 2.5 cm^2 (C) 4 cm^2 (D) 5 cm^2

443 The vertical displacement of the particle of the string at x = 7 cm and t = 0.01 s will be
(A) 0.75 cm (B) 0.5 cm (C) 0.25 cm (D) zero

444. The transverse velocity of the particle at x = 13 cm and t = 0.015 s will be
(A) –250 cm/s (B) –500 cm/s (C) 500 cm/s (D) –1000 cm/s

PASSAGE 03:

The apparent frequency of sound detected by an observer depends on relative motion between source and observer. This effect is well known to you (known as Doppler effect). Consider a police Car moving on a straight road with a speed of 28 m/s emitting sound from its siren at 1200 Hz. This means that the siren is oscillating 1200 times a second. A man 'M' is standing on the road infront of the Car and another person 'N' is on the back side of the car. The siren sound is getting reflected from a building directly infront of the car. Man 'M' is standing between the car and the building. You may assume car, N,M and building all to lie on a straight line and take the speed of sound in air 340 m/s :

445. The wavelength of sound reaching M and N (directly from the car) respectively are :
(A) 0.260 m, 0.306 m (B) 0.260 m, 0.406 m (C) 0.306 m (D) 0.306 m, 0.206 m

446. What is the frequency of the sound reflected from the building as detected by man M ?
(A) 1208 Hz (B) 1111 Hz (C) 1308 Hz (D) 1415 Hz

447. The apparent frequency of the reflected sound as heard by the police in the car is :
(A) 1208 Hz (B) 1111 Hz (C) 1308 Hz (D) 1415 Hz

PASSAGE 04:

A metallic rod of length 1 m is rigidly clamped at its mid–point. Longitudinal stationary waves are set up in the rod in such a way that there are two nodes on either side of the mid–point . The amplitude of an antinode is 2 x 10^{-6}m. [Young's modulus = 2 x 10^{11} Nm^{-2}, density = 8000 Kg m^{-3}). Origin is at O which is mid-point of the rod. P is a point on the rod such that OP = 10 cm.

448. Velocity of wave in the rod is
(A) 5000 cm/s (B) 1000 m/s (C) 500 m/s (D) 5000 m/s.

449. Magnitude of maximum velocity of the point P is
(A) $\frac{\pi}{20}$ m/s (B) $\frac{\pi}{20}$ cm/s (C) $\frac{\pi}{40}$ m/s (D) none of these

450. Equation of constituent waves in the rod (in M.K.S. unit) are
(A) $y_1 = 10^{-6} \sin(5\pi x - 25000\,\pi t)$, $y_2 = 10^{-6} \sin(5\pi x + 25000\,\pi t)$
(B) $y_1 = 10^{-6} \sin(-5\pi x + 25000\,\pi t)$, $y_2 = 10^{-6} \sin(5\pi x + 25000\,\pi t)$
(C) $y_1 = 2 \times 10^{-6} \cos(5\pi x - 25000\,\pi t)$, $y_2 = 2\times10^{-6} \cos(5\pi x + 25000\,\pi t)$
(D) $y_1 = 2\times10^{-6} \sin(5\pi x - 25000\,\pi t)$, $y_2 = 2\times10^{-6} \sin(5\pi x + 25000\,\pi t)$

PASSAGE 05:

Two plane harmonic sound waves are expressed by the equations.

$$y_1(x, t) = A\cos(0.5\,\pi x - 100\,\pi t)$$
$$y_2(x, t) = A\cos(0.46\,\pi x - 92\,\pi t)$$

(All parameters are in MKS) :

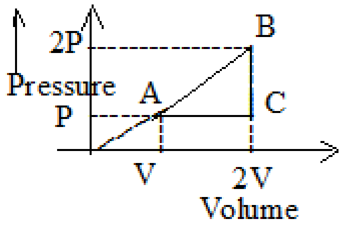
451. How many times does an observer hear maximum intensity in one second ?
(A) 4 (B) 10 (C) 6 (D) 8

452. What is the speed of the sound ?
(A) 200 m/s (B) 180 m/s (C) 192 m/s (D) 96 m/s

453. At x = 0 how many times the amplitude of $y_1 + y_2$ is zero in one second?
(A) 192 (B) 48 (C) 100 (D) 96

PASSAGE 06:

For a thermodynamic process, the work done can be given as the area under the curve on a P-V diagram. One mole of a monatomic idealgas taken along the cycle ABCA as shown in the diagram



454. Select the correct statement:
(a) since area with in the cycle of the graph is always positive so work done is also always positive.
(b) The area with in the cycle gives complete information of work done in a process.
(c) only magnitude of work done is decided by area with in the cycle and complete information can be deduced if cycle is known.
(d) none of the above.

455. The net heat absorbed by the gas in given the cycle is (If cycle is clockwise)
(a) PV (b) PV/2 (c) 2 PV (d) 4P/V

456. The ratio of specific heat (C_1) in the process CA to the specific heat (C_2) in the process BC is]
(a) 2 (b) 5/3 (c) 4 (d) none

PASSAGE 07:

A monoatomic ideal gas sample is given heat Q in a reversible process. One fourth of this heat is used as work done by the gas due to expansion and rest is used for increasing its internal energy.

457. The molar specific heat for the gas in this process is

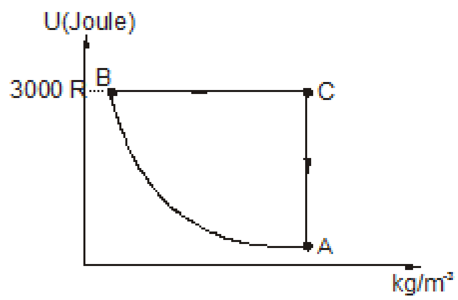
(a) $\frac{3}{2}R$ (b) $\frac{R}{2}$ (c) 2R (d) 3R

458. The equation of process in terms of volume and temperature is

(a) $\frac{V}{T} =$ constant (b) $\frac{V}{\sqrt{T}} =$ constant (c) VT = constant (d) $V\sqrt{T} =$ constant

PASSAGE 8:

The figure shows the variation of potential energy (U) of a 2 moles of Argon gas with its density in a cyclic process ABCA. The gas was initially in the state A whose pressure and temperature are $P_A = 2$ atm $T_A = 300$ K respectively. It is also stated that the path AB is a rectangular hyperbola and the internal energy of the gas at state C is 3000 R. Based on the above information answer the following questions.

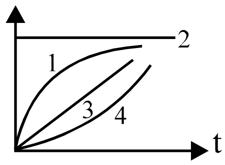


459. The heat supplied to the gas in the process AB is
(a) 700 R (b) 3500 R (c) 4400 R (d) 1600 R

460. Heat supplied in the process CA is
(a) -1400 R (b) 1400 R (c) 2100 R (d) -2100 R

SECTION : (D) - Matrix Match

461. A ball has surface temperature T initially at time t = 0, that is less than surrounding constant temperature T_0. On the vertical axis of the graph shown has either thermal energy radiated/absorbed per unit time or total energy radiated/absorbed till time t by the ball. Correctly match the curves marked in the graph :



(A)	Thermal energy emitted per unit time	(P)	1
(B)	Thermal energy absorbed per unit time	(Q)	2
(C)	Total energy emitted till time t	(R)	3
(D)	Total energy absorbed till time t	(S)	4

462. Match the information given in **Column-I** with that given in **Column-II**. Note that any information in column-I may have more than one matching options in column-II

	Column I (Nature of wave)		**Column II (Properties)**
(A)	Transverse progressive wave	(P)	Amplitude of all particles are same
(B)	Longitudinal progressive wave	(Q)	Phase of all particles may be same
(C)	Transverse standing wave	(R)	May occur in gases
(D)	Longitudinal standing wave	(S)	KE and PE of a small element may be maximum simultaneously

463. A source is emitting a sound of frequency fs and moving with a velocity vs. An observer at some distance from the source has velocity v0 and heard sound of frequency f0. Velocity of medium is vm,, velocity of sound w.r.t to medium is V.

	COLUMN-I		**COLUMN-II**
P)	vs = v0 = 0 and vm has direction from source to observer	1)	fs > f0
Q)	vm has direction from source to observer and vs is towards observer v0 = 0	2)	fs < f0
R)	vm has direction from source to observer and at the time of generating pulse vs is perpendicular to the line joining source and observer v0 = 0	3)	fs = f0
S)	If vm = 0 and v0 & vs are moving in the same direction joining of the source and observer	4)	Whether f0 is greater than fs or less than fs will depend on magnitude of vs and v0 along the line

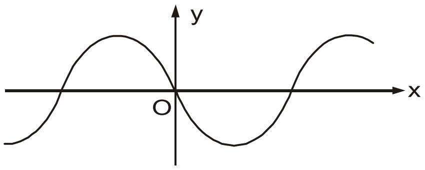
464.

	Column I		**Column II**
(A)	Pitch	(p)	Number of harmonics present in the sound
(B)	Loudness	(q)	Intensity
(C)	Quality	(r)	Frequency
(D)	wave front	(s)	Wave form
		(t)	locus of points vribrating in a phase

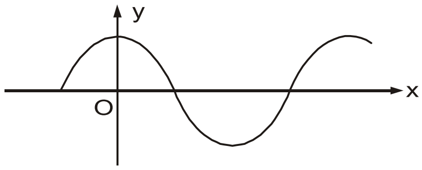
465.

	Column-I		**Column II**
(A)	$y = 4\sin(5x - 4t) + 3\cos(4t - 5x + \pi/6)$	(p)	Particles at every position are performing SHM
(B)	$y = 10\cos\left(t - \frac{x}{330}\right)\sin(100)\left(t - \frac{x}{330}\right)$	(q)	Equation of travelling wave
(C)	$y = 10\sin(2\pi x - 120t) + 10\cos(120t + 2\pi x)$	(r)	Equation of standing wave
(D)	$y = 10\sin(2\pi x - 120t) + 8\cos(118t - 29/30\pi x)$	(s)	Equation of Beats

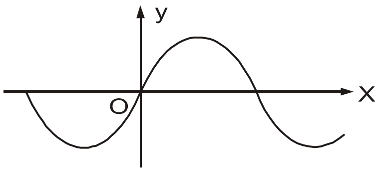
466. For four sine waves, moving on a string along positive x direction, displacement-distance curves (y – x curves) are shown at time t = 0. In the right column, expressions for y as function of distance x and time t for sinusoidal waves are given. All terms in the equations have general meaning. Correctly match y – x curves with corresponding equations.

(A) 

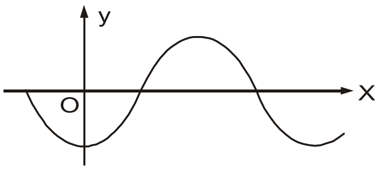
(p) $y = A\cos(\omega t - kx)$

(B) 

(q) $y = -A\cos(kx - \omega t)$

(C) 

(r) $y = A\sin(\omega t - kx)$

(D) 

(s) $y = A\sin(kx - \omega t)$

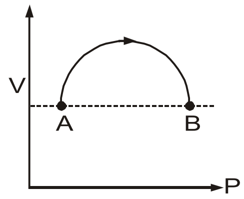
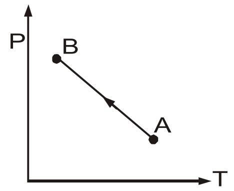
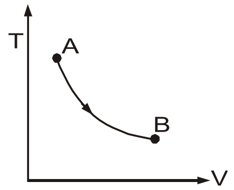
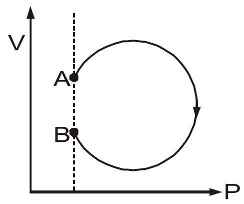
467.

	Column-I		**Column-II**
(A)	Atight string is fixed at both ends and sustaining standing wave	(p)	At the middle, antinode is formed in odd harmonic
(B)	Atight string is fixed at one end and free at the other end	(q)	At the middle, node is formed in even harmonic
(C)	A tight string is fixed at both ends and vibrating in four loops	(r)	The frequency of vibration is 300% more than its fundamental frequency
(D)	A tight string is fixed at one end and free at the other end, vibrating in 2nd mode of vibration.	(s)	Phase difference between SHMs of any two particles will be either π or zero.
		(t)	The frequency of vibration is 400% more than fundamental frequency.

468. An ideal monoatomic gas undergoes different types of processes which are described in column–I. Match the corresponding effects in column–II, The letters have usual meaning.

	Column-I		**Column-II**
(A)	$P = 2V^2$	(p)	If volume increases then temperature will also increase.
(B)	PV^2 = Constant	(q)	If volume increases then temperature will decrease.
(C)	$C = C_V + 2R$	(r)	For expansion, heat will have to be supplied to the gas.
(D)	$C = C_V - 2R$	(s)	If temperature increases then work done by gas is positive.

469. A sample of gas goes from state A to state B in four different manners, as shown by the graphs. Let W be the work done by the gas and ΔU be change in internal energy along the path AB. Correctly match the graphs with the statements provided.

Column–I		Column–II	
(A)		(p)	Both W and ΔU are positive
(B)		(q)	Both W and ΔU are negative
(C)		(r)	W is positive whereas ΔU is negative
(D)		(s)	W is negative whereas ΔU is positive

470. Match the information given in **Column-I** with that given in **Column-II**. Note that any information in column-I may have more than one matching options in column-II

	Column I (Nature of wave)		**Column II (Properties)**
(A)	Transverse progressive wave	(P)	Amplitude of all particles are same
(B)	Longitudinal progressive wave	(Q)	Phase of all particles may be same
(C)	Transverse standing wave	(R)	May occur in gases
(D)	Longitudinal standing wave	(S)	KE and PE of a small element may be maximum simultaneously

471. Match the phenomenon to its property

(A)	Beats	(P)	apparent change in pitch
(B)	Standing wave	(Q)	frequencies are in unison
(C)	Doppler effect	(R)	modification of intensities occur periodically when the waves interfere in same direction
(D)	Resonance	(S)	superposition of two similar waves travelling in opposite direction

472. For a cuboid made of a certain material, coefficient of expansion along it's length was found to be $2 \times 10^{-5}\ K^{-1}$. Then match the columns appropriately:

	COLUMN I		**COLUMN II**
(A)	Coefficient of expansion along breadth	(P)	$1 \times 10^{-5}\ K^{-1}$
(B)	Coefficient of expansion along height	(Q)	$2 \times 10^{-5}\ K^{-1}$
(C)	Coefficient of areal expansion	(R)	$4 \times 10^{-5}\ K^{-1}$
(D)	Coefficient of volume expansion	(S)	$6 \times 10^{-5}\ K^{-1}$
		(T)	$8 \times 10^{-5}\ K^{-1}$

473. Bowling point of water in a certain scale was found to be 300 units. Melting point of water in the same scale was found to be -200 units. Then match the temperatures in in celcius scale (in column I) with corresponding temperatures in that scale (in column II)

	COLUMN I		**COLUMN II**
(A)	20°C	(P)	50
(B)	75°C	(Q)	-100
(C)	60°C	(R)	175
(D)	50°C	(S)	100

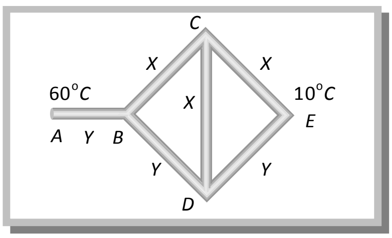
474. One mole of an ideal monoatomic gas undergoes a process $PV^{1/2} = C$, where C is a constant.

	COLUMN I		**COLUMN II**
(A)	Heat given to the gas to increase the temperature of the gas by 100°C	**(i)**	150 R
(B)	Work done on the gas to change the volume from V_0 to $2V_0$	**(ii)**	$\frac{7R}{2}$
(C)	Molar heat capacity of the gas	**(iii)**	350 R
(D)	Change in internal energy of gas when the temperature of the gas is increased by 100°C	**(iv)**	$2c\sqrt{v_0}[\sqrt{2}-1]$

475. For one mole of a monoatomic ideal gas match the following:

	COLUMN I		**COLUMN II**
(A)	Isothermal bulk modulus	**(i)**	$-\frac{RT}{V^2}$
(B)	Adiabatic bulk modulus	**(ii)**	$-\frac{5P}{3V}$
(C)	Slope of P–V graph in isothermal process	**(iii)**	$\frac{T}{V}$
(D)	Slope of PV graph in adiabatic process	**(iv)**	None of these

476. Three rods of material *X* and three rods of material *Y* are connected as shown in figure. All are identical in length and cross-sectional area. If end *A* is maintained at 60°*C*, end *E* at 10°*C*, thermal conductivity of *X* is 0.92 *cal/sec-cm-°C* and that of *Y* is 0.46 *cal/sec-cm-°C*, then match the columns appropriately (Assume steady state):



	COLUMN I		**COLUMN II**
(A)	T_B (in °C)	(P)	30
(B)	T_C (in °C)	(Q)	20
(C)	T_D (in °C)	(R)	10
(D)	Heat through CD (in Joules)	(S)	0

477. Column I gives some devic4es and column II gives some processes on which the functioning of these devices depend. Match the columns:

	Column I		Column II
(A)	Bimetallic Strip	(P)	Radiation from a hot body
(B)	Steam Engine	(Q)	Energy conversion
(C)	Incandescent lamp	(R)	Melting
(D)	Electric fuse	(S)	Thermal expansion of solids

478.

	Column I		Column II
(A)	A perfect reflecting body	(P)	absorbs radiation
(B)	A perfect black body	(Q)	reflects radiation
(C)	An ordinary smooth body	(R)	emits radiations
(D)	An ordinary rough body	(S)	transfers heat

479. When a wave is transmitted from denser to rarer medium:

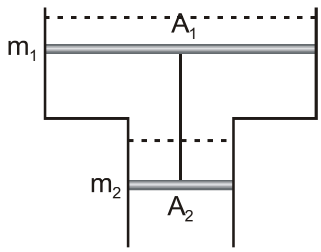
	Column I		Column II
(A)	amplitude of wave	(P)	will remain same
(B)	wavelength of wave	(Q)	will increase
(C)	speed of wave	(R)	will decrease
(D)	frequency	(S)	may increase or decrease

480. From state A (pressure P, volume V) an ideal gas is taken to the state B (pressure P, volume 2V) along a straight line path in P-V diagram.

	Column I		Column II
(A)	Workdone by gas in process A to B exceeds the workdone by it if system was taken from A to B along	(P)	isobar
(B)	In T-V diagram, path AB becomes part of	(Q)	parabola
(C)	In moving from A to B temperature T first	(R)	decreases and then increases
(D)	In moving B to A along an isotherm, temperature T first	(S)	none of these

SECTION : (E) - Integer Type

481. Consider a vertical tube open at both ends. The tube consists of two parts, each of different cross-sections and each part having a piston which can move smoothly in respective tubes. The two pistons are joined together by an inextensible wire. The combined mass of the two piston is 5 kg and area of cross-section of the upper piston is 10 cm^2 greater than that of the lower piston. Amount of gas enclosed by the pistons is one mole. When the gas is heated slowly, pistons move by 50 cm. Find rise in the temperature of the gas, in the form $\frac{X}{R}$K where R is universal gas constant. Use g = 10 m/s^2 and outside pressure = 10^5 N/m^2). Fill value of X in the answer sheet.



482. A straight line source of sound of length L = 10m, emitts a pulse of sound that travels radially outward from the source. What sound energy (in mW) is intercepted by an acoustic cylindrical detector of surface area 2.4 cm^2, located at a perpendicular distance 7m from the source. The waves reach perpendicularly at the surface of the detector. The total power emitted by the source in the form of sound is 2.2 × 10^4 W.
(Use π = 22/7)

483. A tuning fork produces 4 beats per second with another tuning fork of frequency 512 Hz. The first one is now loaded with a little wax and the beat frequency is found to increase to 6 per second. What was the original frequency of the tuning fork ?

484. Consider the three waves represented by

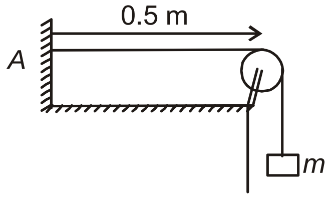
$$y_1 = 3\sin(kx - \omega t)$$

$$y_2 = 3\sin\left(kx - \omega t + \frac{2\pi}{3}\right)$$

and $y_3 = 3\sin\left(kx - \omega t + \frac{4\pi}{3}\right)$

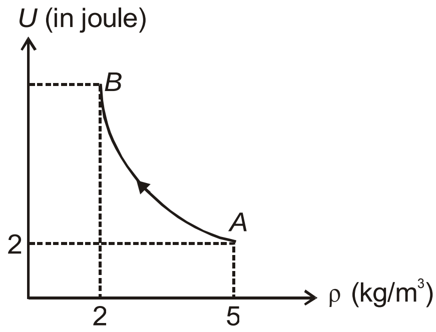
What is the amplitude of resultant of waves at x = 0?

485. A string having length 0.5 m fixed at one end and other end is connected to a block of mass m = 2 kg as shown in figure. The string is set into vibrations which is represented by $Y = 4\sin\left(\frac{\pi x}{5}\right)\cos(50\pi t)$ where x and y are in cm and t is in second. The number of antinodes between point A and the fixed pully is 2n, find n.



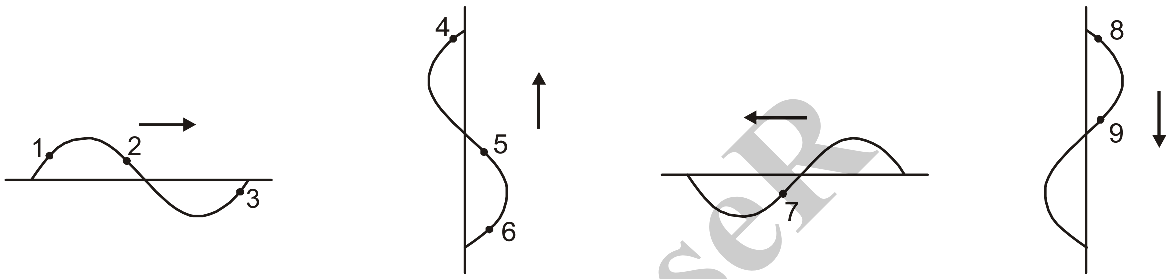
486. Figure shows the variation of the internal energy U with the density ρ of an ideal monoatomic gas for a thermodynamic process AB. Process AB is a part of rectangular hyperbola. Find the work done

(in joule) by gas in process AB.



487. The equation of a standing wave propagating along a string fixed at both ends is given by $y = (4\text{ cm})\sin(0.314\text{ cm}^{-1})x \cdot \cos(3.14\text{ s}^{-1})t$. The linear mass density of the string is 10 g/cm. The average power transmitted through the string is $n \times 10^{-4}$ W. Find n.

488. Consider the transverse mechanical waves shown in the figure. Points are marked on the wave. Arrow shows the direction of waves motion.

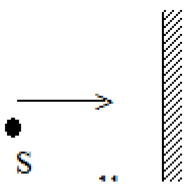


Out of the nine particles marked how many are moving towards their respective mean positions at this instant?

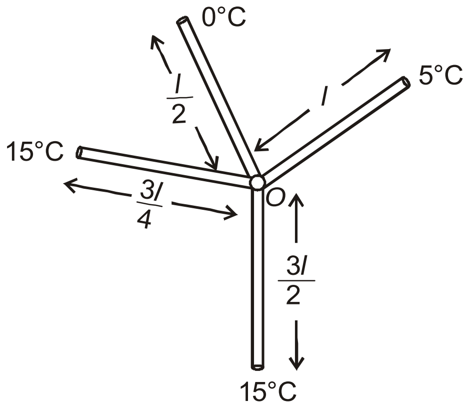
489. Three resonance frequencies of an organ pipe are at 1190, 1360 and 1530 Hz. If velocity of sound in the air is 340 m/s, Find the length of the pipe

490. One end of a uniform rod of length 1m is placed in bolling water while its other end is placed in melting ice. A point P ont he rod is maintained at a constant temperature of 800^{0}C . The mass of steam produced per second is equal to the mass of ice melted per second. If specific latent heat of steam is 7 times the specific latent heat of ice, the distance of P from the steam chamber must be 1/x mtr.find the value of x.

491. A source s having a detector D moving towards a wall with a certain velocity detects 9 beats/s. On doubling the velocity of the source, thedetector D detects 20 beats /s . What is the original frequency (in Hz) of the sound emitted by thesource ? (Take speed of sound as 330 m/s)



492. All the rods in given arrangement are made up of same material and have same cross-section. Find the temperature of junction *O* (in °C). (Neglect the transfer of heat through radiations and convection)



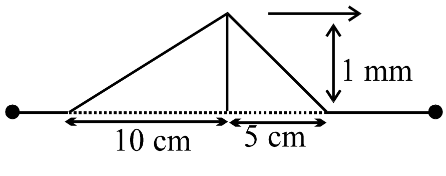
493. A source of sound emitting sound of frequency 10 kHz is moving towards a moving wall with speed 32 m/s. The wall is moving towards source with speed 64 m/s. Find the approximate wavelength (in cm) of wave reflected from the wall. Take speed of sound in air equal to 332 m/s

494. One mole of an ideal monoatomic gas is taken from state A to state B through the process $P = \frac{3}{2}T^{1/2}$. It is found that its temperature increases by 100 K in this process. Now it is taken from state B to C through a process for which internal energy is related to volume as $U = \frac{1}{2}V^{1/2}$. Find the total work performed by the gas (in Joule), if it is given that volume at B is 100 m^3 and at C it is 1600 m^3. [Use R = 8.3 J/mol-K]

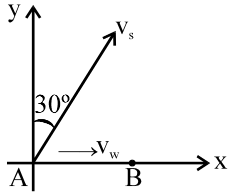
495. Assuming a particle to have the form of a sphere and to absorb all incident light, the radius (in μm) of a particle for which its gravitational attraction to the Sun is counterbalanced by the force that light exerts on it is X. Find 10X. The power of light radiated by the Sun equals $P = 4 \times 10^{26}$ W and the density of the particle is $\rho = 1.0$ g/cm^3 .

[Use $G = \frac{20}{3} \times 10^{-11}$ Nm^2/kg^2 ; $\pi = \frac{25}{8}$ and mass of the Sun $= 2 \times 10^{30}$ kg]

496. The kinetic energy of pulse travelling in a taut string is K mJ. Find the value of 100K. Given T = 10 N and μ = 0.1 kg/m.



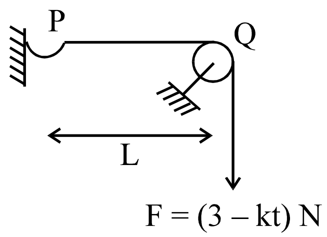
497. In the figure shown a source of sound of frequency 510 Hz moves with constant velocity v_s = 20 m/s in the direction shown. The wind is blowing at a constant velocity v_w = 20 m/s towards an observer who is at rest at point B. Find the frequency (in Hz) detected by the observer corresponding to the sound emitted by the source at initial position A.
[Speed of sound relative to air = 330 m/s]



498. We would like to increase the length of a 15 cm long copper rod of cross-section 4 mm^2 by 1 mm. The energy absorbed by the rod if it is heated is E_1. The energy absorbed by the rod if it is stretched slowly is E_2. Then find E_1/E_2.
[Various parameters of Copper are : Density = 9×10^3 kg/m^3 ; Thermal co-efficient of linear expansion = 16×10^{-6} K^{-1}, Young's modulus = 135×10^9 Pa , Specific heat = 400 J/kg-K]

499. A long uniform string of mass density 0.1 kg/m is stretched with a force of 40 N. One end of the string (x = 0) is oscillated transversely (sinusoidally) with an amplitude of 0.02 m and a period of 0.1 sec, so that travelling waves in the + x direction are set up.
(a) What is the velocity of the waves ?

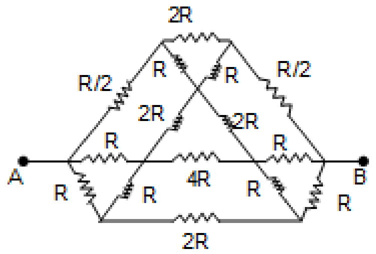
500. In the given figure, a string of linear mass density 3×10^{-2} kg/m and length L = 1 m, is stretched by a force F = (3 – kt) N, where 'k' is a constant and 't' is time in sec. At the time t = 0, a pulse is generated at the end P of the string. Find the value of k (in N/s) if the value of force becomes zero as the pulse reaches point Q.



PHYSICS
PART - III
TOPIC: ELECTRICITY & MAGNETISM
EXERCISE # 01

SECTION : (A) - Single Correct Options

501. For the shown circuit the effective resistance between the points A and B will be

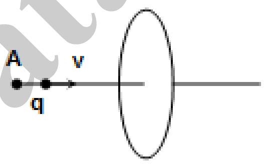


(A) 2R (B) 4R (C) R (D) R/2

502. An n sided regular polygon is inscribed in a circular region of radius R using a wire of cross sectional radius r and resistivity ρ. If the magnetic field in the circular region has a time rate of change given by $\frac{dB}{dt}$ then the current in the wire loop at any time is given by

(A) $\dfrac{\pi R \frac{dB}{dt} r^2 \cos(r/n)}{2\rho}$ (B) $\dfrac{\pi R \frac{dB}{dt} r^2 \cos(r/n)}{\sqrt{2}\rho}$ (C) $\dfrac{\pi R \frac{dB}{dt} r^2 \sin(r/n)}{2\rho}$ (D) $\dfrac{\pi R \frac{dB}{dt} r^2 \cos(r/n)}{3\rho}$

503. A charged particle carrying a charge q is moving along the axis of a conducting ring as shown in the figure. Then:



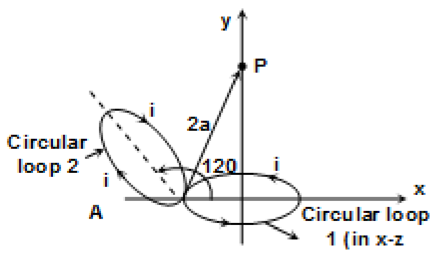
(A) No current will be induced in the loop.

(B) When viewed from A, a clockwise current will flow in the ring.

(C) Current induced will be anticlockwise when viewed from A

(D) Current induced will be initially clockwise, till the time the particle reaches the centre of the ring and then the current will be anticlockwise.

504. Planes of 2 circular loops each of radius a are inclined at an angle of 120° as shown in the figure. Each loop carries a current i as shown in the figure. The magnetic field at point P is

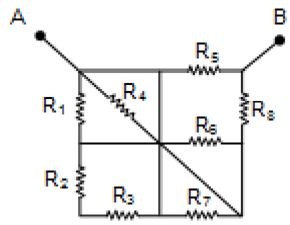


(A) $\frac{\mu_0 i}{8a}$ (B) $\frac{\mu_0 i}{16a}$ (C) $\frac{\mu_0 i}{16\pi a}$ (D) $\frac{\mu_0 i}{8\pi a}$

505. A coil has an ohmic resistance r, inductance L and reactance $X_L = 2\pi fL$. When an AC of RMS value I flows in the coil, the average power consumed in the coil will be

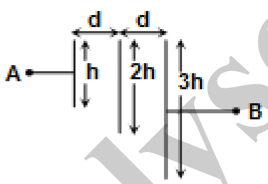
(A) I^2r (B) $I^2(r + X_L)$ (C) I^2X_L (D) $I^2\sqrt{r^2 + X_L^2}$

506. The equivalent resistance between A and B depend on the value of resistance



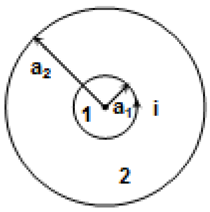
(A) R_1, R_2, R_3 and R_4 (B) R_5 and R_8 (C) R_1, R_6 and R_7 (D) R_2 and R_4

507. Three square conducting plates of side h, 2h and 3h are arranged as shown in the figure. The separation between two adjacent plates is d (see the figure). Find the equivalent capacitance between A and B if $d \ll h$



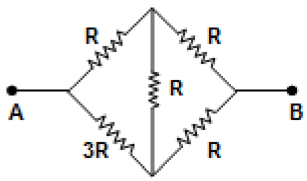
(A) $\frac{\varepsilon_0 h^2}{2d}$ (B) $\frac{36}{13}\frac{\varepsilon_0 h^2}{d}$ (C) $\frac{4}{5}\frac{\varepsilon_0 h^2}{d}$ (D) $\frac{\varepsilon_0 h^2}{d}$

508. Two circular loops 1 and 2 whose centre coincide lie in a plane. The radii of the loops are a_1 and a_2 ($a_1 \ll a_2$). A current i flows in loop 1. Find the magnetic flux ϕ_2 embraced by loop 2.



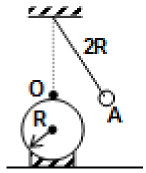
(A) $\frac{\mu_0 i}{2a_2}\pi a_1^2$ (B) $\frac{\mu_0 i}{2a_1}\pi a_2^2$ (C) $\frac{\mu_0 i}{2\pi a_1}a_2^2$ (D) None of these

509. The equivalent resistance between points A and B will be



(A) $\frac{6R}{5}$ (B) $\frac{9R}{7}$ (C) R (D) $\frac{101R}{67}$

510. A uniformly charged non conducting sphere of radius R and total charge Q is fixed on a non-conducting plane. A very small bob having negative charge –q and mass m is hung from a fixed point through a non-conducing and inextensible string of length 2R as shown in the figure. The whole system is placed in a gravity free space. The bob is displaced by a very small angle from the vertical, then find the time required to reach from point A to O. (angular displacement of the bob is so small that it can be assumed, the distance between the bob and the sphere is always nearly equal to R

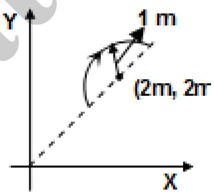


(A) $2\pi\sqrt{\dfrac{4\pi\varepsilon_0 R^3 m}{3Qq}}$ (B) $\dfrac{\pi}{2}\sqrt{\dfrac{4\pi\varepsilon_0 R^3 m}{3Qq}}$ (C) $2\pi\sqrt{\dfrac{8\pi\varepsilon_0 R^3 m}{3Qq}}$ (D) $\dfrac{\pi}{2}\sqrt{\dfrac{8\pi\varepsilon_0 R^3 m}{3Qq}}$

511. An infinite collection of current carrying infinite conductors each carrying a current I outwards perpendicular to papers are placed at x = α_0, $-2\alpha_0$, $3\alpha_0$, $-4\alpha_0$ … ad infinite on the x-axis. Another infinite collection of current carrying conductors each carrying a current I inwards perpendicular to the papers are placed at x = $-\alpha_0$, $2\alpha_0$, $-3\alpha_0$, $4\alpha_0$ … ad infinite here α_0 is a positive constant. Then the magnetic field at the origin due to the above collection of current carrying conductor is

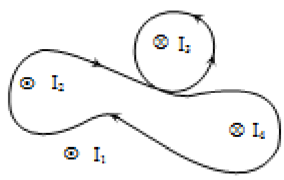
(A) zero (B) $\dfrac{\mu_0 I}{\pi\alpha_0 \ell n2}$ (C) $\dfrac{\mu_0 I \ell n2}{\pi\alpha_0}$ (D) infinite

512. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j} + \hat{k})$ Tesla exists in a region of space. A semicircular wire of radius 1 m carrying current 1 A having its centre at (2, 2, 0) m is placed on the x-y plane as shown in the figure. The force on the semi-circular wire will be



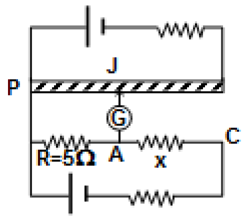
(A) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$ N (B) $\sqrt{2}(\hat{i} - \hat{j} + \hat{k})$ N (C) $\sqrt{2}(\hat{i} + \hat{j} - \hat{k})$ N (D) $\sqrt{2}(-\hat{i} + \hat{j} + \hat{k})$ N

513. Four wires of current I_1 = 2A, I_2 = 4A, I_3 = 6A and I_4 = 8A cut the page perpendicularly at the points a, b, c and d respectively. The value of $\int \vec{B}.d\vec{\ell}$ for the shown loop would be



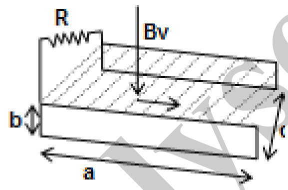
(A) + 2 μ_0-wb/m (B) – 2 μ_0-wb/m (C) + 10 μ_0-wb/m (D) – 10 μ_0-wb/m

514. Circuit for the measurement of resistance by potentiometer is shown. The galvanometer is first connected at point A and zero deflection is observed at length PJ = 10 cm. In second case it is connected at point C and zero deflection is observed at a length 25 cm from point P. Then find the value of the unknown resistance x.



(A) 5Ω (B) 7.5Ω (C) 12.5Ω (D) none of the above

515. The figure shows an apparatus suggested by Faraday to generate electric current from a flowing river. Two identical conducting planes of length a and width b are placed parallel facing one another on opposite sides of the river at a distance d apart. The river is flowing with a velocity v, vertical component of the magnetic field produced by earth is B_v and the resistivity of river water is ρ. Now both the plates are connected by a load resistance R. Find the current through the load R.

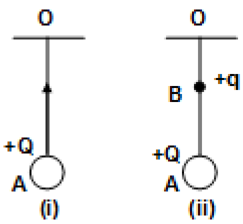


(A) $\frac{B_v vb}{R}$ (B) $\frac{B_v vb}{R+\frac{\rho d}{ab}}$ (C) $\frac{B_v vd}{R+\frac{\rho d}{ab}}$ (D) none of the above

516. A parallel plate capacitor C is equally filled with parallel layers of materials of dielectric constants K_1 and K_2. Then the ratio of new capacitance to the previous capacitance is:

(A) $\frac{2K_1K_2}{K_1+K_2}$ (B) K_1+K_2 (C) $\frac{K_1K_2}{K_1+K_2}$ (D) none of the above

517. The time period of simple pendulum of charged bob is T as shown in the figure. Now a massless charge q is placed at point B, and time period of oscillation is T′, then

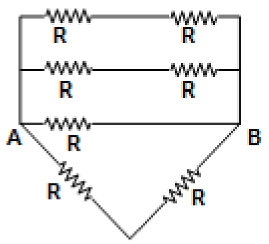


(A) T′ > T (B) T′<T (C) T′ = T (D) can't say

518. A long string with a charge of λ per unit length passes through an imaginary cube of edge a. The maximum flux of the electric field through the cube will be

(A) $\frac{\lambda a}{\varepsilon_0}$ (B) $\frac{\sqrt{2}\lambda a}{\varepsilon_0}$ (C) $\frac{6\lambda a^2}{\varepsilon_0}$ (D) $\frac{\sqrt{3}\lambda a}{\varepsilon_0}$

519. In the network shown below, the equivalent resistance between A and B is



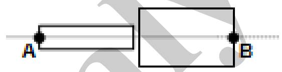
(A) R/2 (B) (2/5)R (C) 2R (D) (5/2)R

520. A charged particle begins to move from the origin in a region which has a uniform magnetic field in the x-direction and a uniform electric field in the y-direction. Its speed is V when it reaches the point (x, y, z). V will depend

(A) only on x (B) only on y

(C) on both x and y but not z (D) on x, y and z

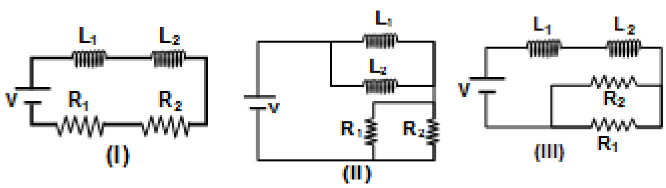
521. A cylindrical element of resistivity ρ, radius r and length ℓ is connected in series with another cylindrical element of conductivity σ, radius 2r and length ℓ, such that axis of both the cylinders is along the same line as shown. The net resistance across A and B is



(A) $\left(\rho+\frac{1}{4\sigma}\right)\frac{\ell}{\pi r^2}$ (B) $\left(\rho+\frac{4}{\sigma}\right)\frac{\ell}{\pi r^2}$ (C) $\left(\frac{1}{\rho}+4\sigma\right)\frac{\pi r^2}{\ell}$ (D) $\left(4\rho+\frac{1}{\sigma}\right)\frac{2\ell}{\pi r^2}$

522. Given L_1 = 1 mH $R_1 = 1\Omega$ L_2 = 2 mH $R_2 = 2\Omega$

(Neglect mutual inductance) The time constants (in ms) for the circuits I, II and III are



(A) 1, 1, $\frac{9}{2}$ (B) $\frac{9}{4}$, 1, 1 (C) 1, 1, 1 (D) 1, $\frac{9}{4}$, 1

523. A horizontal ring of radius $r=\frac{1}{2}$m is kept in a vertical constant magnetic field 1T. The ring is collapsed from maximum area to zero area in 1 sec. Then the emf induced in the ring is:

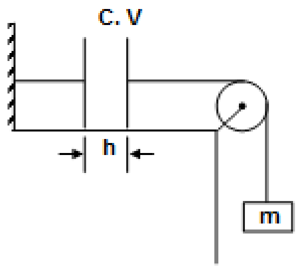
(A) 1 Volt (B) (π/4) Volt (C) (π/2) Volt (D) π Volt

524. A conducting shell of radius a and charge Q is concentric with a solid sphere of charge Q and radius b (b < a). then the electric potential at distance r (b < r < a) from the centre is

(A) $KQ\left(\frac{1}{a}+\frac{1}{b}\right)$ (B) $KQ\left(\frac{1}{a}-\frac{1}{r}\right)$ (C) $KQ\left(\frac{1}{a}+\frac{1}{r}\right)$ (D) $\frac{KQ}{a}$

where, $k=\frac{1}{4\pi\epsilon_0}$

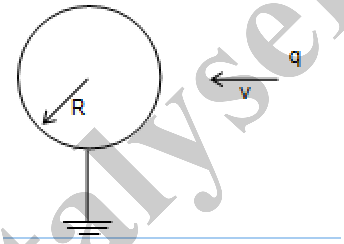
525. A parallel plate capacitor of capacity C tied by non-conducting ropes is charged to a potential difference as shown in the figure. Given $mgh = \frac{1}{2}CV^2$. When released, m will



(A) move up (B) move down C) remain at rest (D) execute SHM

SECTION : (B) - More Than One Correct Options

526. A charged particle having a positive charge q approaches a grounded metallic sphere of radius R with a constant small speed v as shown in the figure. In this situation:



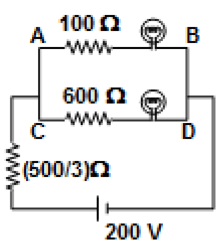
(A) As the charge draws nearer to the surface of the sphere, a current flows in to the ground .

(B) As the charge draws nearer to the surface of the sphere, a current flows out of the ground in to the sphere.

(C) As the charged particle draws nearer, the magnitude of current flowing in the connector joining the shell to the ground increases.

(D) As the charged particle draws nearer, the magnitude of current flowing in the connector joining the sphere to the ground decreases.

527. Two bulbs 25 watt-100 volt and 100 watt-200 volt are connected in the circuit as shown in figure choose the correct answer(s).



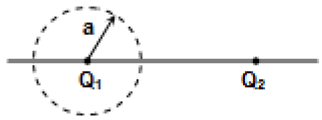
(A) Heat lost per second in the circuit will be 80 J.

(B) Ratio of heat produced per second in bulb will be 1 : 1.

(C) Ratio of heat produced in branch AB to Branch CD will be 2 : 1

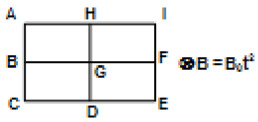
(D) current drawn from the cell is 0.4 amp

528. In the figure shown, there are two charges Q_1 and Q_2 with Q_1 fixed and Q_2 is moving along the line joining the two charges. Considering a spherical surface of radius a centred at Q_1. The electric flux passing through the left half of the surface at the position shown



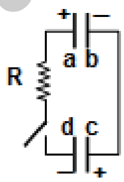
(A) will increase
(B) will decrease
(C) depends on nature of Q_2 and Q_1
(D) depends on velocity of Q_2

529. The loop shown in the figure is kept in varying magnetic field as shown. The loop is in the plane of the page and the magnetic field is into the plane of the page. The loop is made of wire with uniform resistance per unit length.

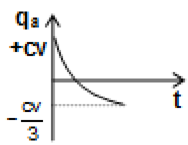


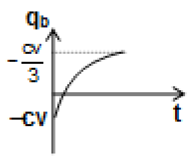
(A) current in branch GH = 0
(B) Current in branch GF = 0
(C) current in branch FI = 0
(D) Current in branch HI = 0

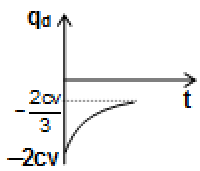
530. A parallel plate capacitor of capacitance C is charged to a potential difference V, such that plate a is positively charged and plate b is negatively charged. Another capacitor of capacitance 2C is charged to a same potential difference V, such that plate c is positively charged and plate d is negatively charged. Now both the capacitors are connected to each other as shown in the figure, such that plate a is connected to plate d and plate b is connected to plate c with the help of a conductor having resistance R.

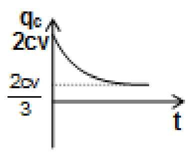


Then choose the correct curve(s) if q_a, q_b, q_c and q_d represents the charge on the plates a, b, c and d respectively

(A)


(B)


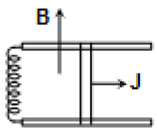
(C)


(D)


531. A uniform disc of radius R lies on the x-y plane with its centre at origin. Its moment of inertia about z-axis is equal to its moment of inertia about an axis along the line $y = K_1x + K_2$ where K_1 and K_2 are constants. Then choose the option(s) having possible values of K_1 and K_2.

(A) $K_1 = 1, K_2 = +\frac{R}{\sqrt{2}}$
(B) $K_1 = 1, K_2 = -\frac{R}{2}$
(C) $K_1 = -1, K_2 = +\frac{R}{2}$
(D) $K_1 = -1, K_2 = -\frac{R}{\sqrt{2}}$

532. Two parallel resistance less rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the correct option(s)



(A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d=\sqrt{\frac{3J^2L}{4B^2\ell^2m}}$.

(B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i=\sqrt{\frac{3J^2}{4Lm}}$

(C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d=\sqrt{\frac{3J^2L}{B^2\ell^2m}}$

(D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i=\sqrt{\frac{3J^2}{mL}}$

533. The plates of a parallel plate capacitor are charged with surface densities σ_1 and σ_2 respectively. The electric field at points.

(A) Inside the region between the plates will be zero.

(B) Inside the metallic plate of the capacitor is zero.

(C) Everywhere in the space will be zero.

(D) Inside the region between the plates will be uniform and nonzero

534. When a charged particle enters a magnetic field then the magnetic field:

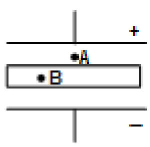
(A) may change the direction of motion of the particle

(B) may not change the direction of motion of the particle

(C) may change the kinetic energy of the particle

(D) will not change the kinetic energy of the particle.

535. A slab S is placed between the plates of a parallel plate charged isolated capacitor as shown. Let E_A and E_B be electric field intensities at A and B respectively. Then,



(A) E_A is opposite to E_B in direction if S is dielectric.

(B) E_A will always be same in direction as E_B.

(C) E_A will be same in direction as E_B only if S is conducting

(D) E_B will be zero if S is a metal

536. Four charges, all of the same magnitude are placed at the four corners of a square. At the centre of the square, the field is E. By suitable choices of the signs of the four charges which of the following can be obtained

(A) $v = 0, E = 0$ (B) $V = 0, E \neq 0$ (C) $v \neq 0, E = 0$ (D) $v \neq 0, E \neq 0$

537. The electric potential in a region along the x-axis varies with x according to the relation $V(x) = 4 + 5x^2$ then

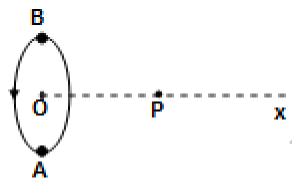
(A) potential difference between the points x = 1 and x = –2 is 15 volt.

(B) Force experienced by a one coulomb charge at x = –1 m will be 10 N.

(C) The force experienced by the above charge will be toward +x-axis

(D) A uniform electric field exists in this region along the x-axis

538. A is a circular loop carrying a current i. P is a point on axis OX, $d\ell$ is an element of length dl on the loop at a point B on it. The magnetic field at P,



(A) due to loop is directed along OX

(B) due to dl is directed along OX

(C) due to dl is perpendicular to OX

(D) due to dl is perpendicular to BP

539. When a positively charged particle is placed in a constant magnetic field then the

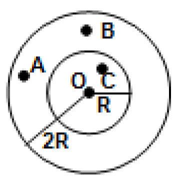
(A) intensity of the magnetic field change

(B) intensity of the magnetic field does not change

(C) direction of the magnetic lines of force changes

(D) direction of the magnetic lines of force does not change

540. A hollow conducting sphere of inner radius R and outer radius 2R is given a charge Q as shown in the figure, then the



(A) potential at A and B is same

(B) potential at O and B is same

(C) potential at O and C is same

(D) potential at A, B, C and O is same

541. An infinite current carrying conductor is placed along the z-axis and a wire loop is kept in the x-y plane. The current in the conductor is increasing with time. Then the

(A) emf induced in the wire loop is zero

(B) magnetic flux passing through the wire loop is zero

(C) emf induced is zero but magnetic flux is not zero

(D) emf induced is not zero but magnetic flux is zero

542. A positively charged thin metal ring of radius R is fixed in x-y plane with its centre at the origin o. A negatively charged particle P is released from rest at the point (O, O, Zo). Then the motion of P is

(A) periodic for all values of Zo

(B) SHM for all values of Zo satisfying O < Zo < R

(C) approximately SHM, provided z >> R

(D) approximately SHM, provided Z << R.

543. A capacitor of capacitance 'C' is connected with a battery of emf ε. After full charging a dielectric of same size of capacitor & dielectric constant k is inserted then choose correct statements.

Capacitor is always connected to battery.

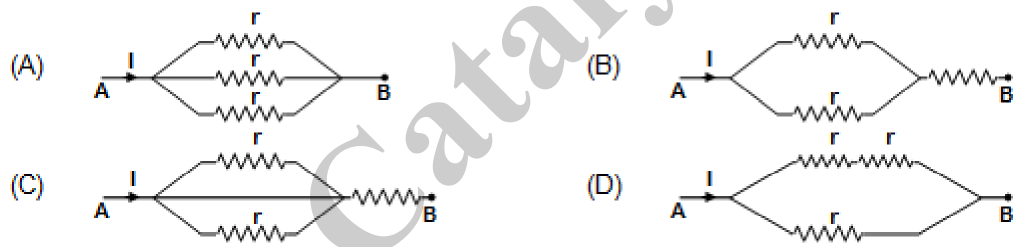
(A) electric field between plates of capacitor remain same

(B) charge on capacitor increased to KC ε

(C) energy on capacitor increased

(D) electric field between plates of capacitor increased

544. If the same current passes through following circuits P and R represents power dissipated and equivalent resistance across A, B points then



(A) $P_1 > P_2 > P_3 > P_4$ (B) $R_1 > R_2 > R_3 > R_4$ (C) $P_2 > P_3 > P_4 > P_1$ (D) $R_2 > R_3 > R_4 > R_1$

545. For induced electric field

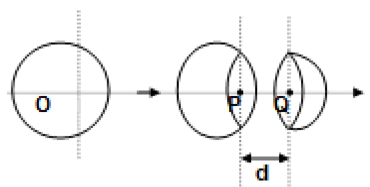
(A) the field lines may not be closed

(B) the field lines are always closed

(C) "potential" is not defined

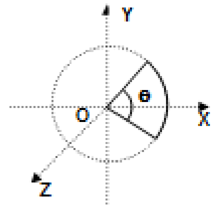
(D) magnetic flux must change

546. An insulating spherical shell of uniform surface charge density is cut into two parts as shown in the figure. $\vec{E}_p$ and $\vec{E}_Q$ denote the electric fields at P and Q respectively. As d (i.e., PQ) ∞



(A) $|\vec{E}_p| > |\vec{E}_Q|$ (B) $|\vec{E}_p| = |\vec{E}_Q|$ (C) $|\vec{E}_p| < |\vec{E}_Q|$ (D) $\vec{E}_p + \vec{E}_Q = 0$

547. A current carrying loop lies in the x–y plane. The magnetic field in the region is given by $\vec{B} = B_0\hat{k} + x\hat{j}$. Then



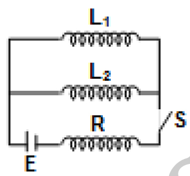
(A) net force acting on the loop is zero for all real x.

(B) initial angular acceleration is minimum then x = 0.

(C) for x = 0, the loop won't rotate.

(D) for x = 0, the loop rotates.

548. In the circuit, a battery of emf E, a resistance R and inductance coil L_1 and L_2 and switch S are connected as shown. Initially the switch is open



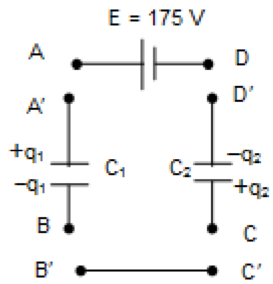
(A) The time constant of the circuit is $\frac{1}{R}\left(\frac{L_1L_2}{L_1+L_2}\right)$

(B) Steady state current in the inductor L_1 is $\frac{EL_2}{R(L_1+L_2)}$

(C) Steady state current in the inductor L_2 is $\frac{EL_1}{R(L_1+L_2)}$

(D) In steady state the total energy stored in the inductor coils is

549. In the figure shown capacitor C_1 = 4μF is charged to a value q_1 = 200 μC and capacitor C_2 = 2μF is charged to a value q_2 = 400μC. The polarities are as shown. Now a source of EMF E = 175 Volt is connected in the circuit by joining terminals A to A′, B to B′, C to C′ and D to D.



(A) After a long time the charges on capacitor C_1 and C_2 are 100 μC and 300 μC respectively.

(B) Internal energy of the capacitor system decreases.

(C) Work done by the battery in the process is -1.75×10^{-2} J.

(D) all of them are incorrect

550. Three concentric spherical metallic shells, A, B and C of radii a, b and c (a<b<c) have surface charge densities σ, $-\sigma$ and σ respectively.

(A) Potential of shell A is $\frac{\sigma}{\varepsilon_0}(a-b+c)$

(B) Potential of shell C is $\frac{\sigma}{\varepsilon_0 C}(a^2-b^2+c^2)$

(C) Potential of C is $\frac{\sigma}{2\varepsilon_0 c}(a^2+b^2-c^2)$

(D) If the shell A and C is same then relation between radii a, b and c is c = a + b.

551. A solenoid is connected to a source of constant emf for a long time. A soft iron piece is increased into it, then

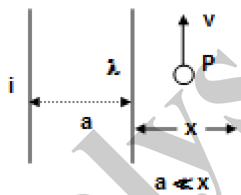
(A) self inductance of solenoid gets increased

(B) flux linked with solenoid increases, hence steady state current gets decreased.

(C) energy stored in the solenoid gets increased

(D) magnetic field due to solenoid get increased.

552. Two long, thin parallel conductors are kept very close to each other, without touching one carries a current I, and the other has charge λ per unit length. A proton moving parallel conductors with velocity v is un deflected. Assume 'C' is the velocity of light. Then



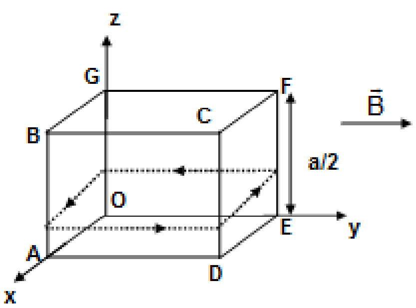
(A) $V = \frac{\lambda C^2}{i}$

(B) $v = \frac{i}{\lambda}$

(C) $c = \frac{i}{\lambda}$

(D) The proton may be at any distance from the conductor

553. A wooden cubical block ABCDEFG of mass m and side a is wrapped by a square wire loop of perimeter 4a, carrying current I. The whole system is placed at frictionless horizontal surface in a uniform magnetic field $\vec{B} = B_0\hat{j}$ as shown in the figure. In this situation, normal given by horizontal surface by a distance x from centre. Choose correct statement(S)



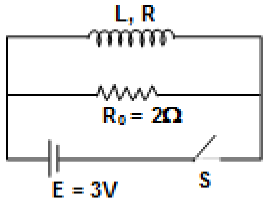
(A) Block must not topple if I < $\frac{mg}{aB_0}$

(B) Block must not topple if I < $\frac{mg}{2aB_0}$

(C) $x = \frac{a}{4}$ if $I = \frac{mg}{2aB_0}$

(D) $x = \frac{a}{4}$ if

554. In inductor if inductance L = 1H and resistance R = 3/2 is connected a circuit as shown in the figure. Initial switch was closed for a long time and switch is opened at t = 0 sec. Now choose correct statement(s)



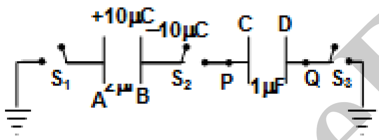
(A) $I = e^{-5t}$, where I is current following through R_0 at t = t sec

(B) Net heat generated in the circuit after switch is opened is 0.3 joule

(C) $I = 3e^{-5t}$ where I is current following through R_0 at t = t sec

(D) Total heat generated in the resistance R after switch is opened is 0.18 joule

555. A system of two capacitor is shown in the figure. If $V_p - V_Q = +20V$, then



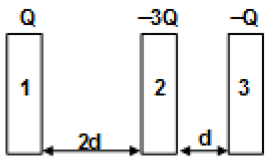
(A) If switch S_2 is closed potential difference between plate A and plate B is 15 volt.

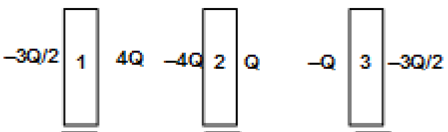
(B) If all switches are closed, $Q_A = -\frac{20}{3}\mu C$ and $Q_C = \frac{10}{3}\mu C$

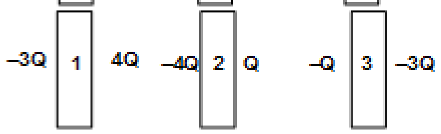
(C) If all switches are closed $V_P - V_Q = \frac{10}{3}$ volt

(D) If all switches are closed, energy loss in the circuit is 75 μj

556. Three identical metal plates with large surface area [A] are kept are given charges $Q, -3Q$ and Q respectively. If plate B is earthed and potential of plate 2 is V then choose the correct statement(s) if $\frac{A\varepsilon_0 V}{d} = Q$ $d \ll A$



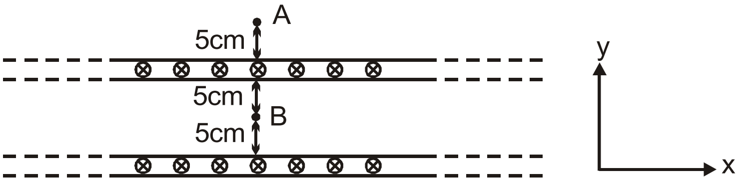
(A) 

(B) 

(C) Potential difference between plates 1 and 2 is 8 V

(D) Magnitude of charge supplied by earth is 5 Q

557. Figure shows cross section of two large parallel metal sheets carrying electric currents along their surfaces. The current in each sheet is $\frac{10}{\pi}$ A/m along the width. Consider two points A and B, as shown in the figure with their positions.



(A) Magnetic field at A is 4μT along x-direction.

(B) Magnetic field at A is 4μT along negative x-direction.

(C) Magnetic field at B is zero.

(D) Magnetic field at B is 2μT along x-direction

558. Two capacitors of capacitance 1 μF and 2 μF are separately charged by a common battery. The two capacitors start discharging through equal resistances separately at $t = 0$.

(A) The current in each of the two discharging circuit is zero at $t = 0$

(B) The currents in the two discharging circuits at $t = 0$ are equal but not zero

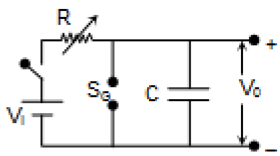
(C) The currents in two discharging circuits are equal at all times

(D) Capacitor of 1 μF capacity loses 50% of its charge earlier than capacitor of capacity 2 μF

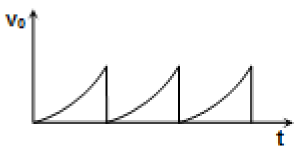
SECTION : (C) -Passage Type Questions

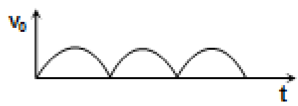
PASSAGE 01:

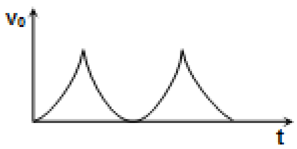
A semi tooth voltage waveform V_0 can be obtained across the capacitor C in figure. R is a variable resistor, V_i is an ideal battery, and SG is a spark plug consisting of two electrodes with an adjustable distance between them, when the voltage across the electrodes exceeds the firing voltage V_s, the air between the electrodes breaks down. Hence the gap becomes a short circuit and remains so until the voltage across the gap becomes very small

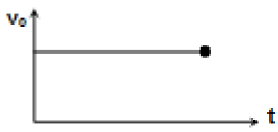


559. The plot of output voltage waveform V_0 versus time t will be

(A)


(B)


(C)


(D)


560. What condition must be satisfactory order to have a almost linearly varying semi tooth

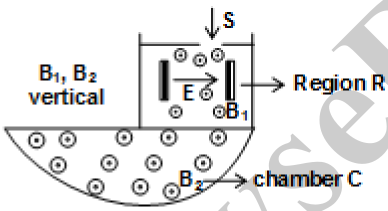
(A) C is very large (B) C is very small (C) R is very small (D) None of these.

561. If condition in previous question is satisfied, a simplified expression for the time period T of the waveform will be

(A) $V_f RC$ (B) $\frac{V_f}{V_{battery}} \times RC$ (C) $\frac{V_{battery}}{V_f RL}$ (D) $\frac{V_{battery}}{V_f} RC$

PASSAGE 02:

Figure shows the plan of a mass spectrometer in which positive ions are produced and accelerated by a suitable voltage towards a plate with a slit S. The ions then pass into an evacuated region R where there is a horizontal electric field E and a vertical magnetic field B_1. Region R acts as velocity selector, such that the ions passes the region un-deflected. Finally, they pass into an evacuated chamber C where there is a vertical magnetic field B_2. In a particular experiment singly charged ions of two isotopes (A) and (B) of neon of masses 20 u and 22 u are respectively accelerated from rest and passed through region R where the electric field is 3×10^4 V/m. The values of the magnetic fields B_1 and B_2 are both equal to 0.4 T



Answer the following questions based on the above passage

562. What is the speed of both type of ions before entering the chamber C.

(A) 7.5×10^4 m/s for both type of ions (B) 7.5×10^4 m/s for A and 6.82×10^4 m/s for B

(C) 6.82×10^4 m/s for A and 7.5×10^4 m/s for B (D) can't be calculated.

563. Find the radius of curvature of the path of the ion A in chamber C

(A) 42.80 mm (B) 38.91 mm (C) 25.70 mm (D) 23.36 mm

564. Calculate the separation of the isotopes after one half revolution inside the chamber C.

(A) 7.8 mm (B) 3.9 mm (C) 42.80 mm (D) 38.91 mm

PASSAGE 03:

Two point charges, of equal charge +q, are kept distance d apart. Another, similar charge is brought from infinity to a point such that the three charges form an equilateral triangle of side d.

565. The electrostatic force between the two charges kept at distance d apart, during the time in which the third charge is brought from infinity, is:

(A) $\frac{1}{4\pi\epsilon_0}\frac{q^2}{d^2}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{2q^2}{d^2}$

(C) $\frac{1}{4\pi\epsilon_0}\frac{q}{d}$ (D) it changes with time and connot be defined.

566. Another charge is to be placed at the centre of the triangle so that the system is in equilibrium, the value of the charge is:

(A) $\frac{q}{\sqrt{3}}$ (B) $\frac{q}{3}$ (C) $\frac{-q}{3}$ (D) $\frac{-q}{\sqrt{3}}$

567. Due to the presence of this charge at the centre, the electrostatic force between the charges kept at the corners:

(A) increases in magnitude

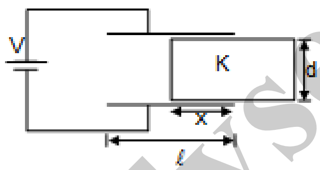
(B) decreases in magnitude

(C) does not change in magnitude and direction.

(D) only changes in direction but remains same in magnitude.

PASSAGE 04:

In the situation shown in figure the area of the square plates is A. The dielectric slab is released from rest and mass of the slab is m



568. What is the charge on the capacitor when dielectric slab is released

(A) $\frac{\varepsilon_0 \ell V}{2d}[\ell + x(k-1)]$ (B) $\frac{\varepsilon_0 \ell V}{d}[\ell + x(k-1)]$ (C) $\frac{\varepsilon_0 \ell V}{d}[2\ell + x(k-1)]$ (D) None of these

569. Motion of the dielectric slab after releases is

(A) SHM (B) Oscillatory (C) Uniform (D) Cannot be predicted

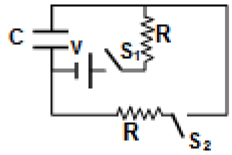
570. The time period of the slab is

(A) $8\sqrt{\frac{(\ell - x)m\ell d}{\varepsilon_0 AV^2(k-1)}}$ (B) $8\sqrt{\frac{2(\ell - x)m\ell d}{\varepsilon_0 AV^2(k-1)}}$ (C) $4\sqrt{\frac{(\ell - x)m\ell d}{\varepsilon_0 AV^2(k-1)}}$ (D) $3\sqrt{\frac{(\ell - x)m\ell d}{\varepsilon_0 AV^2(k-1)}}$

PASSAGE 05:

A capacitor of capacitance C can be charged (with the help of a resistance R) by a voltage source of emf V, by closing switch S_1 while keeping switch S_2 open. Initially the capacitor is uncharged.

571. At time t = 0 switch S_1 is closed and S_2 is left open and if τ is the time constant for this circuit then:



(A) τ = 2RC and charge on the capacitor at t = τ is CV $\left(1-e^{-1}\right)$

(B) τ = RC and charge on the capacitor at t = 2τ is CV $\left(1-e^{-1}\right)$

(C) τ = RC and charge on the capacitor at t = 2τ is CV $\left(1-e^{-2}\right)$

(D) $\tau = \frac{RC}{2}$ and charge on the capacitor at t = τ/2 is CV

572. The rate of increase of charge on the capacitor is

(A) $\frac{V}{R}e^{-tRC}$ (B) $2\frac{V}{R}e^{-t/RC}$ (C) $\frac{V}{R}e^{-t/RC}$ (D) none of the above

573. When the capacitor has reached steady state then switch S_1 is opened and S_2 is closed. The ratio of RC to the charge on the capacitor after the first RC seconds is:

(A) R/v (B) Re/v (C) Rv/e (D) R/ve

PASSAGE 06:

A student constructs a series RLC circuit. While operating the circuit at a frequency *f* she uses an AC voltmeter and measures the potential difference across each device as $(\Delta V_R)_{max}$ = 8.8 V, $(\Delta V_L)_{max}$ =2.6V, , and $(\Delta V_c)_{max}$=7.4V.

574. The circuit is constructed so that the inductor is next to the capacitor. What result should the student expect for a measurement of the combined potential difference $(\Delta V_L + \Delta_C)_{max}$ across the inductor and capacitor ?

(A) 10.0V (B) 7.8V (C) 7.4V (D) 4.8V

575. What result should the student expect for a measurement of the amplitude E_m of the potential difference across the power supply?

(A) 18.8V (B) 13.6V (C) 10.0V (D) 4.0V

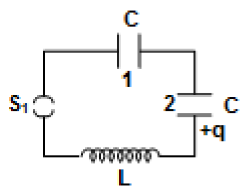
576. How should the frequency of this circuit be changed to increase the current i_m through the circuit?

(A) Increase *f*. (B) Decrease *f*.

(C) The current is already at a maximum. (D) There is not enough information to answer the question.

PASSAGE 07:

In the figure shown, capacitor 1 has a charge Q and 2 has no charge. At t = 0, the switch S_1 is closed. L is the inductance of the inductor.



577. At what time the charges will be equally divided between the two capacitors for the first time?

(A) $\frac{\pi\sqrt{LC}}{2\sqrt{2}}$ (B) $\frac{\pi\sqrt{LC}}{\sqrt{2}}$ (C) $\frac{\pi\sqrt{LC}}{2}$ (D) $\frac{\pi\sqrt{LC}}{4}$

578. What will be the current in the circuit at the time mentioned above?

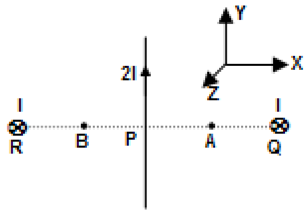
(A) $\frac{Q}{\sqrt{LC}}$ (B) $\frac{Q}{\sqrt{2LC}}$ (C) $\frac{Q\sqrt{2}}{\sqrt{LC}}$ (D) $\frac{2Q}{\sqrt{LC}}$

579. When the capacitor 2 will be fully charged for the first time? Here ω is the angular frequency of LC oscillation.

(A) $\frac{2\pi}{\omega}$ (B) $\frac{\pi}{\omega}$ (C) $\frac{\pi}{2\omega}$ (D) $\frac{4\pi}{\omega}$

PASSAGE 08:

Three infinitely long straight conductors are at P, Q and R arranged as shown in the figure. Here RB = BP = PA = AQ = d



580. The direction of magnetic field at A due to wire at R is

(A) $\hat{j}$ (B) $+\hat{k}$ (C) $-\hat{j}$ (D) $-\hat{k}$

581. The net magnetic field at A due to all wires is

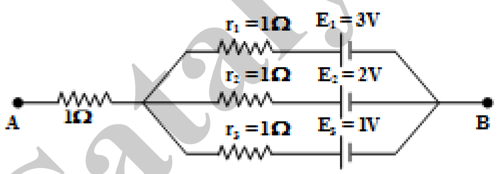
(A) $\frac{\mu_0 I}{2\pi d}\left[\frac{2}{3}\hat{j}-2\hat{k}\right]$ (B) $\frac{\mu_0 I}{2\pi d}\left[\frac{2}{3}\hat{j}+2\hat{k}\right]$ (C) $\frac{\mu_0 I}{2\pi d}\left[-\frac{2}{3}\hat{j}+2\hat{k}\right]$ (D) none of these

582. The net magnetic field at B due to all wires is

(A) $\frac{\mu_0 I}{2\pi d}\left[\frac{2}{3}\hat{j}-2\hat{k}\right]$ (B) $\frac{\mu_0 I}{2\pi d}\left[\frac{2}{3}\hat{j}+2\hat{k}\right]$ (C) $\frac{\mu_0 I}{2\pi d}\left[-\frac{2}{3}\hat{j}+2\hat{k}\right]$ (D) none of these

PASSAGE 09:

In the circuit shown in the figure E_1 = 3V, E_2 = 2V, E_3 = 1 V, R = r_1 = r_2 = r_3 = 1 Ω.



583. Potential difference between the points A and B is

(A) 3 V (B) 2 V (C) 1.5 V (D) 2.5 V

584. Current in the branch containing E_2

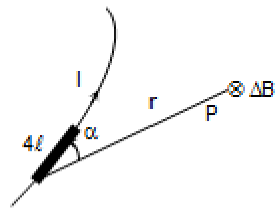
(A) 1 A (B) 1.5 A (C) 0 A (D) 2 A

585. If r_2 is short circuited and point A is connected to point B. Then current through resistance R is

(A) 1 A (B) 1.5 A (C) 0 A (D) 2 A

PASSAGE 10:

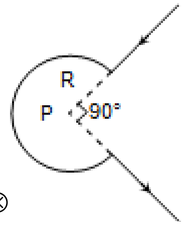
To calculate B for any shape of conductor, Boit and Savart gave a law which can now stated as follows. The flux density B at a point P due to a small element $\Delta\ell$ of a conductor carrying current is given by



$$\Delta B = \frac{\mu_0}{4\pi}\frac{I\Delta\ell \sin\alpha}{r^2}$$

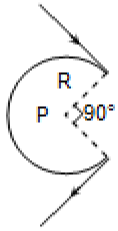
where r is the distance from the point to a element and α is the angle between the element and line joining it to point P

586. Calculate magnetic field at point P.



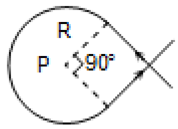
(A) $\frac{\mu_0}{4\pi}\frac{I}{R^2}\left(\frac{\pi}{2}\right)\odot$ (B) $\frac{\mu_0 3I}{8R}\odot$ (C) $\frac{\mu_0 3I}{8R}\otimes$ (D) $\frac{\mu_0 3I}{8R^2}\otimes$

587. Calculate magnetic field at point P.



(A) $\frac{\mu_0}{4\pi}\frac{I}{R}\left(\frac{3\pi}{2}-2\right)\odot$ (B) $\frac{\mu_0}{4\pi}\frac{I}{R}\left(\frac{3\pi}{2}+2\right)\otimes$ (C) $\frac{\mu_0}{4\pi}\frac{I}{R}\left(\frac{\pi}{2}-2\right)\odot$ (D) $\frac{\mu_0}{4\pi}\frac{I}{R}\left(\frac{\pi}{2}+2\right)\otimes$

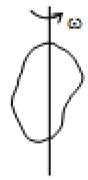
588. Calculate magnetic field at point P.



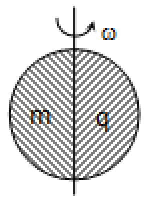
(A) $\frac{\mu_0}{4\pi}\frac{I}{R}\left(\frac{3\pi}{2}+2\right)\odot$ (B) $\frac{\mu_0}{4\pi}\frac{I}{R}\times\frac{3\pi}{2}\otimes$ (C) $\frac{\mu_0}{2\pi}\frac{I}{R}\odot$ (D) $\frac{\mu_0 I}{2R}\otimes$

PASSAGE 11:

Consider a non-conducting homogeneous rigid body of mass m which has a total charge q distributed similar to mass distribution. The mass is rotated about an axis passing through the body with angular speed ω. The magnetic moment of the body M and the angular momentum L of the body about the same axis is given by

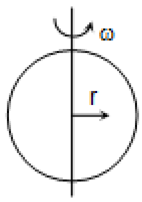


589. Consider a hypothetic spherical body. The body is cut into two parts about the diameter. One of hemispherical portion has mass distribution m while the other portion has identical charge distribution q. The body is rotated about the axis with constant speed ω. Then the ratio of M to L is



(A) $\frac{q}{2m}$ (B) $>\frac{q}{2m}$ (C) $<\frac{q}{2m}$ (D) Cannot be calculated.

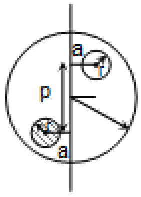
590. Consider a solid sphere whose mass and charge distribution is proportional to r^2, where r is the radial distance from the axis of rotation. The solid sphere is rotated about the diameter of the sphere. Then the ratio of M to L is



(A) $\frac{q}{2m}$ (B) $>\frac{q}{2m}$

(C) $<\frac{q}{2m}$ (D) Cannot be calculated

591. Consider a solid non-conducting sphere of radius r and mass m. Initially the mass and the charge q is uniformly distributed over the entire volume. A small sphere of charge is picked from a distance a from the axis of rotation, keeping the mass distribution unchanged and embedded at the opposite side by same distance from the axis as shown in the figure. Again the system is rotated about the given axis by constant angular speed ω. Then ratio of M to L is



(A) $\frac{q}{2m}$ (B) $>\frac{q}{2m}$ (C) $<\frac{q}{2m}$ (D) Cannot be calculated.

SECTION : (D) - Matrix Match

592. A charged particle having charge q and mass m is to be subjected to a combination of constant uniform magnetic field ($\vec{B}$) and a constant uniform gravitational field ($\vec{E}$). Apart from these field forces there exists no other force. Now match the column.

	Column I		Column II
(A)	The charged particle moves without change in its direction.	(1)	It is possible that both $\vec{B}$ and $\vec{E}$ are zero.
(B)	The charged particle moves without change in its velocity.	(2)	It is possible that both $\vec{B}$ and $\vec{E}$ are non zero
(C)	The charged particle takes a circular path	(3)	It is possible that $\vec{B}$ is zero and $\vec{E}$ is not zero
(D)	The charged particle take a parabolic path	(4)	It is possible that $\vec{B}$ is non zero and $\vec{E}$ is zero

593. Match the physical quantities with their appropriate dimensions

	Column I		Column II
(A)	Inductance	(1)	$MLT^{-2}A^{-2}$
(B)	Capacitance	(2)	$ML^2T^{-2}A^{-1}$
(C)	Magnetic flux	(3)	$M^{-1}L^{-2}T^4A^2$
(D)	Permeability of free space	(4)	$ML^2T^{-2}A^{-2}$

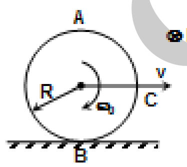
594.

	Column – I		Column – II
(A)	Inside a conducting charged sphere.	(i)	constant field
(B)	Inside a uniformly charged sphere.	(ii)	constant potential
(C)	For a cavity in a conductor.	(iii)	Linearly varying field
(D)	Spherical cavity inside a uniformly charged sphere.	(iv)	Linear variation of potential.

595.

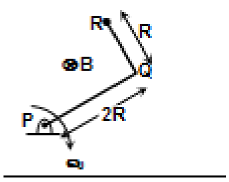
Column – I

(A) A ring or radius R is rolling on a horizontal surface with angular velocity ω_0 as shown. emf induced across AB is



(B) Consider the situation in part (a). The emf induced across AC will be

(C) Consider an L shaped rod shown in the figure. The rod rotates about P with an angular velocity ω_0. The emf induced across PR is



(D) Consider the situation shown in part C. The emf induced across QR will be

Column – II

(i) $B\omega_0R^2$

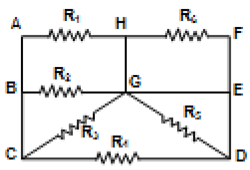
(ii) $2\,B\omega_0R^2$

(iii) $2.5\,B\omega_0R^2$

(iv) $0.5\,B\omega_0R^2$

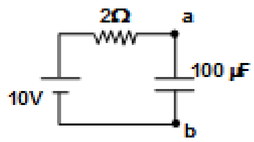
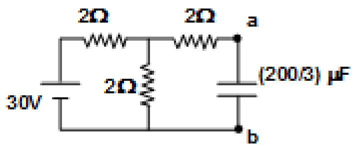
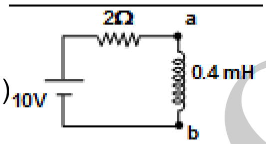
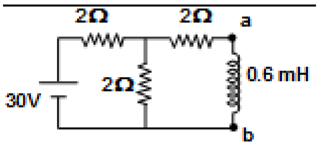
(v) $1.5\,B\omega_0R^2$

596. For the circuit shown in the figure



Column – I		**Column – II**	
(A)	Equivalent resistance between G and C is	(i)	dependent on R_1
(B)	Equivalent resistance between BG is	(ii)	dependent on R_2
(C)	Equivalent resistance between EA is	(iii)	dependent on R_3
(D)	Equivalent resistance between DG is	(iv)	Independent of R_1, R_2, R_3, R_4, R_5, R_6

597. Choose the correct equation of current in the List II as a function of t through the circuit element 'ab' of the circuits in the List – I

List – I		**List – II**	
(A)		(i)	$i = (5A)\left[1 - e^{-\frac{t}{(2\times10^{-4}\ \text{sec})}}\right]$
(B)		(ii)	$i = (5A)\left[1 - e^{-\frac{t}{(4\times10^{-4}\ \text{sec})}}\right]$
(C		(iii)	$i = (5A)\, e^{-\frac{t}{(2\times10^{-4}\ \text{sec})}}$
(D)		(iv)	$i = (5A)\, e^{-\frac{t}{(4\times10^{-4}\ \text{sec})}}$

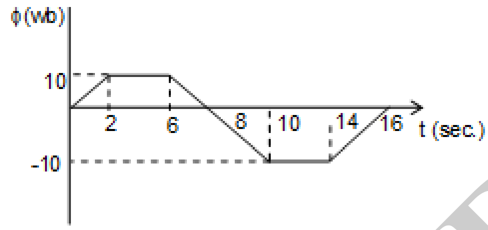
598. Four bulbs of 25W, 40W, 60W and 100W are connected in series and their combination is connected across a main power supply. Match the following list I with the potential difference across each bulb in list II

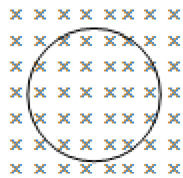
List I		**List II**	
(A)	25 W	(i)	Highest potential differences
(B)	100 W	(ii)	Second highest potential difference
(C)	60 W	(iii)	Third highest potential difference
(D)	40 W	(iv)	Lowest potential difference

599. For information given in list I, match it with that of list II

	List – I		List - II
(A)	The force on a point charge due to other point charge $[F_{12}]$is	(i)	Medium independent
(B)	The net force on a point charge $[F_1]$ is	(ii)	Affected by the presence of another point charge
(C)	The electric field lines at a point are	(iii)	Not affected by the presence of another point charge
(D)	The electric field intensity at a point is	(iv)	Medium dependent

600. Magnetic flux in a circular coil of resistance 10 Ω changes with time as shown in the figure. Direction indicates a direction perpendicular to paper inwords. Match the following





	List – I		List – II
(A)	At 1 sec induced current is	(i)	Clock wise
(B)	At 5 sec induced current is	(ii)	Anticlockwise
(C)	At 9 sec induced current is	(iii)	Zero
(D)	At 15 sec induced current is	(iv)	2A
		(v)	None

601. Match the type of field lines of the fields given in list I with the possible geometrical shapes, they can acquire, in list II

	List – I		List – II
(A)	Gravitational field	(i)	Circle
(B)	Electrostatic field	(ii)	Parallel straight lines
(C)	Magnetic field	(iii)	Straight lines diverging from a point
(D)	non-conservative electric field	(iv)	Straight lines converging on to a point

602. $\mu_0 \to$ R Permeability, $\tau_0 \to$ permittivity

$L \to$ Inductance, $R \to$ Resistance, $C \to$ Capacitance

$\eta \to$ coefficient of viscosity

	List – I(Terms)		List – II(Dimension Formula)
A.	$\frac{1}{\mu_0 \varepsilon_0}$	i.	L^2T^{-2}
B.	L	ii.	$[ML^{-1}T^{-1}]$
C.	$\frac{R}{LC}$	iii.	$[ML^2T^{-2}A^{-2}]$
D.	η	iv.	$[ML^2T^{-4}A^{-2}]$

603. There is a small dipole of dipole moment $\vec{P}$ is placed parallel to an internal electric field $\vec{E}_0$ at origin along x-axis

	List – I		List – II
(A)	Radius of equipotential sphere centre at dipole	(i)	$y = \pm\sqrt{2}x$
(B)	Electric field at a point $(x_0, 0)$ on x-axis line in which dipole lies	(ii)	$\left(\frac{k\lvert\vec{P}\rvert}{\lvert\vec{E}_0\rvert}\right)^{1/3}$
(C)	Electric field due to dipole is along y-axis on the line.	(iii)	$\left(\vec{E}_0 + \frac{2k\vec{P}}{x_0^3}\right)$
(D)	Work done to move a unit charge from origin to $(x_0, 0)$	(iv)	$-\left(E_0x_0 + \frac{kP}{x_0^2}\right)$

604.

	List – I		List – II
(A)	Pair production when high energy photons enter a thin walled box in vacuum.	(i)	Charge is invariant.
(B)	Charge of a body is independent of its speed	(ii)	Conservation of mass and energy
(C)	Spherical charged conductor for an internal point	(iii)	Equipotential surface
(D)	Uniformly charged disc	(iv)	Electric field component in the plane is in radially outward direction

605.

	List – I		List - II
(A)	Electrostatic field	(i)	non-conservative
(B)	Induced electric field	(ii)	conservative
(C)	Gravitational field	(iii)	lines always closed
(D)	Induced Magnetic field	(iv)	lines are open

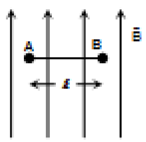
606. Figure A shows a straight current carrying conductor AB of length 'ℓ' placed in the plane of paper. Considering the plane of paper to be in the XY-plane a uniform magnetic field exists along the positive Y-direction as shown. The conductor experience a force is 10 N.

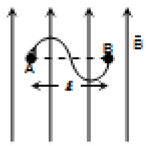
Figure B shows a curved current carrying conductor AB straight line distance between A and B is ℓ'. The conductor is in the plane of paper and in same magnetic field as in figure A.

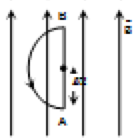
Figure C shows a semicircular closed conductor that carries current and is placed in the plane of paper and in same magnetic field as in figure A. Radius of the given semi-circle is $\ell/2$.

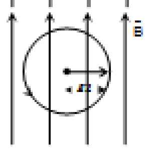
The system shown in figure D is a circular current carrying conductor placed in the plane of paper and in the same magnetic field as in figure A. Radius of conductor is $\ell/2$.

Assuming current in each conductor has same value

(A) 

(B) 

(C) 

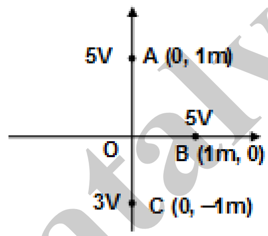
(D) 

	List – I		List – II
(A)	Force experienced by the conductor in figure B	(i)	zero
(B)	Force experienced by the conductor in figure C.	(ii)	10 N
(C)	Force experienced by the curved part of the conductor in figure C.	(iii)	slightly less than 10 N
(D)	Force experienced by the conductor in figure D.	(iv)	slightly more than 10 N

607.

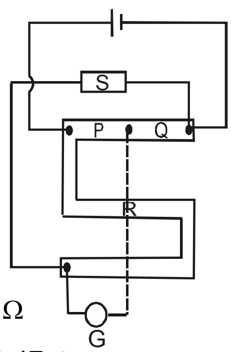
	List – I		List - II
(A)	Magnetic flux density due to a current carrying circular coil	(i)	zero
(B)	Magnetic flux density at a point on a current carrying thin wire	(ii)	Maximum at the centre
(C)	Electric field strength due to an uniform charged ring	(iii)	Continuously decreases as we move away from the centre along the axis.
(D)	Electric potential due to an uniform charged ring	(iv)	Continuously increases as we move away from the centre upto a definite distance along the axis.

608. An uniform electric field exists in straight line in the X–Y plane. The potential at different point in region are shown in the figure. Now match the following list if charge particles P (mass = 10^{-6} kg, charge $\sqrt{2}$ μC) and θ (mass = 10^{-6} kg. Charge = $-2\sqrt{2}$ μC are released from origin)



	List I		List II
(A)	Co-ordinate of position of particle P at time t = $2\sqrt{2}$ sec	(i)	($4\sqrt{2}$ m, $4\sqrt{2}$ m)
(B)	Co-ordinate of position of particle at time t = 2 sec	(ii)	($-4\sqrt{2}$ m, $-4\sqrt{2}$m)
(C)	Distance travelled by particle Q in 2 sec is	(iii)	8 m
(D)	Distance travelled by particle P in 2 sec is	(iv)	4 m

609. Connections made in a post office box are shown in figure.R$\nearrow$ =10W denotes that when R= 10W the pointer in the galvanometer is as $\nearrow$.



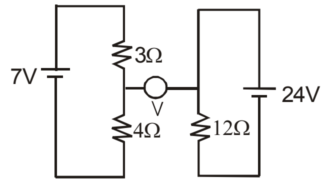
(a) P=100Ω, Q=10Ω , R$\nearrow$ = 400 Ω, R$\nwarrow$ = 500 Ω (P) 46 Ω < S < 47 Ω

(b)P=100Ω, Q=1 Ω , R$\nearrow$ = 460 Ω, R$\nwarrow$ = 470 Ω (Q) 0.46 Ω < S < 0.47 Ω

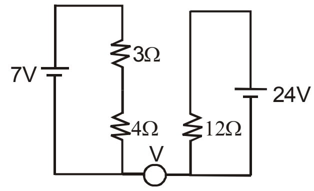
(c)P=100Ω, Q=10Ω , R$\nearrow$ = 460 Ω, R$\nwarrow$ = 470 Ω (R) 40 Ω < S < 50 Ω

(d) P=1000Ω, Q=1 Ω , R$\nearrow$ = 460 Ω, R$\nwarrow$ = 470 Ω (S) 4.6 Ω < S < 4.7 Ω

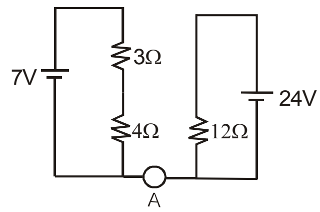
610. Match the readings of the voltmeter and ammeter respectively shown in the figures.

(a) 

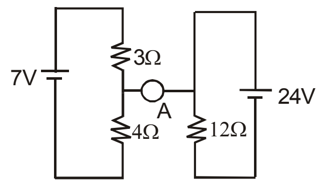
(P) 0V

(b) 

(Q) 20A

(c) 

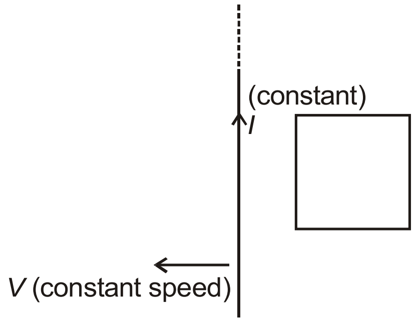
(R) 0A

(d) 

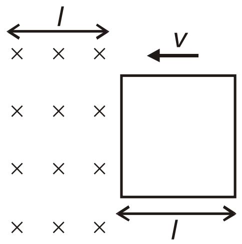
(S) 20 V

611. **Column I** **Column II**

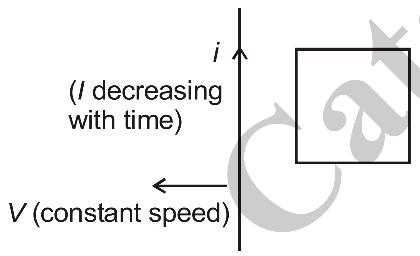
(A) In the figure shown the wire starts moving towards left with constant speed



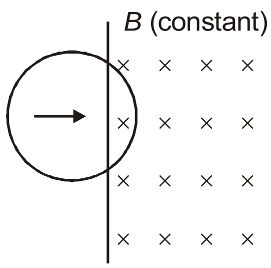
(B) Loop is having with constant speed towards the region of magnetic field for $t < \frac{l}{v}$



(C) In the figure shown the wire starts moving towards left with constant speed



(D) Loop is moving out of magnetic field region with constant speed



(p) Induced emf is clockwise

(q) Induced emf is anti-clockwise

(r) Induced emf can be clockwise and anti-clockwise

(s) emf is increasing in magnitude

(t) emf is decreasing in magnitude

612.

	Column I		Column II
(a)	Consider an infinitely long line charge with a constant positive linear charge density. The electric field at a point which is at a distance 'r' from the line charge is	(p)	Inversely proportional to r^2 when r<R/2.
(b)	Consider a uniformly charged non-conducting ring of radius 'R'. The electric field at a point 'P' located on the axis of the ring at a distance 'r' from its centre is	(q)	Radially outward and in versely proportional to r.
(c)	Consider a spherical charged conductor of a radius 'R' with a concentric spherical cavity of radiusR/2. The electric field at a point 'P' at a distance 'r' from the centre is	(r)	Inversely proportional to r^2 when r > R.
(d)	Consider a spherical uncharged conductor of radius 'R' with a is placed at the centre of the cavity. The electric field at a point 'P' at a distance 'r' from the centre of the sphere is	(s)	Increases with r, reaches a maximum and then de creases.

613. A 1C charge of mass 1kg is projected in a magnetic field of 2T with velocity $2\sqrt{2}$ at angle 45^0, Match the following:

	Table -1		Table -2
(a)	Radius of helical path	(P)	π SI units
(b)	Pitch of helical path	(Q)	1 SI units
(c)	Time period	(R)	2π SI units
(d)	Numbe of rotations in 10 seconds	(s)	$\frac{10}{\pi}$ SI units

614. In L-C oscillations, match the following:

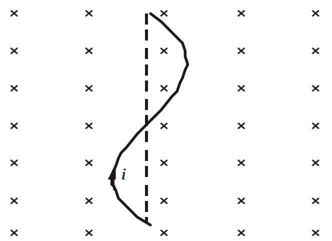
	Table-1		Table -2
(a)	Maximum charge	(P)	Maximum rate of change of current
(b)	Zero charge	(Q)	Maximum electrostatic energy
(c)	Maximum current	(R)	Maximum magnetic energy
(d)	zero current	(S)	zero rate of change of current

615. .

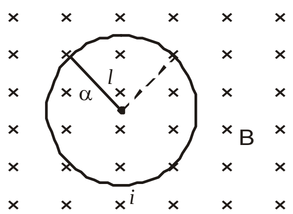
	Column A		Column B
A)	Consider an infinitely long line charge with a constant positive linear charge density. The electric field at a point which is at a distance 'r' from the line charge is	(P)	Inversely proportional to r^2 when $r < \frac{R}{2}$.
B)	Consider a uniformly charged non-conducting ring of radius 'R'. The electric field at a point 'P' located on the axis of the ring at a distance 'r' from its centre is	(Q)	Radially outward and inversely proportional to r.
C)	Consider a spherical charged conductor of a radius 'R' with a concentric spherical cavity of radius $\frac{R}{2}$. The electric field at a point 'P' at a distance 'r' from the centre is	(R)	Inversely proportional to r^2 when r > R.
D)	Consider a spherical uncharged conductor of radius 'R' with a concentric spherical cavity of radius $\frac{R}{2}$. A point charge q(q > 0) is placed at the centre of the cavity. The electric field at a point 'P' at a distance 'r' from the centre of the sphere is	(S)	Increases with r, reaches a maximum and then decreases.

616.

	Column A		Column B
(a)	Force on a wire made of two semicircular wires each of length $\frac{\pi l}{4}$ placed in an uniform magnetic field B and carrying a current i is as shown in the figure.	(p)	Zero



(b)	Tension in the wire of statement (A)	(q)	iLB
(c)	Force on a ring of radius l carrying a current i placed in a uniform magnetic field	(r)	$\frac{ilB}{2\pi}$

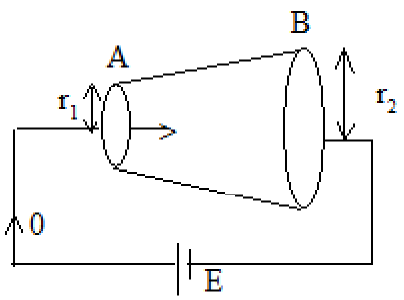


(d)	Tension in the ring of statement (C)	(s)	Can not be determined

617. An electric dipole in uniform electric field is rotated from 30^0 to 120^0. Match the following:

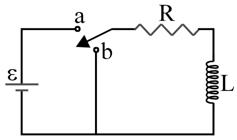
	Table -1		Table -2
(a)	Force	(P)	increases
(b)	Torque	(Q)	decreases
(c)	Potential energy	(R)	remain constant
(d)	stability	(s)	increases then decreases

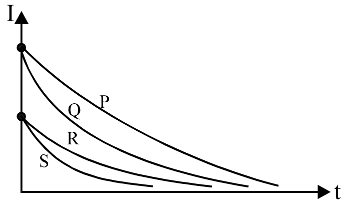
618. A battery of emf E is connected across a conductor as shown as one observe from A to B. Match the following



	Column-I		**Column-II**
(a)	Current	(p)	increases
(b)	Drift velocity of electron	(q)	decreases
(c)	Electric field	(r)	remains same
(d)	Potential drop across the length	(s)	Cannot be determine

619. Figure shows an RL circuit. The switch in the circuit has been closed on **a** for a very long time when it is then thrown to **b**. The resulting current through the inductor is indicated in figure for four sets of values for the resistance R and inductance L in Column I. Which set goes with which curve?





	Column I		*Column II*
(a)	R_0 and L_0	(P)	Curve P
(b)	$2R_0$ and L_0	(Q)	Curve Q
(c)	R_0 and $2L_0$	(R)	Curve R
(d)	$2R_0$ and $2L_0$	(S)	curve S

620.

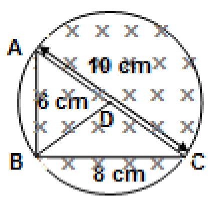
(a)	A circular conducting loop rotating about its axis in a uniform and constant magnetic field parallel to its axis	(P)	Conservation of energy.
(b)	Lenz's law explains	(Q)	Direction of magnetic field.
(c)	If a current in the loop decreases.	(R)	No emf is induced in the loop.
(d)	Flemming's rule explains	(S)	emf is induced in the loop.

621.

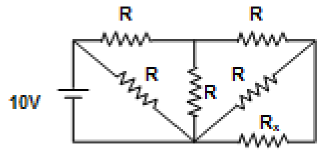
(a)	Gauss Theorem is applicable	(P)	perpendicular to the equipotential surface
(b)	Surface charge density of a conducting body	(Q)	non-conservative field
(c)	Electric lines of forces	(R)	maximum at corner of the body
(d)	Induced electric field	(S)	for any shape of charged body

SECTION : (E) - Integer Type

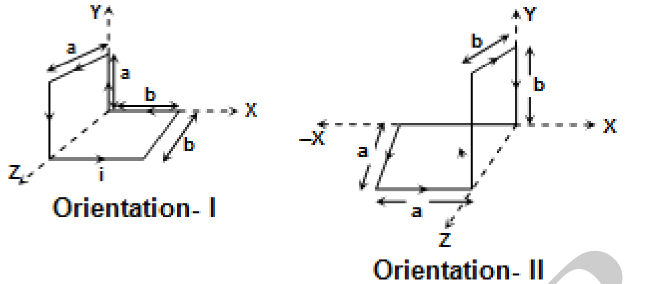
622. ABC is a triangular frame made out of a metallic wire. BD is a median of the triangular frame joining the vertex B to the side AC, (Also made of the same metallic wire). It is known that AB = 6 cm, BC = 8 cm and AC = 10 cm. The cross sectional area of the wire is 1 mm^2 and its resistivity is 24 nΩm. The triangle ABC lies in a cylindrical region of magnetic field such that the intersection of the surface of the cylinder with the plane containing the frame forms the circumcircle of triangle ABC. The magnetic field in the region varies at the rate of 0.263 T/s. What is the magnitude of the induced current in mA in the median BD of the frame?



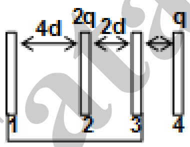
623. If R = 10 Ω, find the value of R_x (in ohm) for which the heat generated per unit time in R_x is maximum.



624. Find the work done (in joule) in slowly rotating the loop shown in the figure from orientation 1 to orientation 2. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{j})$ Tesla exists in the region and a current of 1A flow in the loop in the direction shown in the figure. Given a = b = 1 m.



625. Four identical metal plates of large area A are arranged as shown in the diagram. The plates 1 and 3 are connected by a conducting wire. The charge on plate 2 is 2q and an plate 4 is q. Find the potential difference (in volts) between the plate 2 and 3 if $\frac{qd}{\varepsilon_0 A} = 2$.

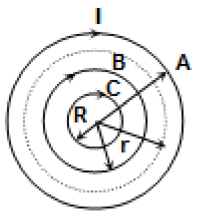


626. A long solenoid A contains another two co-axial solenoids B and C. Radii of solenoids A, B and C are 4 cm, 2 cm and 1 cm respectively. All the solenoids have same number of coils per unit length and current through the coils varies with time t as given.

I_A = 3 Kt

I_B = 2 Kt

I_C = 19 Kt,



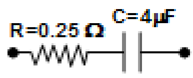
where K is a positive constants and I_A, I_B and I_C are the currents through the solenoid A, B and C respectively. Directions of all the current are same. As a result of the increasing currents a charged particle initially at rest between the solenoids A and B, starts moving along a circular trajectory as shown in the figure. Find the radius r of the circular trajectory, give your answer in C.G.S unit.

627. A battery of emf 12 V and internal resistance 0.5 Ω is charged by a battery charger which supplies a 132 V do supply using a series resistance of 11.5 Ω. What is the terminal voltage in volt of the battery during charging

628. In a car spark coil, an emf of 40,000 volts is induced in its secondary coil when the current in its primary changes from 4A to zero in 10 μs. The mutual inductance between the primary and secondary windings (in Henry) of the spark coil is X/10. Find the value of X.

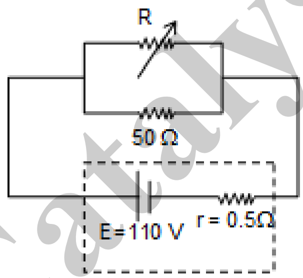
629. A square loop of each side 2 m is kept horizontally in a vertical constant magnetic field 10^{-2} T. The alignment of the loop is such that maximum flux passes through the loop. In 1 second, the loop is collapsed to have zero area, then the magnitude of the emf induced (in v) in the loop is X/100. The value of X is:

630. A 2μF capacitor is charged to 1 μC and connected in parallel to a resistance capacitor combination which has resistor of 0.25 Ω and a 4 μF capacitor (initially uncharged). The magnitude of charge on the uncharged capacitor varies with time as $q = \frac{2}{n}\left(1 - e^{-3t}\right)$

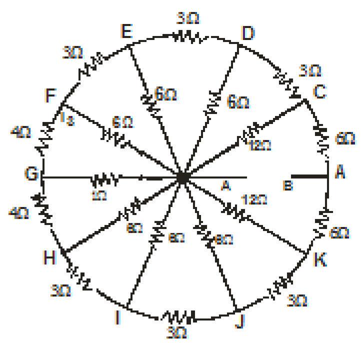


Then the value of n is:

631. An electric circuit consists of a battery emf E = 110 V, and internal resistance is 0.5 Ω and two resistors connected in parallel to the source as shown in figure. Resistance R is chosen so that power liberated in resistance R is maximum. Determine the value of R corresponding to maximum power



632. Find the equivalent resistance of shown circuit across A & B (in Ω).

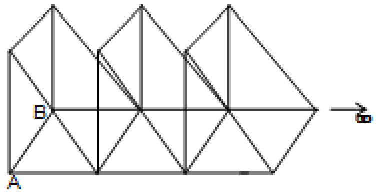


633. In an A.C. series C-R circuit, the voltage applied and the current is given as follows

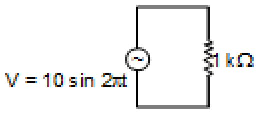
v(t) = 170 sin[6280t + (π/3)] volts

i(t) = 8.5 sin[6280t + (π/2)] volts, then RC is $\frac{\sqrt{K}}{6280}$ sec, then the value of k is ....

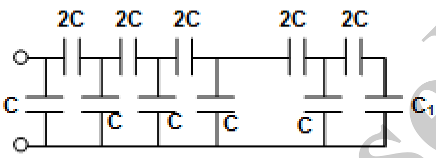
634. If the resistance of each branch is R, find the equivalent resistance between A & B if R = $\sqrt{6}\Omega$



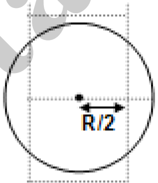
635. A voltage source v = 10 sin 2πt volt is connected to the circuit as shown in the figure. The energy dissipated through the resistor in Joule in first 0.5 sec is 1/Y. The value of Y is:



636. The figure shows a network of capacitance of several units of capacitance. Find the value of C_1 in μF so that equivalent capacitance between A and B is independent of number of units. [given C = 10μF]

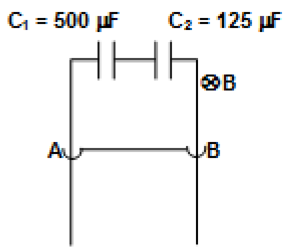


637. A charge Q is uniformly distributed over a solid spherical ball of radius R. The flux of electric field in SI unit across cuboid of dimension (R × 2R × 2R) is .... $\left[\text{if } \frac{Q}{\varepsilon} = 32 \text{ SI} - \text{unit}\right]$ [The front view is shown in the figure.]

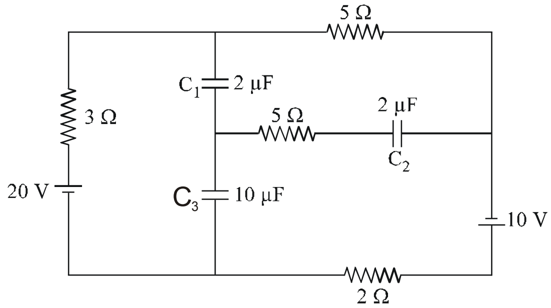


638. A wire of (AB) mass m = 5 milligram and length 1 m can slide freely on a pair of smooth, vertical rails. A magnetic field of 1T exists in the region in the direction perpendicular to the plane of the rails. The rails are connected at the top by two capacitors C_1 = 500 μF and C_2 = 125 μF of breakdown voltage 750 volts and 600 volts respectively.

The wire AB is released at t = 0, calculate the time in sec at which the circuit breaks down

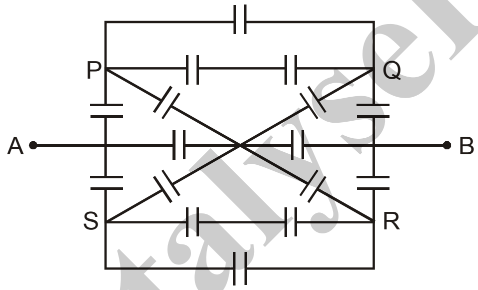


639. In the circuit shown in figure, find the charge on capacitor C_2 in steady state in μC.

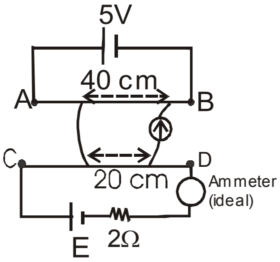


640. A dipole of dipole moment $\vec{P} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at point A (2, –3, 1). The electric potential due to this dipole at the point B (4, – 1, 0) is (ab) × 10^9volt here 'a' represents sign (for negative answer select 0 for positive answer select 1 as first digit of your answer in the OMR Sheet) and 'b' is a single digit that will appear in your answer (For Example : If your answer is – 2 × 10^9 volt then in the OMR sheet you should fill $\begin{array}{|c|}\hline 0 \\ \hline 2 \\ \hline \ \\ \hline \end{array}$). All the parameters specified here are in S.I. units.

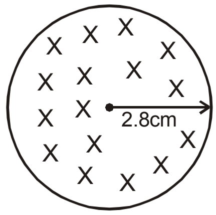
641. In the circuit shown in figure if each capacitor is of capacitance 10 μF, find the equivalent capacitance between points *A* and *B* in μF.



642. AB and CD are two uniform resistance wires of lengths 100 cm and 80 cm respectively . The connections are shown in the figure. The cell of emf 5 V is ideal while the other cell of emf E has internal resistance 2 Ω.A length of 20 cm of wire CD is balanced by 40 cm of wire AB. Find the emf E in volt, if the reading of the ideal ammeter is 2 A. The other connecting wires have negligible resistance.



643. The magnetic field of a cylindrical magnet that has a pole-face radius 2.8 cm can be varied sinusoidally between minimum value 16.8 T and maximum value 17.2 T at a frequency of $\frac{60}{\pi}$ Hz. Cross section of the magnetic field created by the magnet is shown. At a radial distance of 2cm from the axis find the amplitude of the electric field (in mN/C) induced by the magnetic field variation.



644. Given that $C_A = 1\,\mu F$, $C_B = 2\,\mu F$ and $C_C = 2\,\mu F$. Initially each capacitor was charged to potential differences of $V_A = 10$ v, $V_B = 40$ v and $V_C = 60$ v separately and are kept as shown in figure (a). Now they are connected as shown in figure (b). The + and - sign shown in figure (b) represent initial polarities. Find total amount of heat produced in μJ by the time steady state is reached.

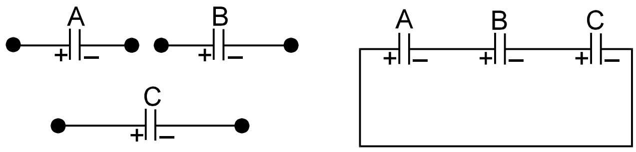


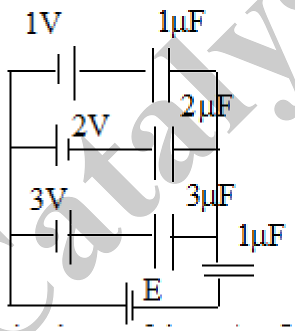
Fig. (a) Fig. (b)

645. *In an LRC series circuit at resonance current in the circuit $10\sqrt{2}$ A. If now frequency of the source is changed such that now current lags by 45° than applied voltage in the circuit. Find the new current in the circuit. Add 1 to your answer if the frequency is to be increased (from resonant frequency) and subtract 1 from your answer if the frequency is to be decreased (from resonant frequency) to get the desired result.*

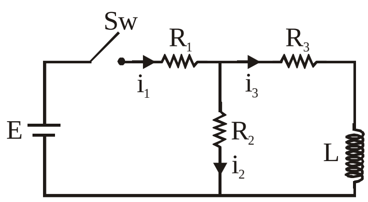
646. The bob of a simple pendulum has a mass of 40 g and a positive charge of 4.0×10^{-6} C. It makes 20 oscillations in 45 s. A vertical electric field pointing upward and of magnitude 2.5×10^4 N/C is switched on. How much time (in seconds) will it now take to complete 20 oscillations ?

647. Consider a LC series circuit.At time t=0, charge in the capacitor is 4C and it is decreasing at a rate of $\sqrt{5}$ C/s.C=1F,L=4H.Find theMaximum charge in the capacitor can be:

648. In the figure shown, find the emf E (in volts) for which charge on 2 μF is capacitor is 4 μC



649. In the circuit shown in the figure E = 15 V, $R_1 = 1\Omega$, $R_2 = 2\Omega$, $R_3 = 2\Omega$ and L = 1.5 H. The currents flowing through R_1, R_2 and R_3 are i_1, i_2 and i_3 , respectively.find the value of i_1 just after closing the switch



650. A 200 km long telegraph wire has capacity of 0.014 $\mu F / Km$ and carries an alternating current of frequency 5 KHz. What should be the value of an inductance required to be connected in series so that impedance is minimum. [in 10^{-5}]

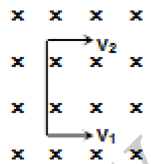
651. A 750 Hz, 20 V source is connected to a resistance of 100 Ω , an inductance of 0.1803 H and a capacitance of 10 μF all in series. The time in which the resistance (thermal capacity 2 J/^{0}C) will get heated by 10^0C is X/10 minutes. The value of X is:

EXERCISE 02

SECTION : (A) - Single Correct Options

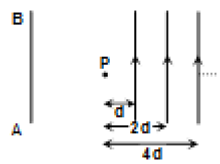
652. In a current carrying wire, the area of cross section (A) varies continuously. For a constant voltage across the wire, which of the following plots best represents the variation of current density (J)

653. In the figure magnetic field points into the plane of paper and the rod of length ℓ is moving in this field such that the bottom most point has a velocity v_1 and the topmost point has the velocity v_2 ($v_2 > v_1$). The emf induced is given by



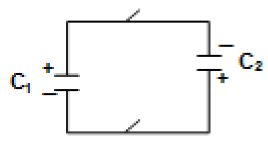
(A) $Bv_1\ell$ (B) $Bv_2\ell$ (C) $\frac{1}{2}B(v_2+v_1)\ell$ (D) $\frac{1}{2}B(v_2-v_1)\ell$

654. Infinite number of wires each having infinite length and carrying current i are placed as shown. A wire AB of infinite length carrying same current i is placed at a distance ℓ from P, the direction of current in wire AB and distance ℓ so that magnetic field at P is zero will be



(A) same direction, d
(B) same direction, d/2
(C) opposite direction, d
(D) opposite direction, d/2

655. Two capacitors $c_1 = 1\ \mu F$ and $c_2 = 3\ \mu F$ are charged to potential difference 20 v and 30 v by batteries. They are then disconnected from the batteries and connected across each other as shown i.e. their plates are connected with opposite polarity, then final potential difference is

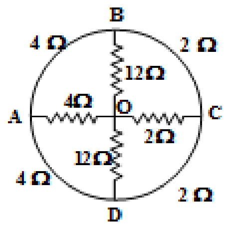


(A) 13.75 V (B) 27.9 V (C) 17.5 V (D) none of these

656. A narrow electron beam passes undeviated through an electric field $E = 3 \times 10^4$ V/m and an overlapping magnetic field $B = 2 \times 10^{-3}$ wb/m^2. Which are perpendicular to each other, then speed of the electron is

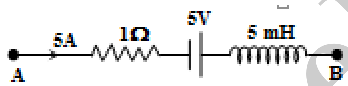
(A) 60 m/s (B) 10.3×10^7 m/s (C) 1.5×10^7 m/s (D) 0.67×10^{-7} m/s

657. The resistance between OA is given by



(A) 1 Ω (B) 2 Ω (C) 3 Ω (D) 4 Ω

658. The circuit segment shown is a part of a network. The current is 5A and is increasing at the rate of 10^3 A/s. The potential difference $V_A - V_B$ is

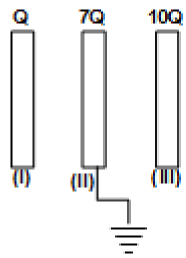


(A) 10 V (B) zero (C) –10 V (D) 5 V

659. Electric field in a region is given by $\vec{E} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ v/m. Another electric field due to a uniformly and positively charged infinite plane is superposed on the given field. The resultant field is observed to be $\vec{E} = (\hat{i} + \hat{j} - 4\hat{k})$ v/m. the surface charge density of the plane is

(A) $2\sqrt{3}\varepsilon_0$ (B) $2\sqrt{5}\varepsilon_0$ (C) $3\sqrt{2}\varepsilon_0$ (D) $\sqrt{5}\varepsilon_0$

660. Three very large identical metal plates are given charges as shown. After earthing the plate II, final charge on the left side of I plate will be

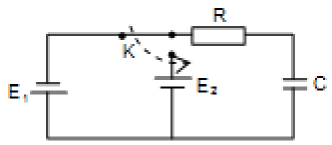


(A) –8Q (B) 9Q (C) Q/2 (D) none of these

661. Ends of two wires A and B having resistivity $\rho_A = 3 \times 10^{-5}$ Ωm and $\rho_B = 6 \times 10^{-5}$ Ωm of same cross section area are joined together to form a single wire. If the resistance of the joined wire does not change with temperature, then find the ratio of their lengths, given that temperature coefficient of resistivity of wire A and B is $\alpha_A = 4 \times 10^{-5}$ / ^{0}C and $\alpha_B = -6 \times 10^{-6}$ /^{0}C. Assume that mechanical dimensions do not change with temperature.

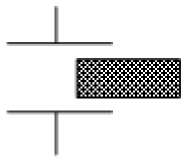
(A) $\frac{3}{7}$ (B) $\frac{10}{3}$ (C) $\frac{3}{10}$ (D) $\frac{1}{2}$

662. Two batteries of emf E_1 and E_2, a capacitor of capacitance C, and a resistor R are connected in a circuit as shown in figure. Then, the amount of heat Q liberated in the resistor after switching the key k will be

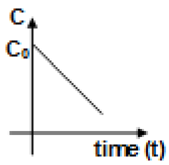


(A) $C(E_1 + E_2)^2$ (B) $\frac{C(E_1 + E_2)^2}{2}$ (C) $\frac{C}{4}(E_1 + E_2)^2$ (D) $\frac{1}{2}CE_1^2$

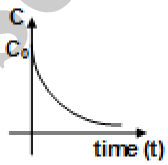
663. A parallel plate capacitor has a dielectric slab of dielectric constant k in it. The slab just fills the space inside the capacitor. The capacitor is charged by a battery and then battery is disconnected. Now the slab is pulled out slowly at t = 0 with constant velocity v. If at time t =0 capacitance of the capacitor is C_0, then the curve between C and time t will be



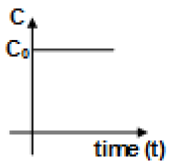
(A)



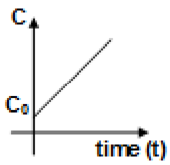
(B)



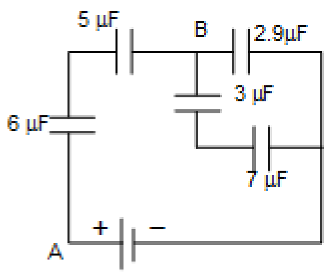
(C)



(D)



664. In the circuit shown if in steady state the potential difference between points A and B is 11V, then potential difference across 7 μ F capacitor is



(A) 1.8 V (B) 2.4 V (C) 3.6 V (D) 3.6 V

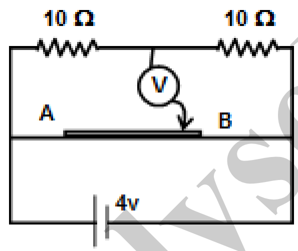
665. Three charges q, q and -2q are fixed on the vertices of an equilateral triangular plate of edge length a. This plate is in equilibrium between two very large plates having surface charge density σ_1 and σ_2 respectively. Find time period of small angular oscillation about an axis passing through its centroid and perpendicular to plane. Moment of inertia of the system about this axis is I.

(A) $2\pi\sqrt{\dfrac{\varepsilon_0 I}{q_a\,|\sigma_1-\sigma_2|}}$ (B) $2\pi\sqrt{\dfrac{\varepsilon_0 I}{2q_a\,|\sigma_1-\sigma_2|}}$ (C) $2\pi\sqrt{\dfrac{2\varepsilon_0 I}{\sqrt{3}q_a\,|\sigma_1-\sigma_2|}}$ (D) $2\pi\sqrt{\dfrac{2\varepsilon_0 I}{q_a\,|\sigma_1-\sigma_2|}}$

666. A resistor of resistance R, capacitor of capacitance C and inductor of inductance L are connected in parallel to AC power source of voltage $\varepsilon_0 \sin \omega t$. The maximum current through the resistance is half of the maximum current through the power source. Then value of R is

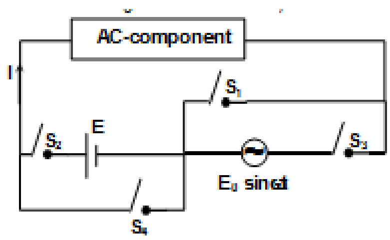
(A) $\dfrac{\sqrt{3}}{\left|\omega C-\dfrac{1}{\omega L}\right|}$ (B) $\sqrt{3}\left|\dfrac{1}{\omega C}-\omega L\right|$ (C) $\sqrt{5}\left|\dfrac{1}{\omega C}-\omega L\right|$ (D) none of these

667. A potentiometer wire AB as shown is 40 cm long of resistance 50 Ω/m free end of an ideal voltmeter is touching the potentiometer wire. What should be the velocity of the jockey as a function of time so that reading in voltmeter is varying with time as (2 sinπt). (Assuming negative terminal of the battery at zero potentially)



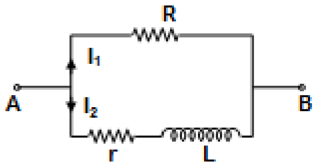
(A) 10 π sin πt cm/s (B) 10 π cos πt cm/s (C) 20 π sin πt cm/s (D) 20 π cos πt cm/s

668. In the given circuit if switches S_3 and S_4 are open, keeping S_1 and S_2 closed, the value of I is I_0. If switches S_1 and S_2 are open, keeping S_3 and S_4 closed, the value of I is $I_0 \sin \omega t$. Now, if switches S_1 and S_4 are open keeping S_2 and S_3 closed, the value of rms current through the AC-component, during one complete cycle, is



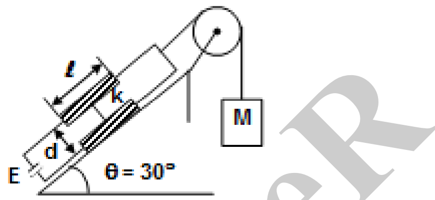
(A) $I_0\left(1+\dfrac{1}{\sqrt{2}}\right)$ (B) $\dfrac{I_0}{\sqrt{2}}$ (C) $I_0\left(1-\dfrac{1}{\sqrt{2}}\right)$ (D) $\sqrt{\dfrac{3}{2}}I_0$

669. A sinusoidal voltage is applied across the point A and B and it is observed that the total current through the circuit lags behind the external voltage by an angle 30°. If the peak value of current I_2 is 2A and it lags behind the external voltage by 45°, calculate the peak value of the total current.



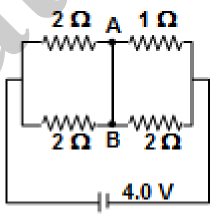
(A) 2A (B) $2\sqrt{2}$A (C) $\sqrt{2}$A (D) none of these

670. The capacitor plates are fixed on an inclined plane and connected to a battery of emf e. The capacitor plates have plate area A, length ℓ and the distance between them is d. A dielectric slab of mass m and dielectric constant K is inserted into the capacitor and tied to a mass M by a massless string as shown in the figure. Find the value of M for which the slab will stay in equilibrium. There is no friction between slab and plates.



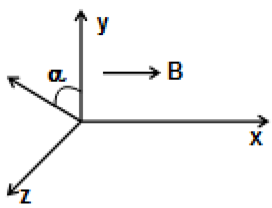
(A) $\frac{m}{2}+\frac{E^2\varepsilon_0 A(k-1)}{2\ell gd}$ (B) $\frac{m}{2}-\frac{E^2\varepsilon_0 A(k-1)}{2\ell gd}$ (C) $\frac{m}{2}+\frac{E^2\varepsilon_0 A(k-1)}{\ell gd}$ (D) $\frac{m}{2}-\frac{E^2\varepsilon_0 A(k-1)}{\ell gd}$

671. In the figure shown, points A and B are connected by a perfectly conducting wire. Calculate the current through AB.



(A) 2A (B) 1A (C) 1.5 A (D) 2.5 A

672. In a region of space, a uniform magnetic field B exists in the x direction. An electron is fired from the origin with its initial velocity u making an angle α with the y – direction in the yz plane. In the subsequent motion of the electron



(A) y – coordinate of the electron never be negative

(B) z – coordinate of the electron never be negative

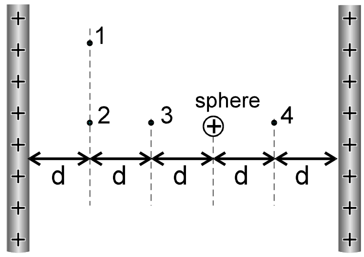
(C) x – coordinate of the electron never be negative.

(D) trajectory of the electron would be helical.

673. A non conducting ring of radius r and mass m has a uniformly distributed charge Q. A magnetic field perpendicular to the plane of the ring changes at the rate $\frac{dB}{dt} = k$. The angular velocity of the ring after a time internal Δt is,

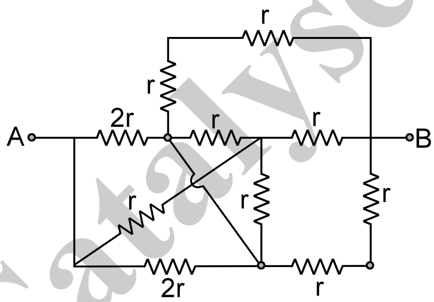
(A) zero (B) $\frac{Qk\Delta t}{m}$ (C) $\frac{Qk\Delta t}{2m}$ (D) $\frac{\pi Qk\Delta t}{m}$

674. The figure shows two large, closely placed, parallel, nonconducting sheets with identical (positive) uniform surface charge densities, and a sphere with a uniform (positive) volume charge density. Four points marked as 1, 2, 3 and 4 are shown in the space in between. If E_1, E_2, E_3 and E_4 are magnitude of net electric fields at these points respectively then :



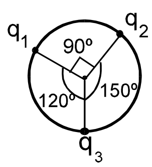
(A) $E_1 > E_2 > E_3 > E_4$ (B) $E_1 > E_2 > E_3 = E_4$ (C) $E_3 = E_4 > E_2 > E_1$ (D) $E_1 = E_2 = E_3 = E_4$

675. The equivalent resistance between A and B in the arrangement of resistances as shown, is :



(A) 4r (B) 3r (C) 2.5r (D) r

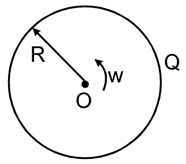
676. Three positive point charges q_1,q_2 and q_3 form an isolated system.Suppose the charges have generated a property due to which like charges attract. The charges are moving along a circle with same speed ,maintaining angles as shown in the figure.The charge q_1 experiences a force f_1 due to other two charges .Similarly q_2 experiences a force f_2 and q_3, a force f_3.The ratio $f_1 : f_2 : f_3$ is



(A) 1 : 1 : 1
(B) $q_1 : q_2 : q_3$
(C) $1 : \sqrt{3} : 2$
(D) this ratio can not be calculated.

SECTION : (B) - More Than One Correct Options

677. A nonconducting disc having uniform positive charge Q, is rotating about its axis with uniform angular velocity ω. The magnetic field at the center of the disc is.



(A) directed outward

(B) having magnitude $\frac{\mu_0 Q\omega}{4\pi R}$

(C) directed inwards

(D) having magnitude $\frac{\mu_0 Q\omega}{2\pi R}$

678. A parallel plate capacitor of capacitance 10 μF is connected to a cell of emf 10 Volt and fully charged. Now a dielectric slab (k=3) of thickness equal to the gap between the plates, is completely filled in the gap, keeping the cell connected. During the filling process :

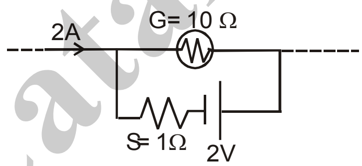
(A) The increase in charge on the capacitor is 200 μC.

(B) The heat produced is zero.

(C) Energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab.

(D) Energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab + heat produced.

679. The galvanometer shown in the figure has resistance 10Ω .It is shunted by a series combination of a resistance S = 1Ω and an ideal cell of emf 2V. A current 2A passes as shown.



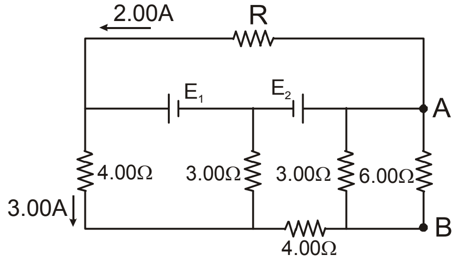
(A) The reading of the galvanometer is 1A

(B) The reading of the galvanometer is zero

(C) The potential difference across the resistance S is 1.5 V

(D) The potential difference across the resistance S is 2 V

680. In the circuit shown in figure, E_1 and E_2 are two ideal sources of unknown emfs. Some currents are shown. Potential difference appearing across 6Ω resistance is $V_A - V_B$ = 10V.



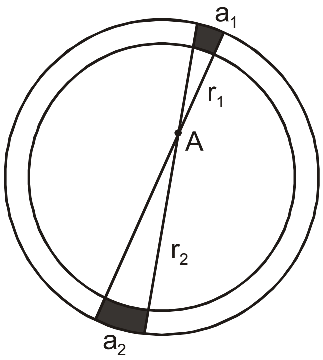
(A) The current in the 4.00 Ω resistor is 5A.

(B) The unknown emf E_1 is 36 V.

(C) The unknown emf E_2 is 54 V.

(D) The resistance R is equal to 9 Ω.

681. A wire having a uniform linear charge density λ, is bent in the form of a ring of radius R. Point A as shown in the figure, is in the plane of the ring but not at the centre. Two elements of the ring of lengths a_1 and a_2 subtend very small same angle at the point A. They are at distances r_1 and r_2 from the point A respectively.



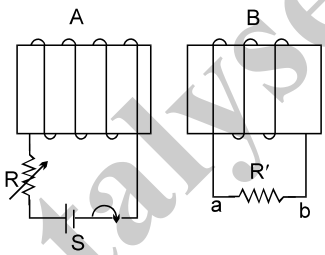
(A) The ratio of charge of elements a_1 and a_2 is r_1/r_2.

(B) The element a_1 produced greater magnitude of electric field at A than element a_2.

(C) The elements a_1 and a_2 produce same potential at A.

(D) The direction of net electric field at A is towards element a_2.

682. For the given electromagnetically coupled circuits: (S is initially in closed state)



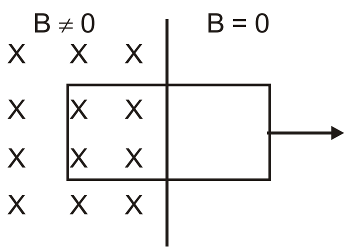
(A) When switch S is opened current in R′ flows from a to b

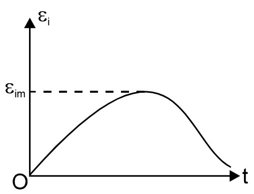
(B) When switch S is opened current in R′ flows from b to a

(C) When coil B is brought closer to coil A current in R′ flows from b to a

(D) When R is decreased then current in R′ flows from b to a

683. A plane rectangular loop is placed in a magnetic field. The emf induced in the loop due to this field is ε_i whose maximum value is ε_{im}. The loop was pulled out of the magnetic field at a variable velocity. Assume the $\vec{B}$ is uniform and constant. ε_i is plotted against time t as shown in the graph. Which of the following are/is correct statement(s):





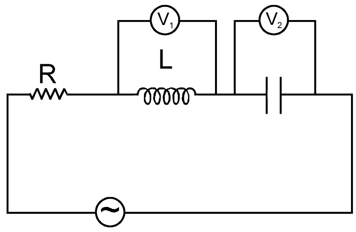
(A) ε_{im} is independent of rate of removal of coil from the field.

(B) The total charge that passes through any point of the loop in the process of complete removal of the loop does not depend on velocity of removal.

(C) The total area under the curve (ε_i vs t) is independent of rate of removal of coil from the field.

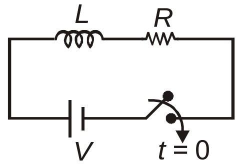
(D) The area under the curve is dependent on the rate of removal of the coil.

684. In the figure shows R = 100 Ω, $L = \frac{2}{\pi}$ H and $C = \frac{8}{\pi}\mu F$ are connected in series with an a.c. source of 200 volt and frequency 'f'. V_1 and V_2 are two hot-wire voltmeters. If the readings of V_1 and V_2 are same then:



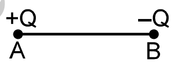
(A) f = 125 Hz
(B) f = 250 π Hz
(C) current through R is 2A
(D*) $V_1 = V_2$ = 1000 volt

685. The switch is closed at $t = 0$, in the LR circuit shown



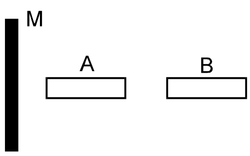
(A) At $t = 0$, voltage across inductor is V
(B) At $t = 0$, energy stored by the inductor is zero
(C) At $t = \frac{L}{R}$, energy stored by the inductor is less than half of the energy stored by the inductor at $t = \infty$
(D) At $t = \frac{L}{R}$, rate of heat dissipation in the circuit is more than half of the rate of heat dissipation at $t = \infty$

686. Two point charges of the same magnitude and opposite sign are fixed at points A and B. A third point charge is to be balanced at point P by the electrostatic force due to these two charges. The point P cannot lie at:



(A) the perpendicular bisector of line AB
(B) the mid point of line AB
(C) the left of A
(D) none of these.

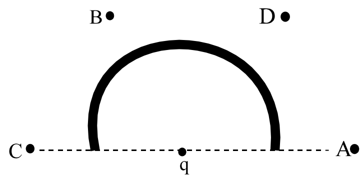
687. A large nonconducting sheet M is given a uniform charge density. Two uncharged small metal rods A and B are placed near the sheet as shown in figure.



(A) M attracts A
(B) M attracts B
(C) A attracts B
(D) B attracts A

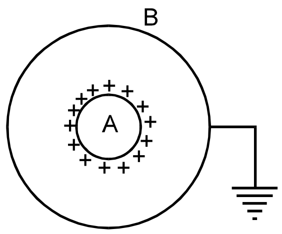
688. At distance of 5cm and 10cm outwards from the surface of a uniformly charged solid sphere, the potentials are 100V and 75V respectively. Then
(A) potential at its surface is 150V.
(B) the charge on the sphere is (5/3) × 10^{-9}C.
(C) the electric field on the surface is 1500 V/m.
(D) the electric potential at its centre is 225V.

689. Figure shows a charge q placed at the centre of a hemisphere. A second charge Q is placed at one of the positions A, B, C and D. In which position(s) of this second charge, the flux of the electric field through the hemisphere remains unchanged ?



(A) A (B) B (C) C (D) D

690. A and B are two concentric spherical shells. A is given a charge Q while B is uncharged. If now B is earthed as shown in Figure.Then:



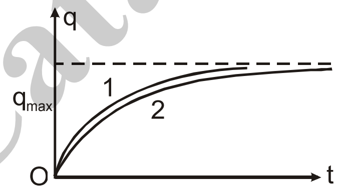
(A) The charge appearing on inner surface of B is -Q

(B) The field inside and outside A is zero

(C) The field between A and B is not zero

(D) The charge appearing on outer surface of B is zero

691. The charge on the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure. Choose the correct statement(s) related to the two circuits.



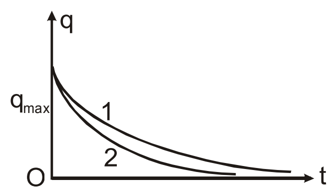
(A) Both the capacitors are charged to the same magnitude of charge

(B) The emf's of cells in both the circuits are equal.

(C) The emf's of the cells may be different

(D) The emf E_1 is more than E_2

692. The instantaneous charge on a capacitor in two discharging RC circuits is plotted with respect to time in figure. Choose the correct statement(s) (where E_1 and E_2 are emf of two DC sources in two different charging circuits).



(A) $R_1C_1 > R_2C_2$ (B) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$ (C) $R_1 > R_2$ if $E_1 = E_2$ (D) $C_2 > C_1$ if $E_1 = E_2$

693. *A parallel plate capacitor of capacitance 10 mF is connected to a cell of emf 10 Volt and fully charged. Now a dielectric slab (k=3) of thickness equal to the gap between the plates, is completely filled in the gap, keeping the cell connected. During the filling process :*

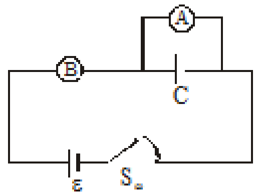
(A) The increase in charge on the capacitor is 200 μC.

(B) The heat produced is zero.

(C) Energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab.

(D) Energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab + heat produced.

694. A capacitor of capacitance C is connected to two voltmeters A and B. A is an ideal voltmeter having infinite resistance, while B has resistance R. The capacitor is uncharged and then the switch S_ω is closed at t = 0,



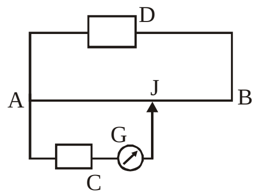
(a) readings of B and A will be ε and zero at t = 0

(b) during time interval ($0 \le t < \infty$) readings of B and A are changing

(c) reading of A and B will be equal at t = RC ln 2

(d) none of these

695. The figure shows a potentiometer arrangement. D is the driving cell while C is the cell whose emf is to be measured. AB is the potentiometer wire and G is a galvanometer. J is the jockey which can touch any point on AB. Which of the following is/are essential conditions for obtaining balance point



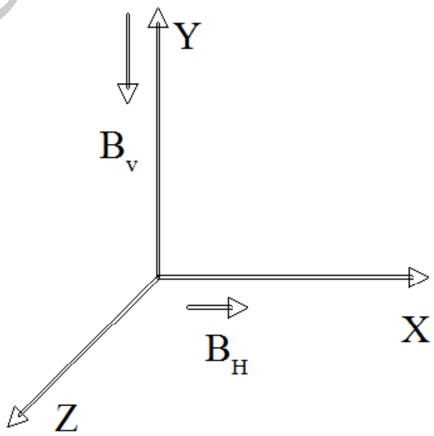
(a) the emf of D must be greater than the emf of C

(b) either the positive terminals of both D and C or the negative terminals of both D and C must be joined to A

(c) the positive terminals of D and C must be joined to A

(d) the resistance of G must be less than the resistance of AB

696. At a given place horizontal and vertical component of earth's magnetic field B_H and B_v are along X and Y axis respectively. The total flux of earth's magnetic field associated with an area S is:



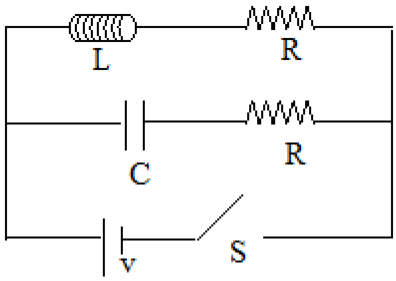
(a) Zero if area S is in X-Y plane.

(b) B_HS if area S is in Y-Z plane.

(c) - B_vS if area S is in X-Z plane.

(d)- B_HS if area S is in X-Z plane.

697. In the circuit shown $R = \sqrt{\frac{L}{C}}$, Switch S is closed at time t =0. The time constant of L-R and C-R, part of the circuit is τ_L and τ_C then :



(a) $\tau_L = \tau_C$

(b) $\tau_L = 2\tau_C$

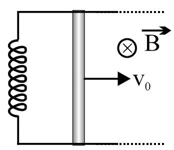
(c) After time t = CR ln 2, the current through capacitor and inductor will be equal.

(d) After time $t = \frac{L}{R \ln 2}$, the current through inductor and capacitor will be equal.

698. Which of the following(s) is/are conservative field(s) ?

(a) $\vec{E} = 4\hat{i} + 5\hat{j} + 6\hat{k}$ (b) $\vec{E} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{(x^2+y^2+z^2)^{3/2}}$ (c) $\vec{E} = 3x\hat{i}$ (d) $\vec{E} = x\hat{j} - y\hat{i}$

699. A loop is formed by two parallel conductors connected by a solenoid with inductance L and a conducting rod of mass M which can freely slide over the conductors. The conductors are located in a uniform magnetic field with induction B perpendicular to the plane of loop. The distance between conductors is *l*. At t = 0, the rod is given a velocity v_0 directed towards right and the current through the inductor is initially zero.



(a) The maximum current in circuit during the motion of rod is $v_0\sqrt{\frac{M}{L}}$.

(b) The rod moves for some distance and comes to permanently rest.

(c) The velocity of rod when current in the circuit is half of maximum is $\frac{\sqrt{3}}{2}v_0$.

(d) The rod oscillates in SHM.

700. Capacitor C_1 of capacitance $1\mu F$ and capacitor C_2 of capacitance $2\mu F$ are separately charged fully by a common battery. The two capacitors are then separately allowed to discharge through equal resistors at time t = 0.

(a) The current in each of the two discharging circuits is zero at t = 0

(b) The currents in the two discharging circuits at t = 0 are equal but not zero

(c) The currents in the two discharging circuits at t = 0 are unequal

(d) Capacitor C_1, loses 50% of its initial charge sooner than C_2 loses 50% of its initial charge

701. In the series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.

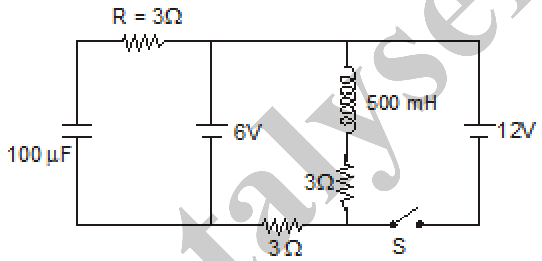
(a) It the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.

(b) If the capacitor is short-circuited, the voltage drop across the inductor will be $\frac{30}{\sqrt{2}}$ V.

(c) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.

(d) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.

702. In the given circuit switch is closed at $t = 0$,



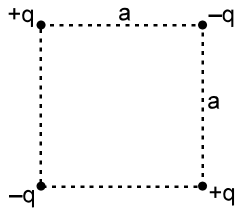
(a) Current in the inductor when the circuit reaches the steady state is 4 A.

(b) The net change in flux in the inductor is 1.5 Wb.

(c) The time constant of the circuit after closing S is 555.55 s.

(d) The charge stored in the capacitor in the steady state is 1.2 mC.

703. In a system of two dipoles placed in the way as shown in figure :



(A) It is possible to consider a spherical surface of radius a and whose centre lies within the square shown, through which total flux is + ve

(B) It is possible to consider a spherical surface of radius a and whose centre lies within the square shown through which total flux is – ve

(C) It is possible to consider a spherical surface of radius a and whose centre lies within the square shown through which total flux is zero

(D) There are two points within the square at which electric field is zero

704. Pick the correct statements.

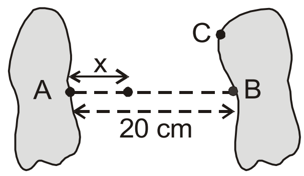
(A) If a point charge is placed off-centre inside an electrically neutral spherical metal shell then induced charge on its inner surface is uniformly distributed.

(B) If a point charge is placed off-centre inside an electrically neutral, isolated spherical metal shell, then induced charge on its outer surface is uniformly distributed.

(C) A non-metal shell of uniform charge attracts or repels a charged particle that is outside the shell as if all the shell's charge were concentrated at the centre of the shell.

(D) If a charged particle is located inside a non-metal shell of uniform charge, there is no electrostatic force on the particle due to the shell

705. Two metallic bodies separated by a distance 20 cm, are given equal and opposite charges of magnitude 0.88 μC. The component of electric field along the line AB, between the plates, varies as $E_x = 3x^2 + 0.4$ N/C, where x (in meters) is the distance from one body towards the other body as shown.



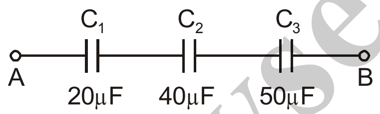
(A) The capacitance of the system is 10μF

(B) The capacitance of the system is 20μF

(C) The potential difference between A and C is 0.088 Volt.

(D) The potential difference between A and C cannot be determined from the given data.

706. In the arrangement shown, a potential difference is applied between points A and B. No capacitor can withstand a potential difference of more than 100 V.



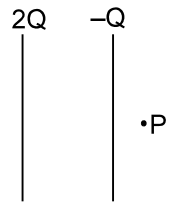
(A) The magnitude of the maximum potential difference that can exist between points A and B is 300 V.

(B) The maximum potential difference that can exist across C_2 is 50 V.

(C) The maximum charge that can be stored by C_3 is 2 mC.

(D) The maximum energy that can be stored in the three capacitor arrangement is 190 mJ

707. In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is 'C'. P is a point outside the capacitor and close to the plate of charge –Q. The distance between the plates is 'd'.



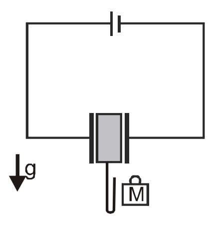
(A) A point charge at point 'P' will experience electric force due to capacitor

(B) The potential difference between the plates will be $\frac{3Q}{2C}$

(C) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$

(D) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\epsilon_0 d^2}$

708. Figure shows a capacitor with its plates vertical and a hook is attached to a light dielectric slab and slab is initially at rest inside the plates of capacitor. Neglect the any frictional forces. Now a small block of mass *M* is brought close to the *U* shaped hook and released suddenly to hang it in the hook :



(A) The dielectric with block may oscillate.

(B) The dielectric with block may be fall down by some distance and will come to permanently rest at equilibrium position.

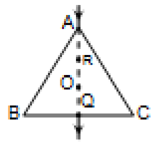
(C) The dielectric with block may continue to fall and come out of the capacitor.

(D) The block may come to rest at the bottom of hook and dielectric is not displaced at all.

SECTION : (C) -Passage Type Questions

PASSAGE 01:

The wire ABC shown in the figure forms an equilateral triangle of side a. The wires are identical. A current i is flowing in at point A and leaving out at point Q, just midway of wire BC as shown in the figure.



709. The magnetic field B at the centroid O of the wire is

(A) zero (B) $\frac{3\mu_0 i}{4\pi e}$ (C) $\frac{\mu_0 i}{2\pi e}$ (D) none of these

710. The magnetic field B at a point P just above the Q is

(A) zero (B) $\frac{\mu_0 i}{\pi e}$ (C) $\frac{\mu_0 i}{3\pi e}$ (D) none of these

711. The magnetic field B at point R, which lies on the line joining AO is

(A) zero (B) $\frac{\mu_0 i}{\pi e}$ (C) $\frac{\mu_0 i}{3\pi e}$ (D) none of these

PASSAGE 02:

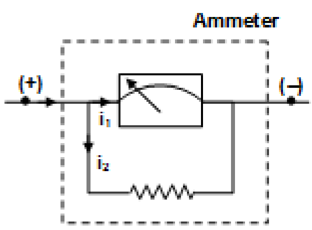
Ammeter is a device to measure an electric current. In order to convert a galvanometer into an ammeter, a resistance having small value is connected in parallel with the galvanometer coil.

This resistor is called the shunt. The current to be measured is passed through the ammeter by connecting it in series with the segment which carries the current

When the ammeter is connected in a segment of a circuit, the resistance of the segment increases. Which reduces the main current to be measured.

To minimize the error, the equivalent resistance of ammeter circuit should be small. This is one reason why the shunt having a small resistance r is connected in parallel to the coil.

A galvanometer of resistance 100 Ω gives a full scale deflection for a current of 1 mA



712. Shunt required to convert it into an ammeter giving full scale deflection for a current of 10 amp is

(A) $\frac{1}{100}\Omega$ (B) $\frac{100}{9999}\Omega$ (C) $\frac{100}{1999}\Omega$ (D) $\frac{100}{10001}\Omega$

713. When the current through ammeter is giving full scale deflection then heat generated by shunt per unit time is

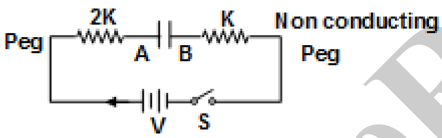
(A) 1 Watt
(B) 0.9999 Watt
(C) 1.0001 Watt
(D) no heat will be dissipated

714. When this ammeter is connected across the terminals of a battery a current of 4A flows through it. The current drops to 1amp when a resistance of 1.5 Ω is connected in series with the ammeter. The internal resistance of the battery is

(A) 5 Ω
(B) 10.1 Ω
(C) 0.49 Ω
(D) 4.9 Ω

PASSAGE 03:

In the given circuit, two identical parallel conducting plates A and B are connected to a 25 V battery by metal springs of spring constant 2k and k, respectively. Initially the switch 's' is open and the plates are uncharged. In fact, the two plates from a capacitor. When the switch 's' is closed, distance between the plates becomes 2 mm and it is one third of initial distance between the plates. Also the electric potential energy stored in the capacitor is found to be 7.5×10^{-4} J.



715. Initially, when the plates are uncharged, capacitance of the capacitor is

(A) 1.2 μF
(B) 1 μF
(C) 0.8 μF
(D) 0.6 μF

716. Spring constant of the spring connected to plate A is

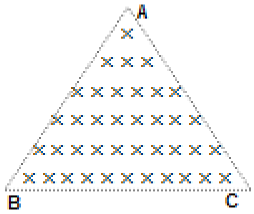
(A) 212.65 N/m
(B) 281.25 N/m
(C) 316.45 N/m
(D) 463.75 N/m

717. Extension of the spring connected plate B is

(A) 2.67 mm
(B) 3.22 mm
(C) 4.33 mm
(D) 1.33 mm

PASSAGE 04:

When a charge particle enters in a uniform magnetic field perpendicular to the magnetic field then its path becomes circular and its radius of curvature is given as $r = \frac{mv}{qB}$, where m is mass of the charge particle, v is the velocity and A uniform magnetic field is present in a region which is in the form of equilateral triangle (ΔABC) of side 'a' The magnetic field is in downwards direction and its intensity is 'B'. A positive point charge of mass 'm' and charge 'q' enters in the magnetic field with a certain speed along line BC. Then (q is the charge and B is magnetic intensity).



718. Maximum possible angular deviation is

(A) 60°
(B) 90°
(C) 120°
(D) 180°

719. Speed of the charge particle for which angular deviation is maximum

(A) $\frac{9qB}{\sqrt{3}m}$ (B) $\frac{9qB}{m}$ (C) $\frac{\sqrt{3}9qB}{m}$ (D) $\frac{9qB}{3m}$

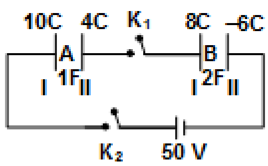
720. The speed of charge particle for which angular deviation is $\pi/2$.

(A) $\frac{qBq(\sqrt{3}-1)}{2m}$ (B) $\frac{\sqrt{3}9qB}{2m}$ (C) $\frac{\sqrt{3}(\sqrt{3}+1)9Bq}{2m}$ (D) $\frac{\sqrt{3}(\sqrt{3}-1)9Bq}{2m}$

PASSAGE 05:

A system of capacitors A and B is shown whose each plate have different charges are shown in the figure.

Initially switch K_1 B closed



721. The charge on inner side of first plate of capacitor A is

(A) –14C (B) –2C (C) +14 C (D) +2C

722. The charge on outer side of second plate of capacitor B is

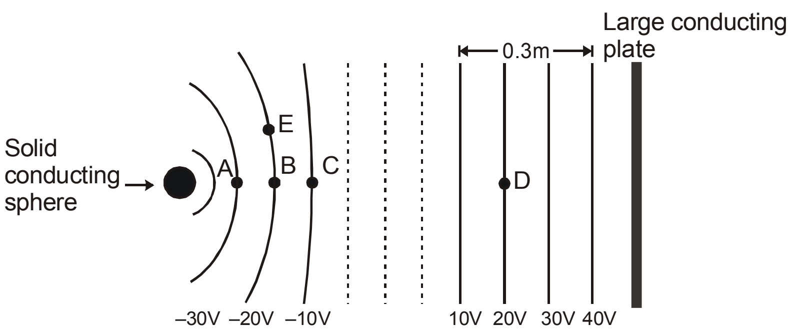
(A) +8 C (B) – 8C (C) +14 C (D) –14C

723. The amount of charge supplied by battery if both switches are closed

(A) +8/3 C (B) – 8/3C (C) +14/3 C (D) none of these

PASSAGE 06:

The sketch below shows cross-sections of equipotential surfaces between two charged conductors that are shown in solid black. Some points on the equipotential surfaces near the conductors are marked as A,B,C,........ . The arrangement lies in air. (Take $\epsilon_0 = 8.85 \times 10^{-12}\ C^2/N\ m^2$]



724. Surface charge density of the plate is equal to

(A) $8.85 \times 10^{-10}\ C/m^2$ (B) $-8.85 \times 10^{-10}\ C/m^2$

(C) $17.7 \times 10^{-10}\ C/m^2$ (D) $-17.7 \times 10^{-10}\ C/m^2$

725. A positive charge is placed at B. When it is released :
(A) no force will be exerted on it.
(B) it will move towards A.
(C) it will move towards C.
(D) it will move towards E.

726. How much work is required to slowly move a – 1μC charge from E to D ?
(A) 2×10^{-5} J
(B) -2×10^{-5} J
(C) 4×10^{-5} J
(D) -4×10^{-5} J

PASSAGE 07:

As a charged particle 'q' moving with a velocity $\vec{v}$ enters a uniform magnetic field $\vec{B}$, it experiences a force $\vec{F} = q(\vec{v} \times \vec{B})$.

For $\theta = 0^\circ$ or 180°, θ being the angle between $\vec{v}$ and $\vec{B}$, force experienced is zero and the particle passes undeflected. For $\theta = 90^\circ$, the particle moves along a circular arc and the magnetic force (qvB) provides the necessary centripetal force $\left(\frac{mv^2}{r}\right)$. For other values of θ ($\theta \neq 0^\circ, 180^\circ, 90^\circ$), the charged particle moves along a helical path which is the resultant motion of simultaneous circular and translational motions.

Suppose a particle, that carries a charge of magnitude q and has a mass 4×10^{-15} kg, is moving in a region containing a uniform magnetic field $\vec{B} = -0.4\,\hat{k}$ T. At some instant, velocity of the particle is $\vec{v} = (8\hat{i} - 6\hat{j} + 4\hat{k}) \times 10^6$ m/s and force acting on it has a magnitude 1.6 N.

Answer the following questions :

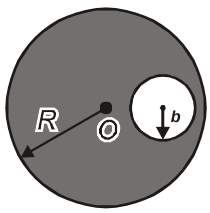
727. Motion of charged particle will be along a helical path with :
(A) A translational component along x-direction and a circular component in the y-z plane
(B) A translational component along y-direction and a circular component in the x-z plane
(C) A translational component along z-axis and a circular component in the x-y plane
(D) Direction of translational component and plane of circular component are uncertain

728. Angular frequency of rotation of particle, also called the 'cyclotron frequency' is :
(A) 8×10^5 rad/s
(B) 12.5×10^4 rad/s
(C) 6.2×10^6 rad/s
(D) 4×10^7 rad/s

729. If the coordinates of the particle at $t = 0$ are (2 m, 1 m, 0), coordinates at a time $t = 3T$, where T is the time period of circular component of motion, will be (take $\pi = 3.14$) :
(A) (2 m, 1 m, 400 m)
(B) (0.142 m, 130 m, 0)
(C) (2 m, 1 m, 1.884 m)
(D) (142 m, 130 m, 628 m)

PASSAGE 08:

A non-conducting sphere of radius R has a uniform volume charge density ρ. A spherical cavity of radius b is created inside the sphere, whose centre lies at a distance a from the centre of sphere. It can be shown that field inside cavity will be uniform

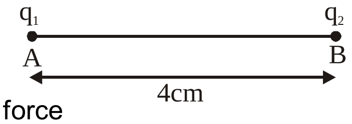


730. What is electric field at any point inside the spherical cavity?
(A) $\frac{\rho a}{3\varepsilon_0}$
(B) $\frac{\rho b}{3\varepsilon_0}$
(C) $\frac{\rho(a-b)}{3\varepsilon_0}$
(D) Zero

731. What is magnitude of electric field at point O?
(A) Zero
(B) $\frac{\rho b^3}{3\varepsilon_0 a^2}$
(C) $\frac{\rho b}{3\varepsilon_0}$
(D) $\frac{\rho a^3}{3\varepsilon_0 b^2}$

PASSAGE 09:

Two points charges q_1 = + 1μC and q_2 = – 2μC are placed at A and B respectively as shown in the figure. The distance between q_1 and q_2 is 4 cm.



732. A line of force emanates from q_1 making an angle 90º with AB. This line of force

(A) enters q_2 at an angle 90º
(B) enters q_2 at an angle 60º
(C) enters q_2 at an angle 45º
(D) does not enter q_2 but goes off to ∞

733. The net electric flux will be zero.

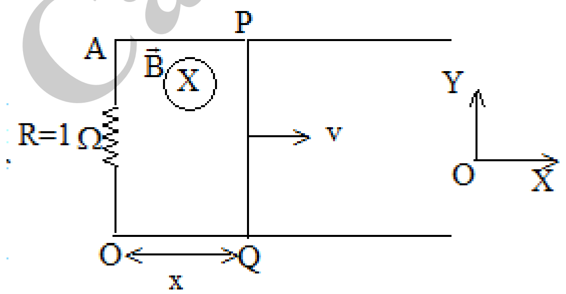
(A) over any surface that encloses a volume including A and B, but having very large radius
(B) over any surface that includes A twice and B once
(C) over any surface that encloses a volume excluding A and B
(D) only over a surface that encloses zero volume

734. The electrostatic potential is zero at

(A) a point on the line AB between q_1 and q_2 but closer to q_2
(B) a point on the line AB but not between A and B
(C) infinitely many point in space
(D) no point in space

PASSAGE 10:

Consider two parallel, conducting frictionles tracks are kept in gravity free space as shown in the figure. A movable conductor PQ, initially kept at OA, given a velocity 10 m/s towards right. If space contains a magnetic field which depends upon the distance moved by conductor PQ from OA line and given by



$$\vec{B} = cx\left(-\hat{k}\right)\left[c = \text{constant} = 1\text{S.I.}-\text{unit}\right]$$

The mass of conductor PQ is 1kg and length of PQ is 1 m. Answer the following questions based on above passage.

735. The distance travelled by conductor when its speed is 5 m/s is

(A) $\left(\frac{15}{2}\right)^{1/3}$
(B) $\left(\frac{5}{2}\right)^{1/3}$
(C) $(10)^{1/3}$
(D) None of the above

736. The heat loss during the time interval t = 0 to time t sec. When the speed of conductor is 5 m/s is

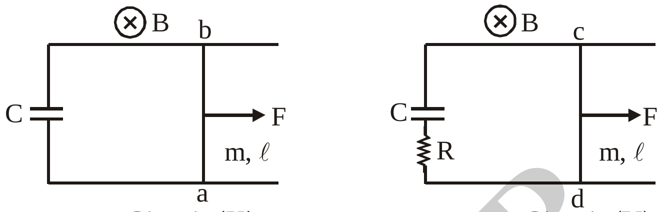
(A) 50 J
(B) 30J
(C) 10J
(D) none of the above

737. The work done by magnetic force acting on the conductor PQ during its motion in the time interval t=0 to t = t sec when the speed of conductor is 5 m/s is

(A) zero (B) 50 J (C) 10J (D) 30 J

PASSAGE 11:

For the two given circuits at t = 0, a constant force F acts at the middle points of the rigid conducting wires ab and cd. At t = 0 both wire are at rest. The electric resistance of the circuit (a) is zero, while for the circuit (b) electrical resistance is R. The electrical resistance of the horizontal rails is zero. There is no friction between rails and rigid wires ab and cd. Both the circuits are placed in a vertical constant magnetic field B (as shown in the figure). The mass and length of the each wire ab and cd is m, ℓ respectively



Circuit (X) Circuit (Y)

(Q. 738 - 740)

At $t=\sqrt{\frac{m\ell}{F}}$, electric current for circuit (X) is

(a) zero (b) $\frac{B\ell CF}{(m+B^2\ell^2C)}$

(c) $\frac{B\ell CF}{m}$ (d) none of these

At t = 0, acceleration of wire cd for circuit Y is

(a) $\frac{F}{m}$ (b) $\frac{F}{(m+B^2\ell^2C)}$

(c) $\frac{F}{B^2\ell^2C}$ (d) none of these

At t = 0, acceleration of wire ab for circuit X is

(a) $\frac{F}{m}$ (b) $\frac{F}{(m+B^2\ell^2C)}$

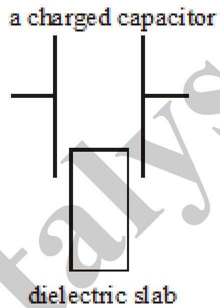
(c) $\frac{F}{B^2\ell^2C}$ (d) none of these

SECTION : (D) - Matrix Match

741. In the series L - C - R circuit fed by a sinusoidal AC source

	Column A		**Column B**
(a)	If the circuit is purely reactive	(p)	Maximum Power loss.
(b)	Purely resistive	(q)	No power loss.
(c)	If the frequency $\frac{1}{2\pi}\sqrt{\frac{1}{LC}}$	(r)	Maximum Current.
(d)	Inductor in a choke coil of fluorescent tube is used because through inductor, there is	(s)	Current reduction

742. A parallel plate air capacitor is charged by connecting its plates to the terminals of a battery. The battery is disconnected and a dielectric slab is introduced partially between the plates, as shown in the figure. Consider the change in the value of each quantity mentioned in the first column below from the time when no dielectric slab was intruduced to the time when it was, and match it wit the nature of change in it as mentioned in column on the right.

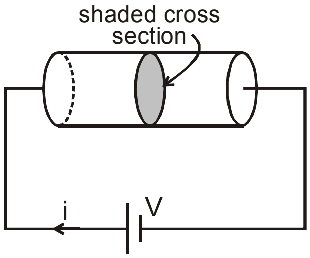


	Column A		**Column B**
(a)	Surface density of charge, σ	(p)	Increases.
(b)	Electric field intensity, E	(q)	Decreases.
(c)	Charge on the capacitor, q	(r)	Remains same.
(d)	Net force acting on either plate, F	(s)	Increase at some points decreases at others.

743. An electric dipole is placed in an electric field. The column I gives the description of electric field and the angle between the dipole moment $\vec{p}$ and the electric field intensity $\vec{E}$ and the Column II gives the effect of the electric field on the dipole. Match the description in Column I with the statements in Column II and indicate your answer by darkening appropriate bubbles in the 4 × 4 matrix given in the ORS.

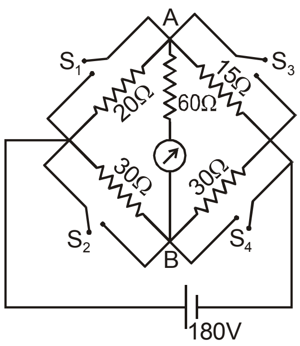
	Column I		**Column II**
(A)	Uniform electric field, $\theta = 0$	(p)	force = 0
(B)	Electric field due to a point charge, $\theta = 0$	(q)	Torque = 0
(C)	Electric field between the two oppositely charged large plates, $\theta = 90°$	(r)	$\vec{p}\cdot\vec{E} = 0$
(D)	Dipole moment parallel to uniformly charged long wire.	(s)	Force $\neq 0$

744. Column I gives physical quantities based on a situation in which an ideal cell of emf V is connected across a cylindrical rod of uniform cross-section area and conductivity (σ) as shown in figure. E, J, ϕ and i are electric field at, current density through, electric flux through and current through shaded cross-section respectively as shown in figure. Physical quantities in column II are related to those in column I. Match the expressions in Column I with the statements in Column II and indicate your answer by darkening appropriate bubbles in the 4 × 4 matrix given in the OMR.



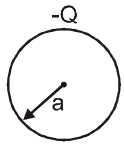
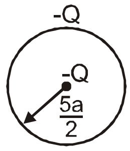
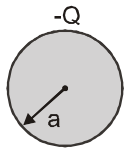
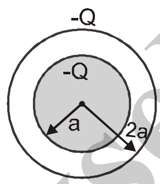
Column I		Column II
(A)	$\frac{\phi}{i}$	(p) Conductivity of rod
(B)	$\frac{E}{J}$	(q) Resistance of rod
(C)	$\sigma\phi V$	(r) Resistivity of rod
(D)	$\frac{V}{\sigma\phi}$	(s) Power delivered to rod

745. Consider the circuit shown. The resistance connected between the junction A and B is 60 Ω including the resistance of the galvanometer. The switches have no resistance when shorted and infinite resistance when opened. All the switches are initially open and they are closed as given in column I. Match the condition in column I with the direction of current through galvanometer and the value of the current through the battery in column II and indicate your answer by darkening appropriate bubbles in the 4 × 4 matrix given in the OMR.



Column I	Column II
(A) Only switch S_1 is closed	(p) Current from A to B
(B) Only switch S_2 is closed	(q) Current from B to A
(C) Only switch S_3 is closed	(r) current through the battery is 12.0 A
(D) Only switch S_4 is closed	(s) current through the battery is 15.6 A

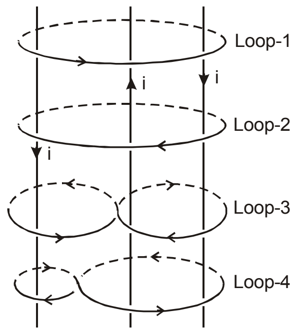
746. In each situation of column-I, some charge distributions are given with all details explained. In column -II The electrostatic potential energy and its nature is given situation in column -II. Then match situation in column-I with the corresponding results in column-II.

Column-I		Column-II
(A) A thin shell of radius a and having a charge – Q uniformly distributed over its surface as shown		(p) $\frac{1}{8\pi\in_0}\frac{Q^2}{a}$ in magnitude
(B) A thin shell of radius $\frac{5a}{2}$ and having a charge – Q uniformly distributed over its surface and a point charge – Q placed at its centre as shown.		(q) $\frac{3}{20\pi\in_0}\frac{Q^2}{a}$ in magnitude
(C) A solid sphere of radius a and having a charge – Q uniformly distributed throughout its volume as shown.		(r) $\frac{2}{5\pi\in_0}\frac{Q^2}{a}$ in magnitude
(D) A solid sphere of radius a and having a charge – Q uniformly distributed throughout its volume. The solid sphere is surrounded by a concentric thin uniformly charged spherical shell of radius 2a and carrying charge –Q as shown		(s) Positive in sign

747. In each situation of column-I some changes are made to a charged capacitor under conditions of constant potential difference or constant charge. Condition of constant potential difference means that the a cell is connected across the capacitor and condition of constant charge means that the capacitor is isolated. Match the conditions in column-I with corresponding results in column-II.

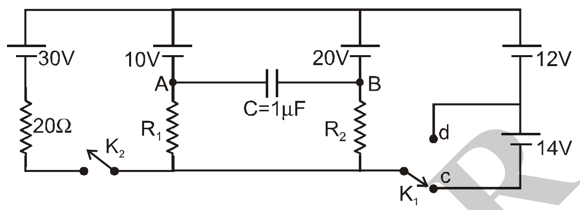
Column I	Column II
(A) For a capacitor maintained at constant potential difference, the separation between plates is increased.	(p) Then electric field inside the capacitor decreases in comparison to what it was before the change.
(B) For a capacitor maintained at constant charge, the separation between the plates is increased	(q) Then electric field inside the capacitor remains same.
(C) For a capacitor maintained at constant potential difference, area of the both the plates is doubled.	(r) Then potential energy stored in the capacitor decreases in comparison to what it was before the change.
(D) For a capacitor maintained at constant charge, area of both plates is doubled	(s) The potential energy stored in the capacitor increases in comparison to what it was before the change

748. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in the figure :



	Column–I	**Column–II**
(A)	Along closed Loop-1	(p) $\oint \vec{B}.d\vec{\ell} = \mu_0 i$
(B)	Along closed Loop-2	(q) $\oint \vec{B}.d\vec{\ell} = -\mu_0 i$
(C)	Along closed Loop-3	(r) $\oint \vec{B}.d\vec{\ell} = 0$
(D)	Along closed Loop-4	(s) net work done by the magnetic force to move along a unit charge the loop is zero.

749. A circuit involving five ideal cells , three resistors (R_1, R_2 and 20Ω) and a capacitor of capacitance C = 1 μF is shown. Match the conditions in column-I with results given in column-II.



	column-I		**column-II**
(A)	K_2 is open (off) and K_1 is in position C	(p)	Potential at point A is greater than potential at B
(B)	K_2 is open (off) and K_1 is in position D	(q)	Current through R_1 is downward
(C)	K_2 is closed (on) and K_1 is in position C	(r)	Current through R_2 is upward
(D)	K_2 is closed (on) and K_1 is in position D	(s)	Charge on capacitor is 10μC.

750.

	List – I		**List - II**
(A)	The force on a point charge due to other point charge $[F_{12}]$ is	(i)	Medium independent
(B)	The net force on a point charge $[F_1]$ is	(ii)	Affected by the presence of another point charge
(C)	The electric field lines at a point are	(iii)	Not affected by the presence of another point charge
(D)	The electric field intensity at a point is	(iv)	Medium dependent

751. **Column-I and Column-II** contains four entries each. Entries of column-I are to be matched with some entries of column-II. One or more than one entries of column-I may have the matching with the same entries of column-II and one entry of column-I may have one or more than one matching with entries of column-II

On a capacitor of capacitance C_0 following steps are performed in the order as given in column-I.

(a) Capacitor is charged by connecting it across a battery of EMF V_0.

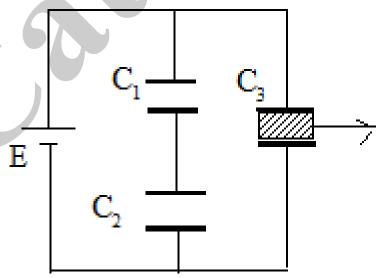
(b) Dielectric of dielectric constant 'k' and thickness 'd' is inserted.

(c) Capacitor is disconnected from battery.

(d) Separation between plates is doubled.

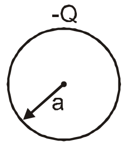
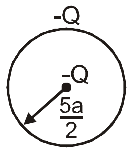
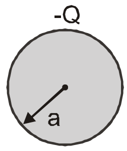
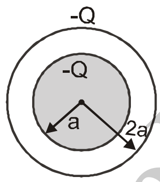
Column-I (Steps performed)		**Column-II** (Final value of Quantity) (Symbols have usual meaning)	
(A)	(a) (d) (c) (b)	(P)	$Q = \dfrac{C_0 V_0}{2}$
(B)	(d) (a) (c) (b)	(Q)	$Q = \dfrac{k C_0 V_0}{k+1}$
(C)	(b) (a) (c) (d)	(R)	$C = \dfrac{k C_0}{k+1}$
(D)	(a) (b) (d) (c)	(S)	$V = \dfrac{V_0 (k+1)}{2k}$

752. Three capacitors C_1, C_2 & C_3 are connected to battery as shown. where $C_1 = C_3 = 1F$ & $C_2 = 2F$ with E= 10V. The capacitor C_3 consists of dielectric slab of di-electric constant K=2.The slab is pulled out of capacitor C_3 by an external agent

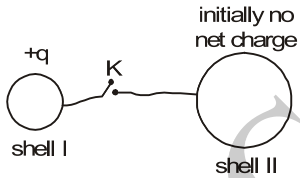


	List-I		**List-II**
A.	work consumed by the battery	p.	50 SI units
B.	work done by an external agent to pull the slab from C_3	q.	100 SI units
C.	heat generated in the circuit	r.	zero
D.	change in electrostatic potential energy of capacitors C_1 & C_2	s.	non-zero

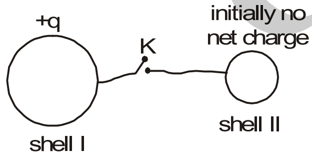
753. In each situation of column-I, some charge distributions are given with all details explained. In column -II The electrostatic potential energy and its nature is given situation in column -II. Then match situation in column-I with the corresponding results in column-II.

Column-I		Column-II
(A) A thin shell of radius a and having a charge – Q uniformly distributed over its surface as shown		(p) $\dfrac{1}{8\pi\in_0}\dfrac{Q^2}{a}$ in magnitude
(B) A thin shell of radius $\dfrac{5a}{2}$ and having a charge – Q uniformly distributed over its surface and a point charge – Q placed at its centre as shown.		(q) $\dfrac{3}{20\pi\in_0}\dfrac{Q^2}{a}$ in magnitude
(C) A solid sphere of radius a and having a charge – Q uniformly distributed throughout its volume as shown.		(r) $\dfrac{2}{5\pi\in_0}\dfrac{Q^2}{a}$ in magnitude
(D) A solid sphere of radius a and having a charge – Q uniformly distributed throughout its volume. The solid sphere is surrounded by a concentric thin uniformly charged spherical shell of radius 2a and carrying charge –Q as shown		(s) Positive in sign

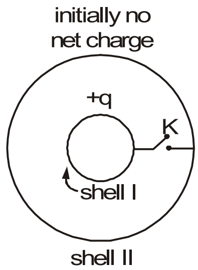
754. Column I gives certain situations involving two thin conducting shells connected by a conducting wire via a key K. In all situations one sphere has net charge +q and other sphere has no net charge. After the key K is pressed, column II gives some resulting effect. Match the figures in Column I with statements in Column II.

(A) 

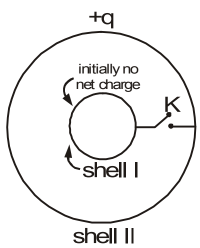
(p) charge flows through connecting wire

(B) 

(q) Potential energy of system of spheres decreases.

(C) 

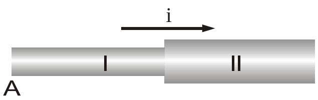
(r) No heat is produced.

(D) 

(s) The sphere I has no charge after equilibrium is reached.

(t) charge does not flows through connecting wire

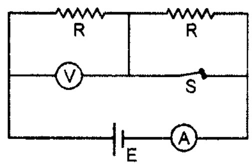
755. Column I gives physical quantities of a situation in which a current i passes through two rods I and II of equal legth that are joined in series. The ratio of free electron density (n, resistivity (p) and cross- section area (A) of both are in ratio $n_1 : n_2 = 2 : 1$, $p_1 : p_2 : = 2 : 1$ and $A_1 : A_2 = 1 : 2$ respectively. Column II gives corresponding results. Match the rations in column I with the values in Column II.



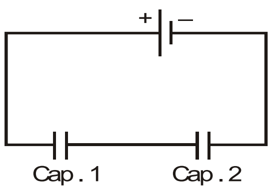
Columnn I	Column II
(A) $\dfrac{\text{Drift velocity of free electron in rod I}}{\text{Drift velocity of free electron in rod II}}$	(p) 0.5
(B) $\dfrac{\text{Electric field in rod I}}{\text{Electric field in rod II}}$	(q) 1
(C) $\dfrac{\text{Potential difference across rod I}}{\text{Potential difference across rod II}}$	(r) 2
(D) $\dfrac{\text{Average time taken by free electron to move from A to B}}{\text{Average time taken by free electron to move from B to C}}$	(s) 4

756 In the circuit shown, batery, ammeter and voltmeter are ideal and the switch S is initially closed as shown. When switch S is opened, match the parmeter of column I with the effects in column II.

Columnn I	Column II
(A) Equivalent resistance across the battery,	(p) Remains same
(B) Power dissipated by left resistance R	(q) Increases
(C) Voltmeter reading	(r) decreases
(D) Ammeter reading	(s) Becomes zero.



757. Two identical capacitors are connected in series, and the combination is connected with a battery, as shown. Some chages in the capacitor 1 are now made independintly after the steady state is achieved, listed in column-I. Some effects which may occur in new steady state due to these changes on the capacitor 2 are listed in column-II. Match the changes one capacitor 1 in column-I with corresponding effect on capacitor 2 in column-II.

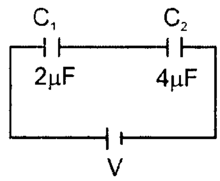


Column I	Column II
(A) A dielectric slab is inserted.	(p) Charge on the capacitor increases.
(B) Separation between plates jincreased	(q) Charge on the capacitor decreases.
(C) A metal plate is inserted connecting both plates	(r) Energy stored in the capacitor increases.
(D) The left plate is grounded.	(s) Energy stored is capactor is decreased
	(t) No change is occured

758. In each situation of column-some changes are made to a charged capacitor under conditions of constant potential difference or constant charge. Condition of constant potential difference means that the a cell is connected across the capacitor and condition of constant charge means that the capacitor is isolated. Match the conditions in conditions in column-I with coresponding results in column-II.

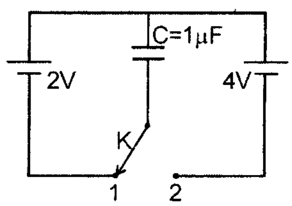
	Column I		Column II
(A)	For a capacitor maintained at constant potential difference, the separation between plates is increased.	(p)	Then electric field inside the capacitor decreases in compatison to what it was before the change.
(B)	For a capacitor maintained at constant charge, the separation between the plates is increased	(q)	Then electric field inside the capacitor remains same.
(C)	For a capacitor maintained at constant potential difference, area of the both the plates is doubled.	(r)	Then potential energy stored in the capacitor decreases in comparison to what it was before the change.
(D)	For a capacitor maintained at constant charge, area of both plates is doubled	(s)	The potential energy stored in the capacitor decreases in comparison to what it was before the change.
		(t)	Capacitance of capacitor decreases

759. In the given figure, the separation between the plates of C_1 is slowly increased to double of its inital value then.



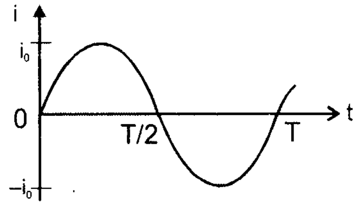
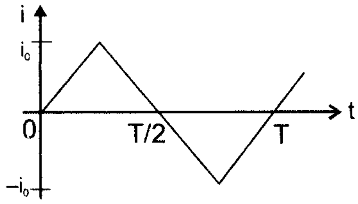
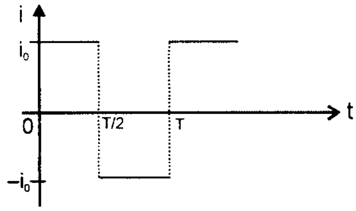
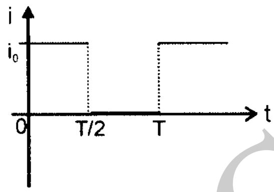
	Column-I		Column-II
(A)	the potential difference across C_1	(p)	increases
(B)	the potential difference across C_2	(q)	decreases
(C)	the energy stored in C_1	(r)	increases by a factor of 6/5
(D)	the energy stored in C_2	(s)	decreases by a factor of 18/25
		(t)	decreases by a factor of 9/35

760. The circuit involves two ideal cells connected to a 1 μF capacitor via a key K. Initially the key K is in position 1 and the capacitor is charged fully by 2V cell. The key is pushed to position 2. Column I gives physical quantities involving the circuit after the key k is pushed from position 1. Column II gives corresponding results. Match the statements in Column I whith the corresponding values in Column II.

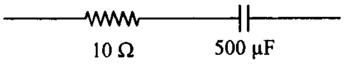
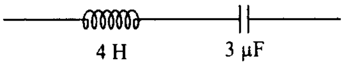
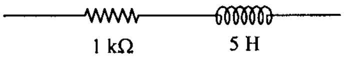


	Column I		Column II
(A)	The net charge crossing the 4volt cell in μC is	(p)	2
(B)	The magnitude of work done by 4Volt cell in μJ is	(q)	6
(C)	The gain in potential energy of capacitor in μJ is	(r)	8
(D)	The net heat produced in circuit in μJ is	(s)	16

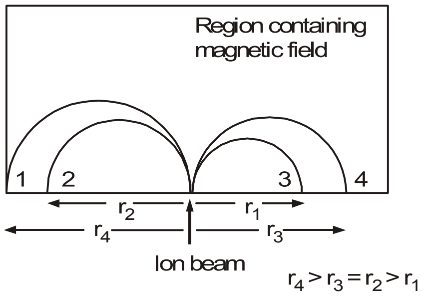
761. In Column I, variation of current i with time t is given in figures. In column II root mean square current i average current is given. Match the column I with corresponding quantities given in Column II

Column I		**Column II**	
(A)		(p)	$i_{ms} = \frac{i_0}{\sqrt{3}}$
(B)		(q)	Average current for positive half cycle is i_0
(C)		(r)	Average current for positive half cycle is $\frac{i_0}{2}$
(D)		(s)	Full cycle average current is zero.
		(t)	Root mean square value of current for positive half cycle is i_0

762. Four fifferent circuit components are given in each situation of column-I and all the components are connected across an ac source of same angular frequency ω = 200rad/sec. The information of phase difference between the current and source voltage in each situation of column-I is given in column-II. Match the circuit components in column-I with corresponding results in column-II.

	Column - I		**Column - II**
(A)		(p)	the magnitude of required phase difference is $\frac{\pi}{2}$.
(B)		(q)	the magnitude of required phase difference is $\frac{\pi}{4}$.
(C)		(r)	the current leads in phase to sourcevoltage.
(D)		(s)	the magnitude of required phase difference is zero
		(t)	the current lags in phase to source voltage.

763. A beam consiting of four types of ions A, B, C and D enters a region that contains a uniform magnetic field as shown. The field is perpendicular to the plane of the paper, but its precise direction is not given. All ions in the beam travel with the same speed. The table below gives the masses and charges of the ions.

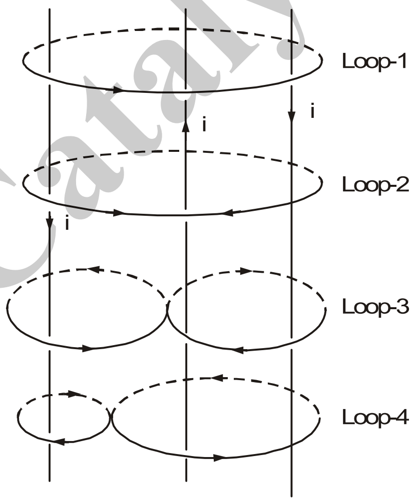


ION	MASS	CHARGE
A	2m	+e
B	4m	−e
C	2m	−e
D	m	+e

The ions fall at different positions 1, 2, 3 and 4 as shown. Correctly match the ions with respective falling positions.

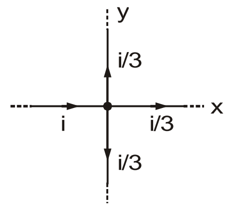
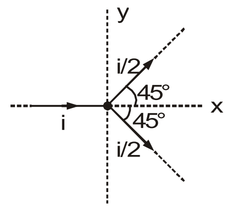
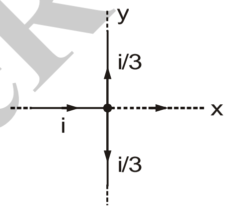
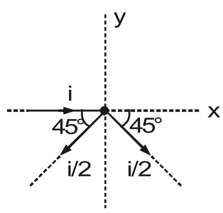
Column - I		**Column - II**	
(A)	a	(p)	1
(B)	b	(q)	2
(C)	c	(r)	3
(D)	d	(s)	4

764. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in the figure:

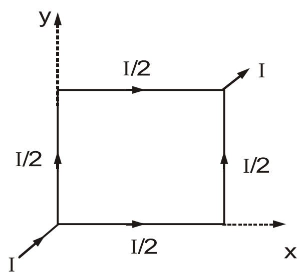


	Column - I		**Column - II**
(A)	Along closed Loop - 1	(b)	$\oint \vec{B}.d\vec{\ell} = \mu_0 i$
(B)	Along closed Loop - 2	(q)	$\oint \vec{B}.d\vec{\ell} = -\mu_0 i$
(C)	Along closed Loop - 3	(r)	$\oint \vec{B}.d\vec{\ell} = 0$
(D)	Along closed Loop - 4	(s)	net work done by the magnetic force to move a unit charge along the loop is zero.

765. Column-II gives four situations in which three or four semi infinite current carrying wires are placed in xy-plane as shown. The magnitude and direction of current is shown each figure. Column-I gives statements regarding the x and y components of magnetic field at a point P whose coordinates are P (0, 0, d). Match the statements in column-I with the corresponding figures in column-II

	Column-I		**Column-II**
(A)	The x component of magnetic field at point P is zero in	(p)	
(B)	The z component of magnetic field at point P is zero in	(q)	
(C)	The magnitude of magnetic field at point P is $\frac{\mu_0 i}{4\pi d}$ in	(r)	
(D)	The magnitude of magnetic field at point P is less than $\frac{\mu_0 i}{2\pi d}$ in	(s)	

766. A square loop of uniform conducting wire is as shown in figure. A current I (in amperes) enters the loop from one end exits the loop from opposite end as shown in figure. The length of one side of square loop is ℓ metre. The wire has uniform cross section area and uniform linear mass density. In four situations of column I, the loop is subjected to four different magnetic field. Under the conditions of column I, match the column I with corresponding results of column II (B_0 in column I is a positive nonzero constant)

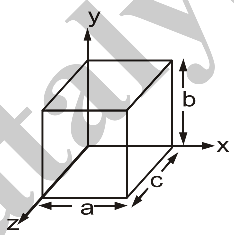


Column I	**Column II**
(A) $\vec{B} = B_o\hat{i}$ in tesla	(p) magnitude of net force on loop is $\sqrt{2}B_o I\ell$ newton
(B) $\vec{B} = B_o\hat{j}$ in tesla	(q) magnitude of net force on loop is zero
(C) $\vec{B} = B_o(\hat{i}+\hat{j})$ in tesla	(r) magnitude of net torque on loop about its centre is zero
(D) $\vec{B} = B_o\hat{k}$ in tesla	(s) magnitude of net force on loop is $B_o I\ell$ newton

767. A particle enters a space where exists uniform magnetic field $\vec{B} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k}$ & uniform electric field $\vec{E} = E_x\vec{i} + E_y\vec{j} + E_z\vec{k}$ with initial velocity $\vec{u} = u_x\vec{i} + u_y\vec{j} + u_z\vec{k}$. Depending on the values of various components the particle selects a path. Match the entries of column A with the entries of column B. The components ohter than specified in column A in each entry are non-zero. Neglect gravity.

Column-I		Column-II	
(A)	$B_y = B_z = E_x = E_z = 0;\ u = 0$	(p)	circle
(B)	$E = 0,\ u_xB_x + u_yB_y \neq -u_zB_z$	(q)	helix with uniform pitch and constant radius
(C)	$\vec{u} \times \vec{B} = 0,\ \vec{u} \times \vec{E} = 0$	(r)	cycloid
(D)	$\vec{u} \perp \vec{B},\ \vec{B} \parallel \vec{E}$	(s)	helix with variable pitch and constant radius
		(t)	straight line

768. The figure shows a metallic solid block, placed in a way so that its faces are parallel to the coordinate axes. Edge lengths along axis x, y and z are a, band c respectively. The block is in a region of uniform magnetic field of magnitude 30mT. One of the edge length of the block is 25 cm. The block is moved at 4 m/s parallel to each axis and in turn, the resulting potential difference V that appears across the block is measured. When the motion is parallel to the y axis, V = 24 mY; with the motion parallel to the z axis, V = 36 mV; with the motion parallel to the x axis, V = 0. Using the givein information, correctly match the dimensions of the block with the values given.



Column I		Column II	
(A)	a	(p)	20 cm
(B)	b	(q)	24 cm
(C)	c	(r)	25 cm
(D)	$\frac{bc}{a}$	(s)	30 cm
		(t)	26 cm

769. Column-I gives situations involving a charged particle which **may be** realised under the condition given in column-II. Match the situations in column-I with the condition in column-II.

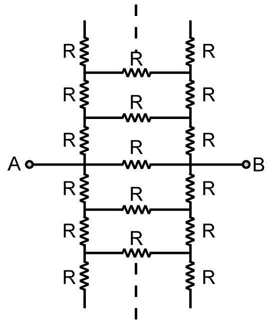
	Column I		Column II
(A)	Increase in speed of a charged particle	(p)	Electric field uniform in space and constant with time
(B)	Exert a force on an electron initially at rest	(q)	Magnetic field uniform in space and constant with time.
(C)	Move a charged particle in a circle with uniform speed	(r)	Magnetic field uniform in space but varying with time.
		(s)	Magnetic field non-uniform in space but constant with time
(D)	Accelerate a moving charged particle	(t)	Electric field non-uniform in space but constant with time

770. A square loop of conducting wire is placed near a long straight current carrying wire as shown. Match the statements in column-I with the corresponding results in column-II.

	Column-I		Column-II
(A)	If the magnitude of current I is increased	(p)	Induced current in the loop will be clockwise
(B)	If the magnitude of current I is decreased	(q)	Induced current in the loop will be anticlockwise
(C)	If the loop is moved away from the wire	(r)	wire will attract the loop
(D)	If the loop is moved towards the wire	(s)	wire will repel the loop
		(t)	Torque about centre of mass of loop is zero due to magnetie force

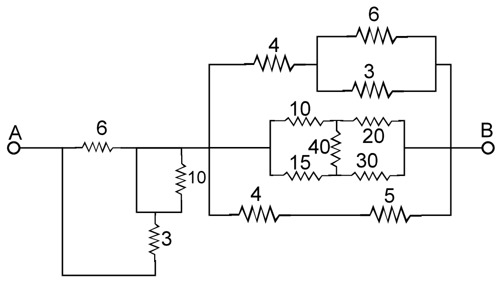
SECTION : (E) - Integer Type

771. Find equivalent resistance (in ohm) between A and B



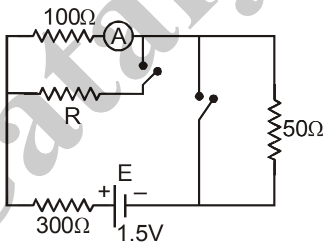
$R = 100\sqrt{3}\ \Omega$

772. Find equivalent resistance (in ohm) between A and B

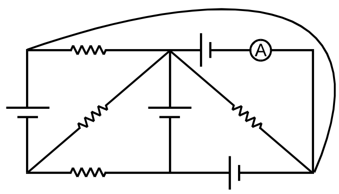


All resistances are in Ω.

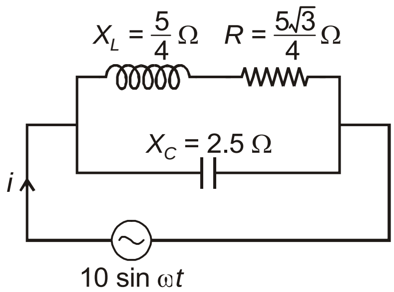
773. In the circuit shown, the reading of the (an ideal) ammeter is the same with both switches open as with both closed. Find the value of resistance R in ohm.



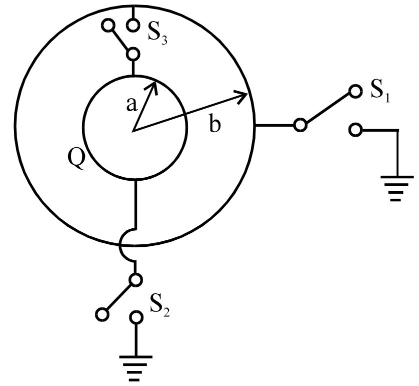
774. In the circuit diagram shown , each battery is ideal having an e.m.f. of 1 volt . Each resistor has a resistance of 1 Ω . Ammeter (A) has a resistance of 1 Ω. Find the total thermal power (in Watt) produced in the circuit.



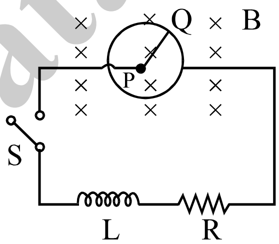
775. Find the peak value of *i* (in ampere) in the following circuit



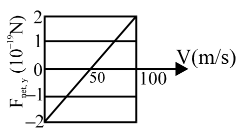
776. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' (> a). Initially switches S_1, S_2 and S_3 are open and sphere 'A' carries a charge Q. First the switch 'S_1' is closed to connect the shell B with the ground and then opened. Now the switch 'S_2' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch 'S_3' is closed to connect the spheres together. The heat (in Joule) which is produced after closing the switch S_3 is X. Find 10X [Consider b = 4 cm, a = 2 cm & Q = 8 μC]



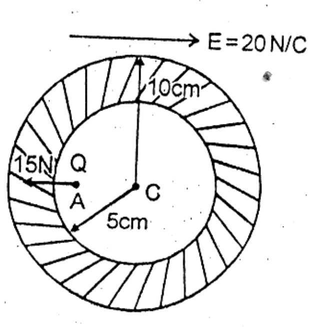
777. The diagram shows a circuit having a coil of resistance R = 10 Ω and inductance L connected to a conducting rod PQ which can slide on a perfectly conducting circular ring of radius 10 cm with its centre at P. Assume that friction and gravity are absent and a constant uniform magnetic field of 5 T exists as shown in figure. At t = 0, the circuit is switched on and simultaneously a time varying external torque is applied on the rod so that it rotates about P with a constant angular velocity 40 rad/s. The magnitude of this torque (in milli Nm) when current reaches half of its maximum value is X/Y. Find X+Y Neglect the self inductance of the loop formed by the circuit.



778. At time t_1, an electron is sent along the positive direction of an x-axis, through both an electric field **E** and a magnetic field **B**, with **E** directed parallel to the y-axis. Graph gives the y-component $F_{net,\,y}$ of the net force on the electron due to the two fields, as a function of the electron's speed V at time t_1. Assuming B_x = 0, find magnitude of electric field **E** in mN/C. (Use e = 1.6 × 10^{-19} C)

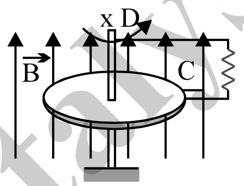


779. In a conducting hollow sphere of inner and outer radii 5 cm and 10 cm respectively, a point change 1μC is placed at point A, that is 3 cm from the centre C of the hollow sphere. An external uniform electric field of magnitude 20 N/C is also applied. Net electric force on the this charge is 15N, away from the centre of the sphere as shown. Find magnitude of force exerted by the induced charge on the charge placed at point A in newtons.



780. In a spark plug which is connected to the secondary coil of transformer an emf 40000 V is induced when in primary coil current changes from 4A to 0 in 10 μs. The self inductance of secondary coil is 1000H, find the minimum value of self inductance of primary coil in μH.

781. A metal disc of radius r = 0.1 m is placed perpendicular to a uniform magnetic field of induction B = 0.50 T. It is capable of rotation about an axis XY parallel to the inducation B, the axis is passing through its centre. Using sliding contacts C and D the disc is connected to a resistance R = 2.5Ω. Determine the mechanical power consumed in mW in rotating the disc if a current of 0.10 A flows through R. Friction can be neglected.



782. A uniform magnetic field of induction B = 1T exists in a circular region of radius $R=\sqrt{\frac{13}{2}}$ m . A loop of radius $R=\sqrt{\frac{13}{2}}$ m encloses the magnetic field at t = 0 and then pulled at a uniform speed v = 1 m/s in the plane of the paper. Find the induced EMF in volts in the loop at t = 1 s.

783. When a box 'A' is connected across an a.c. source of 200 V, 0.4 A of current flows in circuit. When a capacitor of reactance 400 Ω is connected in series with box and connected across a.c. source, the power factor of circuit becomes 1. What is power factor of box 'A'. (Express your answer in the order of 10^{-1})

784. A wire of length L and 3 identical cells of negligible internal resistance are connmected in series, when the temperature of the wire is raised by ΔT in time t due to the current. The same temperature rise is observed in the same time when N similar cells are connected in series with a wire of length 2L but of same material and cross-section. Find the value of N.

785. The distance between the plates of a parallel plate condenser is 0.05 m . A field of 3×10^4 volt/m is established btweeen the plates . It is then disconnected from the battery & an uncharged metal plates of thickness 0.01 m is inserted into the condenser parallel to its plates . Find the potential difference between the plates before the introduction of the metal plates

786. In the previous question,Find the potential difference between the plates before the introduction of the metal plates.

787. In the previous question, what would be the potential difference , if a plate of di-electric constant $\in = 2$ is introduced in place of metal plate?

788. *A coil takes a current of 2 A and 200 W power from an AC source of 220 V , 50 Hz . The resistance (in ohms) of the coil is:*

789. An AC emf $e = 200\sqrt{2}$ sin (100t) is connected to a 1 mF capacitor through an AC ammeter . The reading of the ammeter (in mA) shall be :

790. Magnetic flux through a loop of resistance R = 0.2 W is varying according to the relation $f = 6t^2 + 7t + 1$, where f is in milliwebers and t is in seconds . The magnitude of the emf (in mV) induced in the loop at t = 2s is

791. The equation of an alternating voltage is e = 220 sin (wt + p/6) and the equation of the current in the circuit is i = 10 sin (wt - p/6) . The impedance of the circuit (in ohms) is :

792. An alternating voltage is connected in series with a resistance R and an inductance L . If the potential drop across the resistance is 200V and across the inductance is 150 V , the applied voltage is

793. In a region of uniform magnetic induction $B = 10^{-2}$ T , a circular coil of radius 30 cm and resistance p^2 ohm is rotated about an axis which is perpendicular to the direction of B and which forms a diameter of the coil . If the coil rotates at 200 rpm , the amplitude of the alternating current (in mA) induced in the coil is

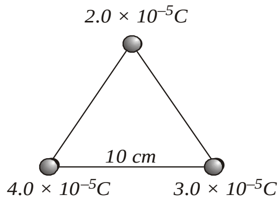
794. A coil of 50 turns & area 10^{-2} m^2 is placed with its plane normal to the field between the poles of a powerful horse shoe magnet . The coil has a resistance of 5 W & is connected to a ballistic galvanometer of resistance 35 W . When the coil is suddenly removed from the field completely , a deflection of 120 divisions is registered in the galvanometer . If the sensitivity of the galvanometer is 15 mC per division , calculate 1000X if the flux density of the field is X Tesla

795. The bob of a simple pendulum has a mass of 40 g and a positive charge of 4.0×10^{-6}C. It makes 20 oscillations in 45 s. A vertical electric field pointing upward and of magnitude 2.5×10^4N/C is switched on. How much time (in seconds) will it now take to complete 20 oscillations?

796. Two identically charged spheres are suspended by strings of equal length. The strings make an angle of 30º with each other, when suspended in a liquid of density 0.8 gm/cc, the angle remains the same. What is the dielectric constant of the liquid? The density of the material of the sphere is 1.6 gm/cc.

797. A proton is released from rest 10cm from a large sheet carrying a surface charge density of -2.21×10^{-9} cm^{-2}. It will strike the sheet after how much time (in microseconds):

798. How much work has to be done in assembling three charged particles at the vertices of an equilateral triangle as shown in figure?



799. A charge *q* is shifted from the surface of a hollow charged sphere with uniformly distributed charge *Q* to a point at a distance *R*/2 from the centre. Now if the same charge is shifted from surface to the same point in a solid uniformly charged sphere of same charge *Q*, the ratio of the two work done is

800. Two identical particles, each having a charge of 2.0×10^{-4}C and mass of 10g, are kept at a separation of 10cm and then released. What would be the speeds of the particles when the separation becomes large?

PHYSICS
PART - IV
TOPIC: OPTICS & MODERN PHYSICS
EXERCISE # 01

801. A regular hexagonal lamina of side a made up of perfectly absorbing material is kept in a region where a parallel beam of light with intensity I having a large aperture falls on it. If the area normal of the hexagon makes an angle of 30° with the beam then the force experienced by the hexagon will be (c is the velocity of light)

(A) $\frac{5a^2I}{4c}$ (B) $\frac{9a^2I}{4c}$ (C) $\frac{a^2I}{c}$ (D) $\frac{6a^2I}{c}$

802. If an endoergic nuclear reaction is brought about by bombarding a stationary nucleus with a projectile then

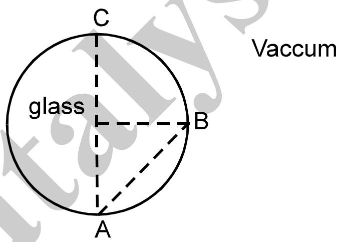
(A) Kinetic energy of the projectile must be equal to the magnitude of Q-value of the reaction

(B) Kinetic energy of the projectile must be less than the magnitude of the Q-value of the reaction

(C) The kinetic energy of the projectile must be more than the magnitude of Q-value of the reaction

(D) The momentum will not be conserved in such a nuclear reaction

803. It is found that all electromagnetic signals sent from A towards B reach point C inside the glass sphere, as shown. The speed of electromagnetic signals in glass can not be



(A) 1.0×10^8 m/s (B) 2.4×10^8 m/s (C) 2×10^7 m/s (D) 4×10^7 m/s

$\therefore$ only (B) is not possible.

804. The energy that should be added to an electron, to reduce its debroglie wavelength from 2×10^{-9} m to 0.5×10^{-9} m will be :

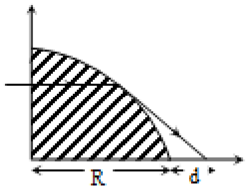
(A) 1.1 MeV (B) 0.56 MeV (C) 0.56 KeV (D) 5.6 eV

805. Hydrogen (H), deuterium (D), singly ionised helium $(He)^+$ and doubly ionised (Li^{++}) all have one electron round the nucleus. Consider n = 2 to n = 1 transition. The wavelength of emitted radiations are λ_1, λ_2, λ_3 and λ_4 respectively. Then approximately

(A) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$ (B) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(C) $\lambda_1 = 2\lambda_2 = 2\sqrt{2}\lambda_3 = 3\sqrt{2}\lambda_4$ (D) $\lambda_1 = {}_2 = 2\lambda_3 = 3\lambda_4$

806. A uniform horizontal light beam is incident upon a prisms (quarter cylindrical shape) as shown in the figure. The radius of the prism is R and the cylinder material has a refractive index $2/\sqrt{3}$. A patch on the table for a distance d from the surface of the cylinder is unilluminated. Find the value of d in terms of R.



(A) R/2 (B)R (C) $\sqrt{3}$ R (D) 2R

807. In Young's double slit experiment the slits are 0.5 mm apart and interference is observed on a screen placed at a distance of 100 cm from the slits. It is found that the 9^{th} bright fringe is at a distance of 9.0 mm from the second dark fringe. Find the wavelength of light used in the experiment

(A) 8000 Å (B) 6000 Å (C) 4000 Å (D) 2000 Å

808. The following data are given for a crown glass prism ;
refractive index for violet light $n_v = 1.521$
refractive index for red light $n_r = 1.510$
refractive index for yellow light $n_y = 1.550$
Dispersive power of a parallel glass slab made of the same material is :

(A) 0.01 (B) 0.02 (C) 0.03 (D) 0

809. Three convex lens L_1, L_2, L_3 have the same radius of curvature but their r.i. are $\mu_1 = 1.2$, $\mu_2 = 1.4$, $\mu_3 = 1.6$. The ratio of their focal length $f_1 : f_2 : f_3$, when placed in air, is:

(A) 1 : 2 : 3 (B) 3 : 2 : 1 (C) 6 : 3 : 2 (D) 2 : 3 : 6

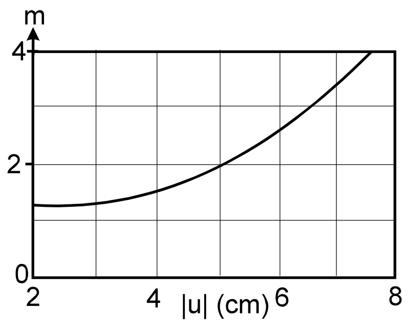
810. A fraction f_1 of a radioactive sample decays in one mean life, and a fraction f_2 decay in one half life

(A) $f_1 > f_2$
(B) $f_1 < f_2$
(C) $f_1 = f_2$
(D) may be A, B, C depending on the values of the man life and half life.

811. A photon of energy 12.1 eV corresponds to light of wave length λ_0. Due to an electron from n = 3 to n =1 in a hydrogen atom, light of wavelength λ is emitted. If we take in to account the recoil of the atom when the photon is emitted

(A) $\lambda = \lambda_0$
(B) $\lambda < \lambda_0$
(C) $\lambda > \lambda_0$
(D) the data is not sufficient to reach a conclusion.

812. An object kept on the principal axis and infront of a spherical mirror, is moved along the axis itself. Its lateral magnification m is measured, and plotted versus object distance |u| for a range of u, as shown below :



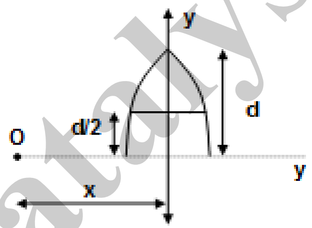
The magnification of the object when it is placed at a distance 20 cm in front of the mirror, is

(A) – 1 (B) 1 (C) 8 (D) 20

813. The refractive index of the material of the lens (see the figure) is given by

$\mu = \mu_0\left(1+\frac{y}{d}\right)$; for $0 \ll y \ll \frac{d}{2}$

$= \frac{3\mu_0}{2}$ for $\frac{d}{2} < y \le d$, where μ_0 is a positive constant. The number images of a point object O formed by the lens is (assume $d \ll x$)

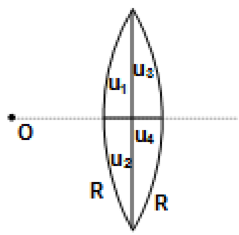


(A) 1 (B) 2 (C) 0 (D) ∞

814. The radioactive samples have decay constants λ_1 and λ_2 ($\lambda_1 > \lambda_2$). Let p_1 = probability that a particular nucleus in the first sample will decay and p_2 probability that a particular nucleus in the second sample will decay. Then

(A) $p_1 > p_2$ (B) $p_1 < p_2$ (C) $p_1 = p_2$ (D) $\frac{p_1}{p_2} = \frac{\lambda_1}{\lambda_2}$

815. A thin lens is made up of four different materials as shown. The number of images for a point object at O is (Given $\mu_1 + \mu_3 \ne \mu_2 + \mu_4$ and aperture is very small compared to the object distance.)



(A) 1 (B) 2 (C) 3 (D) 4

SECTION : (B) - More Than One Correct Options

816. It is desired to produce an achromatic combination of thin lenses having an effective focal length f in air. The two lenses to be used for the combination must have their dispersive powers ω_1 and ω_2 and focal lengths for yellow light as f_1 and f_2 respectively then:(An achromatic combination means that the combination must have same focal length for all the colours)

(A) $f_1 = \left(1 - \frac{\omega_1}{\omega_2}\right)f$ (B) $f_2 = \left(1 - \frac{\omega_2}{\omega_1}\right)f$ (C) $f_1 = \left(1 + \frac{\omega_2}{\omega_1}\right)f$ (D) $f_2 = \left(1 + \frac{\omega_1}{\omega_2}\right)f$

817. Which of the following statements are true about X-rays?

(A) They are deflected by electric field and magnetic field.

(B) They ionise the gas through which they pass.

(C) They exhibit polarization under special conditions.

(D) They travel with a speed of light.

818. When a hydrogen atom is excited from ground state to first excited state.

(A) its kinetic energy increases by 102 eV (B) its kinetic energy decreases by 10.2 eV.

(C) its potential energy increases by 20.4 eV. (D) its angular momentum increases by 1.05×10^{-34} J–s

819. A virtual image larger than the object can be formed by a

(A) Convex mirror (B) Concave mirror (C) Convex lens (D) concave lens

820. Mark the correct statement(s)

(A) To observe interference, two sources of slightly difference frequencies are required.

(B) To observe interference, two coherent sources of same frequency must be placed some distance apart from each other.

(C) To observe interference, two coherent sources must have same frequency and same amplitude.

(D) none of these

821. The penetrating power of X-rays increases with the

(A) increase in its frequency (B) decreases in its frequency

(C) increase in its wavelength (D) decreases in its wavelength

822. When an electron moving at a high speed strikes a metal surface, which of the following are possible?

(A) The entire energy of the electron may be converted into an x-ray photon.

(B) Any fraction of the energy of the electron may get converted to heat

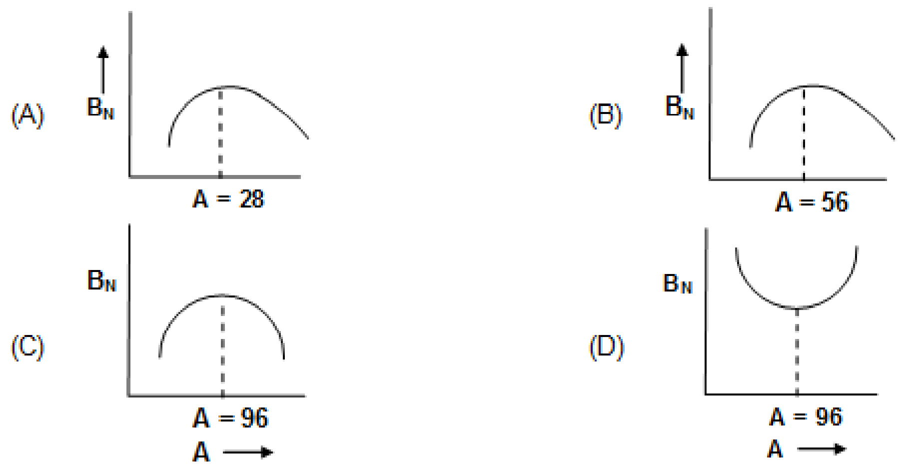
(C) The entire energy of the electron may get converted to heat

(D) The electron may undergo an elastic collision with the target metal

823. Let λ_α, λ_β and λ_α- denote the wavelength of x-rays of the k_α, k_β and L_α lines in the characteristic x-ray spectrum for a metal.

(A) $\lambda_\alpha^{'} > \lambda_\beta > \lambda_\gamma$ (B) $\lambda_\alpha^{'} > \lambda_\alpha > \lambda_\gamma$ (C) $\frac{1}{\lambda_\alpha} + \frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha^{'}}$ (D) $\frac{1}{\lambda_\beta} + \frac{1}{\lambda_\alpha} = \frac{1}{\lambda_\alpha^{'}}$

824. The dependence of binding energy per nucleon, B_N, on the mass number A, is represented by



825. The electric field at a point in vacuum associated with light wave is given by $E = E_0 \sin \omega_1 t \sin \omega_2 t$. If this light is used in ejecting photoelectrons from a metal of work function ϕ, the maximum kinetic energy of the photoelectrons is

(A) $\dfrac{h\omega_1}{2\pi} - \phi$

(B) $\dfrac{h\omega_2}{2\pi} - \phi$

(C) $\dfrac{h(\omega_1 + \omega_2)}{2\pi} - \phi$

(D) not predictable as magnitudes of ω_1 and ω_2 are not known.

826. White light is used to illuminate the two slits in young double slit experiment. The separation between the slits is b and the screen is at a distance d (>>b) from the slits. Certain wavelengths are missing. Some of the missing wavelengths are.

(A) $\lambda = b^2/d$ (B) $\lambda = 2b^2/d$ (C) $\lambda = b^2/3d$ (D) $\lambda = 2b^2/d$

827. An electron with kinetic energy varying from 5 eV to 50 eV is incident on a hydrogen atom in its ground state. The collision

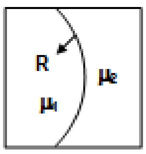
(A) May be elastic

(B) may be partially elastic

(C) must be completely inelastic

(D) from zero to 13.6 eV be elastic and more than 27.2 eV be inelastic

828. A plane glass plate behaves as a lens when made as shown in figure.



(A) For $\mu_1 = \mu_2$, it will behave as a glass slab

(B) For $\mu_1 > \mu_2$, it will behave as a divergent lens.

(C) For $\mu_1 < \mu_2$, it will behave as a convergent lens.

(D) for any value of μ_1 and μ_2 it behave as a lens

829. Photoelectric effect supports quantum nature of light because

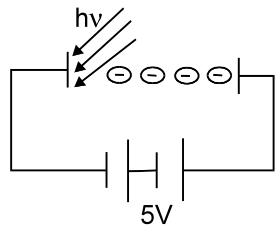
(A) there is a minimum frequency of light below which no photoelectrons are emitted

(B) the maximum kinetic energy of photoelectrons depends only on the frequency of light and not on its intensity

(C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately.

(D) electric charge of photoelectrons is quantised.

830. Photons of energy 5 eV are incident on cathode. Electrons reaching the anode have kinetic energies varying from 6eV to 8eV.



(A) Work function of the metal is 2 eV

(B) Work function of the metal is 3 eV

(C) Current in the circuit is equal to saturation value.

(D) Current in the circuit is less than saturation value.

831. Pick out the correct statement(s) :

(A) A moving electron in isolation cannot spontaneously change into an x-ray photon in free space.

(B) The optical spectrums in the visible region are not sensitive probe of nuclear charge as these involve transitions of outermost electrons for which effective inner charge is almost independent of the atomic number.

(C) In moseley's equation $\sqrt{\nu} = a(z-b)$, constant a has dimension of frequency equal to $\frac{1}{2}$.

(D) X-rays can produce photoelectric effect.

832. A sample of hydrogen atom gas contains 100 atoms. All the atoms are excited to the same n^{th} excited state. The total energy released by all the atoms is $\frac{4800}{49}$ Rch (where Rch = 13.6 eV), as they come to the ground state through various types of transitions.

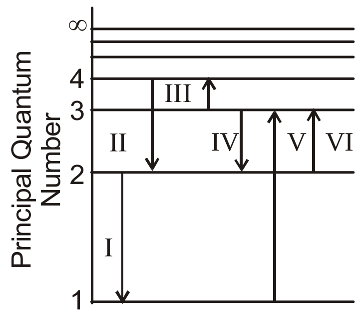
(A) Maximum energy of the emitted photon is $\frac{48}{49}$ Rch

(B) Initially atoms are in 6th excited state.

(C) Initially atoms are in 7th excited state.

(D) Maximum total number of photons that can be emitted by this sample is 100.

833. The figure above shows an energy level diagram for the hydrogen atom. Several transitions are marked as I, II, III, _______ . The diagram is only indicative and not to scale.



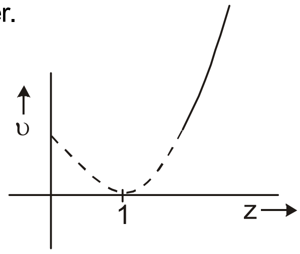
(A) The transition in which a Balmer series photon absorbed is VI.

(B) The wavelength of the radiation involved in transition II is 486 nm.

(C) IV transition will occur when a hydrogen atom is irradiated with radiation of wavelength 103nm.

(D) IV transition will emit the longest wavelength line in the visible portion of the hydrogen spectrum.

834. Figure shows the variation of frequency of a characteristic x-ray and atomic number.



(A) The characteristic x-ray is K_α

(B) The characteristic x-ray is K_β

(C) The energy of photon emitted when this x-ray is emitted by a metal having z = 101 is 204 keV.

(D) The energy of photon emitted when this x-ray is emitted by a metal having z = 101 is 102 keV.

835. A single electron orbits a stationary nucleus of charge + Ze where Z is a constant and e is the magnitude of electronic charge. It releases 47.22 eV energy if it comes from the third orbit to second orbit. [Use ionization energy of hydrogen atom = 13.6 eV]

(A) The value of the Z is 5.

(B) The wavelength of electromagnetic radiation required to remove the electron from first orbit to infinity, is nearly 3653 pm.

(C) The radius of the first orbit is 10.6 pm.

(D) The angular momentum of the electron in first orbit is 1.05×10^{-34} J-s.

836. Two electrons starting from rest are accelerated by equal potential difference.

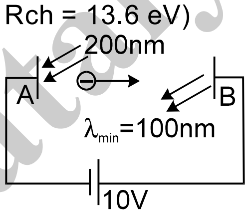
(A) they will have same kinetic energy

(B) they will have same linear momentum

(C) they will have same de Broglie wave length

(D) they will produce x-rays of same minimum wave length when they strike different targets.

837. In the figure shown electromagnetic radiations of wavelength 200nm are incident on the metallic plate A. The photo electrons are accelerated by a potential difference 10V. These electrons strike another metal plate B from which electromagnetic radiations are emitted. The minimum wavelength of the emitted photons is 100nm. (use hc = 12400 eVÅ, use Rch = 13.6 eV)



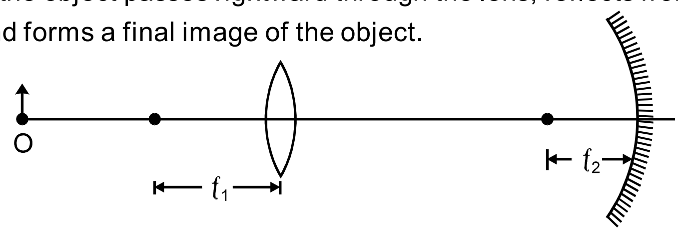
(A) Energy of incident photons is 6.2 eV.

(B) Maximum kinetic energy of ejected electrons from plate A is 2.4 eV

(C) Maximum kinetic energy of electrons striking plate B is 2.4 eV

(D) Work function of the metal A is 4.8 eV

838. An object is placed in front of a converging lens at a distance equal to twice the focal length f_1 of the lens. On the other side of the lens is a concave mirror of focal length f_2 separated from the lens by a distance 2 (f_1 + f_2). Light from the object passes rightward through the lens, reflects from the mirror, passes leftward through the lens, and forms a final image of the object.



(A) The distance between the lens and the final image is equal to $2f_1$.

(B) The distance between the lens and the final image is equal to $2(f_1 + f_2)$.

(C) The final image is real, inverted and of same size as that of the object.

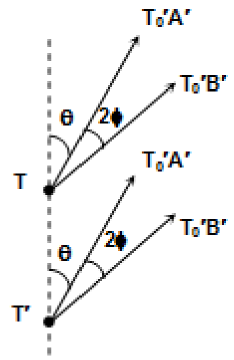
(D) The final image is real, erect and of same size as that of the object.

SECTION : (C) -Passage Type Questions

PASSAGE 01:

A radio transmitter consists of two vertical transmitting aerials T and T′ separated by distance d. It has been constructed so that citizens of city A can receive radio programmes but those at city B are unable to do so. Both aerials transmit radio signals of equal amplitude of S; T′ is leading T in phase by δ. The cities A and B are at a considerable distance from the transmitter.

The angle between T′T and the direction of A is θ, as indicated in the diagram. This angle can be varied by changing the direction of TT′. The angle between the directions of cities A and B is 2ϕ as indicated in the diagram. All angles are expressed in radians.



839. What is the path difference between transmitters T and T′ and city A

(A) d sin θ (B) d cos θ (C) d sin$(\theta + \phi)$ (D) d cos $(\theta + \phi)$

840. The radio signals reaching city A interfere constructively with Nth order interference. Choose the correct relation between the known physical quantities for this occurrence.

(A) $d\cos\theta = N\lambda$ (B) $d\cos\theta - \frac{\lambda\delta}{2\pi} = N\lambda$ (C) $d\cos\theta + \frac{\lambda\delta}{2\pi} = N\lambda$ (D) $\frac{\lambda\delta}{2\pi} = N\lambda$

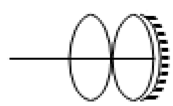
841. Choose the correct relation between the known physical quantities for destructive interference of the radio signals reaching city B, so that no signal is detected at B.

(A) $d\cos\theta - \frac{\lambda\delta}{2\pi} = \left(M + \frac{1}{2}\right)\lambda$ (B) $d\cos(\theta + 2\phi)\ \frac{\lambda\delta}{2\pi} = \left(M + \frac{1}{2}\right)\lambda$

(C) $d\cos\theta + \frac{\lambda\delta}{2\pi} = \left(M + \frac{1}{2}\right)\lambda$ (D) $d\cos(\theta + 2\phi) + \frac{\lambda\delta}{2\pi} = \left(M + \frac{1}{2}\right)\lambda$

PASSAGE 02:

Two convex lenses, each of focal length 40 cm, and a concave mirror of focal length 10 cm are kept as shown. An object is placed at 10 cm from the combination. Assume the lenses to be thin. Then answer the following



842. For a image to be formed, the total number of refractions occurring through the combination is

(A) 4 (B) 9 (C) 1 (D) 8

843. The radius of curvature of the combination is

(A) 10 cm (B) 5 cm (C) 90 cm (D) none of the above

844. The image distance from the combination is

(A) 5 cm (B) 10 cm (C) 20 cm (D) 40 cm

PASSAGE 03:

Number of radioactive nuclei in a sample reduces exponentially with time. The decay rate of a sample is also called its 'activity'. It can be shown that activity of a sample also decreases with time in an exponential manner. SI unit of activity is Becquerel (Bq) and 1 Bq = 1 decay/sec.

Activity of a radioactive sample was measured over a period of 10 hours beginning at t = 0. Results of these observations are given below.

Time(hour)	Decays/minute
1	5000
2	4205
4	2974
6	2103
8	1487
10	1051

845. Half life of the given sample is

(A) 6 hour (B) 4 hour (C) 1.5 hour (D) 7.5 hour

846. Activity of the sample at t = 16 hour will be nearly.

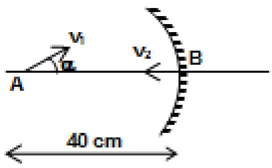
(A) 11.4 Bq (B) 8.6 Bq (C) 6.2 Bq (D) 3.4 Bq

847. Number of radioactive nuclei in the sample at t = 0 is of the order

(A) 1.54×10^8 (B) 2.72×10^4 (C) 1.72×10^5 (D) 2.06×10^6

PASSAGE 04:

A point object of mass m is placed at a distance 40 cm from a concave mirror of mass m and focal length f = 10 cm. At the given instant, object and mirror velocities V_1 and V_2 are as shown in the figure.



848. At the instant shown, what is the speed of the image if V_1 = 6 cm/s, V_2 = 0 and $\alpha = 90°$?

(A) 2 cm/s (B) 6 cm/s (C) 9 cm/s (D) 18 cm/s

849. If v_1 = 6 cm/s, v_2 = 12 cm/s, $\alpha = 0°$, the speed of the image when the object comes back to point A after an elastic collision with the mirror is

(A) 6 cm/s (B) 12 cm/s (C) 18 cm/s (D) 24 cm/s

850. The magnitude of average velocity of the image during the journey after collision described in the previous problem is:

(A) 1 cm/s (B) 2 cm/s (C) 3 cm/s (D) 4 cm/s

PASSAGE 05:

The radionuclide ^{56}Mn is being produced in a cyclotron at a constant rate P by bombarding a maganese target with deutrons. ^{56}Mn has a half life of 2.5 h and the target contains large number of only the stable maganese isotope ^{55}Mn. The reaction that produces ^{56}Mn is :

$$^{55}Mn + d \rightarrow {}^{56}Mn + p$$

After being bombarded for a long time, the activity of the target due to ^{56}Mn becomes constant equal to $13.86 \times 10^{10}\ s^{-1}$.
(Use ln2 = 0.693; Avogadro No = 6×10^{23}; atomic weight ^{56}Mn = 56 gm/mole)

851. At what constant rate P, ^{56}Mn nuclei are being produced in the cyclotron during the bombardment?
(A) 2×10^{11} nuclei/s (B) 13.86×10^{10} nuclei/s (C) 9.6×10^{10} nuclei/s (D) 6.93×10^{10} nuclei/s

852. After a long time, number of ^{56}Mn nuclei present in the target, is equal to
(A) 5×10^{11} (B) 20×10^{11} (C) 1.2×10^{14} (D) 1.8×10^{15}

853. After a long time bombardment, number of ^{56}Mn nuclei present in the target depends upon
(a) the number of ^{56}Mn nuclei present at the start of the process.
(b) half life of the ^{56}Mn
(c) the constant rate of production P.
(A) All (a), (b) and (c) are correct (B) only (a) and (b) are correct
(C) only (b) and (c) are correct (D) only (a) and (c) are correct

PASSAGE 06:

If energy in the ground state is taken as zero then the energy levels of the tungsten atom with an electron knocked out are as follows :

Shell containing vacancy	K	L	M
Energy in keV	69.5	11.3	2.3

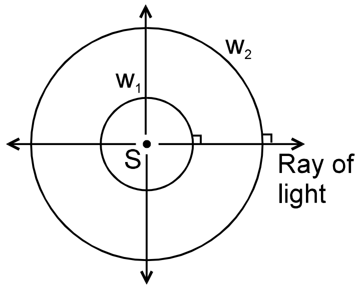
854. The minimum value of the accelerating potential that can result in the production of the characteristic K_β and K_α lines of tungsten, is
(A) 69.5 kV (B) 67.2 kV (C) 58.2 kV (D) 2.3 kV

855. For this same accelerating potential, what is the value of minimum wavelength as observed in the continuous X-rays ?
(A) 18.5 pm (B) 17.8 pm (C) 14.6 pm (D) 10.8 pm

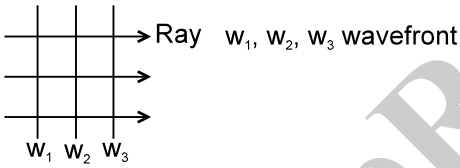
856. The wavelength of the characterstic K_β X-rays will be
(A) 21.3 pm (B) 20.8 pm (C) 18.5 pm (D) 16.2 pm

PASSAGE 07:

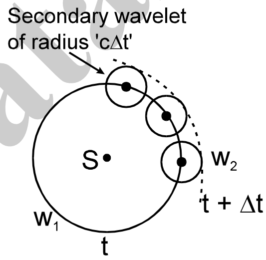
Huygen was the first scientist who proposed the idea of wave theory of light. He said that the light propagates in form of wavefronts. A wavefront is an imaginary surface at every point of which waves are in the same phase. For example the wavefronts for a point source of light is collection of concentric spheres which have centre at the origin. w_1 is a wavefront. w_2 is another wavefront.



The radius of the wavefront at time 't' is 'ct' in this case where 'c' is the speed of light. The direction of propagation of light is perpendicular to the surface of the wavefront. The wavefronts are plane wavefronts in case of a parallel beam of light.

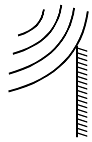


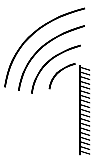
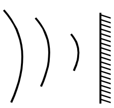
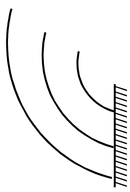
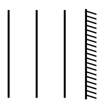
Huygen also said that every point of the wavefront acts as the source of secondary wavelets. The tangent drawn to all secondary wavelets at a time is the new wavefront at that time. The wavelets are to be considered only in the forward direction (i.e. the direction of propagation of light) and not in the reverse direction. If a wavefront w_1 at time t is given, then to draw the wavefront at time $t + \Delta t$ take some points on the wavefront w_1 and draw spheres of radius '$c\Delta t$'. They are called secondary wavelets.



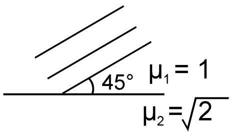
Draw a surface w_2 which is tangential to all these secondary wavelets. w_2 is the wavefront at time '$t + \Delta t$'.
Huygen proved the laws of reflection and laws of refraction using concept of wavefronts.

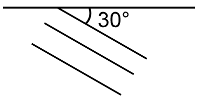
857. Spherical wave fronts shown in figure, strike a plane mirror. Reflected wave fronts will be as shown in

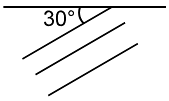


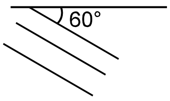
(A)  (B)  (C)  (D) 

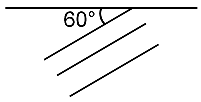
858. Wavefronts incident on an interface between the media are shown in the figure. The refracted wavefronts will be as shown in



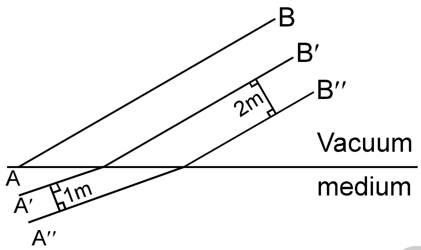
(A) 

(B) 

(C) 

(D) 

859. Certain plane wavefronts are shown in figure. The refractive index of medium is



(A) 2 (B) 4 (C) 1.5 (D) Cannot be determined

PASSAGE 08:

A glass prism with a refracting angle of 60^0 has a refractive index 1.52 for red and 1.6 for violet light. A parallel beam of white light is incident on one face at an angle of incidence, which gives minimum deviation for red light. Find :

[**Use:** sin (50º) = 0.760; sin (31.6º) = 0.520 ; sin (28.4º) = 0.475; sin (56º) = 0.832 ; π = 22/7]

860. The angle of incidence at the prism is :

(A) 30º (B) 40º (C) 50º (D) 60º

861. The angular width of the spectrum is :

(A) 6º (B) 4.8º (C) 9.6º (D) 12º

SECTION : (D) - Matrix Match

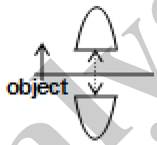
862. Match the Column – I with Column – II.

	Column I		**Column II**
(A)	k_α photon of aluminium	(1)	Will be most energetic among the four.
(B)	k_β photon of aluminium	(2)	Will be least energetic among the four.
(C)	k_α photon from sodium	(3)	Will be more energetic than the k_α photon of Lithium.
(D)	k_α photon from Beryllium	(4)	Will be less energetic than k_α photon of magnesium.

863. Match the atom given in list I with their ionization energy given in list II

	List I		**List II**
(A)	Lithium atom	(i)	54.4 eV
(B)	Helium atom	(ii)	13.6 eV
(C)	Beryllium atom	(iii)	122 eV
(D)	Hydrogen atom	(iv)	217.6 eV

864. When a convex lens is broken into half along the optical axis and symmetrically separated as shown. Then, considering initial and final situation, match the following



	List – I		**List – II**
(A)	Object size	(i)	increases
(B)	Image size	(ii)	decreases
(C)	Intensity	(iii)	remains the same
(D)	Image position	(iv)	insufficient data

865. For photoelectric effect, when the values given in list I are increased individually then match them with the correct option(s) in list II

	List – I		**List – II**
(A)	Frequency of incident light	(i)	Stopping potential increases
(B)	Intensity of incident light	(ii)	Stopping potential decreases
(C)	Work function	(iii)	No of photoelectrons emitted increases.
(D)	Frequency and intensity of incident light	(iv)	No of photoelectrons emitted decreases.

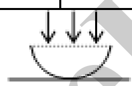
866.

	List – I		**List – II**
(A)	Radioactive nuclei	(i)	Quantization comes into picture
(B)	Bohr H-atom	(ii)	may emit electromagnetic waves
(C)	Photoelectric effect	(iii)	Existence of cut-off wavelength
(D)	X-ray generation	(iv)	Reverse process of photoelectric effect

867. What is the shape of the wavefront in each of the following cases

	List – I		List – II
(A)	Light diverging from a point source	(i)	plane
(B)	Light diverging from a line source	(ii)	spherical
(C)	Light emerging out of a convex lens when a point source is placed at it focus.	(iii)	cylindrical
(D)	Light reflected by concave mirror when point source is placed at it focus	(iv)	conical

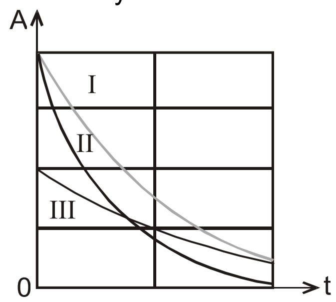
868.

	List – I		List – II
(A)	In young's double slit experiment with monochromatic light	(i)	The shape of interference fringes observed is circular
(B)	In young's double P in hole experiment with monochromatic light	(ii)	The shape of interference fringes observed is straight line
(C)	The phenomena is carried out with a thin film formed by placing a plane convex lens on the top of a flat glass plate with monochromatic light as shown in the figure	(iii)	The shape of interference fringes observed is Hyperbolic
		(iv)	Fringe width for all dark fringes are same
(D)	The phenomena of interference is carried out with Lloyd's mirror arrangement as shown in the figure.		

869. Match the following

	List I		List II
(A)	For a concave mirror, when an object is placed between pole and focus, the image is	(i)	Virtual
(B)	For a convex mirror, when an object is placed between pole and infinity, the image is	(ii)	Enlarged
(C)	For a concave mirror, when an object is placed between the centre of curvature and focus, the image is	(iii)	Erect
(D)	For a convex mirror, when an object is placed at infinity, the image is	(iv)	Formed between pole and focus

870. Figure shows activities A of three different radioactive material's samples (labelled as I, II and III) versus time. Using the given information, correctly match the requisite parameter in the left column with the options given in right column. Consider only their natural decay as the cause of rate of change of no. of parent nuclei



(a)	Disintegration constant has maximum value for the material of sample ...	(P)	I
(b)	Half life is maximum for the material of the sample	(Q)	II
(c)	Initially if samples of all three materials have same number of atoms then number of parent atoms will be maximum after their respective two half-lives in the sample...	(R)	III
(d)	Suppose all the materials decay by emitting α-particles of same energy and initially all three samples contain same amount (in gm) of the materials. Till the end of time span equal to their respective half lives, maximum energy is radiated by the sample	(S)	anyone out of these, as it is not possible to compare.

871.

	COLUMN - I		**COLUMN - II**
(a)	A convex lens in a denser medium will behave like a	(P)	converging lens
(b)	A concave lens in a rarer medium will behave like a	(Q)	diverging lens
(c)	A planoconvex lens silvered on its curved surface and placed in air will behave like a	(R)	concave mirror
(d)	A planoconcave lens silvered on its plane surface and placed in air will behave like a	(S)	convex mirror

872. Match the entries in column I with appropriate entries in column II

h – Planck constant K – Boltzmann constant
λ – Wavelength R – Resistance
C – Capacitance F – Force
V – Potential L – Inductance

	Column I		**Column II**
(A)	$\frac{hK}{\lambda}$	(p)	$\frac{\text{(watt) (metre)}}{\text{kelvin}}$
(B)	$\frac{K\lambda}{RC}$	(q)	$\frac{\text{(ampere) (kelvin)}}{\text{(joule)}^2}$
(C)	$\frac{F}{RC}$	(r)	$\frac{\text{(joule)(second)(newton)}}{\text{(kelvin)}}$
(D)	$\frac{VC}{hK}$	(s)	$\frac{\text{newton}}{\text{second}}$
		(t)	$\frac{\text{(ohm)(joule) (metre)}}{\text{(henry)(kelvin)}}$

873. A real object is being seen by optical component listing in column A and nature of image of this object is listing in column B:

Column A		Column B Nature of image	
(a)	Convex lens.	(p)	Real.
(b)	Convex mirror.	(q)	Virtual.
(c)	Concave lens.	(r)	Erect.
(d)	Concave mirror.	(s)	Inverted.

874.

	Column- I		Column-II
(a)	X-rays	(p)	Davisson and Germer experiment
(b)	Atomic number determination	(q)	Crystal structure determination
(c)	Duality of matter	(r)	Moseley's law
(d)	Duality of radiation	(s)	Bragg's law
		(t)	None

875.

	Column-I		Column-II
(a)	Wave character of radiation	(p)	Photoelectric effect
(b)	Photon character of radiation	(q)	Compton effect
(c)	Interaction of a photon with an electron, such that photon energy is equal to or slightly greater than the binding energy of electron, is more likely to result in	(r)	Diffraction
(d)	Interaction of a photon with an electron, such that photon energy is much greater than the binding energy of electron, is more likely to result in	(s)	Interference

876. In regard to motion of an electron in the atom and using Bohr's theory of atom, match Column - I with Column - II:

	Column - I		Column -II
(a)	Total energy of electron is	(p)	Directly proportional to n
(b)	Angular momentum of electron	(q)	Negative
(c)	Potential energy of electron is	(r)	Directly proportional to $\frac{1}{n}$
(d)	Linear momentum of electron is	(s)	Directly proportional to $-\frac{1}{n^2}$

(here n is the principal quantum number of the shell)

877.

	Column A		Column B
(a)	Characteristic x-rays	(p)	Photons liberated due to reduction in energy of accelerated electrons colliding with the atoms of target material.
(b)	Continuous x-rays	(q)	Accelerated electron creating a vacancy by knocking out inner electron of an atom of the target material and an electron from higher energy level occupying the vacancy and thus producing photons.
(c)	Minimum wavelength of x-rays	(r)	Depends on voltage applied between filament and target material.
(d)	Minimum wavelength of characteristic x-rays	(s)	Depends on target material

878. In hydrogen like atoms, match the following:

Table-1		**Table - 2**	
(a)	v_n	(p)	Proportional to n^2
(b)	r_n	(q)	Proportional to n
(c)	L_n	(r)	Proportional to n^{-2}
(d)	E_n	(s)	Proportional to n^{-1}

879. An unstable nucleus decays by various decay processes. Match each of the decay process in Column I with the relevant statement in Column II.

	Column I		***Column II***
(a)	γ – decay	(P)	accompanied by the emission of a neutrino
(b)	β^- – decay	(Q)	without change of number of nucleons
(c)	β^+ – decay	(R)	followed by X-ray emission
(d)	K – capture	(S)	resulting nuclei has lower energy configuration

880. An object located between the focus and the pole of a concave mirror moves towards the pole with a constant velocity along its principal axis. Consider the image formed by paraxial rays. Let θ_0 and θ_I represent the magnitudes (absolute values) of the angles subtended by the object and its image at the pole of the mirror respectively; and let m be defined as $\frac{\theta_I}{\theta_0}$. Use the New Cartesan Sign Convention.

	Column A		**Column B**
(a)	Velocity of image	(p)	Positive.
(b)	Acceleration of image	(q)	Negative.
(c)	$\frac{d\theta_0}{dt}$, i.e., the rate at which θ_0 changes with time	(r)	Zero.
(d)	$\frac{dm}{dt}$	(s)	Changes from positive to negative.

881.

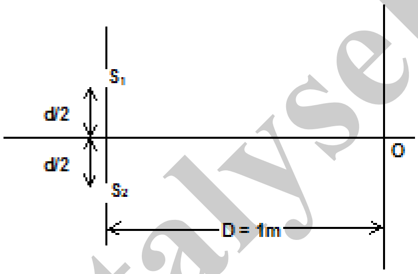
	Column A		**Column B**
(a)	Excited atom of small atomic number with a vacancy in M – shell	(p)	β–rays.
(b)	Excited atom of large atomic number with a vacancy in K – shell	(q)	Infra–red rays.
(c)	Electron – positron annihilation	(r)	X–rays.
(d)	Unstable nuclei	(s)	γ–rays.

SECTION : (E) - Integer Type

882. A man is looking down a 9 m deep tank filled with a transparent liquid of refractive index $\sqrt{3}$. The line of sight of the man makes an angle of 30° with the horizontal. What is the apparent depth of the tank as seen by the man?

883. A point source of power P = $\frac{8hc}{\lambda \ln 1.5}$ is kept on the axis (perpendicular to the plane of the disc and passing through the centre of disc) at a distance a = $\sqrt{2}R$ from the centre of the disc. λ is the wavelength of light emitted by the source, R = radius of disc, h = plank's constant and c = speed of light in vacuum. Assuming that each photon striking the plate ejects one electron, find the number of electrons emitted per unit time from the plate.

884. In a double slit experiment the separation between the slits is 1mm. Light rays fall normally on the plane of the slits and the interference pattern is observed on a screen placed at a distance of 1m from the plane of the slits. The arrangement is shown in the figure. When one of the slits is covered by a transparent strip of thickness 4μm, the central maximum is formed at a distance of 2mm from the point O. When the entire apparatus (one of the slits remaining covered) is immersed in a liquid, the distance between the central maximum and the point O is reduced to 0.5mm. The refractive index of the liquid is of the form $\frac{X}{Y}$. Then the value of X + Y is:



885. In an hydrogen atom, the energy released during transition from 4th to 2nd level is E. Now, light of the same energy E is incident on a metal surface of work function 0.56×10^{-19}J. The maximum KE (in eV) of the emitted electrons is X/10. Find the value of X(Take Ionization energy of hydrogen atom = 13.6 eV)

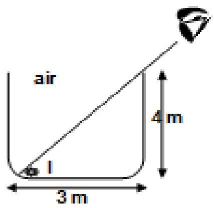
886. In the reaction ${}_1H^1 + {}_1H^3 \rightarrow {}_1H^2 + {}_1H^2$ protons ${}_1H^1$ with kinetic energy 8.03 MeV are incident on ${}_1H^3$ at rest. What is the total kinetic energy of products (both ${}_1H^2$) emitted along the direction of the incident proton in MeV

Give $m({}_1H^1) = 1.007825$ u $m({}_1H^3)$ = 3.016049 u

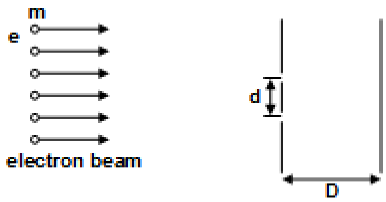
1u = 931.5 meV/C^2

$m({}_1H^2) = 2.014102$ u

887. An insect inside a beaker is just observable from a eye as shown from outside. At t=0 insect starts moving towards 'B' in straight line with constant velocity 0.5 m/s. If simultaneously a liquid of refractive index μ is gently filled inside the beaker with uniform rate $\frac{dy}{dt}$ so that insect is just observed always to the eye till t=4 sec. find refractive index of liquid is $\frac{X\sqrt{17}}{Y}$. Assume insect survives always inside the liquid. Find X + Y.



888. An electron beam is accelerated through a potential difference V and allowed to fall on the slits of a YDSE set up. Calculate the fringe width in mm. Given $V = \frac{50\ D^2h^2}{med^4}$, where h = Planck constant, m = mass of an electron, e = charge on an electron and d = 10 mm.



889. A double convex lens forms a real image of an object on a screen which is fixed. Now the lens is given a constant velocity v = 1ms^{-1} along its axis and away from the screen. For the purpose of forming the image always on the screen, the object is also required to be given an appropriate velocity. Find the velocity in ms^{-1} of the object at the instant its size is double the size of the image towards the screen.

890. A thin rod of length (f/3) is placed along the principle axis of a concave mirror of focal length 'f' such that its image which is real and elongated, just touches the d. The linear magnification is X/Y. The value of X+Y is:

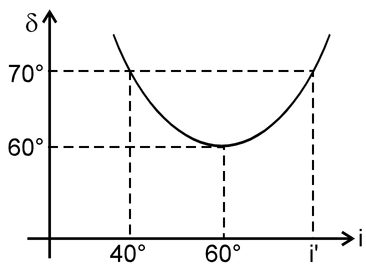
891. A monochromatic of wavelength 400 nm falls on the metal plate of a photoelectric arrangement. The work function of the metal is 2.5 eV. The maximum magnitude of the linear momentum of emitted photoelectron is 4.2×10^{-n} m/sec, then the value of n is ....

892. A neutron beam, in which each neutron has same kinetic energy, is passed through a sample of hydrogen like gas (but not hydrogen) in ground state. Due to collision of neutrons with the ions of the gas, ions are excited and then they emit photons. Six spectral lines are obtained in which one of the lines is of wavelength (6200/51) nm. What is the minimum possible value of kinetic energy of the neutrons for this to be possible. The mass of neutron and proton can be assumed to be nearly same. Find the answer in the form X × 10^{-2} eV and fill value of X.

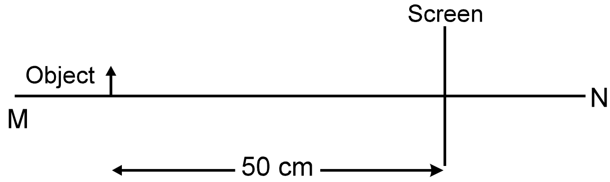
893. A beam of light has power of 144 W equally distributed among three wavelengths of 410 nm, 496 nm and 620 nm. The beam is incident at an angle of incidence of 60° on an area of 1 cm^2 of a clean sodium surface, having a work function of 2.3 eV. Assuming that there is no loss of light by reflection and that each energetically capable photon ejects a photoelectron, find the saturation photocurrent in μA (micro ampere).

894. An extended object of size 2 mm is placed on the principal axis of a converging lens of focal length 10 cm.It is found that when the object is placed perpendicular to the principal axis the image formed is 4 mm in size.The size of image when it is placed along the principal axis is ____________ mm.

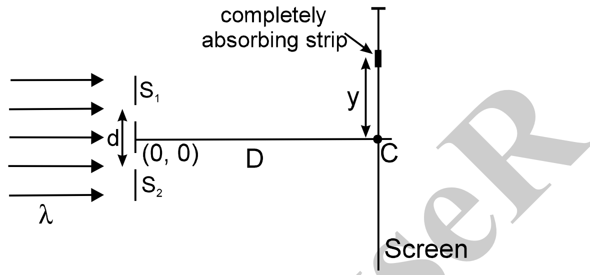
895. The curve of angle of incidence versus angle of deviation shown below has been plotted using a prism. Find the value of refractive index (in multiple of 10^{-2}) of the prism used.



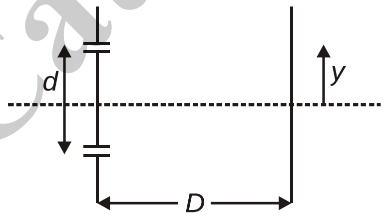
896. An object is placed 50 cm from a screen, as shown. A converging lens is moved such that line MN is its principal axis. Sharp images are formed on the screen in two positions of lens separated by 10 cm. Find the focal length of the lens in cm.



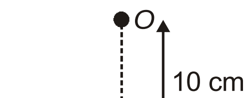
897. Figure shows two, identical narrow slits S_1 and S_2. A very small completely absorbing strip is placed at distance 'y' from the point C. 'C' is the point on the screen equidistant from S_1 and S_2. Assume $\lambda << d << D$ where λ, d and D have usual meaning. When S_2 is covered the force due to light acting on strip is 'f' and when both slits are opened the force acting on strip is 2f. Find the minimum positive 'y' (<< D) coordinate (in μm) of the strip if $\frac{\lambda D}{d}$ = 0.1 cm .



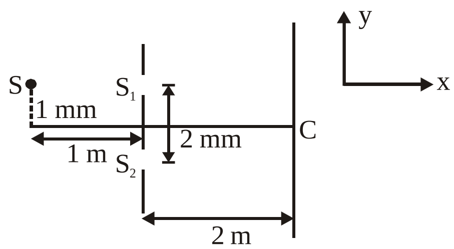
898. In YDSE if incident light consists of two wavelengths λ_1 = 4000 Å and λ_2 = 5600 Å and is parallel to line *SO*. The minimum distance *y* upon screen, measured from point *O*, will be where the bright fringe due to two wavelengths coincide is $\frac{n\lambda_1 D}{d}$. Find *n*.



899. An object is placed at a distance of 10 cm from convex lens $\left(\mu = \frac{3}{2}\right)$ which is placed on plane mirror as shown in figure. In this situation, image is formed on the object itself. Now the space between lens and mirror is filled with water $\left(\mu = \frac{4}{3}\right)$. Find, by how much distance (in cm) object must be moved so, that image again coincides with the object itself?



900. In a Young's double slit experiment set up, source S of wavelength 6000Å illuminates wo slits S_1 and S_2 which act two coherent sources. The source S oscillates about its shown position according to the equation $y = 1 + \cos \pi t$ where y is in millimeters and t in seconds At t = 2 second, the position of central maxima is x mm below C.Find the value of x.



901. In the above question,at t = 1 second, a slab of thickness 2×10^{-3} mm and refractive index 1.5 is placed in just front of S_1. The central maxima is formed at ymm aboce C.Find the value of y.

EXERCISE # 02

SECTION : (A) - Single Correct Options

902. A ^7Li target is bombarded with a proton beam current of 10^{-5} A for 1 hrs to produce ^7Be of activity 1.4×10^8 disintegration per second. If 1000 protons are required to produce one ^7Be radioactive nucleus, then the half life period of ^7Be is (Take ln2 = 0.7)

(A) 1.125×10^6 sec (B) 1.526×10^6 sec (C) 1.625×10^4 sec (D) 15.26×10^6 sec

903. The electron in hydrogen atom makes transition from the M shell to L shell. The ratio of the magnitude of initial to final acceleration of the electron is

(A) 9 : 4 (B) 81 : 16 (C) 4 : 9 (D) 16 : 81

904. A convex lens focuses a distant object 40 cm from it on a screen placed 10 cm away from it. A glass plate (μ = 1.5) and thickness 3 cm inserted between the lens and the screen. Then the position of the object from the lens such that its image is again focused on the screen______.

(A) 72 cm (B) 62 cm (C) 36 m (D) 24 cm

905. An object is kept in front of a concave mirror at a distance of (200 ± 0.04) m and its image is formed in front of the mirror at a distance (3.00 ±0.03) m. then the focal of the mirror is

(A) (1.20 ± 0.04) m (B) (1.4 ± 0.04)m (C) (1.20 ± 0.05) m (D) (1.2 ± 0.03) m

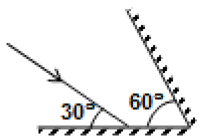
906. The electric field of a light wave at a point is E = (100 N/c) sin $(3 \times 10^{15}t)$ sin $(6 \times 10^{15}t)$ where t is time in seconds. This light falls on a metal surface having work function of 2 eV, then maximum possible kinetic energy of photoelectrons is about

(A) 16 eV (B) 7 eV (C) 6 eV (D) 4 eV

907. Current in a X-ray tube operating at 40 kV is 10 mA. 1% of the total kinetic energy of electrons hitting the target is converted into X-rays. Then the heat produced in the target per second is

(A) 396 J (B) 4J (C) 400 J (D) cannot be determined.

908. Two plane mirrors are at an angle 60° and an incident ray falls on the first mirror as shown in the figure. What is the smallest angle between the incident ray and the ray obtained each from the two mirrors?



(A) 0° (B) 90° (C) 120° (D) 150°

909. The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$ where n-$_1$ and n_2 are the principal quantum numbers of the two states. Assume the Bohr model to be valid. The time period of the electron in the initial state is eight times that in the final state. Then the ratio n_1:n_2 is

(A) 1:2 (B) 2:1 (C) 4:1 (D) 1:4

910. A nitrogen nucleus $^{14}_{7}N$ absorbs a neutron and can transform into lithium nucleus $^{7}_{3}Li$ under suitable conditions after emitting:

(A) 4 protons and 3 neutrons (B) 5 protons and 1 beta particle

(C) 2 alpha particles and 2 gamma photons (D) 2 alpha particles, 4 protons and 2 beta particles.

911. White light is used to illuminate the two slits in a young's double slit experiment. The separation between the slits is b and the screen is at a distance d ($\gg$ b) from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are:

(A) $\lambda = \frac{4b^2}{d}$ (B) $\lambda = \frac{2b^2}{d}$ (C) $\lambda = \frac{b^2}{3d}$ (D) $\lambda = \frac{b^2}{4d}$

912. The minimum separation between a real object and its real image in a convex lens of focal length f is

(A) 4 f (B) 3 f (C) f (D) 2 f

913. When a real image is formed for a real object in a convex lens, then the intensity is I. When the upper half of lens is covered by opaque material then the intensity is I_1 and when the lower half is covered and upper is uncovered then intensity is I_2. Then

(A) $I = I_1 = I_2$ (B) $I > I_1 > I_2$ (C) $I < I_1 < I_2$ (D) $I > I_1 = I_2$

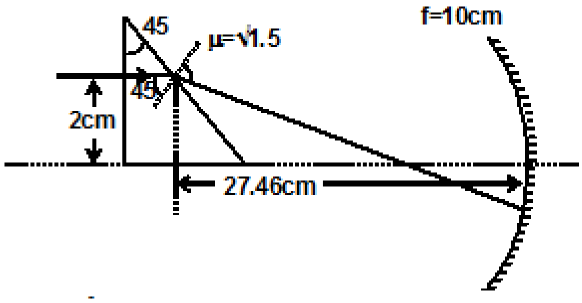
914. When a metallic surface is illuminated with monochromatic light of wavelength l, the stopping potential is $5V_0$. When the same surface is illuminated with light of wavelength 3l, the stopping potential is V_0. The work function of the metallic surface is :

(A) $\frac{hc}{6\lambda}$ (B) $\frac{hc}{5\lambda}$ (C) $\frac{hc}{4\lambda}$ (D) $\frac{2hc}{4\lambda}$

915. An object moves along the axis of a concave mirror of focal length f. Position of object varies as $x = f/2 \sin \omega t$ from a point on x axis at a distance of $\frac{3f}{2}$. Time after which position of image will coincide with position of object is

(A) $\frac{\pi}{2\omega}$ (B) $\frac{2\pi}{\omega}$ (C) $\frac{2\pi}{3\omega}$ (D) $\frac{3\pi}{2\omega}$

916. A ray of light falls normally on a right angled prism whose $\mu = \sqrt{1.5}$ and prism angle is 45^0. After passing through prism it falls on a concave mirror which is at a distance of 27.46 cm from point at of emergence of prism. What is the net deviation if focal length of mirror is 10 cm. ($\sqrt{3} = 1.73$)



(A) 15^0 (B) 60^0 (C) 90^0 (D) 180^0

SECTION : (B) - More Than One Correct Options

917. Which of the following statements is/are correct about the refraction of light from a plane surface when light ray is incident in denser medium. [C is critical angle]

(A) The maximum angle of deviation during refraction is $\frac{\pi}{2}$ – C, when angle of incidence is C.

(B) The maximum angle of deviation for all angle of incidences is π – 2C, when angle of incidence is slightly greater than C.

(C) If angle of incidence is less than C then deviation increases if angle of incidence is also increased.

(D) If angle of incidence is greater than C then angle of deviation decreases if angle of incidence is increased.

918. According to Bohr's atomic model

(A) Orbital speed of electron is independent of electronic mass

(B) The angular momentum of electron in stationary orbit is an integral multiple of $\frac{h}{2\pi}$ (Planck constant)

(C) The magnetic field at the centre is proportional to $\frac{1}{n^4}$ (where n is principal quantum number)

(D) The magnetic field at the centre is proportional to $\frac{1}{n^5}$ (where n is principal quantum number)

919. In which of the following situations, out of deutron and α-particle, the α-particle has smaller de-Broglie wavelength? The two particles are

(A) Move with the same speed
(B) Move with the same linear momentum
(C) Move with the same kinetic energy
(D) Accelerated through same potential

920. Choose the correct statement(s) in case of photoelectric effect

(A) The number of emitted photoelectrons is proportional to the intensity of incident light

(B) Photoelectric current is proportional to frequency of light

(C) Photoelectric emission will take place or not, depends on the intensity of light

(D) The velocity of emmitted photoelectrons does not depends on intensity of light

921. In Young's double slit experiment, slits are arranged in such a way that besides central bright fringe there are only two bright fringes on either side of it. The slit separation d for given condition cannot be (λ is wavelength of light used for experiment)

(A) $\frac{\lambda}{4}$
(B) $\frac{\lambda}{8}$
(C) $\frac{3\lambda}{2}$
(D) $\frac{\lambda}{2}$

922. In Bohr's model of the hydrogen atom, let R, V, T and E represent the radius of the orbit, speed of the electron, time period of revolution of the electron and the total energy of the electron respectively. The quantities proportional the quantum number n are

(a) VR
(b) RE
(c) V/ E
(d) T/R

923. An image of a bright square is obtained on a screen with the aid of a convergent lens. The distance between the square and the lens is 40 cm. The area of the image is nine time larger than that of the square. Select the correct statement(s) :

(a) Image is formed at a distance 120 cm from lens.

(b) Image is formed at a distance 360 cm from lens.

(c) Focal length of lens is 30 cm.

(d) Focal length of lens is 36 cm.

924. The following four statements are regarding well defined observable interference pattern. Select the correct statement(s) :

(a) Condition for sustained interference : The two waves should have same frequency.

(b) Condition for observation : The separation between light sources should be as small as possible.

(c) Condition for good contrast : The amplitudes of interfering waves should be equal or at least very nearly equal.

(d) Condition for good contrast : Slit width must be very narrow (considerably less than the fringe width).

925. When an electron jumps from n = 6 to n =2 in hydrogen like atom:

(a) total ten emission lines will be obtained
(b) angular momentum will increase by three times
(c) kinetic energy will increased nine times
(d) Potential energy will increase three times

926. A neutron collides head on with a stationary hydrogen atom in ground state. Which of the following statement is/are correct ? (Assume that mass of hydrogen atom is nearly equal to mass of neutron and energy of ground state of hydrogen atom is, –13.6 eV).

(a) If kinetic energy of the neutron is less than 13.6 eV, collision must be elastic
(b) If kinetic energy of the neutron is greater than 20.4 eV, collision may be inelastic
(c) Inelastic collision may take place only when kinetic energy of the neutron is 10.2 eV
(d) For perfectly inelastic collision minimum kinetic energy of the neutron should be 10.2 eV.

927. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the x-direction. Assume that all photons emitted due to electron transititons inside the gas emerge in the y-direction. Let A and B denote the lights emerging from the gas in the x- and y-directions respectively.

(a) some of the incident wavelengths will be absent in A.
(b) Only those wavelength will be present in B which are absent in A.
(c) B will contain some visible light.
(d) B will contain some infrared light.

928. At a point on the screen in YDSE experiment 3 rd maxima is observed at t =0. Now screen is slowly moved with constant speed away from the slits is such a way that the centre of slits and centre of screen lie on same line always and at t = 1 sec the intensity at that point is observed (3/4)th of maximum intensity in between 2nd and 3rd maxima. The speed of screen will be (D = separation between the screen and slits d = separation between the slits, d << D, $\lambda = 5000\,\overset{0}{\text{A}}$)

(a) $\frac{5D}{13}$ (b) $\frac{13D}{5}$ (c) $\frac{17D}{5}$ (d) $\frac{D}{17}$

929. In which of the following cases the heavier of the two particle has a smaller de Broglie wavelength ? The two particles

(a) move with the same speed
(b) move with the same linear momentum
(c) move with the same kinetic energy
(d) have the same change of potential energy in a conservative field.

930. In radioactive decay match the following:

Table -1	Table -3
(A) Emission of 2 α and 3 β - particles	(P) No change in atomic number
(B) Emission of 3 α and 2 β -particles	(Q) Atomic number decreases by four
(C) Emission of 2 α and 4 β -Particles	(R) Atomic number decreases by one
(D) Emission of γ -decay	(S) Mass number decreases by eight
	(T) Mass number decreases by twelth

(a) A-(R,S) (b) B-(Q,T) (c) C-(Q) (d) D-(P)

931. The deviation produced by a prism depends on
(a) The angle of incidence of light on its face (b) The refracting angle of prism
(c) The refractive index of the prism (d) The wavelength of the light used.

932. For a certain radioactive substance, it is observed that after 4 hours, only 6.25% of the original sample is left undecayed. If follows that
(a) the half life of the sample is 1 hour
(b) The mean life of the sample is 1/ln2 hour
(c) The decay constant of the sample is ln 2 hour^{-1}
(d) After a further 4 hours, the amount of the substance left over would by only 0.39% of the original amount.

933. When a wave is reflected from a mirror, there is no change in its
(a) amplitude (b) frequency (c) wavelength (d) velocity

934. The focal length of a lens depends on
(a) The object and the image distance (b) The refractive index of the material
(c) The radius of curvature of the surfaces (d) The wavelength of light used

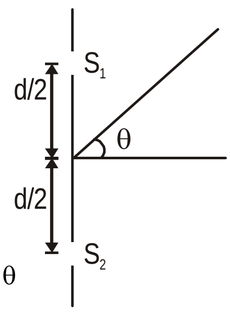
935. An electron in H atom first jumps from second excited state to first excited state and then from first excited state to ground state. Let the ratio of wavelength, momentum and energy of photons emitted in these two cases a, b and c respectively. Then:

(a) $a = \frac{9}{4}$ (b) $b = \frac{5}{27}$ (c) $c = \frac{5}{27}$ (d) $c = \frac{1}{a}$

936. A small mirror is suspended by a thread. A short pulse of monochromatic light rays is incident normally on the mirror and gets reflected. Which of the following statements is/are correct?
(a) Mirror will begin to oscillate
(b) Wavelength of reflected rays will be greater than that of incident rays.
(c) Wavelength of reflected rays may be less than that of incident rays.
(d) There will be no effect of short pulse on mirror.

937. An electron with kinetic energy = EeV collides with a hydrogen atom in the ground state. The collision will be elastic for
(a) no value of E (b) for E < 10.2 eV (c) for E < 13.6 eV (d) only for E < 3.4 eV

938. In an interference arrangement similar to Young's double-slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance d = 150.0 m. The intensity I (θ) is measured as a function of θ, where θ is defined as shown. If I_0 is the maximum intensity, then I (θ) for $0 \leq \theta \leq 90°$ is given by



(a) I (θ) = I_0/ 2 for θ = 30° (b) I (θ) = I_0 / 4 for θ = 0°
(c) I (θ) = I_0 for θ = 0° (d) I (θ) is constant for all values of θ

SECTION : (C) -Passage Type Questions

PASSAGE 01:

While discussing Bohr's atomic model, we consider the nucleus to be at rest. This is true when the mass of the nucleus is very large compared to mass of an electron.

When the nucleus is of finite mass, both the nucleus and electron are in circular motion about their centre of mass. If M and m be the mass of nucleus and electron then force of interaction between nucleus and electron = $F = M\omega^2 r_1 = m\omega^2 r_2$

Or $$F\left(\frac{M+m}{Mm}\right) = \omega^2 (r_1 + r_2)$$

Or $$F = \frac{Mm}{M+m}\omega^2 (r_1 + r_2) = \mu\omega^2 r$$

where μ is the reduced mass and $r = r_1 + r_2$. Therefore, the nucleus of finite mass may be supposed to be infinite mass provided the electron mass is replaced by its reduced mass.

939. In the finite mass model

(A) Angular momentum of electron and nucleus is same

(B) Linear momentum of electron and nucleus is same

(C) Energy and momentum of electron and nucleus is same

(D) Time period and momentum of electron and nucleus is different

940. If E_1 is the energy of 'electron + nucleus' in Bohr's model when nucleus is stationary and E_2 is the energy when nucleus is moving then E_1/E_2 is

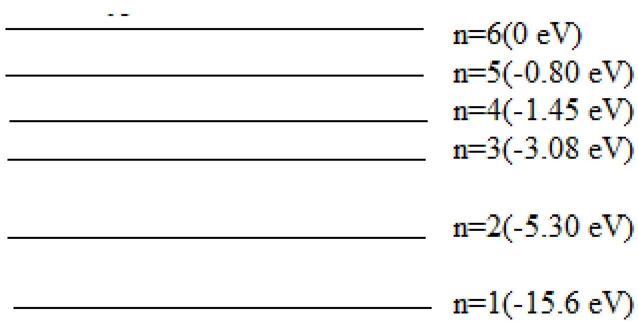
(A) $\frac{M+m}{M}$ (B) $\frac{M}{m}$ (C) $\frac{m}{M}$ (D) $\frac{M-m}{M}$

941. In previous if R_1 and R_2 are the Rydberg's constant then R_1/R_2 is

(A) $\frac{M+m}{M}$ (B) $\frac{M}{m}$ (C) $\left(\frac{M+m}{M}\right)^2$ (D) $\frac{M-m}{M}$

PASSAGE 02:

The energy levels of a hypothetical one electron atom are shown in figure



942. Find the ionization potential of this atom

(a) 15.6 eV (b) 13.5eV (c) 12.2 eV (d) 10.7 eV

943. Find the short wavelength limit of the series terminating at n=2

(a) 2339 $\overset{0}{A}$ (b) 3224 $\overset{0}{A}$ (c) 3452 $\overset{0}{A}$ (d) 7685 $\overset{0}{A}$

944. Find the excitation potential for the state n=3

(a) 10.2 V (b)12.52 V (c) 13.7 V (d) 15.3 V

945. Find the wave number of the photon emitted for the transition n=3 to n=1

(a)1.009 $\times$ 10^{-7} m^{-1} (b) 2.409 $\times$ 10^{-7} m^{-1} (c) 3.098 $\times$ 10^{-7} m^{-1} (d)5.009 $\times$ 10^{-7} m^{-1}

946. What is the minimum energy that an electron will have after interacting with this atom in the ground state if the initial kinetic energy of the electron is 6 eV

(a) 2eV (b) 8eV (c) 4eV (d) 6eV

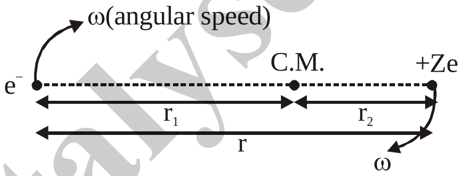
947. What is the minimum energy that an electron will have after interacting with this atom in the ground state if the initial kinetic energy of the electron is 11 eV

(a)0.2eV (b) 0.4eV (c) 0.7eV (d) 0.5eV

PASSAGE 03:

A hydrogen–like ion is formed when an atom loses all but one of its electrons so that it consists of a single electron moving around a nucleus of charge +Ze. Bohr's model of atom, though it is far from being adequate, is still quite useful and successful in giving an account of hydrogen atom and hydrogen–like ions (uni–electron system). Using Bohr's theory, energy of electron moving in the n^{th} orbit of a orbit of a hydrogen like ion having a nuclear charge +Ze can be expressed as

$$E_n = -\frac{mZ^2e^4}{3\varepsilon_0^2h^2n^2}$$



(m, e – mass and charge of electron; $_0$ – absolute permittivity of space; h – Planck's constant)

This expression is valid if the nucleus is assumed to be at rest. In deriving the energy expression given above, we use an approximation that nucleus being extremely massive than the electron, the center of mass of the system will be extremely close to the nucleus, so that, motion of electron can then be visualized as taking place around the nucleus. However, it is more appropriate to visualize the motion of both – the electron and the nucleus. In fact, the electron and the nucleus, both orbit around their center of mass under the mutual electric force, as shown in the figure. If we consider the motion of nucleus also, the energy given by the expression above, has to be modified and thus needs some correction. With m as the mass of electron and M that of nucleus, consider the situation shown in the figure and answer the following questions

948. Distance of electron from the centre of mass, r_1 is

(a) $\frac{r}{2}$ (b) $\frac{m}{M+m}r$ (c) $\frac{M}{M+m}r$ (d) $\frac{mM}{m+M}$

949. Kinetic energy of the nucleus will be

(a) $\frac{Mr^2\omega^2}{2}$ (b) $\frac{Mm}{2(M+m)}r^2\omega^2$ (c) $\frac{M^2m}{2(M+m)^2}r^2\omega^2$ (d) $\frac{Mm^2}{2(M+m)^2}r^2\omega^2$

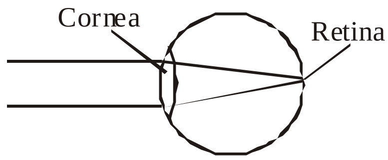
950. Total kinetic energy of the system (nucleus + electron) is

(a) $\frac{Mm}{2(M+m)}r^2\omega^2$ (b) $\frac{(M+m)r^2\omega^2}{2}$ (c) $\frac{Mm^2}{2(M+m)^2}r^2\omega^2$ (d) $\frac{M^2m}{2(M+m)^2}r^2\omega^2$

PASSAGE 4:

The figure shows a simplified model of the eye that is based on the assumption that all the refraction of the light entering the eye occurs at the location of the cornea. In this model, the converging lens of the eye (eye–lens), located at the cornea, has a fixed focal length of 2 cm. Parallel rays coming from a very distant object are refracted by this lens to produce a focused image on the retina. The retina, then, transmits electrical impulses to the brain through the optic nerve

Two common defects of vision are myopia and hyperopia. Myopia, sometimes referred to as nearsightedness, occurs when the lens of the eye focusses the image of a distant object in front of the retina. Hyperopia, sometimes referred to as farsightedness, occurs when the cornea focusses the image of a nearby object behind the retina. Both these problems can be corrected by introducing an artificial lens in front of the eye so that the two lens system produces a focussed image on the retina.



If an object is so far away from the lens system that its distance may be taken as infinite, then the following relationship holds : $\frac{1}{f_e}+\frac{1}{f_1-x}=\frac{1}{v}$, where f_e is the focal length of the eye–lens, f_1 is the focal of the correcting artificial lens; x is the distance from the correcting lens to the cornea, and v is the image distance measured from the cornea. (Note : The index of refraction is 1.0 for air and 1.5 for glass which is used for the correcting lens)

951. How far away should the retina be from the cornea (eye–lens) for normal vision ?

(a) 0.5 cm (b) 1.0 cm (c) 2.0 cm (d) 4.0 cm

952. For a distant object, the image produced by the cornea is :

(a) real and inverted (b)real and upright (c) virtual and inverted (d)virtual and upright

953. What kind of lens would be suitable to correct myopia and hypermetropia respectively? (Note: Assume that the correctiong lens is at the focal point of the cornea so that $x = f_c$).

(a) Converging, Converging (b) Converging, Diverging

(c) Diverging, Diverging (d) Diverging, Converging.

PASSAGE 5:

A point object is placed on the optic axis of an equiconvex glass lens of focal length 7.5 cm. The distance of object from the lens is 20 cm as shown in figure.

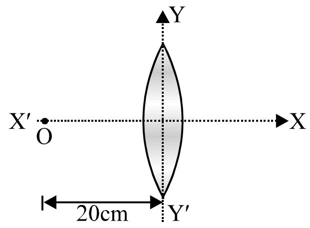


Fig. I

954. The lens is split into two halves along the plane YY′ and right half is shifted towards right by 30 cm keeping the left half fixed at its position as shown in figure II. What is the shift in the location of the final image?

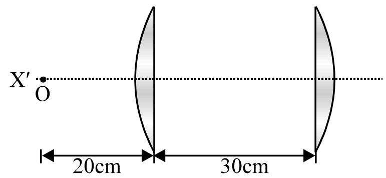


Fig. II

(a) 28 cm (b) 48 cm (c) 62 cm (d) 10cm

955. The right half lens is further displaced perpendicular to line XX′ by 6 mm as shown in figure III. What is further shift of image from its earlier position (position in figure II)?

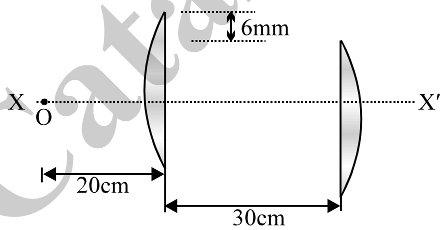


Fig. III

(a) 6 mm (b) 4 mm (c) 3 mm (d) 2mm

956. The two halves are grouped together as shown in figure II and the space between the lenses is filled with water as shown in figure IV. What is shift in position of image from its position of figure (II)
(μ_g = 3/2, μ_w = 4/3) ?

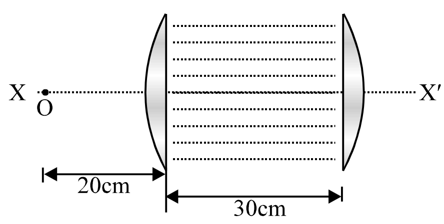
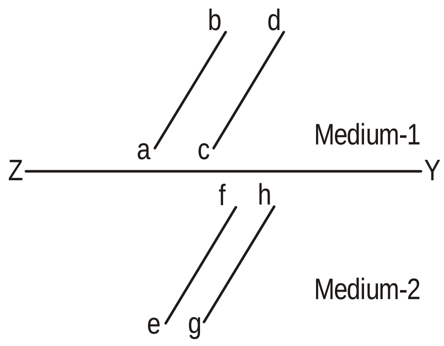


Fig. IV

(a) 7.5 cm (b) 10.7 cm (c) 0.7 cm (d) 17.5 cm

PASSAGE 06:

The figure shows a surface XY separating two transparent media. medium-1 and medium-2. The lines *ab* and *cd* represent wavefronts of a light wave travelling in medium-1 and incident on XY. The lines *ef* and *gh* represent wavefronts of the light wave in medium-2 after refraction.



957. Light travels as a

(a) parallel beam in each medium

(b) convergent bem in each medium

(c) divergent beam in each medium

(d) divergent beam in one medium and covergent beam in the other medium

958. The phases of the light wave at *c*, *d*, *e* and *f* are ϕ_c , ϕ_d , ϕ_e and ϕ_f respectively. It is given that $\phi_c \neq \phi_f$

(a) ϕ_c cannot be equal to ϕ_d

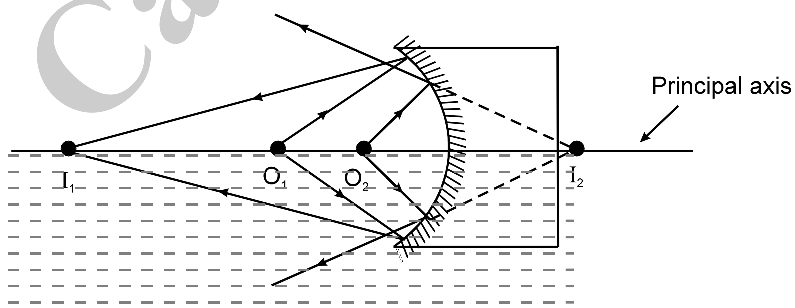
(b) ϕ_d can be equal to ϕ_e

(c) $(\phi_d - \phi_f)$ is equal to $(\phi_c - \phi_e)$

(d) $(\phi_d - \phi_c)$ is not equal to $(\phi_f - \phi_e)$

PASSAGE 07:

A block with a concave mirror of radius of curvature 1m attached to one of its sides floats, with exactly half of its length immersed in water and the other half exposed to air.



Any ray originating from an object O_1 and O_2 (as shown in figure) floating on the surface of water first gets reflected by the mirror. This then gets refracted by the water surface if the image formed by reflection is real (I_1). If the image is virtual (I_2), then the reflected ray never encounters the air-water interface and hence there is no refraction. The image for the next three questions refers to the final image, formed after both the reflection and refraction (if it occurs at all) has taken place.

959. The final image formed is unique (i.e. only one image is formed) if the point object floats on the surface of water at a distance x in front of the mirror where

(A) x is less than 50 cm

(B) x is less than 1m

(C) x is between 50 cm and 1m

(D) for any value of x

960. There is no refraction of light rays reflected from the mirror if the point object floats on the surface of water at a distance x in front of the mirror where

(A) x is less than 50 cm

(B) x is less than 1m

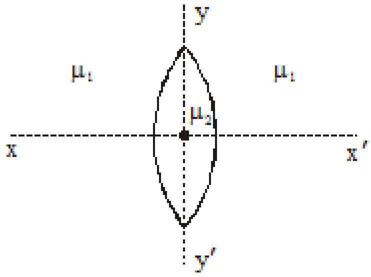
(C) x is between 50 cm and 1m

(D) for any value of x

SECTION : (D) - Matrix Match

961.

A equi convex lens of refractive index μ_2 and focal length f (in air) is kept in medium of refractive index μ_1.



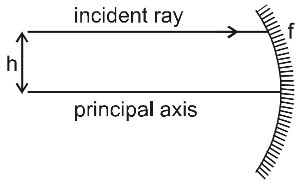
Column A		Column B	
a)	If lens is cut in two equal parts by a plane yy'. ($\mu_1 = 1$)	(p)	Lens will act as a glass slab.
b)	If lens is cut in two equal parts by a plane xx'. ($\mu_1 = 1$)	(q)	Lens will be converging and focal length will change.
c)	If $\mu_1 = \mu_2$.	(r)	Lens will be converging and focal length will remain same.
d)	If $\mu_1 > \mu_2$.	(s)	Lens will be diverging and focal length will change.

962. Select the correct characteristics of the image of a real object formed by a lens of focal length f from the choices given below :

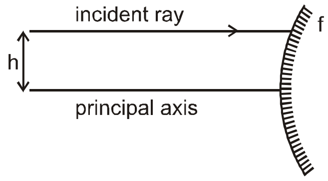
Nature of lens and Position		Characteristics of the image observed :	
(i)	Lens is converging and 2f away from the object	(A)	virtual, erect, diminished
(ii)	Lens is converging and between f and 2f from the object	(B)	virtual, erect, magnified.
(iii)	Lens is diverging and distant f from the object.	(C)	real, inverted, diminished
(iv)	Lens is converging and less than f from the object.	(D)	real, inverted, magnified
		(E)	real, inverted, same size.

963. A ray is parallel to principal axis and at a distance h from principal axis as shown in each situation of column-I. The focal length of mirror or lens in each situation of column-I is f (h < < f). Match each situation in column-I with the ***magnitude of deviation of incident ray produced*** in column-II.

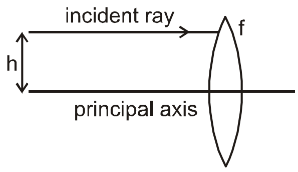
Column-I | **Column-II**

(A) 

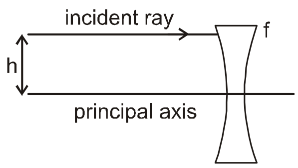
(p) $180^\circ - \dfrac{h}{f}$

(B) 

(q) $180^\circ - \dfrac{2h}{f}$

(C) 

(r) $\dfrac{h}{f}$

(D) 

(s) $\dfrac{2h}{f}$

964. In each situation of column-I, an incident wavefront and its corresponding reflected or refracted wavefront is shown. In column-II the optical instrument used for reflection or refraction is given. Always take the optical instrument to the right of incident wavefront. The incident wavefront is moving towards right. Match each pair of incident and reflected/refracted wavefront in column-I with the correct optical instrument given in column-II.

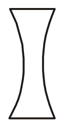
Column-I

Incident wavefront | **Reflected/Refracted Wavefront**

Column-II

Optical instrument used

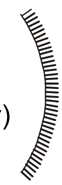
(A) 

(p) 

(B) 

(q) 

(C) 

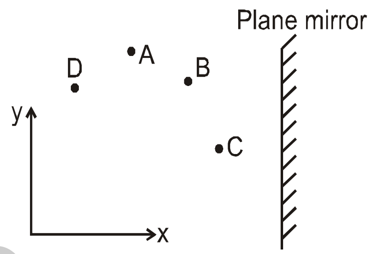
(r) 

(D) 

(s) 

965. Four particles are moving with different velocities in front of stationary plane mirror (lying in y-z plane). At t = 0, velocity of A is $\vec{v}_A = \hat{i}$, velocity of B is $\vec{v}_B = -\hat{i} + 3\hat{j}$, velocity of C is $\vec{v}_C = 5\hat{i} + 6\hat{j}$, velocity of D is $\vec{v}_D = 3\hat{i} - \hat{j}$. Acceleration of particle A is $\vec{a}_A = 2\hat{i} + \hat{j}$ and acceleration of particle C is $\vec{a}_C = 2t\hat{j}$. The particle B and D move with uniform velocity (Assume no collision to take place till t = 2 seconds). All quantities are in S.I. Units. Relative velocity of image of object A with respect to object A is denoted by $\vec{V}_{A',A}$. Velocity of images relative to corresponding objects are given in column I and their values are given in column II at t = 2 second. Match column I with corresponding values in column II and indicate your answer by darkening appropriate bubbles in the 4 × 4 matrix given in OMR.

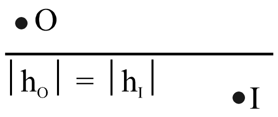
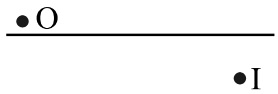
Column I	**Column II**
(A) $\vec{V}_{A',A}$	(p) $2\hat{i}$
(B) $\vec{V}_{B',B}$	(q) $-6\hat{i}$
(C) $\vec{V}_{C',C}$	(r) $-12\hat{i} + 4\hat{j}$
(D) $\vec{V}_{D',D}$	(s) $-10\hat{i}$



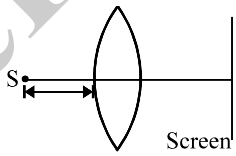
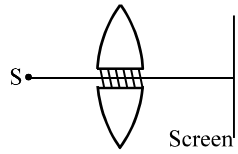
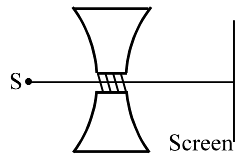
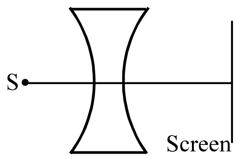
966. Match the column :

	Column-I		**Column-II**
(A)	A transparent piece of glass used for the dispersion of white light	(p)	Angle of prism
(B)	The angle between incident ray and emergent ray, when refraction takes place through a prism.	(q)	Spectrum
(C)	The angle between two refracting surfaces of a prism.	(r)	Dispersion
(D)	A band of colours formed on the screen when white light passes through prism	(s)	Prism
(E)	The phenomenon due to which white light splits into seven colours on passing through the prism.	(t)	Angle of deviation

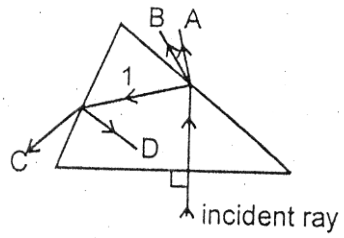
967. In **Column I** are shown four diagrams of real object point O, image point I and principal axis have been sketched. Select the proper optical system from **Column II** which can produce the required image. (Image may be real or virtual)

	Column I		**Column II**
(A)		(P)	Diverging lens
(B)		(Q)	Converging lens
(C)		(R)	Concave mirror
(D)		(S)	Convex mirror

968. Light from source S ($|u| < |f|$) falls on lens and screen is placed on the other side. The lens is formed by cutting it along principal axis into two equal parts and are joined as indicated in column II.

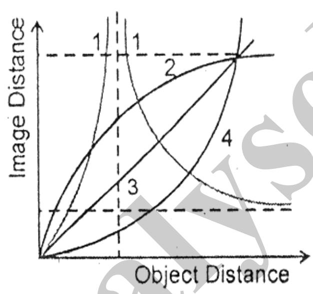
	Column I		**Column II**
(A)	Plane of image move towards screen if $\|f\|$ is increased	(P)	 Small portion of each part near pole is removed. The remaining parts are joined
(B)	Images formed will be virtual	(Q)	 The two parts are separated slightly The gap is filled by opaque material
(C)	Separation between images increase if $\|u\|$ decreases	(R)	 The two parts are separated slightly The gap is filled by opaque material
(D)	Interference pattern can be obtained if screen is suitably positioned	(S)	 Small portion of each part near pole is removed. The remaining parts are joined

969. A white light ray is incident on a glass prism, and it create four refracted rays A,B,C and D. Match the refracted rays with the colours given (1 & D are rays due to total internal reflection)



	Ray		Colour
(A)	A	(P)	Red
(B)	B	(Q)	greeen
(C)	C	(R)	yellow
(D)	D	(S)	blue

970. A small particle is placed at the pole of a concave mirror and then moved along the principal axis to a large distance. During the motion, the distance between the pole of the mirror and the image is measured. The procedure is then repeated with a convex mirror, a concave lens and a convex lens. The graph is plotted between image distance versus object distance. Match the curves shown in the graph with the mirror of lens that is corresponding to it. (Curve 1 has two segments)



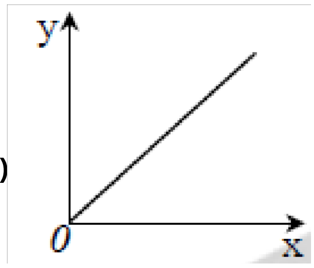
	Lens/Mirror		Curve
(A)	Converging lens	(P)	1
(B)	Converging Mirror	(Q)	2
(C)	Diverging Lens	(R)	3
(D)	Diverging Mirror	(S)	4

971. A nucleus X, initially at rest, undergoes alpha-decay according to the equation. ${}_{92}^{A}X \rightarrow {}_{Z}^{238}Y + \alpha$ .The α particle produced in the above process is found to move in a circular track of radius 0.11m in a uniform magnetic field of 3 Tesla. Given that m(Y) = 228.03u; $m({}_0^1n) = 1.009u$; $m({}_2^4He) = 4.003u$ & $m({}_1^1H) = 1.008u$

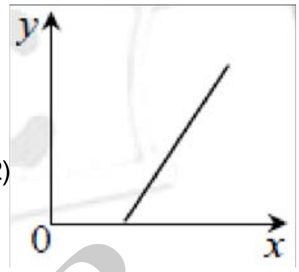
	List-I		**List-II**
A.	value of A	p.	90
B.	value of Z	q.	232
C.	the energy(in MeV) released during the process	r.	1823.2
D.	the binding energy of the parent nucleus(in MeV)	s.	5.3

972. The graphs given apply to a convex lens of focal length *f*, producing a real image at a distance v from the optical centre when self, luminous object is at distance u from the optical centre. The modulus of magnification is *m*. Identify the following graphs (Assume best fit approximation) with the first named quantity being plotted along *y*-axis.

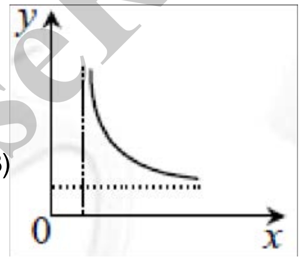
(P) $|v|$ against $|u|$

(1) 

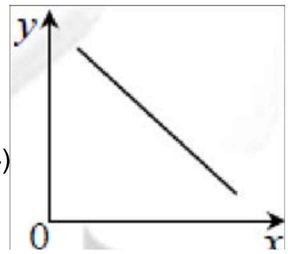
(Q) $\dfrac{1}{|v|}$ against $\dfrac{1}{|u|}$

(2) 

(R) m against $|v|$

(3) 

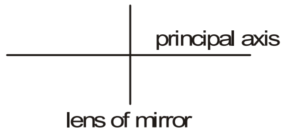
(S) $(m+1)$ against $\dfrac{|v|}{f}$

(4) 

973.

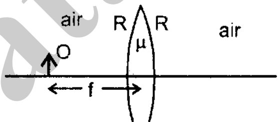
Column-I		**Column-II**	
(A)	In refraction	(p)	Speed of wave does not change
(B)	In reflection	(q)	Wavelength is decreased
(C)	In refraction from rarer to denser medium	(r)	Frequency does not change
(D)	In reflection from a denser medium	(s)	Phase change os π takes place

974. Column - I gives certain situations regarding a point object and its image formed by an optical instrument. The possible optical instruments are concave and convex mirrors or lenses as given in Column - II . Same side of principal axis means both image and object should either be above the principal axis or both should be below the principal axis a shown in figure. Same side of optical instrument means both image and object should be either left of the optical instrument or both should be on right of optical instrument or both should be on right of the optical instrument as shown in figure. Match the statements in column - I with corresponding statements in column - II



	Column I	Column II
(A)	If point object and its image are on same side of principal axis and opposite sides of the optical instrument then the optical instrument is	(p) Concave mirror
(B)	If point object and its image are on opposite side of principal axis and same sides of the side of principal axis and same sides of the optical instrument then the optical instrument is	(q) Convex mirror
(C)	If point object and its image are on same side of principal axis and same sides of the optical instrument then the optical instrument is	(r) Concave lens
(D)	If point object and its image are on opposite side of principal axis and apposite sides of the optical instrument then the optical instrument is	(s) Convex lens

975. An object O (real) is placed at focus of an equi-biconvex lens as shown in figure 1. The refractive index is $\mu = 1.5$ and the radius of curvature of either surface of lens is R. The lens is surrounded by air. In each statement of column - 1 some changes are made to situation given above and information regarding final image of column - 1. Match the statements in column 1 with resulting image in column- II



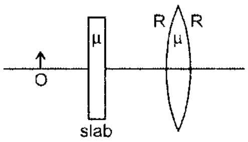
(fig.1)

Column - I

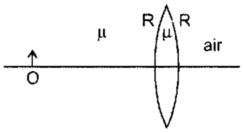
(A) If the refractive index of the lens is doubled (that is, made 2μ) then

(B) If the radius of curvature is doubled (that is, made 2R) then

(C) If a glass slab of refractive index μ = 1.5



(D) If the left side of lens is filled with a medium of refractive index m = 1.5 as shown, then



Column - II

(p) final image is real

(q) final image is virtual

(r) final image becomes smaller in size in comparison to size of image before the change was made

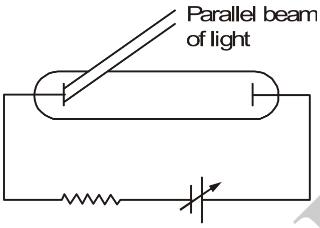
(s) final image is of same size of object.

(t) final image is of large size of object

976 An object O is kept perpendicular to the principal axis of a spherical mirror. Each situation (A, B, C, and D) gives object coordinate u in centimeters with sign the type of mirror, and then the distance (centimeters, without sign) between the focal point and the pole of the mirror, On the right side information regarding the image is given. Correctly match the situations on the left side with the images described on the right side.

Situation	**u**	**Mirror**	**Image**
A	–18	Concave, 12	(P) Real, Errect, Enlarged
B	–12	Concave, 18	(Q) Virtual, Errect, Diminished
C	–8	Convex, 10	(R) Real Inverted, Enlarged
D	–10	Convex, 8	(S) Virtual, Erract, Enlarged

977. In the shown experimental setup to study photoelectric effect, two conducting electrodes arcenclosed in an evacuated glass-tube as shown. A parallel beam of monochromatic light, falls on photosensitive electrodes. The emf of battery shown is high enough such that all photoelectrons ejected from left electrode will reach the right electrode. Under initial conditions photoelectrons are emitted. As changes are made in each situation of column I ; Match the statements in column I with with results in column II.

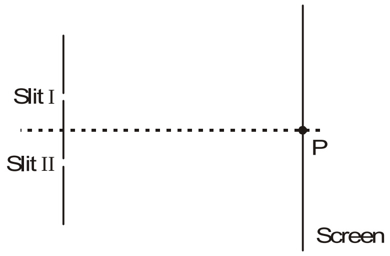


	Column I		**Column II**
(A)	If frequency of incident light is increased keeping number of photons per second constant	(p)	magnitude of stopping potential will increase
(B)	If frequency of incident light is increased and number of photons per second is decreased	(q)	current through circuit may stop
(C)	If work function of photo sensitive electrode is increased	(r)	maximum kinetic energy of ejected photoelectrons will increase
(D)	If number of photons per second of incident light is increased keeping its frequency constant	(s)	saturation current will increase
		(t)	saturation current will decrease

978. In column-I, consider each process just before and just after it occurs. Initial system is isolated from all other bodies. Consider all product particles (even those having rest mass zero) in the system. Match the system in colum in column-I with the result they produce in column-II.

	Column-I		**Column - II**
(A)	Spontaneous radioactive decay of an uranium nucleus initially at rest as given by reaction ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He + ...$	(p)	Number of protons is increased
(B)	Frsion reaction of two hydrogen nuclei as given by reaction ${}^{1}_{1}H + {}^{1}_{1}U \rightarrow {}^{2}_{1}H...$	(q)	Momentum is conserved
(C)	Fission of U^{235} nucleus initiated by a thermal neutron as given by reaction ${}^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n + ...$	(r)	Mass is converted to energy or vice versa
(D)	β–decay (negative beta decay)	(s)	Charge is conserved

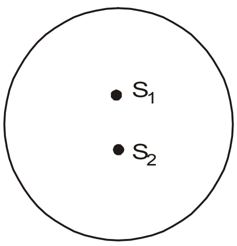
979. A double slit interference pattern is produced on a screen as shown in the figure, using monochromatic light of wavelength 500nm. Point P is the location of the central bright fringe, that is produced when light waves arrive in phase without any path difference. A choice of three strips A, B and C of transparent materials with different thicknesses and refractive indices is available, as shown in the table, these are placed over one or both of the slits, singularly or in conjunction, causing the interference pattern to be shifted across the screen from the original pattern. In the column-I, how the strips have been placed, is mentioned whereas in the column-II, order of the fringe at point P on klthe screen that will be produced due to the placement of the strip (s), is shown. Correctly match both the column.



Film	A	B	C
Thickness (in μm)	5	1.5	0.25
Refractive Index	1.5	2.5	2

Column - I		Column - II	
(A)	Only strip B is placed over slit-I	(p)	First Bright
(B)	Strip A is placed over slit-I and strip C is placed over slit-II	(q)	Fourth Dark
(C)	Strip A is placed over the slit-I and strip B and strip C are placed over the slit-II in conjunction.	(r)	Fifth Dark
(D)	Strip A and strip C are placed over slit-I (in conjuction) and strip B is placed over sli $\frac{\sqrt{3}}{2}$ t-II.	(s)	Central Bright

980. Two coherent point sources of light having wavelength λ are seperated by a distance d. A circle is drawn in space surrounding both the point sources as shown. The plane of circle contains both the point sources. The distance d between both the sources is given in coulmn - I and the total number of corresponding points of maximum intensity and minimum intensity on the periphery of the shown circle are given in column-II. Match each situation of column-I with the results in column-II.



	Column - I	Column-II
(A)	d = 99.4 λ	(p) 398 points of maximum intensity
(B)	d = 99.6 λ	(q) 400 points of maximum intensity
(C)	d = 100 λ	(r) 396 points of maximum intensity
(D)	d = 100.4 λ	(s) 400 points of maximum intensity

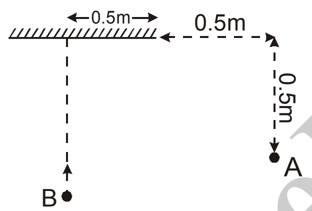
SECTION : (E) - Integer Type

981. A single electron orbits around a stationary nucleus of charge +Ze, where Z is a constant and *e* is the magnitude of the electronic charge. It requires 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find the value of Z

982. In a sample of rock, the ratio of ^{206}Pb to ^{238}U nuclei is found to be 0.5. The age of the rock is

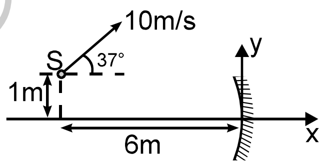
$4.5\times10^{9}\,\dfrac{\ln\frac{X}{Y}}{\ln 2}$ year. The value of X + Y is: (Assume that all the ^{206}Pb in the rock was produced due to the decay of ^{238}U present, when the rock was formed initially and $T_{1/2}(^{238}U) = 4.5\times10^{9}$ year)

983. A man 'A' stands at the position shown in the figure and a second man 'B' approaches the mirror along the line perpendicular to it which passes through its centre. At what distance (in cm) from the mirror will 'B' be at the moment when 'A' and 'B' first see each other in the mirror.

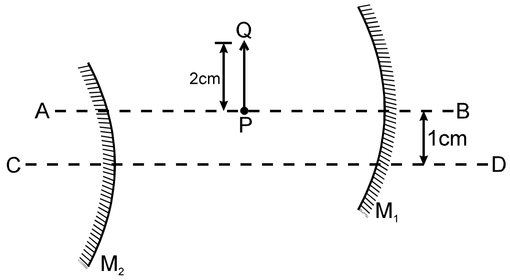


984. A point object is 10 cm away from a plane mirror, while the eye of an observer (pupil diameter 5.0 mm) is 20 cm away. Assuming both the eye and point to be on the same line perpendicular to the mirror, the area of the mirror used in observing the reflection of the point is π/X cm^2. **The value of X is:**

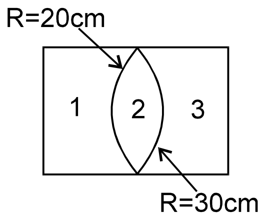
985. A point source S is moving with a speed of 10 m/s in x-y plane as shown in the figure. The radius of curvature of the concave mirror is 4m. The velocity vector of the image formed by paraxial rays is $X\hat{i}+Y\hat{j}$. The value of |X+Y| is:



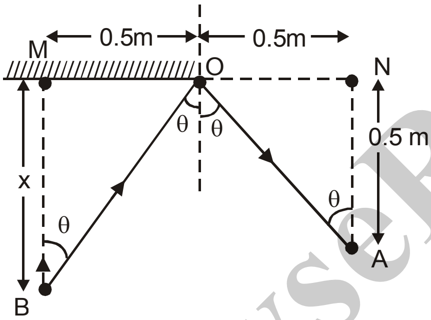
986. In the figure shown M_1 and M_2 are two spherical mirrors of focal length 20 cm each. AB and CD are their principal axes respectively which are separated by 1 cm. PQ is an object of height 2 cm and kept at distance 30 cm from M_1. The separation between the mirrors is 50 cm. Consider two successive reflections first on M_1 then on M_2. Find the size of the 2^{nd} image.



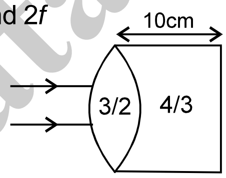
987. $n_1 = n_3 = 3/2$ and $n_2 = 4/3$. The lenses are thin and the whole arrangement is placed in air. The magnitude of focal length (in cm) of the system is __________



988. A neutron is scattered through ($\equiv$ deviation from its old direction) θ degree in an elastic collision with an initially stationary deutron. If the neutron loses $\frac{2}{3}$ of its initial K.E. to the deutron then find the value of θ. (In atomic mass unit, the mass of a neutron is 1u and mass of a deutron is 2u).



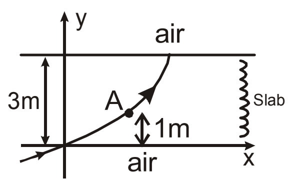
989. A parallel paraxial beam of light is incident on the arrangement as shown. $\mu_A = 3/2$, $\mu_B = 4/3$, the two spherical surfaces are very close and each has a radius of curvature 10 cm. The point where the rays are focussed. is at a distance "f" cm from point of entry. Find $2f$



990. Two plane mirrors are placed with reflecting surfaces parallel and facing to each other. A point object is placed between them at a distance 5cm from first mirror and 3 cm from second mirror. Find the distance (in cm) between the 3rd image behind first mirror and the 3rd image behind the second mirror.

991. An interference pattern is obtained by using a Fresnel's biprism. If the fringe width is 4 mm when air is the surrounding medium, then find the fringe width (in mm) if water is the surrounding medium. Keeping the same source. Assume $n_{glass} = 1.5$, $n_{water} = 4/3$, $n_{air} = 1$.

992. A ray of light travelling in air is incident at grazing angle on a slab of 3m thickness with variable refractive index, $n(y) = [k\,y^{3/2} + 1]^{1/2}$ where $k = 1\ m^{-3/2}$ and follows path as shown. What is the total deviation produced by the slab when the ray is at A (y = 1m), in degrees ?



993. A bomb at rest explodes into three fragments with pieces have masses in the ratio 2 : 3 : 6. It is observed that pieces other than lightest piece fly perpendicular to each other and have same de Broglie wavelengths. If kinetic energy of the heaviest and lightest pieces are k_1 and k_2, what is value of $\frac{k_2}{k_1}$?

994. To generate a power of 3.2 MW, the number of fissions of U^{235} per minute is 6×10^x. The Value of X is:(Energy released per fission = 200 MeV, 1 eV = 1.6×10^{-19} J)

995. If in nuclear reactor using U^{235} as fuel, the power output is 4.8 MW, the number of fissions per second is $1.5K \times 10^Y$(Energy released per fission of U^{235} = 200 MeV watts, e eV = 1.6×10^{-19} J). Value of K + Y:

996. Half-life of a substance is 20 minutes. What is the time(in minutes) between 33% decay and 67% decay?

997. In an experiment to determine e/m using Thomson's method, electrons from the cathode accelerate through a potential difference of 1.5 kV. The beam coming out of the anode enters crossed electric and magnetic field of strengths 2×10^4 V/m and 8.6×10^{-4} T respectively. The value of $\frac{e}{m}$ of electron will be $Y \times 10^{10}$ C/kg. The value of Y is:

998. Power of a convex lens is + 5D (m_g= 1.5). When this lens is immersed in a liquid of refractive index m, it acts like a divergent lens of focal length 100 cm. Then refractive index of the liquid will be 5/X. The value of X is:

999. Eye piece of an astronomical telescope has focal length of 5 cm. If angular magnification in normal adjustment is 10, then distance (in cm) between eye piece and objective should be

1000. An achromatic convergent doublet has power of +2D. If power of convex lens is + 5D, then ratio of the dispersive powers of a convergent and divergent lenses will be X/Y. The value of X + Y is:

END OF MODULE

JEE ADVANCED REVISION PACKAGE - AnswerKey (PHYSICS)

Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
1	B	51	AC	101	A-(iv),B-(ii),C-(i),D-(iii)	151	4
2	A	52	AD	102	A-(iv),B-(ii),C-(i),D-(ii,iii,iv)	152	B
3	C	53	BCD	103	A-(iii),B-(ii),C-(i),D-(iv)	153	B
4	A	54	ACD	104	A-(iii,iv),B-(iii,iv),C-(ii),D-(ii)	154	D
5	A	55	AC	105	A-(ii),B-(i),C-(i),D-(ii)	155	D
6	B	56	BC	106	A-(i),B-(i),C-(iii),D-(iv)	156	B
7	C	57	AB	107	A-(ii),B-(ii),C-(iv),D-(iii)	157	B
8	C	58	BCD	108	A-(i,ii,iii,iv),B-(i,ii,iii,iv),C-(ii),(i,ii,iii,iv)	158	A
9	C	59	D	109	A-(ii,iv),B-(i,iv),C-(ii,iv),D-(iii,iv)	159	A
10	B	60	A	110	A-(ii),B-(iii),C-(i),D-(iv)	160	B
11	B	61	D	111	A-(iii,iv),B-(i),C-(iii,iv),D-(i,ii,iii,iv)	161	A
12	D	62	A	112	A-(iv),B-(iii),C-(i),D-(ii)	162	A
13	B	63	A	113	A-(iii,iv),B-(I,ii,iv),C-(iii,iv),D-(iii,iv)	163	A
14	C	64	C	114	A-(iv),B-(iii),C-(ii),D-(i)	164	A
15	A	65	A	115	A-(Q),B-(S),C-(R),D-(P)	165	A
16	A	66	A	116	A-(Q),B-(R),C-(S),D-(P)	166	B
17	D	67	D	117	A-(Q),B-(P),C-(S),D-(R)	167	B
18	A	68	C	118	A-(R),B-(S),C-(P),D-(Q)	168	D
19	C	69	D	119	A-(P,Q,R),B-(S),C-(R),D-(S)	169	B
20	B	70	A	120	A-(Q),B-(P),C-(S),D-(R)	170	B
21	A	71	D	121	A-(Q),B-(R),C-(S),D-(P)	171	A
22	A	72	C	122	2	172	B
23	D	73	A	123	20.0ms-1	173	C
24	B	74	A	124	3kgm2	174	B
25	D	75	C	125	1	175	C
26	AD	76	C	126	2	176	C
27	BCD	77	B	127	4	177	ABCD
28	AD	78	C	128	37	178	AC
29	AC	79	B	129	24	179	CD
30	ABD	80	C	130	25	180	ABC
31	AD	81	B	131	20	181	BD
32	AB	82	B	132	5	182	AD
33	ABD	83	A	133	10	183	BC
34	ABC	84	B	134	4	184	BC
35	AD	85	C	135	2	185	AB
36	AC	86	B	136	9	186	BC
37	BCD	87	A	137	600	187	ABC
38	BCD	88	C	138	5	188	B
39	AC	89	A	139	1	189	B
40	AB	90	C	140	2	190	AC
41	ABCD	91	C	141	5	191	ABD
42	AB	92	A-(i),B-(I,ii,iii),C-(I,ii,iii,iv),D-(I,ii,iii,iv)	142	26	192	BCDE
43	ABC	93	A-(ii),B-(iv),C-(iii,iv),D-(I,iv)	143	2007	193	CD
44	ABC	94	A-(ii),B-(i),C-(i),D-(iv)	144	200	194	ABC
45	BCD	95	A-(iv),B-(i),C-(ii),D-(iii)	145	20	195	CD
46	C	96	A-(ii),B-(i),C-(iv),D-(iii)	146	55	196	BCD
47	BCD	97	A-(iv),B-(i),C-(ii),D-(ii,iii)	147	3125	197	ABCD
48	AB	98	A-(iv),B-(I,iii),C-(iii),D-(i)	148	5	198	BC
49	CD	99	A-(iv,v),B-(iv),C-(v),D-(iv,v)	149	2	199	BD
50	CD	100	A-(1,2,3,4),B-(4),C-(4),D-(3)	150	2	200	BCD

AnswerKey							
Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
201	ACD	251	A-(R),B-(Q),C-(S),D-(P)	301	A	351	A
202	ABD	252	A-(R,S),B-(Q),C-(P),D-(Q)	302	A	352	D
203	ABC	253	A-(r),B-(p),C-(s),D-(qs)	303	B	353	B
204	ABC	254	A-(q),B-(q),C-(r),D-(s)	304	A	354	A
205	AB	255	A-(r),B-(q),C-(s),D-(p)	305	C	355	A
206	AC	256	A-(p),B-(p),C-(q),D-(rs)	306	A	356	D
207	ABD	257	A-(P),B-(T),C-(S),D-(R)	307	A	357	C
208	AD	258	A-(p,q),B-(r,s),C-(p,q,s),D-(p,q,s)	308	B	358	A
209	D	259	A-(r),B-(p),C-(q),D-(qs)	309	D	359	B
210	B	260	A-(S),B-(R),C-(R),D-(R)	310	A	360	A
211	A	261	A-(r),B-(p),C-(q),D-(q)	311	B	361	B
212	B	262	A-(p,q),B-(p,q),C-(q,r),D-(q,r)	312	C	362	A-(i),(ii),(iii),(iv),B-(i),(ii),(iii),(iv),C-(i),(ii),D-
213	D	263	A-(r,s),B-(r,s),C-(p,q,s),D-(r,s)	313	D	363	A-(iv),B-(i),C-(i),D-(iii)
214	A	264	A-(p),B-(q),C-(q,s),D-(r)	314	A	364	A-(iv),B-(iv),C-(i),D-(ii)
215	A	265	A-(Q,R,S),B-(P,Q,S),C-(P,Q),D-	315	C	365	A-(iii),B-(iv),C-(ii),D-(i)
216	B	266	A-(p),B-(q,r,s),C-(q,r,s),D-(p)	316	AD	366	A-(iii),B-(iii),C-(iv),D-(ii)
217	B	267	A-(r),B-(p),C-(s),D-(q)	317	AC	367	A-(iv),B-(I,iv),C-(ii,iii,iv)
218	B	268	A-(p),B-(r,s),C-(q,s),D-(r,s)	318	AD	368	A-(iii),B-(i),C-(ii),D-(iv)
219	C	269	A-(Q),B-(Q,R,S),C-(Q,R,S),D-(P)	319	ABD	369	A-(i),(ii),(iii),(iv),B-(i),(ii),(iii),(iv),C-(i),(ii),D-
220	A	270	A-(Q),B-(S),C-(P),D-(R)	320	CD	370	A-(iii),B-(iv),C-(i)
221	A	271	0	321	A	371	A-(I,iv),B-(iv),C-(i,iv),D-(ii,iii)
222	C	272	5	322	ABD	372	A-(i,ii),B-(i,ii),C-(iii),D-(iv)
223	B	273	7690J	323	BD	373	A-(i,ii),B-(i,ii),C-(iii,iv),D-(iv)
224	D	274	465W	324	BD	374	A-(p,q),B-(p,q),C-(r,s),D-
225	A	275	1000N/m	325	BC	375	A-(q,s),B-(p,s),C-(p,q),D-
226	C	276	1860	326	CD	376	A-(s),B-(r),C-(q),D-(p)
227	C	277	50	327	BC	377	A-(s),B-(p),C-(q),D-(r)
228	C	278	2	328	ABC	378	A-(Q),B-(P,R),C-(P,S),D-(S)
229	A	279	9(X-2,Y-11)	329	A	379	A-(p),B-(r),C-(q),D-(s)
230	C	280	6	330	BC	380	A-(ii),B-(i),C-(iii)
231	D	281	9	331	AC	381	A-(R),B-(P),C-(P),D-(R)
232	D	282	2	332	ABC	382	273
233	A	283	6	333	ABC	383	24
234	C	284	5	334	AC	384	5
235	B	285	5	335	AC	385	55
236	A	286	3	336	ABCD	386	10
237	C	287	3	337	CD	387	959
238	C	288	1	338	CD	388	145
239	D	289	4	339	C	389	192
240	D	290	1	340	B	390	10
241	A-(S),B-(P),C-(Q),D-(R)	291	125	341	A	391	8
242	A-(r),B-(p),C-(p),D-(q)	292	1	342	B	392	0
243	A-(R,A),B-(S),C-(P),D-(R)	293	5	343	A	393	4
244	A-(R,S),B-(R,S),C-(P,Q),D-(P,Q)	294	1	344	C	394	2
245	A-(P,R),B-(PS),C-(QS),D-(QR)	295	1	345	A	395	82
246	A-(S),B-(R),C-(P),D-(Q,S)	296	4	346	C	396	488
247	A-(S),B-(R),C-(P),D-(P)	297	5	347	C	397	4
248	A-(S),B-(P),C-(P),D-(QR)	298	2	348	B	398	2
249	A-(R),B-(S),C-(P),D-(Q)	299	25	349	B	399	5
250	A-(Q),B-(S),C-(R),D-(P)	300	4	350	D	400	12

AnswerKey							
Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
401	14	451	A	501	C	551	ACD
402	A	452	A	502	A	552	AD
403	C	453	C	503	A	553	BD
404	A	454	C	504	B	554	ABD
405	B	455	A	505	A	555	BCD
406	C	456	B	506	B	556	BC
407	A	457	C	507	C	557	AC
408	B	458	B	508	A	558	BD
409	B	459	B	509	B	559	A
410	D	460	D	510	D	560	A
411	A	461	A-(P),B-(Q),C-(S),D-(R)	511	C	561	B
412	B	462	A-(PS),B-(PRS),C-(CQ),D-(QR)	512	B	562	A
413	B	463	A-(3),B-(2),C-(3),D-(4)	513	B	563	B
414	A	464	A-(r),B-(q),C-(qs),D-(t)	514	B	564	A
415	A	465	A-(p,q),B-(s),C-(p,r),D-(s)	515	C	565	A
416	C	466	A-(r),B-(p),C-(s),D-(q)	516	A	566	D
417	ACD	467	A-(p,q,s),B-(r,s),C-(q,r,s),D-(r,s,t)	517	C	567	C
418	BC	468	A-(p,r,s),B-(q),C-(p,r,s),D-(q,r)	518	D	568	B
419	abcd	469	A-(s),B-(q),C-(r),D-(q)	519	B	569	B
420	ABCD	470	A-(P,S),B-(P,R,S),C-(Q),D-(Q,R)	520	C	570	A
421	ABD	471	A-(R),B-(S),C-(P),D-(Q)	521	A	571	C
422	AD	472	A-(Q),B-(Q),C-(R),D-(S)	522	A	572	C
423	ABCD	473	A-(Q),B-(R),C-(S),D-(P)	523	B	573	B
424	BD	474	A-(R),B-(P),C-(P),D-(R)	524	C	574	D
425	BC	475	A-(iv),B-(iv),C-(i),D-(ii)	525	C	575	C
426	D	476	A-(P),B-(Q),C-(Q),D-(S)	526	AC	576	A
427	AB	477	A-(S),B-(Q),C-(P,Q),D-(Q,R)	527	ACD	577	A
428	ABD	478	A-(Q,S),B-(P,R,S),C-(P,Q,R,S),D-(P,Q,R,S)	528	CD	578	B
429	AD	479	A-(Q),B-(Q),C-(Q),D-(P)	529	AC	579	B
430	ACD	480	A-(S),B-(Q),C-(S),D-(S)	530	ABCD	580	C
431	ABCD	481	75	531	AD	581	A
432	BD	482	12	532	AB	582	C
433	AC	483	508HZ	533	BD	583	B
434	C	484	0	534	ABD	584	C
435	ABD	485	5	535	BD	585	D
436	ABCD	486	2	536	ABCD	586	B
437	AD	487	2	537	ABC	587	A
438	AC	488	4	538	AD	588	A
439	D	489	1	539	BD	589	A
440	B	490	9	540	ABCD	590	A
441	C	491	41	541	AB	591	A
442	B	492	7	542	AD	592	A-(1,2,3),B-(1,2,4),C-(4),D-(3)
443	C	493	2	543	ABC	593	A-(4),B-(3),C-(2),D-(1)
444	A	494	435	544	CD	594	A-(i,ii),B-(iii),C-(I,ii),D-(i)
445	A	495	6	545	BCD	595	A-(ii),B-(i),C-(iii),D-(iv)
446	C	496	15	546	BD	596	A-(i,ii,iii),B-(i,ii,iii),C-(i,ii,iii),D-(iv)
447	D	497	525	547	ABC	597	A-(iii),B-(iii),C-(i),D-(i)
448	D	498	500	548	ABCD	598	A-(i),B-(iv),C-(iii),D-(ii)
449	A	499	20	549	ABC	599	A-(iii,i),B-(ii,iv),C-(ii,v),D-(ii,iv)
450	A	500	20	550	ABD	600	A-(ii),B-(iii),C-(i),D-(ii)

AnswerKey							
Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
601	A-(ii,iii),B-(ii,iii,iv),C-(i,ii),D-(i)	651	58	701	BC	751	A-(P,R,S),B-(P,R,S),C-(R),D-(Q,R)
602	A-(i),B-(iii),C-(iv),D-(ii)	652	B	702	AB	752	A-(Q,S),B-(P,S),C-(R),D-(R)
603	A-(ii),B-(iii),C-(i),D-(iv)	653	C	703	ABC	753	A-(P,S),B-(Q,S),C-(Q,S),D-(S)
604	A-(ii),B-(i),C-(iii),D-(iv)	654	B	704	BCD	754	A-(P,Q),B-(P,Q),C-(P,Q,S),D-
605	A-(ii,iv),B-(iii,i),C-(ii,iv),D-(ii,iii)	655	C	705	AC	755	A-(Q),B-(S),C-(S),D-(Q)
606	A-(ii),B-(i),C-(i),D-(i)	656	C	706	BCD	756	A-(Q),B-(R),C-(R),D-(R)
607	A-(ii,iii),B-(i),C-(iv),D-(ii,iii)	657	B	707	ABC	757	A-(P,R),B-(QS),C-(PR),D-(T)
608	A-(ii),B-(i),C-(iii),D-(iv)	658	D	708	AC	758	A-(P,R,T),B-(Q,S,T),C-(Q,S),D-
609	A-(R),B-(S),C-(P),D-(Q)	659	B	709	A	759	A-(P,R),B-(Q),C-(QS),D-(QT)
610	A-(P),B-(P),C-(R),D-(R)	660	D	710	A	760	A-(P),B-(R),C-(Q),D-(P)
611	A-(p,t),B-(q),C-(p,t),D-(q,t)	661	C	711	A	761	A-(S),B-(P,R,S),C-(Q,S,T),D-(Q,T)
612	A-(Q),B-(S),C-(R),D-(P,R)	662	B	712	B	762	A-(QR),B-(PT),C-(),D-(QT)
613	A-(Q),B-(R),C-(P),D-(S)	663	A	713	B	763	A-(S),B-(Q),C-(Q),D-(R)
614	A-(P,Q),B-(R,S),C-(R,S),D-(P,Q)	664	A	714	C	764	A-(Q,S),B-(P,S),C-(Q,S),D-(P,S)
615	A-(Q),B-(S),C-(R),D-(PR)	665	C	715	C	765	A-(P,Q,R),B-(P,Q,R,S),C-(R),D-
616	A-(q),B-(p),C-(p),D-(q)	666	A	716	B	766	A-(R,S),B-(R,S),C-(Q,R),D-(P,R)
617	A-(R),B-(S),C-(P),D-(Q)	667	D	717	A	767	A-(R),B-(QT),C-(T),D-(S)
618	A-(r),B-(q),C-(q),D-(q)	668	D	718	C	768	A-(R),B-(S),C-(P),D-(Q)
619	A-(Q),B-(S),C-(P),D-(R)	669	B	719	A	769	A-(P,R,T),B-(P,R,T),C-(Q,S),D-
620	A-(S),B-(R),C-(P),D-(Q)	670	A	720	D	770	A-(Q,S),B-(P,R),C-(P,R),D-(Q,S)
621	A-(R),B-(P),C-(S),D-(Q)	671	B	721	D	771	100
622	10mA	672	C	722	A	772	5
623	6	673	C	723	D	773	600
624	6	674	C	724	A	774	22
625	4	675	D	725	B	775	4
626	3	676	C	726	D	776	18
627	17V	677	AD	727	C	777	9
628	1	678	ABC	728	D	778	1250
629	4	679	BD	729	C	779	15
630	3	680	ABCD	730	A	780	10
631	55	681	ABC	731	B	781	25
632	5	682	ACD	732	B	782	5
633	3	683	BC	733	C	783	6
634	4	684	AC	734	C	784	6
635	40	685	ABD	735	D	785	1500
636	20	686	ABC	736	B	786	1200
637	11	687	CD	737	C	787	1350
638	90	688	BCD	738	B	788	50
639	10	689	AC	739	A	789	40
640	2	690	ACD	740	B	790	31
641	13	691	AC	741	A-(Q,R),B-(P,R),C-(P,R),D-(Q,S)	791	22
642	12	692	AC	742	A-(S),B-(Q),C-(R),D-(Q)	792	250
643	240mN/C	693	ABC	743	A-(P,Q),B-(Q,S),C-(P,R),D-(P,R)	793	6
644	3025	694	BC	744	A-(R),B-(R),C-(S),D-(Q)	794	144
645	11	695	AB	745	A-ps,B-qr,C-q,D-p	795	52
646	52	696	ABC	746	A-(P,S),B-(Q,S),C-(Q,S),D-(S)	796	2
647	6	697	AC	747	A-(P,R),B-(Q,S),C-(Q,S),D-(P,R)	797	4
648	34	698	ABC	748	A-(Q,S),B-(P,S),C-(Q,S),D-(P,S)	798	234
649	5	699	ACD	749	A-(P,Q,S),B-(P,Q,R,S),C-(P,Q,S),D-(P,Q,R,S)	799	0
650	36	700	BD	750	A-(3,1),B-(2,4),C-(2,5),D-(2,4)	800	600

AnswerKey							
Qs.	Ans.	Qs.	Ans.	Qs.	Ans.	Qs.	Ans.
801	B	851	B	901	1	951	C
802	C	852	D	902	A	952	A
803	B	853	C	903	D	953	D
804	D	854	A	904	A	954	A
805	A	855	B	905	C	955	B
806	B	856	C	906	D	956	C
807	B	857	C	907	A	957	A
808	B	858	B	908	C	958	C
809	C	859	A	909	B	959	D
810	A	860	C	910	C	960	A
811	C	861	A	911	C	961	A-(Q),B-(R),C-(P),D-(S)
812	A	862	A-(3),B-(1,3),C-(3,4),D-(2,3,4)	912	A	962	i-E,ii-D,iii-A,iv-B
813	D	863	A-(3),B-(1),C-(4),D-(2)	913	D	963	A-(P),B-(P),C-(R),D-(R)
814	C	864	A-(3),B-(1),C-(2),D-(3)	914	A	964	A-(PR),B-(QS),C-(QR),D-(PS)
815	B	865	A-(1),B-(3),C-(2),D-(1,3)	915	D	965	A-(S),B-(P),C-(S),D-(Q)
816	AB	866	A-(2),B-(1,2,3),C-(1,3),D-(3,4)	916	D	966	A-s,B-t,C-p,D-q,E-r
817	BCD	867	A-(2),B-(3),C-(1),D-(1)	917	ABCD	967	A-(Q),B-(R),C-(P),D-(S)
818	BCD	868	A-(2,4),B-(3,4),C-(1),D-(2,4)	918	ABD	968	A-(PQ),B-(PQRS),C-(RS),D-(P)
819	BC	869	A-(1,2,3),B-(1,3,4),C-(2),D-(1,3)	919	ACD	969	A-(P),B-(R),C-(Q),D-(S)
820	B	870	A-(Q),B-(R),C-(R),D-(S)	920	AD	970	A-(P),B-(P),C-(Q),D-(Q)
821	AD	871	A-(Q),B-(Q),C-(R),D-(S)	921	ABCD	971	A-(Q),B-(P),C-(S),D-(R)
822	BC	872	A-(R),B-(P,T),C-(S),D-(Q)	922	ACD	972	A-(3),B-(4),C-(2),D-(1)
823	BD	873	A-(P,Q,R,S),B-(Q,R),C-(Q,R),D-(P,Q,R,S)	923	AC	973	A-(R),B-([R),C-(QR),D-(PRS)
824	B	874	A-(Q,R,S),B-(R),C-(P),D-(T)	924	ABCD	974	A-(PQ),B-(PQ),C-(RS),D-(RS)
825	C	875	A-(R,S),B-(P,Q),C-(P),D-(Q)	925	AC	975	A-(PR),B-(QR),C-(QR),D-(QR)
826	AC	876	A-(Q,S),B-(P),C-(Q,S),D-(R)	926	AB	976	A-(R),B-(S),C-(Q),D-(Q)
827	ABD	877	A-(Q),B-(P),C-(R),D-(S)	927	ACD	977	A-(PR),B-(PRT),C-(Q),D-(S)
828	ABC	878	A-(S),B-(P),C-(Q),D-(R)	928	AD	978	A-(PRS),B-(QRS),C-(QRS),D-(PQRS)
829	ABCD	879	A-(Q,S),B-(Q,S),C-(P,Q,S),D-(P,Q,R,S)	929	ACD	979	A-(R),B-(R),C-(S),D-(P)
830	AD	880	A-(Q),B-(P),C-(P),D-(R)	930	ABD	980	A-(PR),B-(PS),C-(QS),D-(S)
831	ABCD	881	A-(Q),B-(R),C-(S),D-(PS)	931	ABCD	981	5
832	AB	882	1	932	ABCD	982	5
833	ABD	883	2	933	BCD	983	50
834	AD	884	7	934	BCD	984	144
835	ABCD	885	22	935	BCD	985	6
836	ACD	886	4	936	AB	986	8
837	AB	887	8	937	BC	987	72
838	AC	888	1	938	AC	988	90
839	B	889	3	939	B	989	35
840	B	890	5	940	A	990	48
841	B	891	25	941	A	991	12
842	D	892	6375	942	A	992	45
843	D	893	1760	943	A	993	6
844	B	894	8	944	B	994	18
845	B	895	173	945	A	995	18
846	C	896	12	946	D	996	20
847	D	897	250	947	C	997	18
848	A	898	7	948	C	998	3
849	D	899	5	949	D	999	55
850	B	900	4	950	A	1000	8